RECURRENCE COEFFICIENTS OF DISCRETE MEASURES WITH POWER-LAW ATOMS: POINTWISE ASYMPTOTICS AND THE $1/n$ TERM

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ABSTRACT. Let $\sigma = \sum_{k\ge 1} k^{-s}\ell(k)\,\delta_{k^{-p}}$ with $p > 0$ and $s > 1$, so that its Jacobi matrix is a compact operator with eigenvalues k^{-p}. A previous paper established that the recurrence coefficients satisfy $n^p b_n \to c(p)$ in a double Cesàro sense, with an explicit constant $c(p)$ that comes from a constrained equilibrium problem. Here we prove the pointwise law $n^p b_n = c(p)(1 + O(1/n))$ and $a_n/(2b_n) = 1 + O(1/n)$ for every $p > 0$ and $s > 1$, both for pure powers ($\ell \equiv 1$) and for every factor ℓ that extends to a slowly varying analytic function in a sector $|\arg t| \le \vartheta$, however small $\vartheta > 0$ is. For pure powers with $p > 1$ we also find the $1/n$ term: $n^p b_n = c(p)(1 - (s-1)/(2n) + O(n^{-2}))$ and $n^p a_n = 2c(p)(1 - (s-1+p)/(2n) + O(n^{-2}))$. The proof is a Riemann–Hilbert analysis of discrete orthogonal polynomials whose nodes accumulate at a hard edge. The outer scalar function depends on whether p is larger than, equal to or smaller than 1. At the hard edge a non-analytic slowly varying factor is handled by a comparison, in the manner of Conway and Deift, with an explicitly solvable Bessel model; this needs the Bessel order $(s-1-p)/p$ to lie in $(-1, 1)$. A Christoffel transformation, which shifts s by p and preserves the law together with its rate, removes the restriction.

1. INTRODUCTION

A compact Jacobi operator is determined by its spectral measure. If its eigenvalues are k^{-p} and its spectral weights decay like k^{-s}, how do its matrix entries behave? For $p > 0$, $s > 1$ and a sequence $\ell\colon \mathbb{N} \to (0, \infty)$ we study the discrete measure

$$\sigma_{p,s,\ell} := \sum_{k\ge 1} k^{-s}\ell(k)\,\delta_{k^{-p}}, \qquad \sigma_{p,s} := \sigma_{p,s,1}. \tag{1.1}$$

It lives on $(0, 1]$, and its atoms accumulate only at 0. Let P_n be its monic orthogonal polynomials and $h_n := \int P_n^2\,\mathrm{d}\sigma$. They satisfy the three-term recurrence

$$xP_n = P_{n+1} + a_n P_n + b_n^2 P_{n-1} \qquad (n \ge 0,\ b_0 := 0,\ b_n > 0), \tag{1.2}$$

so that $b_n^2 = h_n/h_{n-1}$; the numbers a_n and b_n are the entries of the Jacobi matrix of σ. With

$$I(p) := \frac{\sqrt{\pi}}{2}\,\frac{\Gamma(\frac{1}{p} + \frac{1}{2})}{\Gamma(\frac{1}{p} + 1)}, \qquad c(p) := \frac{1}{4}\Big(\frac{\pi}{2I(p)}\Big)^p, \tag{1.3}$$

for example $c(1) = \frac{1}{2}$ and $c(2) = \pi^2/16$, the question is whether

$$n^p b_n \longrightarrow c(p) \qquad \text{and} \qquad n^p a_n \longrightarrow 2c(p) \qquad (n \to \infty), \tag{1.4}$$

and at what rate. We call (1.4) the *pointwise law*. The constant $c(p)$ comes from the first paper of this series [15], which proves the first half of (1.4) in an averaged sense (see (1.6) below).

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1.1. **Results.** The slowly varying factors we allow are those that extend analytically to a sector, written in the logarithmic variable $\xi = \log t$.

Definition 1.1 (Analytic slowly varying factors)**.** Let $\vartheta > 0$. A sequence $\ell\colon \mathbb{N} \to (0,\infty)$ is *analytic of angle* ϑ if $\ell(k) = \exp\Lambda(\log k)$ for all $k \geq 1$, where, for some $c_0 > 0$, the function Λ is analytic on $S_\vartheta := \{\xi \in \mathbb{C} : \ \operatorname{Re}\xi \geq -c_0,\ |\operatorname{Im}\xi| \leq \vartheta\}$, real on $S_\vartheta \cap \mathbb{R}$, bounded on $S_\vartheta \cap \{\operatorname{Re}\xi \leq 0\}$, and

$$\varepsilon_\Lambda(T) := \sup\bigl\{|\Lambda'(\xi)| :\ \xi \in S_\vartheta,\ \operatorname{Re}\xi \geq T\bigr\} \longrightarrow 0 \qquad (T \to \infty). \tag{1.5}$$

We then also say that Λ is analytic of angle ϑ. If $\vartheta' < \vartheta$, a sequence that is analytic of angle ϑ is also analytic of angle ϑ'.

Such an ℓ is slowly varying. The angle may be arbitrarily small, because the Riemann–Hilbert analysis evaluates the weight only in a sector whose opening we are free to shrink (Lemma 4.7 and Remark 4.8). For example, $\exp\sqrt{\log(k+e)}$ and $(\log(k+e))^\beta$, $\beta \in \mathbb{R}$, are analytic of every angle $\vartheta < \pi$, while $\Lambda(\xi) = \sqrt{\xi+1} + (\xi+2)^{-1/2}\log(1 - q\cos 2\xi)$ with $q = 1/\cosh(0.6)$ has logarithmic branch points at $\xi = \pi m \pm 0.3i$, $m \in \mathbb{Z}$, and is analytic only of the angles $\vartheta < 0.3$ (Section 10).

Theorem 1.2. *Let $p > 0$ and $s > 1$, and let ℓ be analytic of angle ϑ for some $\vartheta > 0$. Then the recurrence coefficients of $\sigma_{p,s,\ell}$ satisfy, as $n \to \infty$,*

$$n^p b_n = c(p)\bigl(1 + O(1/n)\bigr), \qquad \frac{a_n}{2b_n} = 1 + O(1/n),$$

with constants depending on p, s and ℓ.

The constant sequence $\ell \equiv 1$ is analytic of every angle, with $\Lambda \equiv 0$.

Corollary 1.3 (Pure powers)**.** *For every $p > 0$ and $s > 1$ the measure $\sigma_{p,s}$ satisfies $n^p b_n = c(p)(1 + O(1/n))$ and $a_n/(2b_n) = 1 + O(1/n)$. In particular, for real $q > 0$ and r with $2q + r > 1$, the measure $\sigma_{q,2q+r}$ has the moments $\zeta(qj + 2q + r)$, $j \geq 0$, and Hankel determinants $\det_{1\leq i,j\leq n}\zeta(q(i+j)+r)$ ([15, Corollary 1.4 and its proof]), and its recurrence coefficients satisfy $n^q b_n = c(q)(1 + O(1/n))$ and $a_n/(2b_n) = 1 + O(1/n)$.*

For pure powers with $p > 1$ we identify the next term.

Theorem 1.4. *Let $p > 1$ and $s > 1$. Then the recurrence coefficients of $\sigma_{p,s}$ satisfy, as $n \to \infty$,*

$$n^p b_n = c(p)\Bigl(1 - \frac{s-1}{2n} + O(n^{-2})\Bigr), \qquad n^p a_n = 2c(p)\Bigl(1 - \frac{s-1+p}{2n} + O(n^{-2})\Bigr).$$

Consequently $a_n/(2b_n) = 1 - p/(2n) + O(n^{-2})$ and $b_n = c(p)\bigl(n + \frac{s-1}{2p}\bigr)^{-p}\bigl(1 + O(n^{-2})\bigr)$.

Here, as everywhere in this paper, $a_n/(2b_n)$ pairs the diagonal entry at P_n with the off-diagonal entry b_n between P_{n-1} and P_n, as in (1.2).

The proofs use the following external results, each stated with its hypotheses in §§2.4–2.7. For Theorem 1.2: the formulation of the discrete Riemann–Hilbert problem by Wu, Lin, Xu and Zhao [19], as a pattern only (none of their theorems is applied); the explicit Bessel model of Kuijlaars, McLaughlin, Van Assche and Vanlessen [10, §6], including its freedom in the choice of angle; Theorem 4.4 of Conway and Deift [2], together with the argument of their Theorem 4.7 and Proposition 4.9 and one direction of their Proposition 4.6, which we re-prove; and special-function facts from [8]. Theorem 1.4 uses in addition the classical left-endpoint Bessel parametrix of [10, (6.42)–(6.53)] and the expansions [8, §§9.7, 10.17, 10.40]. The Laurent coefficients of the first corrections that enter the proof of Theorem 1.4 (Section 9

and Appendix E) were computed by exact symbolic series expansion; the coefficient at the hard edge was also computed by hand.

Remark 1.5 (Scope at order $1/n$). Theorem 1.2 bounds the deviation from $c(p)$ at order $1/n$ but does not identify the $1/n$ term. The proof compares the true problem with an exactly solvable comparison problem: relative to it, the non-analytic part of ℓ moves the first two coefficients at infinity by $o(1/n)$ (Section 6), but the comparison problem itself depends on ℓ at order $1/n$. So the $O(1/n)$ is absolute, not relative: we do not claim that $b_n(\sigma_{p,s,\ell})/b_n(\sigma_{p,s}) - 1 = O(\varepsilon_\Lambda(\log n)/n)$. Conjectures for the $1/n$ term when $p \le 1$ or $\ell \not\equiv 1$ are formulated in Section 11 (Conjectures 11.1 and 11.2).

Remark 1.6 (The case $(p,s) = (2,2)$). For $(p,s) = (2,2)$ the coefficients are explicit:

$$b_n = \frac{\pi^2}{(4n+1)\sqrt{(4n-1)(4n+3)}}, \qquad a_n = \frac{2\pi^2}{(4n+1)(4n+5)} \qquad (n \ge 1).$$

They come from the orthonormal Lommel polynomials of order $\frac{3}{2}$ [18, §3] by a quadratic transformation; the derivation is given at the end of Remark 9.6. Thus $n^2 b_n/c(2) = 1 - \frac{1}{2n} + O(n^{-2})$ and $a_n/(2b_n) = 1 - \frac{1}{n} + O(n^{-2})$, in agreement with Theorem 1.4 at $(p,s) = (2,2)$. The closed form does not use the Riemann–Hilbert analysis, so it is an independent test of the $1/n$ term.

1.2. **Context.** *From the averaged law to the pointwise law.* By [15, Corollary 1.2], specialised to the atoms $x_k = k^{-p}$, if the weights $w_k \ge 0$ satisfy $\sum_k w_k < \infty$ and $w_k \ge c_1 k^{-s'}$ for some constants $c_1 > 0$ and $s' \in \mathbb{R}$, then the recurrence coefficients of $\sum_k w_k \delta_{k^{-p}}$ satisfy

$$\frac{2}{n^2} \sum_{j=1}^{n-1} (n-j) \log\bigl(j^p b_j\bigr) \longrightarrow \log c(p). \tag{1.6}$$

The constant comes from an explicitly solvable constrained equilibrium problem [15, Theorem 1.3], whose solution μ^* is the zero distribution that a Jacobi matrix with entries $a_n \approx 2b_n \approx 2c(p)n^{-p}$ would produce. But (1.6) is a double Cesàro mean of $\log(j^p b_j)$, that is, a statement about the logarithm of the Hankel determinant D_n of σ at order n^2. Multiplying the weights by factors in $[e^{-\varepsilon}, e^{\varepsilon}]$ changes $\log D_n$ by at most εn, so the averaged law cannot see such perturbations. Passing from it to (1.4) is a Tauberian step, and the regularity of $j^p b_j$ that this step needs has to come from the weights. That this is not a technicality is shown in the third paper of the series [16]: bounded rough weights can break the pointwise law although the averaged law holds for them. That paper also uses the theorems proved here, as black boxes, to treat general slowly varying weights.

Two difficulties. A Riemann–Hilbert proof of (1.4) meets two obstacles. First, the atoms accumulate at 0, which is a hard edge of μ^* (its density behaves like $x^{-1/2}$ there). After the rescaling $z = n^p x$ used below, the band $(0, 4c(p))$ in which the polynomials oscillate contains infinitely many nodes near its left end, a situation not covered by the Riemann–Hilbert theory of discrete orthogonal polynomials on finite node sets. Second, a slowly varying factor enters the rescaled weight through $r_n(z) := \ell(nz^{-1/p})/\ell(n)$, which is not analytic at the hard edge $z = 0$, so the standard local parametrix does not exist there.

Discrete orthogonal polynomials. The Riemann–Hilbert approach to discrete orthogonal polynomials was developed by Baik, Kriecherbauer, McLaughlin and Miller [1] for measures on finite node sets in a bounded interval, with real-analytic node density and potential; their analysis includes saturated regions, which also appear here. To our knowledge, the only Riemann–Hilbert analysis of discrete orthogonal polynomials on infinitely many nodes accumulating at

a finite point is that of Wu, Lin, Xu and Zhao [19], for the Tricomi–Carlitz polynomials, an observation also made by Dai and Yao [3, §1]; our frame follows theirs.

Non-analytic weights. Conway and Deift [2] met the second difficulty for the weight $\log \frac{2k}{1-x}$ on $(-1,1)$, which carries a slowly varying factor at the endpoint $x=1$; they write that "there is no known parametrix for the Riemann–Hilbert problem in a neighborhood of the logarithmic singularity at $x=1$", and they compare the problem with the Legendre problem through the operator theory behind Riemann–Hilbert problems instead of through a quotient. The comparison argument of Section 6 is theirs [2, Theorem 4.7 and Proposition 4.9]. What we add is a comparison model built from the Szegő function of the full weight (an explicit parametrix rather than a second orthogonality problem), a uniform resolvent bound for a Bessel edge of any order $|\alpha|<1$ (rather than the Legendre value $\alpha=0$), and the treatment of an arbitrary analytic slowly varying factor (rather than one explicit logarithm). Deift and Piorkowski [6] extended the analysis of [2] to the weight $\log \frac{2}{1-x}$ and write, about a comparison with a weight that has a simple zero at the endpoint: "the crucial Theorem 4.7 in [4] requires a different proof in this case: now one needs to control the behaviour of the Cauchy operator acting on spaces with Muckenhoupt weights" [6, §1], [4] being [2]. This is consistent with our restriction to Bessel orders $|\alpha|<1$ in the comparison step (Remark 6.7). Other approaches to non-analytic weights are Piorkowski's parametrix-free method, which needs an a priori estimate for the exact solution [14]; the $\bar\partial$ methods of McLaughlin and Miller [12] and of Trogdon [17], which assume a finite number of Lipschitz derivatives of the weight (for [17], $C^{k,1}$ regularity up to the endpoints of the support); and the approximation of a smooth symbol by analytic ones in Deift, Its and Krasovsky [5, §6.2].

Recurrence asymptotics without analyticity. For exponential weights $W=e^{-Q}$, Levin and Lubinsky [11] prove $o(1)$ asymptotics of the recurrence coefficients, with constants given by equilibrium quantities, under regularity conditions that do not involve analyticity, and without Riemann–Hilbert methods.

Special cases. Besides the Lommel case (Remark 1.6), we mention the Hankel determinants of zeta values studied by Monien [13] and by Haynes and Zudilin [9]; for $q=1$ in the notation of Corollary 1.3, Monien reports a full asymptotic expansion of the determinants, obtained numerically by D. Zagier, which goes beyond Corollary 1.3 but is not proved. The Tricomi–Carlitz polynomials are orthogonal with respect to a measure on the points $\pm 1/\sqrt{k+t_0}$, for a parameter $t_0>0$ [18, §3]; in the squared variable this is an instance of $p=1$ with weights that are not pure powers. Their recurrence coefficients are explicit [18, §3], and [19] obtain uniform asymptotics of the polynomials.

1.3. **Method.** The proof of Theorem 1.2 has an elementary part and a Riemann–Hilbert part. The elementary part is the Christoffel shift, Theorem 3.1: the pointwise law passes from σ to $x\sigma$ together with any regular rate, and $x\sigma_{p,s,\ell}=\sigma_{p,s+p,\ell}$. What is new there is the rate; the statement without rate follows from a proposition of Dai, Piao and Qiao [4, Proposition 2.1] and the classical transfer formulas [7, §1]. The shift reduces Theorem 1.2 to the window $1<s<1+2p$, where the Bessel order $\alpha:=(s-1-p)/p$ of the hard edge lies in $(-1,1)$ (Theorem 7.1). For the Riemann–Hilbert part we work in the rescaled variable $z=n^p x$. There the equilibrium measure μ^* of [15] has the band $(0,4c(p))$, where the polynomials oscillate, the saturated region $(4c(p),\infty)$, where their zeros follow the nodes, and a soft edge at $4c(p)$, and on it we build one frame for all $p>0$ (Section 4). The three geometries $p>1$, $p=1$ and $p<1$ enter only through the outer scalar function that removes the Gamma-function growth of the node masses (Section 5 and Appendix A): the index function $t(z)=nz^{-1/p}$ maps the slit plane onto a sector that lies inside, exactly fills, or exceeds the sector where Stirling's formula holds.

At the hard edge the classical Bessel parametrix is not available, because r_n is not analytic at 0. Following Conway and Deift, we compare the problem with a Bessel model built from the Szegő function of the whole band weight; the two problems differ on the lips of the lens by $o(1)$ without a rate, and for $|\alpha| < 1$ a uniform resolvent bound and Bessel localisation turn this into an $O(1/n)$ error in the coefficients (Section 6). For $\ell \equiv 1$ and $p > 1$ the classical Bessel parametrix does apply, for every $\alpha > -1$ (Section 8), and it gives the $1/n$ term (Section 9): the band factor is constant up to $O(n^{-p})$, the hard-edge correction is annihilated by both functionals that extract a_n and b_n, and the $1/n$ term comes from the Airy correction at the soft edge. This module cannot handle a non-analytic r_n, it fails at $p = 1$, and we do not use it for $p < 1$, where the band factor is not constant to $o(1/n)$ near 0.

1.4. **Organisation.** Section 2 fixes conventions and states every external result used later, and Section 3 proves the Christoffel shift, independently of the rest. Sections 4 and 5 build the Riemann–Hilbert frame for all $p > 0$ and $\alpha > -1$; Proposition 5.6 collects everything the later sections use from it. Sections 6 and 7 treat the hard edge by comparison for $|\alpha| < 1$ and complete the proofs of Theorem 1.2 and Corollary 1.3; Sections 8 and 9 prove Theorem 1.4, and Section 8 also states a localisation bound for the spectral weights, used in [16] and proved in Appendix F. Section 10 gives numerical illustrations and Section 11 conjectures and open problems. A reader interested only in Theorem 1.2 can stop after Section 7 (with Appendices A–D); one interested only in Theorem 1.4 needs Sections 2, 4, 5, 8 and 9 with Appendices A, B and E; from Section 6 it needs only the normalised Bessel model of §6.2 and the identity (6.7).

Throughout, n is the degree, and C, C' denote positive constants that may change from line to line. They are never written c, which is reserved for $c(p)$; the exceptions are c_0 in Definition 1.1 and c_1 in (1.6). Our atoms are exactly k^{-p}, so the atom constant of [15, Theorem 1.1] does not appear.

2. Preliminaries

This section fixes conventions and collects the inputs of the proofs. The only input from [15] is the explicit equilibrium measure (§2.3). Every external result used later is stated in §§2.4–2.7, with its hypotheses and the equation numbers of the source, so that the later sections can refer to it by subsection and part.

Notation used throughout. The data are $p > 0$, $s > 1$ and ℓ; the function Λ, the modulus ε_Λ and the notion "analytic of angle ϑ" are those of Definition 1.1, and $c = c(p)$ is given by (1.3). The Bessel order of the hard edge is $\alpha := (s - 1 - p)/p$, so that $\alpha > -1$ for every $s > 1$. Sections 4 and 5 hold for every $\alpha > -1$ and every ℓ that is analytic of some angle; Sections 6 and 7 assume $|\alpha| < 1$, that is, $1 < s < 1 + 2p$; and Sections 8 and 9 assume $p > 1$ and $\ell \equiv 1$. The Riemann–Hilbert notation is introduced in Section 4, and Table 5.1 lists the quantities that depend on whether $p > 1$, $p = 1$ or $p < 1$. Symbols used only inside one proof are local to it. We write $\mathbb{D}(z_0, r)$ for the open disc of radius r about z_0, and $\mathbb{C}_\pm$ for the open upper and lower half-planes.

2.1. **Orthogonal polynomials and recurrence coefficients.** Let σ be a positive Borel measure on $[0, \infty)$ with infinite support and finite moments. Its Hankel determinants are positive, so the monic orthogonal polynomials P_n ($\deg P_n = n$) exist for every $n \geq 0$. Put $h_n := \int P_n^2 \, \mathrm{d}\sigma > 0$, $P_{-1} := 0$ and $b_0 := 0$. The polynomials satisfy the three-term recurrence

$$xP_n = P_{n+1} + a_n P_n + b_n^2 P_{n-1} \qquad (n \geq 0), \qquad b_n^2 = \frac{h_n}{h_{n-1}} \quad (n \geq 1), \tag{2.1}$$

with $a_n \in \mathbb{R}$ and $b_n > 0$ for $n \geq 1$. We call a_n and b_n the *recurrence coefficients* of σ. Multiplying σ by a positive constant does not change them. The orthonormal polynomials $p_n := P_n/\sqrt{h_n}$ satisfy

$$xp_n = b_{n+1}p_{n+1} + a_n p_n + b_n p_{n-1} \qquad (n \geq 0), \tag{2.2}$$

as one sees by dividing (2.1) by $\sqrt{h_n}$ and using $b_n^2\sqrt{h_{n-1}/h_n} = b_n$. Since xp_j has degree $j+1$, orthonormality gives

$$a_n = \int xp_n^2 \,\mathrm{d}\sigma, \qquad b_n = \int xp_{n-1}p_n \,\mathrm{d}\sigma, \qquad \int xp_n p_j \,\mathrm{d}\sigma = 0 \quad (j \leq n-2). \tag{2.3}$$

In matrix form, (2.2) says that the Jacobi matrix of σ is the symmetric tridiagonal matrix $\mathcal{J}(\sigma)$ with diagonal $(a_0, a_1, \dots)$ and off-diagonal $(b_1, b_2, \dots)$. The monic form (2.1) corresponds to the non-symmetric matrix $J = J(\sigma)$ with $J_{n,n} = a_n$, $J_{n,n+1} = 1$ and $J_{n+1,n} = b_{n+1}^2$; this is the form used in Section 3. The ratio $a_n/(2b_n)$ always pairs the diagonal entry at P_n with the off-diagonal entry b_n between P_{n-1} and P_n; both are read off from the same Riemann–Hilbert matrix of degree n (Proposition 5.6(viii)).

If ℓ is analytic of angle ϑ for some $\vartheta > 0$, then $\ell(k) \leq C_\eta k^\eta$ for every $\eta > 0$ by the Potter bound of Lemma 4.9(i). Taking $\eta < s - 1$ shows that $\sigma_{p,s,\ell}$ is a finite measure on $(0, 1]$ with infinitely many atoms, so everything above applies to it.

2.2. **Regular rates.**

Definition 2.1. A sequence $(\eta_n)_{n\geq 1}$ of positive numbers is a *regular rate* if η_n is non-increasing, $n\eta_n$ is non-decreasing, and $\eta_n \to 0$.

The basic example is $\eta_n = 1/n$. Every regular rate satisfies

$$\frac{1}{n} \leq \frac{\eta_n}{\eta_1}, \qquad \eta_{n-1} \leq 2\eta_n \quad (n \geq 2), \qquad \sum_n \eta_n = \infty. \tag{2.4}$$

The first holds because $n\eta_n \geq \eta_1$; the second because $(n-1)\eta_{n-1} \leq n\eta_n$; the third follows from the first.

Remark 2.2 (Regular majorants). Every non-increasing sequence $\mathsf{v}_n > 0$ with $\mathsf{v}_n \to 0$ is dominated by a regular rate, namely $\eta_n := \max_{m\leq n}(m\mathsf{v}_m)/n$. Here $\eta_n \geq \mathsf{v}_n$ and $n\eta_n$ is non-decreasing by construction; $\eta_{n+1} \leq \max\bigl(\eta_n\, n/(n+1), \mathsf{v}_{n+1}\bigr) \leq \eta_n$ because $\mathsf{v}_{n+1} \leq \mathsf{v}_n \leq \eta_n$; and $\eta_n \to 0$, since $m\mathsf{v}_m/n \leq \mathsf{v}_1/\sqrt{n}$ for $m \leq \sqrt{n}$ and $m\mathsf{v}_m/n \leq \mathsf{v}_{\lceil\sqrt{n}\rceil}$ for $\sqrt{n} < m \leq n$. So Theorem 3.1 applies to any monotone rate. In this paper we use it only with $\eta_n = 1/n$.

2.3. **The equilibrium measure.**

The Riemann–Hilbert analysis of Sections 4–9 is built on the equilibrium measure of the constrained problem solved in [15]. We use only its explicit form and the equality of its potential on the band; the optimality statement [15, Theorem 1.3(iii)] is not used in this paper.

Let $c = c(p)$, let $\gamma_p := \log c(p) + p$, and let $\lambda_p(x) := \frac{1}{p}x^{-1-1/p}$ be the density of the nodes k^{-p} (the image of Lebesgue measure on $(0,\infty)$ under $u \mapsto u^{-p}$). For $y \in [0, 1]$ and $a', b' > 0$ write $\mathcal{B}_y(a', b') := B(a', b')^{-1}\int_0^y r^{a'-1}(1-r)^{b'-1}\,\mathrm{d}r$ for the regularised incomplete beta function, where B is Euler's Beta function. For $\beta > 0$ let arc_β be the arcsine law on $[0, \beta]$, with density $1/(\pi\sqrt{y(\beta - y)})$, and put $\beta_\tau := 4c\tau^{-p}$ for $0 < \tau \leq 1$.

Proposition 2.3. *Let $p > 0$ and let $\mu^* := \int_0^1 \mathrm{arc}_{\beta_\tau}\,\mathrm{d}\tau$, a probability measure on $(0,\infty)$.*

(i) μ^* *has the density*

$$\psi^*(x) = \lambda_p(x)\,\mathcal{B}_{\min(1,\,x/4c)}\big(\tfrac{1}{p}+\tfrac{1}{2},\tfrac{1}{2}\big), \qquad x>0.$$

In particular $\psi^* = \lambda_p$ *on* $[4c,\infty)$ *(saturation) and* $0<\psi^*<\lambda_p$ *on* $(0,4c)$.

(ii) *The function* $U^*(x) := \int \log|x-y|\,\mathrm{d}\mu^*(y)$ *satisfies* $U^*(x) = \gamma_p$ *for* $0<x\le 4c$, *and* $U^*(x)\ge\gamma_p$ *for all* $x>0$.

(iii) (*Hard edge.*) *As* $x\to 0^+$,

$$\psi^*(x) = C_* x^{-1/2}\big(1+O(x)\big), \qquad C_* := \frac{(4c)^{-1/p-1/2}}{p\,a_*B(a_*,\frac12)}, \quad a_* := \frac1p+\frac12.$$

Proof. Parts (i) and (ii) are [15, Theorem 1.3(i)–(ii)], transported from the variable u to $x = u^{-p}$. There, μ_u^* is the image of μ^* under $x\mapsto x^{-1/p}$ and has density $\rho^*(u) = \mathcal{B}_{\min(1,u^{-p}/4c)}(\frac1p+\frac12,\frac12)$, with $\rho^*=1$ exactly on $(0,(4c)^{-1/p}]$ and $0<\rho^*<1$ on $((4c)^{-1/p},\infty)$. Since $|\,\mathrm{d}u/\,\mathrm{d}x| = \lambda_p(x)$, the density of μ^* in x is $\lambda_p(x)\rho^*(x^{-1/p})$, which is (i). Part (ii) is quoted verbatim, with $\gamma_p = \log c + p$. For (iii), let $y = x/4c\le 1$. Then $\int_0^y r^{a_*-1}(1-r)^{-1/2}\,\mathrm{d}r = \frac{y^{a_*}}{a_*}(1+O(y))$, so $\psi^*(x) = \frac1p x^{-1-1/p}(x/4c)^{1/p+1/2}\big(a_*B(a_*,\frac12)\big)^{-1}(1+O(x))$, which is (iii). □

The point $x=4c$ is the soft edge, $(0,4c)$ is the band, and $x=0$ is the hard edge, where μ^* has the inverse-square-root density of (iii).

2.4. **The discrete interpolation problem of Wu, Lin, Xu and Zhao.** This subsection and the next three state the external results used in Sections 4–9 and the appendices. They are needed only from Section 4 on. Throughout, the Cauchy transform on an oriented contour Σ is

$$\mathcal{C}f(z) := \frac{1}{2\pi i}\int_\Sigma \frac{f(s)\,\mathrm{d}s}{s-z}, \qquad z\in\mathbb{C}\setminus\Sigma,$$

and $\mathcal{C}_\pm f$ are its non-tangential boundary values from the left $(+)$ and the right $(-)$ of Σ, so that $\mathcal{C}_+ - \mathcal{C}_- = 1$. For a jump matrix v on Σ we write $\mathcal{C}_v f := \mathcal{C}_-\big(f(v-I)\big)$. The matrices $\sigma_1,\sigma_2,\sigma_3$ are the Pauli matrices, and $\sigma_3 = \mathrm{diag}(1,-1)$.

Wu, Lin, Xu and Zhao [19] develop the Riemann–Hilbert method for discrete orthogonal polynomials on infinitely many nodes with an accumulation point, for the Tricomi–Carlitz polynomials. We apply none of their theorems to our measures; we use the formulation of the problem and the order of the transformations, and our versions are written out in Sections 4 and 5.

(a) The problem and its solution. Let $\sigma = \sum_{k\ge1} w_k\delta_{x_k}$ with distinct nodes $x_k\in(0,1]$, $w_k>0$ and $\sum_k w_k<\infty$, and fix $n\ge1$. Following [19, (2.14) and Theorem 2.3], put

$$Y(z) := \begin{pmatrix} P_n(z) & \displaystyle\sum_k \frac{w_k P_n(x_k)}{z-x_k} \\[2ex] h_{n-1}^{-1}P_{n-1}(z) & \displaystyle h_{n-1}^{-1}\sum_k \frac{w_k P_{n-1}(x_k)}{z-x_k} \end{pmatrix}. \tag{2.5}$$

Then Y is analytic in $\mathbb{C}\setminus(\{x_k\}_k\cup\{0\})$; at each node it has a simple pole with $\mathrm{Res}_{z=x_k}Y(z) = \lim_{z\to x_k}Y(z)\left(\begin{smallmatrix}0 & w_k\\ 0 & 0\end{smallmatrix}\right)$ (the residue condition [19, (2.14)]); and $Y(z)z^{-n\sigma_3} = I + Y_1/z + Y_2/z^2 + O(z^{-3})$ as $z\to\infty$, with

$$(Y_1)_{12}(Y_1)_{21} = b_n^2, \qquad \frac{(Y_2)_{12}}{(Y_1)_{12}} - (Y_1)_{22} = a_n. \tag{2.6}$$

Proof. The sums converge locally uniformly off $\{x_k\}\cup\{0\}$, which gives analyticity, the simple poles and $\operatorname{Res}_{x_k} Y_{j2} = w_k Y_{j1}(x_k)$. Write $P_m(x) = x^m + \pi_m x^{m-1} + \dots$. For $|z| > 2$, expanding $(z-x)^{-1}$ gives $\sum_k w_k P_m(x_k)/(z-x_k) = \sum_{i\ge 0} z^{-i-1}\int P_m x^i \,\mathrm{d}\sigma$, and $\int P_m x^i\,\mathrm{d}\sigma$ equals 0 for $i < m$, h_m for $i = m$ and $-\pi_{m+1}h_m$ for $i = m+1$. Hence $(Y_1)_{12} = h_n$, $(Y_1)_{21} = h_{n-1}^{-1}$, $(Y_2)_{12} = -\pi_{n+1}h_n$ and $(Y_1)_{22} = -\pi_n$; and $\pi_n = \pi_{n+1} + a_n$ by (2.1). □

(b) Transformations. The transformations of Section 4 follow [19, §§3–5]: removal of the poles by triangular factors built from $\Pi = \sin\theta/\theta'$ and $e^{\pm i\theta}$ [19, (3.9)–(3.15)], normalisation by the g-function [19, (4.1)], the lens and saturated-region factors $V_\pm$, $V_\pm^{\triangle}$ [19, (4.12)–(4.13)], and conjugation by an outer scalar [19, (5.1)]. In [19] these are written in a symmetric variable with nodes $\pm x_k$; we use only the pattern, and every jump is recomputed in Section 4 for our one-sided problem.

(c) The soft edge. Our Airy parametrix (Lemma 5.5) follows the pattern of [19, §6.1, (6.4), (6.10)–(6.12)]; it is constructed from the Airy function directly (§2.7), and no statement of [19] is used.

2.5. **The Bessel model of Kuijlaars, McLaughlin, Van Assche and Vanlessen.** We use the explicit Bessel model of [10, §6] and its algebraic properties. In Sections 6 and 7 we apply none of their theorems on orthogonal polynomials, so their standing hypothesis on the weight [10, (1.1)] plays no role there. Section 8 uses the construction of part (c) at the left endpoint; the hypotheses it needs are checked there.

(a) The model. Let $\alpha > -1$ and let Σ_Ψ consist of the rays $\gamma_1 : \arg\zeta = 2\pi/3$, $\gamma_2 : \arg\zeta = \pi$ and $\gamma_3 : \arg\zeta = -2\pi/3$, all oriented towards the origin, with the $+$ side on the left [10, Figure 3]. The function

$$\Psi_\alpha(\zeta) = \begin{pmatrix} I_\alpha(2\zeta^{1/2}) & \frac{i}{\pi}K_\alpha(2\zeta^{1/2}) \\ 2\pi i\zeta^{1/2}I'_\alpha(2\zeta^{1/2}) & -2\zeta^{1/2}K'_\alpha(2\zeta^{1/2}) \end{pmatrix}, \qquad |\arg\zeta| < 2\pi/3$$

[10, (6.23)], continued to $2\pi/3 < \pm\arg\zeta < \pi$ by the Hankel-function formulas [10, (6.24)–(6.25)], solves the Bessel model problem [10, Theorem 6.3], with jumps

$$\begin{pmatrix} 1 & 0 \\ e^{\alpha\pi i} & 1 \end{pmatrix} \text{ on } \gamma_1, \qquad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \text{ on } \gamma_2, \qquad \begin{pmatrix} 1 & 0 \\ e^{-\alpha\pi i} & 1 \end{pmatrix} \text{ on } \gamma_3$$

[10, (6.16)–(6.18)]. Uniformly as $\zeta\to\infty$ in $\mathbb{C}\setminus\Sigma_\Psi$,

$$\Psi_\alpha(\zeta) = (2\pi\zeta^{1/2})^{-\sigma_3/2}\frac{1}{\sqrt{2}}\begin{pmatrix} 1+O(\zeta^{-1/2}) & i+O(\zeta^{-1/2}) \\ i+O(\zeta^{-1/2}) & 1+O(\zeta^{-1/2}) \end{pmatrix} e^{2\zeta^{1/2}\sigma_3}$$

[10, (6.28)], which follows from the large-argument expansions (§2.7) of the Bessel and Hankel functions in (6.23)–(6.25); this also covers the rays moved by (b) (§6.2).

(b) Angle freedom. By the remark after [10, (6.21)], the angle $2\pi/3$ may be replaced by any angle in $(0,\pi)$; the lens contours in the disc are then the preimages of the rays under $\zeta = n^2 f(z)$, and they do not depend on n. We use rays at $\pm(\pi - \vartheta_{\mathrm{lip}})$.

(c) The left endpoint. Our hard edge has the band to its right, as the endpoint -1 of [10]. There the conformal map reverses orientation, and the reversal is absorbed by conjugation with σ_3: $\tilde\Psi(\zeta) = \sigma_3\Psi(\zeta;\alpha\to\beta)\sigma_3$ [10, (6.51)], with prefactor

$$\tilde E_n(z) = N(z)\tilde W(z)^{\sigma_3}\tfrac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}\tilde f(z)^{\sigma_3/4}(2\pi n)^{\sigma_3/2}$$

[10, (6.53)], where N, $\tilde W$ and $\tilde f$ are the global parametrix, an analytic square root of the weight and the local conformal map of [10, (6.42)–(6.50)] (compare the right-endpoint formulas [10, (6.27), (6.31)] and $\det E_n \equiv 1$ [10, (6.33)]). The prefactor is analytic by removability: its

entries have at most a square-root singularity at the endpoint and it is analytic in a punctured neighbourhood [10, proof of Proposition 6.5]. In Section 6 this argument is applied only to a prefactor that does not involve the weight.

2.6. **The comparison argument of Conway and Deift.** The following are from [2, §4], whose Cauchy operator is normalised as ours. Lebesgue exponents are written r, r' ([2] writes p, q).

(a) Boundedness and the class $\partial\mathcal{C}$ [2, §4.1]. $\mathcal{C}_\pm$ are bounded on $L^2(\Sigma)$ (and on $L^r(\Sigma)$, $1 < r < \infty$) if and only if Σ is Carleson. A pair $A_\pm \in L^r(\Sigma)$ belongs to $\partial\mathcal{C}(L^r(\Sigma))$ if $A_\pm = \mathcal{C}_\pm f$ for some (unique) $f \in L^r(\Sigma)$; we write $A_\pm \in I + \partial\mathcal{C}(L^r)$ if $A_\pm - I \in \partial\mathcal{C}(L^r)$.

(b) Theorem 4.4 of [2]. *Let $v_A, v_B \in L^\infty(\Sigma)$ with $v_B^{-1} \in L^\infty(\Sigma)$. Let $A_\pm \in I + \partial\mathcal{C}(L^r(\Sigma))$ satisfy $A_+ = A_- v_A$ on Σ, and let $B_\pm \in I + \partial\mathcal{C}(L^r(\Sigma))$ satisfy $B_+ = B_- v_B$ on Σ, with $B_\pm^{-1} \in I + \partial\mathcal{C}(L^{r'}(\Sigma))$, $1/r + 1/r' = 1$. If A, B are the analytic extensions of $A_\pm, B_\pm$, then*

$$AB^{-1} = I + \mathcal{C}\big(A_-(v_A v_B^{-1} - I)B_-^{-1}\big).$$

Consequence (moments at infinity). If in addition Σ is bounded and $A = I + A_1/z + A_2/z^2 + O(z^{-3})$, $B = I + B_1/z + B_2/z^2 + O(z^{-3})$ as $z \to \infty$, then with $\mathcal{E} := A_-(v_A - v_B)v_B^{-1}B_-^{-1} \in L^1(\Sigma)$ (take $r = r' = 2$),

$$A_1 - B_1 = -\frac{1}{2\pi i}\int_\Sigma \mathcal{E}(s)\,\mathrm{d}s, \qquad A_2 - B_2 = -\frac{1}{2\pi i}\int_\Sigma s\mathcal{E}(s)\,\mathrm{d}s + (A_1 - B_1)B_1. \tag{2.7}$$

For the proof, write $v_A v_B^{-1} - I = (v_A - v_B)v_B^{-1}$; for $|z| > \max_\Sigma |s|$ expand $1/(s - z) = -\sum_{j\ge 0} s^j z^{-j-1}$ in $\mathcal{C}\mathcal{E}(z)$ and compare with $AB^{-1} = I + (A_1 - B_1)z^{-1} + \big(A_2 - B_2 - (A_1 - B_1)B_1\big)z^{-2} + O(z^{-3})$. This is the mechanism of [2, Proposition 4.9], whose proof identifies $\mu = Q_-$ and $\tilde\mu = \tilde Q_-$. Formula (4.14) of [2] is printed without the factor v_B^{-1}; it agrees with (2.7) when $(v_A - v_B)v_B^{-1} = v_A - v_B$, as in their setting. We use the form derived from Theorem 4.4.

(c) Solution formula [2, §4.2]. If $1 - \mathcal{C}_v$ is invertible on $L^2(\Sigma)$, then $\mu := (1 - \mathcal{C}_v)^{-1}I$ gives the unique solution $X(z) = I + \mathcal{C}(\mu(v - I))(z)$ of the Riemann–Hilbert problem normalised at infinity with jump v.

(d) The Neumann-series step [2, proof of Theorem 4.7, (4.12)–(4.13)]. If $\|(1-\mathcal{C}_{\tilde v})^{-1}\|_{L^2\to L^2} \le K$ and $\|\mathcal{C}_v - \mathcal{C}_{\tilde v}\|_{L^2\to L^2} \le 1/(2K)$, then

$$(1 - \mathcal{C}_v)^{-1} = (1 - \mathcal{C}_{\tilde v})^{-1}\big(1 - (\mathcal{C}_v - \mathcal{C}_{\tilde v})(1 - \mathcal{C}_{\tilde v})^{-1}\big)^{-1}$$

exists and has norm at most $2K$, since the inner factor is 1 minus an operator of norm at most $\frac12$. In [2] the smallness comes from $\|v - \tilde v\|_{L^\infty(\Sigma)} \to 0$.

(e) Proposition 4.6 of [2] (attributed there to Deift and Zhou). The operators $(1 - \mathcal{C}_{h_n})^{-1}$ are bounded uniformly in n if and only if the inhomogeneous problem $m_+ = m_- h_n + g$ on Σ, $m_\pm \in \partial\mathcal{C}(L^2(\Sigma))$, is uniquely solvable for every $g \in L^2(\Sigma)$ with $\|m_\pm\|_{L^2} \le C\|g\|_{L^2}$, C independent of n and g. We use only the direction "solvable with bounds ⇒ bounded inverse", and we re-prove it in Appendix D.

We do not use [2, Theorem 4.5], whose proof is specific to the Legendre comparison problem.

2.7. **Special functions.** All formulas are from [8], with principal branches throughout. "Uniformly in closed subsectors" of a sector means uniformly in every closed sector contained in it.

(a) The Gamma function. (5.5.1) $\Gamma(z + 1) = z\Gamma(z)$; (5.5.3) $\Gamma(z)\Gamma(1 - z) = \pi/\sin(\pi z)$, $z \notin \mathbb{Z}$; (5.11.1) $\operatorname{Ln}\Gamma(z) \sim (z - \frac12)\ln z - z + \frac12\ln(2\pi) + \sum_{k\ge1} \frac{B_{2k}}{2k(2k-1)z^{2k-1}}$ as $z \to \infty$, uniformly in closed subsectors of $|\operatorname{ph} z| < \pi$, where by §5.11(ii) the remainder after the term $k = m - 1$ is

at most $\sec^{2m}(\frac{1}{2}\operatorname{ph} z)$ times the first neglected term; (5.12.3), (5.12.1) $\int_0^\infty t^{a'-1}(1+t)^{-a'-b'}\,\mathrm{d}t = B(a',b') = \Gamma(a')\Gamma(b')/\Gamma(a'+b')$ for $\operatorname{Re} a', \operatorname{Re} b' > 0$.

(b) Airy functions (§§9.2, 9.7). Ai is entire, and $\operatorname{Ai}(z)$, $\operatorname{Ai}(ze^{\mp 2\pi i/3})$ solve $w'' = zw$; (9.2.8) $\mathcal{W}\{\operatorname{Ai}(z), \operatorname{Ai}(ze^{\mp 2\pi i/3})\} = e^{\pm \pi i/6}/(2\pi)$; (9.2.12) $\operatorname{Ai}(z)+e^{-2\pi i/3}\operatorname{Ai}(ze^{-2\pi i/3})+e^{2\pi i/3}\operatorname{Ai}(ze^{2\pi i/3}) = 0$. By (9.7.5)–(9.7.6), as $z \to \infty$ uniformly in closed subsectors of $|\operatorname{ph} z| < \pi$, with $\chi := \frac{2}{3}z^{3/2}$,

$$\operatorname{Ai}(z) \sim \frac{e^{-\chi}}{2\sqrt{\pi}\, z^{1/4}} \sum_{k\ge 0} \frac{(-1)^k u_k}{\chi^k}, \qquad \operatorname{Ai}'(z) \sim -\frac{z^{1/4} e^{-\chi}}{2\sqrt{\pi}} \sum_{k\ge 0} \frac{(-1)^k v_k}{\chi^k},$$

where, by (9.7.2), $u_0 = v_0 = 1$, $u_k = (2k+1)(2k+3)\cdots(6k-1)/(216^k k!)$ and $v_k = \frac{6k+1}{1-6k}u_k$; in particular $u_1 = \frac{5}{72}$ and $v_1 = -\frac{7}{72}$.

(c) Modified Bessel functions. (10.25.2) $I_\nu(z) = (\frac{1}{2}z)^\nu \sum_{k\ge 0} (\frac{1}{4}z^2)^k/(k!\,\Gamma(\nu+k+1))$, analytic in $\mathbb{C}\setminus(-\infty, 0]$ (§10.25(ii)); (10.27.4) $K_\nu = \frac{1}{2}\pi(I_{-\nu} - I_\nu)/\sin(\nu\pi)$ for $\nu \notin \mathbb{Z}$; (10.31.2) $K_0(z) = -(\ln(\frac{1}{2}z) + \gamma_{\mathrm{E}})I_0(z) + \frac{1}{4}z^2 + \cdots$, γ_{E} being Euler's constant and the dots an entire function of z^2; (10.28.2) $\mathcal{W}\{K_\nu, I_\nu\}(z) = 1/z$, which with (6.23) of §2.5 gives $\det \Psi_\alpha = 1$.

(d) Large argument (§§10.17, 10.40, ν fixed). The coefficients are, by (10.17.1) and (10.17.8),

$$a_k(\nu) = \frac{(4\nu^2-1^2)(4\nu^2-3^2)\cdots(4\nu^2-(2k-1)^2)}{k!\,8^k},$$
$$b_k(\nu) = \frac{(4\nu^2-1^2)\cdots(4\nu^2-(2k-3)^2)\,(4\nu^2+4k^2-1)}{k!\,8^k},$$

with $a_0(\nu) = b_0(\nu) = 1$; for instance $a_1(\nu) = (4\nu^2-1)/8$ and $b_1(\nu) = (4\nu^2+3)/8$. Let $\mathsf{s}_\pm(z) := \sum_k (\pm 1)^k a_k(\nu) z^{-k}$. The expansions (10.40.1), (10.40.2) and (10.40.5) read

$$I_\nu(z) \sim e^z(2\pi z)^{-1/2}\mathsf{s}_-(z), \qquad |\operatorname{ph} z| < \tfrac{1}{2}\pi,$$
$$K_\nu(z) \sim (\pi/2z)^{1/2}e^{-z}\mathsf{s}_+(z), \qquad |\operatorname{ph} z| < \tfrac{3}{2}\pi,$$
$$I_\nu(z) \sim (2\pi z)^{-1/2}\big(e^z\mathsf{s}_-(z) \pm ie^{\pm\nu\pi i}e^{-z}\mathsf{s}_+(z)\big), \qquad -\tfrac{1}{2}\pi < \pm\operatorname{ph} z < \tfrac{3}{2}\pi,$$

uniformly in closed subsectors; and (10.40.3)–(10.40.4) are the expansions of I_ν', K_ν' in the sectors of (10.40.1)–(10.40.2), with $b_k(\nu)$ in place of $a_k(\nu)$ and an overall minus sign for K_ν'. For the Hankel functions, with $\varpi := z - \frac{1}{2}\nu\pi - \frac{1}{4}\pi$ (10.17.2) and $\delta > 0$ arbitrarily small, (10.17.5) $H_\nu^{(1)}(z) \sim (2/(\pi z))^{1/2}e^{i\varpi}\sum_k i^k a_k(\nu) z^{-k}$ for $-\pi+\delta \le \operatorname{ph} z \le 2\pi - \delta$, and (10.17.6) $H_\nu^{(2)}(z) \sim (2/(\pi z))^{1/2}e^{-i\varpi}\sum_k (-i)^k a_k(\nu) z^{-k}$ for $-2\pi+\delta \le \operatorname{ph} z \le \pi - \delta$; (10.17.11)–(10.17.12) give $H_\nu^{(1,2)\prime}(z) \sim \pm i(2/(\pi z))^{1/2}e^{\pm i\varpi}\sum_k (\pm i)^k b_k(\nu) z^{-k}$ in the same two sectors. None of these expansions restricts $|\nu|$.

3. The Christoffel shift

This section proves that the pointwise law, together with its rate, passes from a measure σ on $[0,\infty)$ to the measure $x\sigma$, $\mathrm{d}(x\sigma) := x\,\mathrm{d}\sigma(x)$. For our measures, multiplication by x shifts the exponent s by exactly p: for $x_k = k^{-p}$,

$$x\,\sigma_{p,s,\ell} = \sum_{k\ge 1} k^{-p}\,k^{-s}\ell(k)\,\delta_{k^{-p}} = \sigma_{p,s+p,\ell}, \tag{3.1}$$

with the same ℓ, and it raises the Bessel order $\alpha = (s-1-p)/p$ by 1. This is why the Riemann–Hilbert analysis of Sections 4–7 only has to cover the window $|\alpha| < 1$.

Theorem 3.1 (Christoffel shift: law and rate)**.** *Let σ be a positive Borel measure on $[0,\infty)$ with infinite support and finite moments, let $p > 0$ and $c > 0$, and let η_n be a regular rate (Definition 2.1). If*

$$|n^p a_n(\sigma) - 2c| + |n^p b_n(\sigma) - c| = O(\eta_n),$$

then the measure $x\sigma$ satisfies the same estimate, with the same c and rate $O(\eta_n)$. Without a rate: if $n^p a_n(\sigma) \to 2c$ and $n^p b_n(\sigma) \to c$, then the same holds for $x\sigma$.

The Jacobi matrix of $x\sigma$ is obtained by factoring the Jacobi matrix of σ as a product LU of bidiagonal matrices and multiplying the factors in the opposite order; these transfer formulas are classical [7, §1]. The entries of the factors are not known in advance, since they involve the values $P_n(0)$. The point of the proof is that no information about $P_n(0)$ is needed beyond its sign: the factors satisfy a recursion whose limiting map has a double (parabolic) fixed point, and positivity alone forces them to converge to it. The factors form a chain sequence, and the statement without rate follows from a proposition of Dai, Piao and Qiao [4, Proposition 2.1] together with the transfer formulas. The part with a rate is the one we need, since it lets the rate $O(1/n)$ survive the shift; we have not found the rate statement in the literature. The parameters of the chain converge only at the rate $O(\sqrt{\eta_n})$ in general, and this is sharp (Remark 3.6); the coefficients of $x\sigma$ involve only their increments, which are $O(\eta_n)$. Theorem 3.1, for discrete measures $\sum_k w_k \delta_{x_k}$ with $w_k > 0$, $\sum_k w_k < \infty$ and bounded, strictly decreasing atoms $x_k \geq 0$ (a class that contains every $\sigma_{p,s,\ell}$), and Lemmas 3.4 and 3.5 below have been formally verified in the Lean proof assistant.

Throughout, a_n, b_n are the recurrence coefficients of σ (§2.1), and $\tilde{a}_n, \tilde{b}_n$ are those of $x\sigma$.

3.1. The transfer formulas.

Lemma 3.2. *Let σ be a positive Borel measure on $[0,\infty)$ with infinite support and finite moments. Then for $n \geq 1$ the polynomial P_n has n simple zeros, all in $(0,\infty)$. In particular $(-1)^n P_n(0) > 0$ for all $n \geq 0$.*

Proof. Let $y_1 < \cdots < y_m$ be the zeros of P_n in $(0,\infty)$ of odd multiplicity, and put $\pi(x) := \prod_j (x - y_j)$ ($\pi := 1$ if $m = 0$). Suppose $m < n$. Then $\int P_n \pi \, d\sigma = 0$ by orthogonality. On the other hand, every remaining factor of P_n is of the form $(x-y)^{2i}$ with y real, or $(x-y)$ with $y \leq 0$, or $(x-z)(x-\bar{z}) = |x-z|^2$ with $z \notin \mathbb{R}$; each is non-negative on $[0,\infty)$. Hence $P_n\pi \geq 0$ on $[0,\infty)$. Since $P_n\pi$ is a non-zero polynomial and σ has infinite support, $\int P_n \pi \, d\sigma > 0$, a contradiction. So $m = n$, the n zeros are simple and positive, and $P_n(0) = \prod_j (-y_j)$ has the sign $(-1)^n$. □

By Lemma 3.2 we may define

$$u_n := -\frac{P_{n+1}(0)}{P_n(0)} > 0 \quad (n \geq 0), \qquad l_0 := 0, \qquad l_n := \frac{b_n^2}{u_{n-1}} > 0 \quad (n \geq 1).$$

Lemma 3.3. *Let σ be as in Lemma 3.2. Then:*

(i) *$a_n = u_n + l_n$ for $n \geq 0$, and $b_n^2 = u_{n-1} l_n$ for $n \geq 1$. Equivalently, $u_0 = a_0$ and $u_n = a_n - b_n^2/u_{n-1}$ for $n \geq 1$.*

(ii) *The measure $x\sigma$ has infinite support, its monic orthogonal polynomials are $\widetilde{P}_n(x) := \big(P_{n+1}(x) + u_n P_n(x)\big)/x$, and its recurrence coefficients are*

$$\tilde{a}_n = u_n + l_{n+1} \quad (n \geq 0), \qquad \tilde{b}_n^2 = u_n l_n \quad (n \geq 1). \tag{3.2}$$

In matrix form: $J(\sigma) = LU$, with L unit lower bidiagonal with subdiagonal $(l_1, l_2, \dots)$ and U upper bidiagonal with diagonal $(u_0, u_1, \dots)$ and superdiagonal 1; and $J(x\sigma) = UL$. This

is the classical Christoffel (or Darboux) transformation; see Derevyagin and Derkach [7, §1], where it is written $J = LU \mapsto UL$. We include the short proof.

Proof. (i) Evaluate (2.1) at $x = 0$: $0 = P_{n+1}(0) + a_n P_n(0) + b_n^2 P_{n-1}(0)$. Dividing by $P_n(0) \neq 0$ gives $a_n = u_n - b_n^2 P_{n-1}(0)/P_n(0) = u_n + b_n^2/u_{n-1} = u_n + l_n$ for $n \geq 1$, and $a_0 = u_0$ since $P_{-1} = 0$. The identity $b_n^2 = u_{n-1} l_n$ is the definition of l_n.

(ii) The support of $x\sigma$ is $\operatorname{supp}\sigma \setminus \{0\}$, which is infinite, and $x\sigma$ has finite moments. The polynomial $P_{n+1} + u_n P_n$ vanishes at 0 by the definition of u_n, so $\widetilde{P}_n$ is a monic polynomial of degree n. For $j \leq n-1$, $\int \widetilde{P}_n x^j \,\mathrm{d}(x\sigma) = \int (P_{n+1} + u_n P_n) x^j \,\mathrm{d}\sigma = 0$. Hence the $\widetilde{P}_n$ are the monic orthogonal polynomials of $x\sigma$.

Next, for $n \geq 1$ the polynomial $P_n - \widetilde{P}_n$ has degree at most $n-1$, and for $j \leq n-2$

$$\int (P_n - \widetilde{P}_n) x^j \,\mathrm{d}(x\sigma) = \int P_n x^{j+1} \,\mathrm{d}\sigma - \int \widetilde{P}_n x^j \,\mathrm{d}(x\sigma) = 0.$$

So $P_n - \widetilde{P}_n = l'_n \widetilde{P}_{n-1}$ for some constant l'_n; put $l'_0 := 0$, consistent with $P_0 = \widetilde{P}_0 = 1$. Then

$$x P_n = x\widetilde{P}_n + l'_n \, x\widetilde{P}_{n-1} = (P_{n+1} + u_n P_n) + l'_n (P_n + u_{n-1} P_{n-1}).$$

Comparing with (2.1) (the P_j are linearly independent) gives $a_n = u_n + l'_n$ and $b_n^2 = l'_n u_{n-1}$, so $l'_n = l_n$ by (i). Finally,

$$x\widetilde{P}_n = P_{n+1} + u_n P_n = (\widetilde{P}_{n+1} + l_{n+1}\widetilde{P}_n) + u_n(\widetilde{P}_n + l_n \widetilde{P}_{n-1}),$$

which is the recurrence of $x\sigma$ with $\tilde{a}_n = u_n + l_{n+1}$ and $\tilde{b}_n^2 = u_n l_n$. □

3.2. **The chain sequence converges.** Assume now that σ satisfies the pointwise law at scale n^p with constant $c > 0$. We work with the rescaled quantities

$$\hat{a}_n := n^p a_n, \quad \hat{b}_n := n^p b_n, \quad \hat{u}_n := n^p u_n, \quad \hat{l}_n := n^p l_n, \quad \varsigma_n := \Big(\frac{n-1}{n}\Big)^p.$$

Multiplying the identities of Lemma 3.3(i) by n^p, and using $n^p \cdot (n-1)^p = n^{2p}\varsigma_n$, we get

$$\hat{u}_n + \hat{l}_n = \hat{a}_n, \qquad \hat{u}_{n-1}\hat{l}_n = \varsigma_n \hat{b}_n^2, \qquad \text{so} \qquad \hat{u}_n = \hat{a}_n - \frac{\varsigma_n \hat{b}_n^2}{\hat{u}_{n-1}} \qquad (n \geq 1). \tag{3.3}$$

If $\hat{a}_n \to 2c$, $\hat{b}_n \to c$ and $\hat{u}_{n-1} \to c$, the last identity tends to

$$\hat{u}_n \approx F(\hat{u}_{n-1}), \qquad F(t) := 2c - \frac{c^2}{t}, \qquad F(t) - t = -\frac{(t-c)^2}{t}, \qquad F(c+y) = c + \frac{cy}{c+y}. \tag{3.4}$$

The map F is increasing on $(0, \infty)$, has the single fixed point c, and this fixed point is double: $F(t) \leq t$ for every $t > 0$. An orbit that settles below c is therefore pushed down towards 0, and one above c is pulled down to c. Positivity of u_n rules out the first possibility.

Lemma 3.4. *Let σ be as in Lemma 3.2, $p > 0$ and $c > 0$. If $\hat{a}_n \to 2c$ and $\hat{b}_n \to c$, then $\hat{u}_n \to c$ and $\hat{l}_n \to c$.*

Lemma 3.4 is a special case of [4, Proposition 2.1]: in the variables $\mathsf{r}_n := b_n^2/(a_{n-1}a_n)$ and $\mathsf{g}_n := l_n/a_n$ one has $\mathsf{r}_n = (1 - \mathsf{g}_{n-1})\mathsf{g}_n$ with $\mathsf{g}_0 = 0$ and $0 < \mathsf{g}_n < 1$, so (r_n) is a chain sequence with parameter sequence (g_n); the hypothesis gives $\mathsf{r}_n = \varsigma_n \hat{b}_n^2/(\hat{a}_{n-1}\hat{a}_n) \to \frac{1}{4}$, and [4, Proposition 2.1] states that then every parameter sequence tends to $\frac{1}{2}$. We include the short proof, because the rate version, Lemma 3.5, follows the same lines.

Proof. A priori bounds. By Lemma 3.2, $u_n, l_n > 0$, so $\hat{u}_n < \hat{a}_n$ and $\hat{l}_n < \hat{a}_n$. Hence $\hat{u}_{n-1} = \varsigma_n \hat{b}_n^2/\hat{l}_n > \varsigma_n \hat{b}_n^2/\hat{a}_n \to c/2$. So there is n_1 with $\hat{u}_n \in [c/3, 3c]$ for $n \geq n_1$.

Recursion. For $n > n_1$, (3.3) reads $\hat{u}_n = F(\hat{u}_{n-1}) + e_n$ with

$$e_n := (\hat{a}_n - 2c) - \frac{\varsigma_n \hat{b}_n^2 - c^2}{\hat{u}_{n-1}}, \qquad |e_n| \leq |\hat{a}_n - 2c| + \frac{3}{c}\,|\varsigma_n \hat{b}_n^2 - c^2| \longrightarrow 0.$$

Fix $\delta \in (0, c/3)$ and put $d := \delta^2/(3c)$. By (3.4), $F(t) \leq t - d$ for every $t \in [c/3, 3c]$ with $|t - c| \geq \delta$. Choose $n_2 > n_1$ with $|e_n| < \min(d/2, \delta)$ for $n \geq n_2$.

Lower bound. Suppose $\hat{u}_m < c - \delta$ for some $m \geq n_2$. Then $\hat{u}_{m+1} \leq \hat{u}_m - d + |e_{m+1}| < \hat{u}_m - d/2 < c - \delta$, and by induction $\hat{u}_{m+j} < \hat{u}_m - jd/2$ for all $j \geq 1$. This contradicts $\hat{u}_n \geq c/3$. So $\hat{u}_n \geq c - \delta$ for all $n \geq n_2$.

Upper bound. Let $n \geq n_2$. If $\hat{u}_n \geq c + \delta$, then $\hat{u}_{n+1} < \hat{u}_n - d/2$. If $\hat{u}_n < c + \delta$, then, F being increasing, $\hat{u}_{n+1} < F(c + \delta) + \delta < c + 2\delta$, because $F(c + \delta) < c + \delta$. The first alternative cannot hold for all $n \geq n_2$, since $\hat{u}_n$ would then decrease by $d/2$ at every step while staying above $c/3$. So $\hat{u}_m < c + \delta$ for some $m \geq n_2$. From then on $\hat{u}_n < c + 2\delta$: if $\hat{u}_n < c + \delta$ this was just shown, and if $c + \delta \leq \hat{u}_n < c + 2\delta$ then $\hat{u}_{n+1} < \hat{u}_n < c + 2\delta$.

Hence $c - \delta \leq \liminf \hat{u}_n \leq \limsup \hat{u}_n \leq c + 2\delta$ for every small $\delta > 0$, so $\hat{u}_n \to c$, and $\hat{l}_n = \hat{a}_n - \hat{u}_n \to c$. □

3.3. **Rates.** With a rate, the same argument gives a rate for $\hat{u}_n$, but only the square root of the input rate. What saves the transfer is that the coefficients of $x\sigma$ depend on $\hat{u}_n$ only through its increments, which converge at the full rate.

Lemma 3.5. *Let σ be as in Lemma 3.2, $p > 0$, $c > 0$, and let η_n be a regular rate with $|\hat{a}_n - 2c| + |\hat{b}_n - c| \leq M\eta_n$ for $n \geq 1$. Then*

$$|\hat{u}_n - c| = O(\sqrt{\eta_n}) \qquad \text{and} \qquad |\hat{u}_n - \hat{u}_{n-1}| = O(\eta_n).$$

Proof. We keep the notation of the previous proof; in particular $\hat{u}_n \in [c/3, 3c]$ for $n \geq n_1$, and $\hat{u}_n \to c$ by Lemma 3.4. We first bound the perturbation e_n. For every $p > 0$ and $n \geq 2$ we have $0 \leq 1 - \varsigma_n \leq \max(p, 1)/n$: for $p \geq 1$ by Bernoulli's inequality $(1 - 1/n)^p \geq 1 - p/n$, and for $p < 1$ because $(1 - 1/n)^p \geq 1 - 1/n$. By (2.4), $1/n \leq \eta_n/\eta_1$. Also $\hat{b}_n$ is bounded, so $|\hat{b}_n^2 - c^2| = O(\eta_n)$. Hence $|\varsigma_n \hat{b}_n^2 - c^2| \leq (1 - \varsigma_n)\hat{b}_n^2 + |\hat{b}_n^2 - c^2| = O(\eta_n)$, and there is M_1 with $|e_n| \leq M_1\eta_n$ for $n > n_1$.

Put $y_n := \hat{u}_n - c$. By (3.4),

$$y_{n+1} = \frac{cy_n}{c + y_n} + e_{n+1} = y_n - \frac{y_n^2}{\hat{u}_n} + e_{n+1} \qquad (n \geq n_1).$$

Lower bound. Let $\lambda_- := \sqrt{6cM_1}$. Suppose $y_m \leq -\lambda_-\sqrt{\eta_m}$ for some $m \geq n_1$. Then $y_m^2/\hat{u}_m \geq \lambda_-^2\eta_m/(3c) = 2M_1\eta_m$, so $y_{m+1} \leq y_m - 2M_1\eta_m + M_1\eta_{m+1} \leq y_m - M_1\eta_m \leq -\lambda_-\sqrt{\eta_m} \leq -\lambda_-\sqrt{\eta_{m+1}}$. By induction $y_{m+j} \leq y_m - M_1 \sum_{i<j} \eta_{m+i}$, which tends to $-\infty$ because $\sum \eta_n = \infty$. This contradicts $\hat{u}_n > 0$. Hence $y_n > -\lambda_-\sqrt{\eta_n}$ for all $n \geq n_1$.

Upper bound. Choose $n_3 \geq n_1$ with $|y_n| \leq c/2$ for $n \geq n_3$; then $\hat{u}_n \leq 2c$ there. Choose $\lambda_+ > 0$ with $\lambda_+^2/(2c) \geq 2M_1 + \lambda_+/(2\sqrt{\eta_1})$ (possible, since the left side is quadratic in λ_+), and $n_4 \geq n_3$ with $\lambda_+\sqrt{\eta_n} \leq c$ for $n \geq n_4$.

Step. Let $n \geq n_4$ and $y_n \leq t := \lambda_+\sqrt{\eta_n}$. The map $y \mapsto cy/(c + y)$ is increasing on $(-c, \infty)$ and $y_n \geq -c/2$, so

$$y_{n+1} \leq \frac{ct}{c + t} + M_1\eta_{n+1} \leq t - \frac{t^2}{2c} + M_1\eta_n = t - \Big(\frac{\lambda_+^2}{2c} - M_1\Big)\eta_n \leq t - \frac{\lambda_+\eta_n}{2\sqrt{\eta_1}},$$

using $t \le c$ and the choice of λ_+. On the other hand, $\eta_{n+1} \ge \eta_n n/(n+1)$ gives $t - \lambda_+\sqrt{\eta_{n+1}} \le \lambda_+\sqrt{\eta_n}\big(1 - \sqrt{n/(n+1)}\big) \le \lambda_+\sqrt{\eta_n}/(2n)$, where the last step is $\sqrt{n(n+1)} \ge n - 1$. Since $1/n \le \eta_n/\eta_1 \le \sqrt{\eta_n/\eta_1}$, this is at most $\lambda_+\eta_n/(2\sqrt{\eta_1})$. Together, $y_{n+1} \le \lambda_+\sqrt{\eta_{n+1}}$.

Entry. If $y_n > \lambda_+\sqrt{\eta_n}$ for some $n \ge n_4$, then $y_n^2/\hat{u}_n \ge y_n^2/(2c) \ge 2M_1\eta_n$, so $y_{n+1} \le y_n - M_1\eta_n$. While y stays above $\lambda_+\sqrt{\eta} \ge 0$ it decreases by the non-summable amounts $M_1\eta_n$, which cannot go on for ever. So $y_m \le \lambda_+\sqrt{\eta_m}$ for some $m \ge n$, and by the step $y_k \le \lambda_+\sqrt{\eta_k}$ for all $k \ge m$.

Hence $|y_n| = O(\sqrt{\eta_n})$ (the finitely many earlier terms only change the constant). Finally, $|\hat{u}_n - \hat{u}_{n-1}| = |y_n - y_{n-1}| \le y_{n-1}^2/\hat{u}_{n-1} + |e_n| = O(\eta_{n-1}) + O(\eta_n) = O(\eta_n)$ by (2.4). □

3.4. **Proof of Theorem 3.1.** By Lemma 3.3(ii), $x\sigma$ is again a positive measure on $[0,\infty)$ with infinite support and finite moments, and by (3.2) and (3.3),

$$n^{2p}\tilde{b}_n^2 = \hat{u}_n\hat{l}_n = \frac{\varsigma_n\hat{b}_n^2\,\hat{u}_n}{\hat{u}_{n-1}},$$
$$n^p\tilde{a}_n = \hat{u}_n + \Big(\frac{n}{n+1}\Big)^p\hat{l}_{n+1} = \hat{a}_{n+1} - (\hat{u}_{n+1} - \hat{u}_n) + \Big(\Big(\frac{n}{n+1}\Big)^p - 1\Big)\hat{l}_{n+1}.$$

Without a rate. By Lemma 3.4, $\hat{u}_n \to c$ and $\hat{l}_n \to c$, so $n^p\tilde{b}_n \to c$ and $n^p\tilde{a}_n \to 2c$.

With a regular rate η_n. By Lemma 3.5, $\hat{u}_n/\hat{u}_{n-1} = 1 + O(\eta_n)$, since $\hat{u}_{n-1} \ge c/3$ for large n. Also $\varsigma_n = 1 + O(1/n) = 1 + O(\eta_n)$ and $\hat{b}_n = c + O(\eta_n)$. Hence $n^p\tilde{b}_n = \hat{b}_n\big(\varsigma_n\hat{u}_n/\hat{u}_{n-1}\big)^{1/2} = c + O(\eta_n)$. For the diagonal, $\hat{a}_{n+1} = 2c + O(\eta_{n+1})$, $\hat{u}_{n+1} - \hat{u}_n = O(\eta_{n+1})$, $\hat{l}_{n+1}$ is bounded and $(n/(n+1))^p - 1 = O(1/n)$. Since $\eta_{n+1} \le \eta_n$ and $1/n \le \eta_n/\eta_1$, this gives $n^p\tilde{a}_n = 2c + O(\eta_n)$. □

Remark 3.6 (Only increments enter). The bound $|\hat{u}_n - c| = O(\sqrt{\eta_n})$ of Lemma 3.5 cannot be improved under the hypotheses of the lemma. Take $\eta_n = 1/n$, any $\theta_0 > 0$, and define $u_n := n^{-p}(c + \theta_0 n^{-1/2})$ for $n \ge 1$, $u_0 := 1$, $b_n := cn^{-p}$, $l_n := b_n^2/u_{n-1}$ $(n \ge 1)$, $l_0 := 0$ and $a_n := u_n + l_n$. All u_n, l_n are positive. The symmetric tridiagonal matrix with diagonal a_n and off-diagonal b_n equals $\mathcal{L}\mathcal{L}^{\mathsf{T}}$, where $\mathcal{L}$ is lower bidiagonal with diagonal $\sqrt{u_n}$ and subdiagonal $\sqrt{l_{n+1}}$. It is therefore a bounded positive semi-definite operator, so it is the Jacobi matrix of a unique positive measure with infinite compact support in $[0,\infty)$. By Lemma 3.3(i), the u_n of that measure are the ones just defined, because $u_0 = a_0$ and $u_n = a_n - b_n^2/u_{n-1}$ determine them. Here $\hat{b}_n = c$. Since $\hat{l}_n = n^p b_n^2/u_{n-1} = \varsigma_n c^2/(c + \theta_0(n-1)^{-1/2})$, $\varsigma_n = 1 - p/n + O(n^{-2})$ and $(n-1)^{-1/2} = n^{-1/2} + O(n^{-3/2})$, we get $\hat{l}_n = c - \theta_0 n^{-1/2} + (\theta_0^2/c - pc)n^{-1} + O(n^{-3/2})$; the terms $\pm\theta_0 n^{-1/2}$ cancel in $\hat{a}_n = \hat{u}_n + \hat{l}_n$, so $\hat{a}_n - 2c = (\theta_0^2/c - pc)/n + O(n^{-3/2})$. So the law holds with rate $1/n$, while $\hat{u}_n - c = \theta_0\, n^{-1/2}$: the parameters $\hat{u}_n$ themselves converge only at the rate $\sqrt{\eta_n}$, and the transfer keeps the full rate only because $\tilde{a}_n$ and $\tilde{b}_n$ involve $\hat{u}_n$ through the increments $\hat{u}_n - \hat{u}_{n-1}$ and the ratio $\hat{u}_n/\hat{u}_{n-1}$.

4. The Riemann–Hilbert problem and the common frame

This section and the next set up the Riemann–Hilbert analysis on which Theorems 7.1 and 1.4 rest, for every $p > 0$, every $s > 1$ and every ℓ analytic of angle $\vartheta > 0$. We encode the discrete measure by a meromorphic interpolation problem of Fokas–Its–Kitaev type, rescale it by n^p, and normalise it with the g-function of the equilibrium measure μ^* of [15]. After rescaling, the nodes fill $(0, n^p]$. On the band $(0, 4c)$ the orthogonal polynomials oscillate; on the saturated region $(4c, n^p)$ the nodes are as dense as the equilibrium measure allows, and the problem is close to a pure interpolation problem; near 0 the band meets infinitely many nodes, and 0 is a hard edge. The frame is that of Wu, Lin, Xu and Zhao for the Tricomi–Carlitz polynomials (§2.4), written in the one-sided variable x.

The matrices of the analysis are related by

$$Y \xrightarrow{z\mapsto n^p z} U \xrightarrow{\text{poles}} R \xrightarrow{g} T \xrightarrow{\text{lens}} S \xrightarrow{F} V \xrightarrow{D_W} Q.$$

§§4.2–4.7 carry out the steps up to V for any outer scalar F that is admissible in the sense of Definition 4.10. Section 5 constructs F (the only step that depends on the geometry), passes to Q, and collects in Proposition 5.6 everything that the later sections need. Proposition 5.6 is the only result of Sections 4 and 5 that is used in Sections 6–9, and a reader who is willing to take the frame on trust may go to it directly. The slowly varying factor ℓ is evaluated at complex points only in a sector of opening proportional to ϑ around the positive axis (Lemma 4.7); this is why every $\vartheta > 0$ suffices.

4.1. **Standing assumptions.** Throughout Sections 4 and 5 and Appendix A we fix $p > 0$ and $s > 1$, and we put

$$\alpha := \frac{s-1-p}{p} > -1. \tag{4.1}$$

No upper bound on s is assumed. None of the lemmas of these two sections and of Appendix A uses $\alpha < 1$; the one item of Proposition 5.6 that needs $|\alpha| < 1$ says so. We fix $\vartheta > 0$ and $\ell(k) = e^{\Lambda(\log k)}$ analytic of angle ϑ (Definition 1.1), with the constant $c_0 > 0$ of that definition, and we use the same letter for the analytic function $\ell(t) := e^{\Lambda(\log t)}$ of a complex variable t with $\log t \in S_\vartheta$. We write $\sigma := \sigma_{p,s,\ell} = \sum_{k\ge1} w_k \delta_{x_k}$, with $x_k = k^{-p}$ and $w_k = k^{-s}\ell(k)$, and $c = c(p)$, $\gamma_p = \log c + p$. The degree n is large, $n \ge n_0$, where n_0 may increase from line to line. Constants $C, C', \dots$ depend only on p, s, ℓ (including ϑ) and the contour parameters fixed in §4.6; they do not depend on n unless they carry the subscript n. We put

$$\begin{aligned} &\vartheta_{\mathrm w} := \mathsf m\,\vartheta^0_{\mathrm w}, \qquad \vartheta_{\mathrm{lip}} := \mathsf m\,\vartheta^0_{\mathrm{lip}}, \qquad \mathsf m := \min\Big(1, \frac{p\vartheta}{\vartheta^0_{\mathrm w}}\Big), \\ &(\vartheta^0_{\mathrm w}, \vartheta^0_{\mathrm{lip}}) := \begin{cases} (7\pi/16,\ \pi/3), & p \ge 1, \\ (3\pi p/8,\ \pi p/4), & 0 < p < 1. \end{cases} \end{aligned} \tag{4.2}$$

Then $\mathsf m \in (0,1]$, and

$$0 < \vartheta_{\mathrm{lip}} < \tfrac54\vartheta_{\mathrm{lip}} < \vartheta_{\mathrm w} < \tfrac{\pi}{2}, \qquad \vartheta_{\mathrm w} < \tfrac{\pi p}{2} \ \text{ if } p < 1, \qquad \vartheta_{\mathrm w} \le p\vartheta, \tag{4.3}$$

since the first two chains hold for the default angles $\vartheta^0_{\mathrm w}, \vartheta^0_{\mathrm{lip}}$ and are preserved by the common factor $\mathsf m$, and $\mathsf m\vartheta^0_{\mathrm w} \le p\vartheta$ by the choice of $\mathsf m$. The angle $\vartheta_{\mathrm w}$ bounds the arguments at which the weight is ever evaluated (Lemma 4.7); ϑ_{lip} is the angle at which the lens leaves the hard edge. Only the relations (4.3) are used below, never the particular values; see Remark 4.8.

4.2. **The interpolation problem.** Let P_n be the monic orthogonal polynomials of σ and $h_n = \int P_n^2 \,\mathrm d\sigma$. For a 2×2 matrix function X with an expansion $X(z) = I + X_1/z + X_2/z^2 + O(z^{-3})$ at ∞ we write

$$\mathsf b(X) := (X_1)_{12}(X_1)_{21}, \qquad \mathsf a(X) := \frac{(X_2)_{12}}{(X_1)_{12}} - (X_1)_{22} \quad \text{(the latter when } (X_1)_{12} \ne 0). \tag{4.4}$$

Lemma 4.1 (Interpolation problem)**.** *Let Y be the matrix (2.5) of §2.4(a), built from the monic orthogonal polynomials P_n, P_{n-1} of σ and $h_{n-1} = \int P_{n-1}^2 \,\mathrm d\sigma$. Then $\det Y \equiv 1$. The rescaled matrix*

$$U(z) := n^{pn\sigma_3}\, Y(zn^{-p}) \tag{4.5}$$

is analytic off $\{X_k\}_{k\ge1} \cup \{0\}$, where

$$X_k := n^p x_k = (n/k)^p \quad (k \ge 1), \qquad X_1 = n^p \ \textit{(the top node)}; \tag{4.6}$$

it has simple poles at the X_k in its second column only, with $\operatorname{Res}_{z=X_k} U_{j2} = n^p w_k\, U_{j1}(X_k)$ *($j=1,2$); and* $U(z)z^{-n\sigma_3} = I + U_1/z + U_2/z^2 + \dots$ *for $|z| > n^p$, with*

$$\mathsf{b}(Uz^{-n\sigma_3}) = n^{2p} b_n^2, \qquad \mathsf{a}(Uz^{-n\sigma_3}) = n^p a_n. \tag{4.7}$$

Proof. For the determinant, write $h_{n-1}\det Y$ as

$$\int \frac{P_n(z)P_{n-1}(x) - P_{n-1}(z)P_n(x)}{z-x}\,\mathrm{d}\sigma(x) = \int \frac{P_n(z)-P_n(x)}{z-x} P_{n-1}(x)\,\mathrm{d}\sigma(x) - \int \frac{P_{n-1}(z)-P_{n-1}(x)}{z-x} P_n(x)\,\mathrm{d}\sigma(x).$$

The first difference quotient is a polynomial in x of degree $n-1$ with leading coefficient 1, so the first integral is h_{n-1}; the second has degree $n-2$ in x, so the second integral vanishes. Hence $\det Y \equiv 1$. The statements for U follow from those of §2.4(a) by the substitution $z = n^p\zeta$: the residue picks up the factor n^p from $\mathrm{d}z = n^p\,\mathrm{d}\zeta$, and $U_j = n^{jp}\, n^{pn\sigma_3} Y_j n^{-pn\sigma_3}$ ($j=1,2$). The constant conjugation by $n^{pn\sigma_3}$ does not change a and b (Lemma 4.2 below with k constant), and the factors n^{jp} give $\mathsf{b}(Uz^{-n\sigma_3}) = n^{2p}\mathsf{b}(Yz^{-n\sigma_3})$ and $\mathsf{a}(Uz^{-n\sigma_3}) = n^p\mathsf{a}(Yz^{-n\sigma_3})$. Now (2.6) gives (4.7). □

Every transformation below acts, near infinity, by a constant diagonal matrix on the left and a scalar diagonal factor on the right. The extraction functionals do not see such factors.

Lemma 4.2 (Invariance of the extraction functionals)**.** *Let $X = I + X_1/z + X_2/z^2 + O(z^{-3})$, and let k be a scalar function, analytic and non-zero near infinity, with $k(z) = k_\infty\big(1 + k_1/z + k_2/z^2 + O(z^{-3})\big)$, $k_\infty \neq 0$. Then $\hat X := k_\infty^{-\sigma_3} X k^{\sigma_3}$ satisfies $\hat X \to I$, $\mathsf{b}(\hat X) = \mathsf{b}(X)$ and $\mathsf{a}(\hat X) = \mathsf{a}(X)$. Conversely, if C is a constant matrix and $CXk^{\sigma_3} \to I$, then $C = k_\infty^{-\sigma_3}$.*

Proof. Put $Z_j := k_\infty^{-\sigma_3} X_j k_\infty^{\sigma_3}$; then $(Z_j)_{12} = k_\infty^{-2}(X_j)_{12}$, $(Z_j)_{21} = k_\infty^2 (X_j)_{21}$, and the diagonal entries are unchanged. Since $k^{\sigma_3} = k_\infty^{\sigma_3}\operatorname{diag}\big(1 + k_1 z^{-1} + k_2 z^{-2},\, 1 - k_1 z^{-1} + (k_1^2 - k_2) z^{-2}\big) + O(z^{-3})$,

$$\hat X_1 = Z_1 + k_1\sigma_3, \qquad \hat X_2 = Z_2 + Z_1\operatorname{diag}(k_1, -k_1) + \operatorname{diag}(k_2, k_1^2 - k_2).$$

Hence $(\hat X_1)_{12}(\hat X_1)_{21} = (Z_1)_{12}(Z_1)_{21} = \mathsf{b}(X)$, and with $(\hat X_2)_{12} = (Z_2)_{12} - k_1(Z_1)_{12}$ and $(\hat X_1)_{22} = (X_1)_{22} - k_1$,

$$\mathsf{a}(\hat X) = \frac{(Z_2)_{12}}{(Z_1)_{12}} - k_1 - (X_1)_{22} + k_1 = \frac{(X_2)_{12}}{(X_1)_{12}} - (X_1)_{22} = \mathsf{a}(X).$$

If $CXk^{\sigma_3} \to I$ then $Ck_\infty^{\sigma_3} = I$. □

We never use that Y is the *unique* solution of an interpolation problem, and no existence theorem for a Riemann–Hilbert problem is needed: every object below is built from Y explicitly.

4.3. **The index function and the band weight.** The node set $\{X_k\}$ is the set of points where the *index function*

$$t(z) := nz^{-1/p} \qquad (z \in \mathbb{C}\setminus(-\infty, 0],\ \text{principal power}) \tag{4.8}$$

takes positive integer values: $t(X_k) = k$. We put

$$\theta(z) := \pi t(z), \qquad \Pi(z) := \frac{\sin\theta(z)}{\theta'(z)}, \qquad \theta'(z) = -\frac{\pi}{p}\,\frac{t(z)}{z}, \tag{4.9}$$

so that Π has simple zeros at the nodes. On $\mathbb{C}\setminus(-\infty, 0]$ we have $\arg t = -\arg z/p$, hence

$$\pm\operatorname{Im}\theta(z) < 0 \quad \text{for } 0 < \pm\arg z < \min(\pi, \pi p), \qquad |e^{\mp 2i\theta(z)}| = \exp\big(-2\pi n|z|^{-1/p}\sin(|\arg z|/p)\big). \tag{4.10}$$

The weight enters through the log variable $\xi := \log t = \log n - (\operatorname{Log} z)/p$. Let

$$\mathcal{Z}_n := \{z : \ 0 < |z| < e^{pc_0} n^p, \ |\arg z| < \vartheta_{\mathrm{w}}\}. \tag{4.11}$$

For $z \in \mathcal{Z}_n$ we have $|\operatorname{Im}\xi| < \vartheta_{\mathrm{w}}/p \le \vartheta$ and $\operatorname{Re}\xi > -c_0$, so $\xi \in S_\vartheta$ and the *residue weight* and the *slowly varying factor*

$$w(z) := n^p\, t(z)^{-s}\, \ell(t(z)), \qquad r_n(z) := \frac{\ell(t(z))}{\ell(n)} = e^{\Lambda(\xi)-\Lambda(\log n)} \tag{4.12}$$

are analytic on $\mathcal{Z}_n$. By Lemma 4.1, $\operatorname{Res}_{X_k} U_{j2} = w(X_k)U_{j1}(X_k)$. The combination that appears in all jumps is

$$W(z) := 2i\theta'(z)w(z) \qquad (z \in \mathcal{Z}_n). \tag{4.13}$$

Lemma 4.3 (Band weight)**.** *On $\mathcal{Z}_n$,*

$$W(z) = \mathsf{w}_n\, z^\alpha\, r_n(z), \qquad \mathsf{w}_n := -\frac{2\pi i}{p}\, n^{1+p-s}\ell(n), \tag{4.14}$$

with the principal power. In particular $|W(x)| = \frac{2\pi}{p} n^{1+p-s}\ell(n)\, x^\alpha r_n(x)$ and $\arg W(x) = -\pi/2$ for $x \in (0, e^{pc_0}n^p)$, where $r_n(x) > 0$. The weight has a hard edge at 0 with Bessel order α, and $\alpha > -1$ if and only if $s > 1$.

Proof. By (4.9), $2i\theta' w = -\frac{2\pi i}{p}\frac{t}{z} n^p t^{-s}\ell(t)$, and $t^{1-s} = n^{1-s} z^{(s-1)/p}$. The exponent of z is $(s-1)/p - 1 = \alpha$. Finally $\ell(t) = \ell(n) r_n$, and $r_n > 0$ on $(0,\infty)$ because Λ is real on the real axis. □

4.4. **The g-function.** We use the equilibrium measure $\mu^* = \int_0^1 \operatorname{arc}_{\beta_\tau}\, \mathrm{d}\tau$, its density ψ^*, the node density λ_p and the potential U^* exactly as in Proposition 2.3, and we recall $\int_x^\infty \lambda_p = x^{-1/p} = t(x)/n$. Put

$$g(z) := \int \log(z-y)\, \mathrm{d}\mu^*(y) \quad (z \in \mathbb{C}\setminus\mathbb{R}), \qquad \varphi := \gamma_p - g, \qquad \tilde\varphi := \varphi \pm \frac{i\theta}{n} \ \text{ in } \mathbb{C}_\pm, \tag{4.15}$$

with principal logarithms. Nothing here depends on ℓ: the external field of the weight is $\frac{1}{n}\log w = O(\frac{\log n}{n})$ on the contour (Lemma 4.9) and is not put into g. Since $\theta/n = \pi z^{-1/p}$, the function $\tilde\varphi$ does not depend on n either.

Lemma 4.4 (g-function)**.** (i) *g has continuous boundary values $g_\pm$ on $\mathbb{R}\setminus\{0\}$, $g_+ - g_- = 2\pi i$ on $(-\infty, 0)$ and $g_+ - g_- = 2\pi i \mu^*((x,\infty))$ on $(0,\infty)$. In particular $\varphi_+ + \varphi_- = 0$ on $(0, 4c)$, and $\tilde\varphi$ has no jump on $(4c,\infty)$.*

(ii) *$g(z) - \operatorname{Log} z \to 0$ as $z \to \infty$, uniformly in $\mathbb{C}\setminus\mathbb{R}$.*

(iii) *U^* is strictly increasing on $[4c,\infty)$, and $U^*(x) = \log x + o(1)$ as $x \to \infty$. Hence $\operatorname{Re}\tilde\varphi(x) = \gamma_p - U^*(x) < 0$ for $x > 4c$, and $\operatorname{Re}\tilde\varphi(x) \le -\frac{1}{2}\log x$ for $x \ge x_0$.*

Proof. (i) For $y > 0$, $\log(z-y)$ jumps by $2\pi i$ across $(-\infty, y)$. Integrating against μ^* gives the jumps; continuity of the boundary values follows from the continuity of ψ^* on $(0,\infty)$. On $(0,4c)$, $g_+ + g_- = 2U^* = 2\gamma_p$, so $\varphi_+ + \varphi_- = 0$. On $(4c,\infty)$, $\mu^*((x,\infty)) = \int_x^\infty \lambda_p = t(x)/n$, so $g_+ - g_- = 2i\theta/n$, and $\tilde\varphi_+ - \tilde\varphi_- = -(g_+ - g_-) + 2i\theta/n = 0$.

(ii) For $z \in \mathbb{C}_+$ and $y > 0$ we have $\operatorname{Log}(z-y) = \operatorname{Log} z + \operatorname{Log}(1 - y/z)$, because $\arg(z-y) - \arg z \in (0,\pi)$; the case $z \in \mathbb{C}_-$ is the mirror image. So $g(z) - \operatorname{Log} z = \int \operatorname{Log}(1-y/z)\, \mathrm{d}\mu^*(y)$. On $\{y \le |z|/2\}$ we use $|\operatorname{Log}(1-y/z)| \le 2y/|z|$ and $\int_{y\le R} y\, \mathrm{d}\mu^* \le 1 + \int_1^R y\lambda_p(y)\, \mathrm{d}y = o(R)$ (as $\psi^* \le \lambda_p$). On $\{y > |z|/2\}$ we use $\psi^* \le \lambda_p \le C|z|^{-1-1/p}$ there and $|\operatorname{Log}(1-y/z)| \le \pi + \log(1+u) + |\log|1-u||$ with $u = y/|z|$, which follows from $|1-u| \le |1-y/z| \le 1+u$. The resulting integral is $O(|z|^{-1/p})$. Both parts tend to 0 uniformly.

(iii) For $\beta > 0$ put $\mathfrak{g}_\beta(z) := 2\log\big((\sqrt{z} + \sqrt{z-\beta})/2\big)$. It is analytic off $(-\infty, \beta]$, equals $\log z + o(1)$ at ∞, and $\mathfrak{g}'_\beta(z) = 1/\sqrt{z(z-\beta)} = \int \mathrm{d}\,\mathrm{arc}_\beta(y)/(z-y)$, the Stieltjes transform of the arcsine law. Hence $\mathfrak{g}_\beta(z) = \int \log(z-y)\,\mathrm{d}\,\mathrm{arc}_\beta(y)$. Taking real parts, $\mathfrak{u}_\beta(x) := \int \log|x-y|\,\mathrm{d}\,\mathrm{arc}_\beta(y)$ equals $\log(\beta/4)$ on $[0,\beta]$ (there $|\sqrt{x} + i\sqrt{\beta - x}|^2 = \beta$), and it is strictly increasing on $[\beta, \infty)$. So $U^* = \int_0^1 \mathfrak{u}_{\beta_\tau}\,\mathrm{d}\tau$ is non-decreasing on $(0,\infty)$. For $x > 4c$ the set of τ with $\beta_\tau < x$ is the interval $((4c/x)^{1/p}, 1]$, of positive length, on which $\mathfrak{u}_{\beta_\tau}$ is strictly increasing at x. So U^* is strictly increasing on $[4c, \infty)$. For the asymptotics, $U^*(x) - \log x = \int \log|1 - y/x|\,\mathrm{d}\mu^*(y)$, and the argument of (ii) on the real axis shows that this tends to 0. Finally, $\operatorname{Re}\tilde{\varphi} = \operatorname{Re}\varphi = \gamma_p - U^*$ on $(4c,\infty)$, because θ is real there; $U^*(4c) = \gamma_p$, which gives the sign, and $U^*(x) \geq \frac{1}{2}\log x + \gamma_p$ for large x. □

4.5. **Signs and the local variables at the two edges.** The lens opening needs $\operatorname{Re}\varphi < 0$ off the band, and the triangular factors that appear later need $\operatorname{Re}\tilde{\varphi} > 0$ just above and below the band. Near the edges we need the local variables of the Bessel and Airy models.

Lemma 4.5 (Signs and local variables)**.** (i) $\operatorname{Re}\varphi(z) < 0$ *for every* $z \in \mathbb{C}\setminus[0,4c]$ *(for* z *on* $(4c,\infty)$ *and* $(-\infty,0)$ *this refers to the common real part of the two boundary values), and* $\operatorname{Re}\varphi_\pm = 0$ *on* $(0,4c)$.

(ii) *For every compact* $K \subset (0,4c)$ *there is* $y_K > 0$ *such that* $\operatorname{Re}\tilde{\varphi}(x \pm iy) > 0$ *for* $x \in K$ *and* $0 < y \leq y_K$. *Moreover* $\partial_y \operatorname{Re}\tilde{\varphi}(x+iy)|_{y=0^+} = \pi(\lambda_p - \psi^*)(x) > 0$ *on* $(0,4c)$.

(iii) *(Hard edge.) There are* $\delta_0 > 0$ *and a function* f, *analytic and univalent on* $\mathbb{D}(0,2\delta_0)$, *real on the real axis, with* $f(0) = 0$ *and* $f'(0) < 0$, *such that* $(\varphi \pm i\pi)^2 = 4f$ *on* $\mathbb{D}(0,2\delta_0)\cap\mathbb{C}_\pm$. *More precisely* $\varphi = \mp i\pi - 2f^{1/2}$ *in* $\mathbb{C}_\pm$, *with the principal root. In particular* f *does not depend on* n, $|\operatorname{Re}\varphi(z)| \leq C|z|^{1/2}$ *on* $\mathbb{D}(0,\delta_0)$, *and*

$$\operatorname{Re}\tilde{\varphi}(z) \geq |\arg z|\Big(\frac{2}{p}|z|^{-1/p} - C|z|^{1/2}\Big) \qquad (z \in \mathbb{D}(0,\delta_0),\ 0 < |\arg z| \leq \vartheta_{\mathrm{w}}). \tag{4.16}$$

(iv) *(Soft edge.) There are* $\varepsilon_0 > 0$ *and a function* f_r, *analytic and univalent on* $\mathbb{D}(4c, 2\varepsilon_0)$, *real on the real axis, with* $f_r(4c) = 0$ *and* $f_r'(4c) > 0$, *such that* $\tilde{\varphi}^2 = \frac{4}{9}f_r^3$ *on* $\mathbb{D}(4c,2\varepsilon_0)$, *and* $\tilde{\varphi} = -\frac{2}{3}f_r^{3/2}$ *(principal branch, cut along* $f_r^{-1}((-\infty,0])$*). Hence, for* $z \in \mathbb{D}(4c,2\varepsilon_0)\setminus\mathbb{R}$,

$$\operatorname{Re}\tilde{\varphi}(z) > 0 \iff \tfrac{\pi}{3} < |\arg f_r(z)| < \pi, \qquad \operatorname{Re}\tilde{\varphi}(z) < 0 \iff |\arg f_r(z)| < \tfrac{\pi}{3},$$

and $\operatorname{Re}\tilde{\varphi} < 0$ *on* $(4c, 4c+2\varepsilon_0)$. *Shrinking* ε_0, *we may assume* $|\arg f_r(z) - \arg(z-4c)| \leq \pi/12$ *on* $\mathbb{D}(4c,2\varepsilon_0)\setminus\mathbb{R}$; *in particular* $\operatorname{Re}\tilde{\varphi} > 0$ *on the vertical segments* $\{4c \pm iy : 0 < y < 2\varepsilon_0\}$.

Proof. (i) We have $\operatorname{Re}\varphi(z) = \gamma_p - \int \log|z-y|\,\mathrm{d}\mu^*(y)$. If $\operatorname{Re} z > 0$ and $\operatorname{Im} z \neq 0$, then $|z-y| > |\operatorname{Re} z - y|$ for all y, so $\operatorname{Re} g(z) > U^*(\operatorname{Re} z) \geq \gamma_p$ (Proposition 2.3 and Lemma 4.4(iii)). On $(4c,\infty)$, $\operatorname{Re} g_\pm = U^* > \gamma_p$ by Lemma 4.4(iii). If $\operatorname{Re} z \leq 0$ and $z \neq 0$, choose $0 < x_0 \leq \min(|z|/2, 4c)$. Then $|z-y| \geq y > y - x_0$ for $y \geq x_0$, and $|z-y| \geq |z| > x_0 \geq |x_0 - y|$ for $0 < y < x_0$; so $|z-y| > |x_0 - y|$ for every $y > 0$, and integrating against μ^* gives $\operatorname{Re} g(z) > U^*(x_0) = \gamma_p$. On $(0,4c)$, $\operatorname{Re} g_\pm = U^* = \gamma_p$.

(ii) In $\mathbb{C}_+$, $\tilde{\varphi} = \varphi + i\theta/n = \gamma_p - g + i\pi z^{-1/p}$. Since ψ^* is real-analytic on $(0,4c)$ (it is λ_p times a regularised incomplete beta function of $x/4c$), $g|_{\mathbb{C}_+}$ continues analytically across $(0,4c)$, and so does $\tilde{\varphi}$. By the Cauchy–Riemann equations, $\partial_y \operatorname{Re} g(x+i0) = -\operatorname{Im} g'(x+i0) = \pi\psi^*(x)$ and $\partial_y \operatorname{Re}(i\pi z^{-1/p})|_{z=x} = -\pi\,\frac{\mathrm{d}}{\mathrm{d}x}x^{-1/p} = \pi\lambda_p(x)$. Since $\operatorname{Re}\tilde{\varphi} = 0$ on $(0,4c)$, a Taylor expansion with a uniform remainder on K gives the claim. $\mathbb{C}_-$ is the mirror image.

(iii) On $(0,4c)$, $\varphi_+(x)=\gamma_p-U^*(x)-i\pi\mu^*((x,\infty))=-i\pi+i\pi\mu^*((0,x])$. By Proposition 2.3, $\psi^*(x)=\lambda_p(x)\,\mathcal{B}_{x/4c}(\frac1p+\frac12,\frac12)$ near 0, and $\mathcal{B}_y(a',\frac12)=y^{a'}\mathfrak{a}_0(y)$ with $\mathfrak{a}_0$ analytic near 0 and $\mathfrak{a}_0(0)>0$. Since $\lambda_p(x)(x/4c)^{1/p+1/2}=Cx^{-1/2}$, we get $\mu^*((0,x])=x^{1/2}\mathfrak{a}(x)$ with $\mathfrak{a}$ analytic near 0 and $\mathfrak{a}(0)>0$ (in fact $\mathfrak{a}(0)=2C_*$ by Proposition 2.3(iii)). Put $f(z):=-\frac{\pi^2}{4}z\mathfrak{a}(z)^2$. Then f is analytic and real on the real axis, $f(0)=0$, $f'(0)=-\pi^2\mathfrak{a}(0)^2/4<0$, so f is univalent on a disc $\mathbb{D}(0,2\delta_0)$. The functions $(\varphi+i\pi)^2$ (on $\mathbb{C}_+$) and $4f$ are analytic in $\mathbb{D}(0,2\delta_0)\cap\mathbb{C}_+$ and have the same continuous boundary values $-\pi^2x\mathfrak{a}(x)^2$ on $(0,2\delta_0)$; by the reflection principle they coincide. The same holds in $\mathbb{C}_-$ with $\varphi-i\pi$. In $\mathbb{D}(0,2\delta_0)\cap\mathbb{C}_+$ the two analytic functions $f^{1/2}$ (principal) and $-\frac{i\pi}{2}z^{1/2}\mathfrak{a}(z)$ have the same square, so they differ by a constant sign; on the band both equal $-\frac{i\pi}{2}x^{1/2}\mathfrak{a}(x)$ (there $f(x+i0)$ lies on the negative axis approached from below), so they are equal, and $\varphi_++i\pi=-2f^{1/2}$ on the band gives $\varphi=-i\pi-2f^{1/2}$ in $\mathbb{D}(0,2\delta_0)\cap\mathbb{C}_+$. The bound on $\operatorname{Re}\varphi$ follows from $|f(z)|\le C|z|$. For (4.16) let $z=re^{i\phi}$ with $0<\phi\le\vartheta_{\mathrm w}$ (the lower half-disc is the mirror image). Then

$$\operatorname{Re}\tilde\varphi(z)=\operatorname{Re}\bigl(-2f^{1/2}+i\pi z^{-1/p}\bigr)=\pi r^{-1/p}\sin(\phi/p)-\pi\operatorname{Im}\bigl(z^{1/2}\mathfrak{a}(z)\bigr).$$

Since $z^{1/2}\mathfrak{a}(z)$ is real for $\phi=0$ and $|\partial_\phi(z^{1/2}\mathfrak{a}(z))|=|z(z^{1/2}\mathfrak{a})'(z)|\le Cr^{1/2}$, its imaginary part is at most $Cr^{1/2}\phi$; and $\sin(\phi/p)\ge\frac{2}{\pi}\frac{\phi}{p}$ because $\phi/p\le\vartheta_{\mathrm w}/p<\pi/2$.

(iv) On $(4c-2\varepsilon_0,4c)$ we have $\tilde\varphi_++\tilde\varphi_-=\varphi_++\varphi_-=0$ and, using Lemma 4.4(i), $\tilde\varphi_+-\tilde\varphi_-=2\pi i\int_x^\infty(\lambda_p-\psi^*)=2\pi i\int_x^{4c}(\lambda_p-\psi^*)$. Near $y=1$, $1-\mathcal{B}_y(a',\frac12)=(1-y)^{1/2}\mathfrak{a}_1(y)$ with $\mathfrak{a}_1$ analytic and $\mathfrak{a}_1(1)>0$, so $\tilde\varphi_+(x)=i\pi(4c-x)^{3/2}\mathfrak{a}_2(x)$ with $\mathfrak{a}_2$ analytic near $4c$ and $\mathfrak{a}_2(4c)>0$. As in (iii), $\tilde\varphi^2$ and $\mathfrak{h}(z):=\pi^2(z-4c)^3\mathfrak{a}_2(z)^2$ are analytic in $\mathbb{D}(4c,2\varepsilon_0)\setminus[4c-2\varepsilon_0,4c+2\varepsilon_0]$ with equal boundary values on $(4c-2\varepsilon_0,4c)$, and $\tilde\varphi$ has no jump on $(4c,4c+2\varepsilon_0)$ (Lemma 4.4(i)). So $\tilde\varphi^2=\mathfrak{h}$. Put $f_r:=(9\mathfrak{h}/4)^{1/3}=(\frac{3\pi}{2})^{2/3}(z-4c)\mathfrak{a}_2(z)^{2/3}$. On $(4c,\infty)$, $\tilde\varphi$ is real and negative (Lemma 4.4(iii)), which fixes the branch $\tilde\varphi=-\frac23 f_r^{3/2}$. Then $\operatorname{Re}\tilde\varphi=-\frac23|f_r|^{3/2}\cos(\frac32\arg f_r)$, which is positive exactly when $\frac32|\arg f_r|\in(\frac{\pi}{2},\frac{3\pi}{2})$. Finally $f_r(z)=f_r'(4c)(z-4c)(1+O(z-4c))$ with $f_r'(4c)>0$ gives the statement about $\arg f_r$, and on the vertical segments $|\arg f_r|\in[\frac{5\pi}{12},\frac{7\pi}{12}]$. □

4.6. **The contour.** The lens is bounded by two arcs, the lips, on which the triangular factors of the lens are exponentially small. We first show that such arcs exist, with all the properties we need.

Lemma 4.6 (Lips)**.** *There is $\delta_1>0$ such that for every $\delta\in(0,\delta_1]$ there is $h_1=h_1(\delta)>0$ with the following property. For every $h_0\in(0,h_1]$ there are C^2 arcs $\Sigma^\pm_{\mathrm{lip}}\subset\mathbb{C}_\pm$ from 0 to $4c\pm ih_0$, mirror images of each other and independent of n, such that:*

- (a) *$\Sigma^\pm_{\mathrm{lip}}\cap\mathbb{D}(0,\delta)$ is the preimage under f (equivalently, under $\zeta=n^2f(z)$) of a segment of the ray $\arg\zeta=\mp(\pi-\vartheta_{\mathrm{lip}})$; on it $\frac34\vartheta_{\mathrm{lip}}\le|\arg z|\le\frac54\vartheta_{\mathrm{lip}}$;*
- (b) *on $\Sigma^\pm_{\mathrm{lip}}\setminus\{0,4c\pm ih_0\}$: $|\arg z|<\vartheta_{\mathrm w}$, $\operatorname{Re}\varphi<0$ and $\operatorname{Re}\tilde\varphi>0$;*
- (c) *$\Sigma^\pm_{\mathrm{lip}}\subset\{\operatorname{Re}z\le4c\}\setminus\mathbb{D}(4c,h_0)$, and $\Sigma^\pm_{\mathrm{lip}}$ meets the line $\operatorname{Re}z=4c$ only at $4c\pm ih_0$.*

Proof. We construct the upper lip from four pieces and round the three corners inside the open set $G_+:=\{z\in\mathbb{C}_+:\ |\arg z|<\vartheta_{\mathrm w},\ \operatorname{Re}\tilde\varphi(z)>0\}$, which contains every piece except its two end points, and outside $\overline{\mathbb{D}(0,\delta)}$; the lower lip is the mirror image. By Lemma 4.5(i), $\operatorname{Re}\varphi<0$ on all of $\mathbb{C}_+$, so only $\operatorname{Re}\tilde\varphi>0$ and $|\arg z|<\vartheta_{\mathrm w}$ have to be checked.

Choose $r_*\in(0,\delta_0)$ with $\frac2p r^{-1/p}>Cr^{1/2}$ for $0<r\le r_*$; then (4.16) gives $S_0:=\{0<|z|\le r_*,\ 0<\arg z<\vartheta_{\mathrm w}\}\subset G_+$. Since $f(z)=f'(0)z(1+O(z))$ with $f'(0)<0$ and f is univalent, the preimage of the ray $\arg\zeta=-(\pi-\vartheta_{\mathrm{lip}})$ near 0 is an analytic arc leaving 0 at the angle ϑ_{lip},

and for $\delta_1 \le r_*/4$ small its part in $\mathbb{D}(0,2\delta)$ satisfies $\frac{3}{4}\vartheta_{\mathrm{lip}} \le \arg z \le \frac{5}{4}\vartheta_{\mathrm{lip}}$ (as $\frac{5}{4}\vartheta_{\mathrm{lip}} < \vartheta_{\mathrm{w}}$, it lies in S_0). This is the first piece; let z_1 be its end point on $|z| = 2\delta$, so that the corner at z_1 can be rounded in $\{\delta < |z| < 2\delta\}$ and (a) is not affected. The second piece runs in S_0: along the ray $\arg z = \arg z_1$ to $|z| = \frac{3}{4}r_*$, and then along the circle $|z| = \frac{3}{4}r_*$ down to height $y_1 := h_0 \sin\frac{\pi}{12}$.

Let $K := [\frac{1}{2}r_*, 4c - \frac{1}{2}\varepsilon_0]$ and let y_K be as in Lemma 4.5(ii); choose $h_1 \le \min(y_K, \frac{1}{2}\varepsilon_0)$ so small that the angle ϕ with $\frac{3}{4}r_* \sin\phi = y_1$ satisfies $\phi < \arg z_1$ and $\frac{3}{4}r_* \cos\phi \ge \frac{1}{2}r_*$, that $\arctan(2y_1/r_*) < \vartheta_{\mathrm{w}}$, and that $\arcsin(h_0/4c) < \vartheta_{\mathrm{w}}$ (the largest argument on the fourth piece). The third piece is the horizontal segment from the end of the second piece to the point $4c + h_0 e^{i11\pi/12}$, at height y_1. For $\operatorname{Re} z \in K$ it lies in G_+ by Lemma 4.5(ii), as $y_1 < y_K$. For $4c - \frac{1}{2}\varepsilon_0 \le \operatorname{Re} z \le 4c - h_0 \cos\frac{\pi}{12}$ it lies in $\mathbb{D}(4c, \varepsilon_0)$ with $\arg(z - 4c) \ge \frac{11\pi}{12}$, hence $\frac{5\pi}{6} \le \arg f_r(z) < \pi$ and $\operatorname{Re}\tilde{\varphi} > 0$ by Lemma 4.5(iv). The fourth piece is the arc $\{4c + h_0 e^{i\phi} : \ \frac{\pi}{2} \le \phi \le \frac{11\pi}{12}\}$, on which $\frac{5\pi}{12} \le \arg f_r < \pi$, so again $\operatorname{Re}\tilde{\varphi} > 0$ (except at the end point $4c + ih_0$ itself, where $\arg f_r \in [\frac{5\pi}{12}, \frac{7\pi}{12}]$ also gives $\operatorname{Re}\tilde{\varphi} > 0$). All four pieces have arguments below ϑ_{w} by the choice of h_1, and none depends on n, since f, f_r and $\tilde{\varphi}$ do not. Property (c) holds by construction. □

We fix, once and for all and independently of n, numbers $\delta, h_0, \varepsilon, \delta'$ with

$$0 < \delta \le \min\bigl(\delta_1, \tfrac{1}{4}, \tfrac{c}{2}\bigr), \qquad 0 < h_0 \le \min\bigl(h_1(\delta), \tfrac{1}{2}\delta \tan\vartheta_{\mathrm{w}}\bigr),$$
$$0 < \varepsilon < \min(h_0, \varepsilon_0, c\sin\vartheta_{\mathrm{w}}), \qquad 0 < \delta' < 1,$$

(the condition $\varepsilon < c\sin\vartheta_{\mathrm{w}}$ gives $\mathbb{D}(4c, 4\varepsilon) \subset \mathcal{Z}_n$, which Lemma 5.2(vi)(b) uses), and the following arcs (Figure 4.1).

- The *band* $[0, 4c]$, the *saturated segment* $(4c, n^p + \delta')$ and the *top segment* $\Sigma_{\mathrm{top}} := \{n^p + \delta' + iy : \ 0 < |y| \le h_0\}$. The top node $X_1 = n^p$ lies inside, and no node lies on $[n^p + \delta', \infty)$.
- The *lips* $\Sigma^{\pm}_{\mathrm{lip}}$ of Lemma 4.6.
- The *vertical segments* $\Sigma^{\pm}_v := \{4c \pm iy : \ 0 < y \le h_0\}$.
- The *saturated lips* $\Sigma^{\triangle\pm} := \{x \pm ih_0 : \ 4c \le x \le n^p + \delta'\}$.

These arcs bound the *lens* $\Omega^{\pm}_{\mathrm{b}}$ (between the band and $\Sigma^{\pm}_{\mathrm{lip}}$, left of $\Sigma^{\pm}_v$) and the *saturated regions* $\Omega^{\triangle}_{\pm} := \{4c < \operatorname{Re} z < n^p + \delta', \ 0 < \pm\operatorname{Im} z < h_0\}$. The rest of $\mathbb{C}$ is Ω_∞. We write Σ for the union of all these arcs. Only finitely many arcs meet at each vertex, and for fixed n, Σ is bounded. By Lemma 4.6(c) the lips meet the saturated lips only at $4c \pm ih_0$. In $\mathbb{D}(4c, \varepsilon)$ the only arcs are the band, the saturated segment and $\Sigma^{\pm}_v$; at 0 only the band and the two lips meet. Nothing lies on $(-\infty, 0)$ or on $(n^p + \delta', \infty)$.

The closures of $\Omega^{\pm}_{\mathrm{b}}$ and $\Omega^{\triangle}_{\pm}$, minus the point 0, lie in $\mathcal{Z}_n$: the lens lies in $|\arg z| < \vartheta_{\mathrm{w}}$ by Lemma 4.6(b); every point of $\Omega^{\triangle}$ has $\operatorname{Re} z \ge 4c$ and $|\operatorname{Im} z| \le h_0 < 4c\tan\vartheta_{\mathrm{w}}$, so $|\arg z| < \vartheta_{\mathrm{w}}$; and $|z| \le n^p + \delta' + h_0 < e^{pc_0} n^p$ for $n \ge n_0$.

4.7. **The transformations.** We now remove the poles, normalise at infinity and open the lens. The pattern is that of §2.4(b); since we work in a one-sided variable, we write every factor out. Put, in $\mathbb{C}_\pm$,

$$\upsilon_\pm := \frac{1}{1 - e^{\mp 2i\theta}}, \qquad \text{so that} \qquad \upsilon_+ - 1 = e^{-2i\theta}\upsilon_+, \quad \upsilon_- = e^{2i\theta}\upsilon_- + 1. \tag{4.17}$$

By (4.10), $|e^{\mp 2i\theta}| < 1$ in $\mathbb{C}_\pm \cap \mathcal{Z}_n$, so $\upsilon_\pm$ is analytic there. Define, on $\mathcal{Z}_n \cap \mathbb{C}_\pm$,

$$L_\pm := \begin{pmatrix} 1 & \mp W e^{\mp 2i\theta}\upsilon_\pm \\ 0 & 1 \end{pmatrix}, \qquad A_\pm := \begin{pmatrix} 1 & 0 \\ \mp e^{2n\varphi}/W & 1 \end{pmatrix}, \qquad A^{\triangle}_\pm := \begin{pmatrix} 1 & 0 \\ \mp 1/(\upsilon_\pm W) & 1 \end{pmatrix}. \tag{4.18}$$

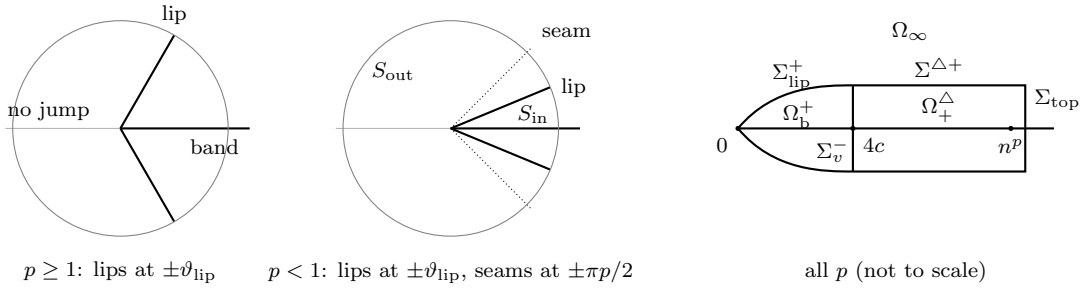


FIGURE 4.1. The contour near the hard edge for $p \geq 1$ (the local pictures for $p > 1$ and $p = 1$ coincide) and for $p < 1$, and the whole contour. The lips are drawn at the default angles $\vartheta_{\text{lip}} = \pi/3$ and $\pi p/4$ (with $p = \frac{1}{2}$); for small ϑ they are rotated towards the band. The negative axis and, for $p < 1$, the seams carry no jump of V or Q. The seams are used only to construct the outer scalar (Appendix A).

Step 1: removal of the poles. Let

$$R := U \text{ in } \Omega_\infty, \qquad R := UL_\pm \text{ in } \Omega^\pm_{\text{b}}, \qquad R := UA^{\triangle}_\pm \upsilon_\pm^{\sigma_3} \text{ in } \Omega^{\triangle}_\pm. \tag{4.19}$$

The restriction of R to each of the four regions $\Omega^\pm_{\text{b}}$, $\Omega^{\triangle}_\pm$ extends continuously to the real axis, including the nodes. Near X_k all factors are meromorphic, and $1 - e^{\mp 2i\theta} = \pm 2i\theta'(X_k)(z - X_k)(1 + O(z - X_k))$ because $e^{2i\theta(X_k)} = 1$. In $\Omega^\pm_{\text{b}}$ the entry $\mp W e^{\mp 2i\theta}\upsilon_\pm$ then has the simple pole $W(X_k)/(2i\theta'(X_k)(z - X_k)) = w(X_k)/(z - X_k)$, and $R_{j2} = U_{j2} - U_{j1}w(X_k)/(z - X_k) + O(1)$ is regular by Lemma 4.1. In $\Omega^{\triangle}_\pm$, $\upsilon_\pm^{-1}$ vanishes simply at X_k, which removes the pole of the second column; the first column is $\upsilon_\pm(U_{j1} \mp U_{j2}/(\upsilon_\pm W))$, and the bracket vanishes at X_k since $U_{j2}/(\upsilon_\pm W) \to \pm U_{j1}(X_k)$, so the first column is regular too.

Step 2: normalisation. $T := e^{-n\gamma_p\sigma_3} R e^{n\varphi\sigma_3}$.

Step 3: the lens. $S := TA_\pm$ in $\Omega^\pm_{\text{b}}$, and $S := T$ elsewhere.

Step 4: the outer scalar. Given a function F with the properties (a) and (b) of Definition 4.10 below and its lens scalar $\tilde{F}$, let

$$V := C_\infty S \tilde{F}^{\sigma_3} \text{ in } \Omega_{\text{b}} \cup \Omega^{\triangle}, \qquad V := C_\infty S F^{\sigma_3} \text{ in } \Omega_\infty, \tag{4.20}$$

with a constant diagonal matrix C_∞ fixed in Lemma 5.3.

Composing the four steps, and using $\upsilon_\pm^{\sigma_3}\tilde{F}^{\sigma_3} = F^{\sigma_3}$ in $\Omega^{\triangle}_\pm$ (definition (4.22) below),

$$V = C_\infty e^{-n\gamma_p\sigma_3} U M, \qquad M := \begin{cases} e^{n\varphi\sigma_3}F^{\sigma_3} & \text{in } \Omega_\infty, \\ L_\pm e^{n\varphi\sigma_3} A_\pm \tilde{F}^{\sigma_3} & \text{in } \Omega^\pm_{\text{b}}, \\ A^{\triangle}_\pm e^{n\varphi\sigma_3} F^{\sigma_3} & \text{in } \Omega^{\triangle}_\pm. \end{cases} \tag{4.21}$$

Since U is analytic off the nodes, the jump of V on any arc of Σ away from the nodes is $M_-^{-1}M_+$.

In [19] the saturated region is treated in two steps, by the factors $\Pi^{-\sigma_3}(\cdot)$ of [19, (3.10)–(3.11)] and $V^{\triangle}_\pm$ of [19, (4.12)–(4.13)] (§2.4(b)). Our factor $A^{\triangle}_\pm \upsilon^{\sigma_3}_\pm$ combines them into one step. In $\Omega^{\triangle}_-$ it carries a minus sign relative to the literal product of the two factors; this is the sign of the factor $-V^{\triangle}_-$ that [19, (4.12)] uses in the regions $\Omega^{\triangle}_{l,+} \cup \Omega^{\triangle}_{r,-}$, which lie in the second and fourth quadrants and are therefore mapped into $\mathbb{C}_-$ by $z \mapsto z^2$. Without this sign, V would have the constant jump $-I$ on the saturated segment.

4.8. **Where the weight is evaluated.** The following bookkeeping explains why an angle $\vartheta > 0$ of any size suffices.

Lemma 4.7 (Where the weight is evaluated)**.** *The functions w, W, r_n and W_0 (defined in (4.24)) are evaluated only at points of $\mathcal{Z}_n$. Specifically, they are evaluated (a) in $\overline{\Omega_{\mathrm{b}} \cup \Omega^{\triangle}} \setminus \{0\}$, which contains every arc of Σ except the top segment and the parts of the lips outside the lens; (b) on the lips and on the top segment; and (c) in the sectors $\{|\arg z| \le \frac{5}{4}\vartheta_{\mathrm{lip}}\} \cap \mathbb{D}(0,\delta)$ used in Section 6 and Appendix B. The region Ω_∞ never uses the weight: there $M = e^{n\varphi\sigma_3}F^{\sigma_3}$ contains neither w nor W. Consequently $\xi(z) = \log t(z)$ always lies in $\{\operatorname{Re}\xi \ge -c_0,\ |\operatorname{Im}\xi| \le \vartheta_{\mathrm{w}}/p\} \subset S_\vartheta$.*

Proof. By (4.21), w and W enter V only in $\Omega_{\mathrm{b}} \cup \Omega^{\triangle}$, whose closure minus $\{0\}$ lies in $\mathcal{Z}_n$ (§4.6); the top segment has $|z| \le n^p + \delta' + h_0 < e^{pc_0}n^p$ and $|\arg z| < \vartheta_{\mathrm{w}}$. The sectors used near 0 lie in $\mathcal{Z}_n$ because $\frac{5}{4}\vartheta_{\mathrm{lip}} < \vartheta_{\mathrm{w}}$. For $z \in \mathcal{Z}_n$, $\operatorname{Re}\xi = \log n - \frac{1}{p}\log|z| > -c_0$ and $|\operatorname{Im}\xi| = |\arg z|/p < \vartheta_{\mathrm{w}}/p$. □

Remark 4.8 (The angles). The angle ϑ of Definition 1.1 enters the proof only through $\vartheta_{\mathrm{w}} \le p\vartheta$, and ϑ_{w} and ϑ_{lip} enter only through (4.3). The angle ϑ_{lip} can be taken as small as we wish because the Bessel model admits rays at $\pm(\pi - \vartheta_{\mathrm{lip}})$ for every $\vartheta_{\mathrm{lip}} \in (0,\pi)$ (§2.5(b)), and the lens near 0 is then the preimage of these rays. Every step that uses the angles needs only a fixed positive lower bound, which (4.3) provides: the decay factors $\sin(\vartheta_{\mathrm{lip}}/2)$ in (6.8) and $\sin(\frac{3}{4}\vartheta_{\mathrm{lip}}/p)$ in Lemma 5.4; the bound $\sin(\phi/p) \ge \frac{2}{\pi}\frac{\phi}{p}$ in (4.16), which needs $\vartheta_{\mathrm{w}}/p < \pi/2$; the separation $|x - z| \ge (x + |z|)/C$ for $|\arg z| \ge \frac{1}{2}\vartheta_{\mathrm{lip}}$ in Lemma 5.2(vi) and Appendix B; and the large-argument expansions of the Bessel functions, which are used in the closed sector $|\arg 2\zeta^{1/2}| \le \frac{1}{2}(\pi - \vartheta_{\mathrm{lip}})$ (§6.2). The order of choices is: ϑ, then $\vartheta_{\mathrm{w}}, \vartheta_{\mathrm{lip}}$ by (4.2), then δ_1, then $\delta \le \delta_1$, which is also taken so small that the angle-dependent conditions in Lemmas 5.4(b), 6.3 and 6.2(i) hold (they depend only on p, s, ℓ, ϑ_{w}, ϑ_{lip}, never on n), then h_0, ε, δ', and n_0 last. Smaller angles make the radii δ, h_0, ε smaller and the threshold n_0 and the constants larger, but they do not change the statements. The upper bounds in (4.3) are what the construction needs: $\vartheta_{\mathrm{w}} < \pi/2$ (convexity of the sector $Z(r)$ in Lemma 5.2, and $|y - z| \ge (|y| + |z|)/\sqrt{2}$ for $y < 0$ in Lemma A.1), $\vartheta_{\mathrm{w}} < \pi p/2$ for $p < 1$ (the lens must lie in the inner sector of the seam construction), and $\frac{5}{4}\vartheta_{\mathrm{lip}} < \vartheta_{\mathrm{w}}$ (the sectors near 0 in which the weight is used lie in $\mathcal{Z}_n$). With the default angles ($\mathsf{m} = 1$) the strip needed is $|\operatorname{Im}\xi| \le 7\pi/(16p)$ for $p \ge 1$ and $|\operatorname{Im}\xi| \le 3\pi/8$ for $p < 1$; it is never necessary to continue Λ to the strip $|\operatorname{Im}\xi| \le \pi/p$ that corresponds to the whole slit plane in z.

Lemma 4.9 (Potter bounds in the sector)**.** *For every $\eta > 0$ there is C_η such that:*

(i) $|\Lambda(\xi)| \le C_\eta + \eta\operatorname{Re}\xi$ *for* $\xi \in S_\vartheta$, $\operatorname{Re}\xi \ge 0$, *and* $|\Lambda| \le C$ *on* $S_\vartheta \cap \{\operatorname{Re}\xi \le 0\}$. *Hence* $|\ell(t)|^{\pm 1} \le C_\eta|t|^\eta$ *for* $|\arg t| \le \vartheta$, $|t| \ge 1$.

(ii) $|r_n(z)|^{\pm 1} \le C_\eta n^{2\eta}(1 + |z|^{-\eta/p})$ *for* $z \in \mathcal{Z}_n$.

(iii) *For* $R \ge 1$, $z \in \mathcal{Z}_n$ *with* $|z| \le R$ *and* $x \in (0, R]$,

$$|\log r_n(z) - \log r_n(x)| \le \frac{\varepsilon_\Lambda(T)}{p}\big(|\arg z| + |\log(x/|z|)|\big),$$

$$|\log r_n(z)| \le \frac{\varepsilon_\Lambda(T)}{p}\big(|\arg z| + |\log|z||\big),$$

with $T := \log n - \frac{1}{p}\log R$. *Here* $\log r_n = \Lambda(\xi) - \Lambda(\log n)$.

Proof. (i) For $\operatorname{Re}\xi \ge T_0$, integrate Λ' along the horizontal segment from $T_0 + i\operatorname{Im}\xi$ to ξ: $|\Lambda(\xi) - \Lambda(T_0 + i\operatorname{Im}\xi)| \le \varepsilon_\Lambda(T_0)\operatorname{Re}\xi$. Choose T_0 with $\varepsilon_\Lambda(T_0) \le \eta$; Λ is bounded on the compact

set $S_\vartheta \cap \{0 \le \operatorname{Re}\xi \le T_0\}$ and, by hypothesis, on $\operatorname{Re}\xi \le 0$. Then $|\ell(t)|^{\pm 1} = e^{\pm \operatorname{Re}\Lambda(\log t)}$. (ii) $|r_n|^{\pm 1} = |\ell(t)|^{\pm 1}\ell(n)^{\mp 1}$ with $|t| = n|z|^{-1/p}$; apply (i) to both factors (for $|t| \le 1$ use the bound on $\operatorname{Re}\xi \le 0$). (iii) Join $\xi(x)$ to $\xi(z)$ by the horizontal segment from $\log n - \frac{1}{p}\log x$ to $\log n - \frac{1}{p}\log|z|$ followed by the vertical segment to $\xi(z)$. Along this path $\operatorname{Re}\xi \ge T$ and $|\operatorname{Im}\xi| \le |\arg z|/p$, and its length is $(|\log(x/|z|)| + |\arg z|)/p$. The second inequality is the case $x = 1$, since $r_n(1) = 1$. □

4.9. **The outer scalar and the jumps of V.** Section 5 constructs, for each geometry, a function F with the following properties. Put $\nu_p := \min(1/p, \frac{1}{2})$.

Definition 4.10 (Admissible outer scalar)**.** A function $F = F_n$, defined for every $n \ge n_0$, is an *admissible outer scalar* if it has the following five properties for every $n \ge n_0$, where the constants C in (c) and (d) do not depend on n.

(a) F is analytic and zero-free in $\mathbb{C} \setminus [0,\infty)$ (in particular, it has no jump on $(-\infty, 0)$), it has continuous boundary values on $(0,\infty)\setminus\{X_k\}$, and $F_+ = F_- e^{2i\theta}$ on $(4c,\infty)\setminus\{X_k\}$.

(b) The *lens scalar*

$$\tilde F := (1 - e^{\mp 2i\theta})F = F/\upsilon_\pm \qquad \text{in } \mathbb{C}_\pm \cap \mathcal{Z}_n \tag{4.22}$$

extends to an analytic, zero-free function on $\mathcal{Z}_n \setminus [4c,\infty)$, with $\tilde F_+ = -\tilde F_-$ on $(4c, e^{pc_0}n^p)$. Thus $\tilde F^2$ is analytic and zero-free on $\mathcal{Z}_n$, and so is the *band factor*

$$H_n(z) := \frac{1}{(4c - z)\tilde F(z)^2} \qquad (z \in \mathcal{Z}_n). \tag{4.23}$$

Moreover $H_n > 0$ on $(0, e^{pc_0}n^p)$, and $H_n(0^+) := \lim_{z\to 0,\, z\in\mathcal{Z}_n} H_n(z)$ exists in $(0,\infty)$.

(c) For $z \in \mathcal{Z}_n$ with $|z| \le 1$: $|\log H_n(z) - \log H_n(0^+)| \le C|z|^{\nu_p}$ and $|(\log H_n)'(z)| \le C|z|^{\nu_p - 1}$.

(d) For $z \in \mathcal{Z}_n$ with $|z| \ge 1$: $|\log H_n(z) - \log H_n(0^+)| \le C\big(1 + \log(1 + |z|)\big)^2$.

(e) (For each fixed n.) $|F|^{\pm 1}$ is bounded on $\mathbb{D}(0,\delta)\setminus(\overline{\Omega_{\mathrm{b}}} \cup [0,\infty))$, and $\tilde F^{\pm 1}$ is bounded on $\mathbb{D}(0,\delta)\cap\Omega_{\mathrm{b}}$. There is N_n with $|F(z)|^{\pm 1} \le C_n|z|^{N_n}$ for $|z| \ge n^p + \delta'$, $z \notin \mathbb{R}$, and $F(iy)$ is bounded and bounded away from 0 as $y \to \pm\infty$.

For such an F the *band weight* is

$$W_0 := \frac{W}{\tilde F^2} = \mathsf{w}_n\, z^\alpha (4c - z) H_n(z) r_n(z) \quad \text{on } \mathcal{Z}_n \tag{4.24}$$

(Lemma 4.3). On $(0, 4c)$ it equals the constant w_n times a positive function.

Lemma 4.11 (Jumps of V)**.** *Let F have the properties (a) and (b) of Definition 4.10. Then V is analytic in $\mathbb{C}\setminus\Sigma$, has continuous boundary values on Σ minus its vertices (including at the nodes), and $V_+ = V_- v_V$ with*

(a) *on the band $(0, 4c)$: $v_V = \begin{pmatrix} 0 & W_0 \\ -1/W_0 & 0 \end{pmatrix}$;*

(b) *on $\Sigma^\pm_{\mathrm{lip}}$ (oriented away from 0), with $\iota_+ = 1$ and $\iota_- = 0$:*

$$v_V = \begin{pmatrix} 1 & 0 \\ e^{2n\varphi}/W_0 & 1 \end{pmatrix} + \begin{pmatrix} \iota_\pm(\upsilon_\pm - 1) & W_0 e^{-2n\tilde\varphi} \\ (\upsilon_\pm - 1)e^{2n\varphi}/W_0 & \iota_\mp(\upsilon_\pm - 1) \end{pmatrix};$$

(c) *on $\Sigma_v^\pm$ (oriented away from the real axis): $v_V = \begin{pmatrix} 1 & -W_0 e^{-2n\tilde\varphi} \\ 0 & 1 \end{pmatrix}$;*

(d) *on the saturated segment $(4c, n^p + \delta')$: $v_V = \begin{pmatrix} 1 & 0 \\ e^{2n\tilde\varphi}/W_0 & 1 \end{pmatrix}$;*

(e) *on* $\Sigma^{\triangle\pm}$ *(oriented from left to right):* $v_V = \begin{pmatrix} 1 & 0 \\ v_\pm e^{2n\varphi}/W_0 & 1 \end{pmatrix}$;

(f) *on* $\Sigma_{\text{top}} \cap \mathbb{C}_\pm$ *(oriented upwards):* $v_V = \begin{pmatrix} 1 & 0 \\ \mp v_\pm e^{2n\varphi}/W_0 & 1 \end{pmatrix}$;

(g) *there is no jump on* $(-\infty, 0)$ *or on* $(n^p + \delta', \infty)$.

Proof. We compute $M_-^{-1}M_+$ from (4.21) arc by arc, using the identities

$$e^{\mp 2i\theta}e^{-2n\varphi} = e^{-2n\tilde{\varphi}}, \qquad F = v_\pm \tilde{F}, \qquad W/\tilde{F}^2 = W_0 \qquad \text{in } \mathbb{C}_\pm. \tag{4.25}$$

For a diagonal matrix $D = \operatorname{diag}(d_1, d_2)$, conjugation $D^{-1}XD$ multiplies the $(1,2)$ entry by d_2/d_1 and the $(2,1)$ entry by d_1/d_2.

(a) $L_-^{-1}L_+ = \left(\begin{smallmatrix} 1 & W \\ 0 & 1 \end{smallmatrix}\right)$: its $(1,2)$ entry is $-W(e^{-2i\theta}v_+ + e^{2i\theta}v_-) = -W\left(\frac{e^{-2i\theta}}{1-e^{-2i\theta}} + \frac{1}{e^{-2i\theta}-1}\right) = W$. Conjugating by $e^{n\varphi_\pm\sigma_3}$ and using $\varphi_- = -\varphi_+$ (Lemma 4.4(i)) gives $\left(\begin{smallmatrix} e^{2n\varphi_+} & W \\ 0 & e^{2n\varphi_-} \end{smallmatrix}\right)$, and

$$A_-^{-1}\begin{pmatrix} e^{2n\varphi_+} & W \\ 0 & e^{2n\varphi_-} \end{pmatrix} A_+ = \begin{pmatrix} 0 & W \\ -1/W & 0 \end{pmatrix}.$$

Since $\tilde{F}$ has no jump on the band (Definition 4.10(b)), the final conjugation gives (a) by (4.25).

(b) Upper lip: the $+$ side is Ω_∞, so $v_V = M_{\Omega_{\text{b}}}^{-1}M_{\Omega_\infty} = \tilde{F}^{-\sigma_3}A_+^{-1}\big(e^{-n\varphi\sigma_3}L_+^{-1}e^{n\varphi\sigma_3}\big)F^{\sigma_3}$. The bracket is $\left(\begin{smallmatrix} 1 & Wv_+e^{-2n\tilde{\varphi}} \\ 0 & 1 \end{smallmatrix}\right)$, and

$$A_+^{-1}\begin{pmatrix} 1 & Wv_+e^{-2n\tilde{\varphi}} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & Wv_+e^{-2n\tilde{\varphi}} \\ e^{2n\varphi}/W & v_+ \end{pmatrix},$$

because $1 + v_+e^{2n\varphi}e^{-2n\tilde{\varphi}} = 1 + v_+e^{-2i\theta} = v_+$. Conjugating with $\tilde{F}^{-\sigma_3}(\cdot)F^{\sigma_3}$, $F = v_+\tilde{F}$, gives $\left(\begin{smallmatrix} v_+ & W_0e^{-2n\tilde{\varphi}} \\ v_+e^{2n\varphi}/W_0 & 1 \end{smallmatrix}\right)$, which is (b). The lower lip (where the $+$ side is the lens) is the mirror computation and gives $\left(\begin{smallmatrix} 1 & W_0e^{-2n\tilde{\varphi}} \\ v_-e^{2n\varphi}/W_0 & v_- \end{smallmatrix}\right)$.

(c) Upper segment, $+$ side $= \Omega_{\text{b}}^+$: $v_V = M_{\Omega^\triangle}^{-1}M_{\Omega_{\text{b}}} = F^{-\sigma_3}e^{-n\varphi\sigma_3}(A_+^\triangle)^{-1}L_+e^{n\varphi\sigma_3}A_+\tilde{F}^{\sigma_3}$. One checks $(A_+^\triangle)^{-1}L_+ = \left(\begin{smallmatrix} 1 & -We^{-2i\theta}v_+ \\ 1/(v_+W) & 1-e^{-2i\theta} \end{smallmatrix}\right)$, and after the conjugation by $e^{n\varphi\sigma_3}$ and right multiplication by A_+ the $(2,1)$ entry vanishes and the diagonal becomes (v_+, v_+^{-1}); the final conjugation with $F = v_+\tilde{F}$ removes the diagonal and gives (c). The lower segment, whose $+$ side is $\Omega_-^\triangle$, is the mirror image.

(d) On $(4c, n^p + \delta')$ both sides lie in $\Omega^\triangle$. Since $v_\pm^{-1} = 1 - e^{\mp 2i\theta}$,

$$(A_-^\triangle)^{-1}A_+^\triangle = \begin{pmatrix} 1 & 0 \\ -(v_+^{-1} + v_-^{-1})/W & 1 \end{pmatrix}, \qquad v_+^{-1} + v_-^{-1} = 2 - 2\cos 2\theta = v_+^{-1}v_-^{-1}.$$

Conjugating by $e^{n\varphi\sigma_3}F^{\sigma_3}$ with boundary values, the diagonal becomes

$$(e^{n(\varphi_+ - \varphi_-)}F_+/F_-)^{\sigma_3} = (e^{-2i\theta}e^{2i\theta})^{\sigma_3} = I$$

by Lemma 4.4(i) and Definition 4.10(a), and the $(2,1)$ entry becomes

$$-e^{n(\varphi_+ + \varphi_-)}F_+F_-/(v_+v_-W) = -e^{2n\tilde{\varphi}}\tilde{F}_+\tilde{F}_-/W.$$

By Definition 4.10(b), $-\tilde{F}_+\tilde{F}_- = \tilde{F}^2$, which gives (d).

(e), (f) Here $v_V = (F^{-\sigma_3}e^{-n\varphi\sigma_3})(A_\pm^\triangle)^{\pm 1}(e^{n\varphi\sigma_3}F^{\sigma_3})$ with the exponent fixed by the orientation, and the $(2,1)$ entry $\mp 1/(v_\pm W)$ of $A_\pm^\triangle$ becomes $\mp F^2e^{2n\varphi}/(v_\pm W) = \mp v_\pm e^{2n\varphi}/W_0$.

(g) On $(-\infty, 0)$: U has no jump, $e^{n\varphi}$ jumps by $e^{-2\pi in} = 1$ (Lemma 4.4(i)) and F has no jump (Definition 4.10(a)). On $(n^p + \delta', \infty)$ there are no nodes, and $M_-^{-1}M_+ = (e^{n(\varphi_+ - \varphi_-)}F_+/F_-)^{\sigma_3} = I$ as in (d).

Continuity at the nodes: in (a) and (d) all entries are analytic across the real axis; together with Step 1 this shows that V has continuous boundary values there. □

5. The outer scalar and the properties of the frame

The frame of Section 4 still lacks an admissible outer scalar F (Definition 4.10), and constructing it is the only step of the Riemann–Hilbert analysis that depends on the geometry. This section constructs F and then carries out the steps common to all $p > 0$: the Szegő function of the band weight, the normalisation at infinity, the estimates on the lips near the hard edge, and the Airy parametrix at the soft edge. Proposition 5.6 collects the outcome; it is the only result of Sections 4 and 5 used later. The standing assumptions of §4.1 hold, so $s > 1$ is arbitrary; only item (vii) of Proposition 5.6 requires $|\alpha| < 1$.

The natural candidate for F comes from Stirling's formula. The node masses carry the factor $1/\Gamma(t)$ through $\Pi = \sin\theta/\theta'$, and

$$\tilde{D}(z) := e^{-j(t(z))}, \qquad j(t) := \operatorname{Ln}\Gamma(t) - \left(t - \tfrac{1}{2}\right)\ln t + t - \tfrac{1}{2}\ln(2\pi),$$

removes this growth. Here j is Binet's function, $\operatorname{Ln}\Gamma$ is the branch analytic on $\mathbb{C} \setminus (-\infty, 0]$ and real on $(0, \infty)$, and $\ln t$ is principal; so $\tilde{D} = \sqrt{2\pi}\, t^{t-1/2} e^{-t}/\Gamma(t)$, and this sign convention is used throughout. Then

$$D := \upsilon_{\pm}\tilde{D} = \frac{\tilde{D}}{1 - e^{\mp 2i\theta}} \quad \text{in } \mathbb{C}_{\pm} \tag{5.1}$$

satisfies $D_+/D_- = (1 - e^{2i\theta})/(1 - e^{-2i\theta}) = -e^{2i\theta}$ on $(0, \infty)$ off the nodes, and $K(z) := (4c - z)^{-1/2}$ (principal branch, cut $[4c, \infty)$) supplies the missing factor -1, so that DK has the jump required in Definition 4.10(a).

The difficulty is the cut of $t(z)$ on $(-\infty, 0)$, and how it is handled depends on how far the sector $|\arg t| < \pi/p$ reaches (Table 5.1): for $p > 1$ the jump of DK on $(-\infty, 0)$ is removed by a Cauchy integral, the Binet-cut scalar B; at $p = 1$ the reflection formula shows that D has no jump there at all; for $p < 1$ the poles of $\Gamma(t)$ enter, and D is used only in the inner sector $|\arg z| < \pi p/2$ and glued to an elementary function outside along two seams. Lemma 5.1 states the result; the constructions for $p \neq 1$ are in Appendix A. The rest of the section (§5.2) is common to all geometries.

Two facts about j are used in all three geometries. By DLMF 5.11.1 with the remainder bound of DLMF §5.11(ii), first term $1/(12t)$ (§2.7), for every $\phi < \pi$,

$$|j(t)| \le \frac{\sec^2(\phi/2)}{12|t|}, \qquad |j'(t)| \le \frac{C_\phi}{|t|^2} \qquad (|\arg t| \le \phi,\ |t| \ge 1), \tag{5.2}$$

the second by Cauchy's estimate on a disc of radius $|t|/C_\phi$ in a slightly larger sector. And for $|t| \le 1$, $\operatorname{Re} t \ge 0$, $t \neq 0$, DLMF 5.5.1 gives $\operatorname{Ln}\Gamma(t) = \operatorname{Ln}\Gamma(1 + t) - \ln t$, hence

$$j(t) = -\tfrac{1}{2}\ln t - \tfrac{1}{2}\ln(2\pi) + O(|t|\log(2/|t|)) \qquad (t \to 0,\ \operatorname{Re} t \ge 0). \tag{5.3}$$

5.1. The outer scalar.

Lemma 5.1 (Outer scalar). *Define*

$$F := \begin{cases} DBK, & p > 1, \text{ with the Binet-cut scalar } B \text{ of (A.1)}, \\ DK, & p = 1, \\ DB_{\mathrm{in}}Ke^{\chi} \text{ in } S_{\mathrm{in}}, \quad B_{\mathrm{out}}Ke^{\chi} \text{ in } S_{\mathrm{out}}, & 0 < p < 1, \text{ with } B_{\mathrm{in}}, B_{\mathrm{out}}, \chi \text{ of §A.2}. \end{cases}$$

Then F is an admissible outer scalar (Definition 4.10). The band factor is $H_n = (\tilde{D}B)^{-2}$, $\tilde{D}^{-2}$, $(\tilde{D}B_{\mathrm{in}}e^{\chi})^{-2}$ respectively, with $H_n(0^+) = B(0)^{-2}$, 1, $e^{-2\chi(0)}$. At $p = 1$,

$$D(z) = \frac{e^{-j(t)}}{1 - e^{\mp 2\pi i t}} = e^{j(-n/z)} \qquad (z \in \mathbb{C}_{\pm}), \tag{5.4}$$

so D extends analytically to $\mathbb{C} \setminus [0, \infty)$, it is real and positive on $(-\infty, 0)$, and $H_n = e^{2j(n/z)}$.

	$p > 1$	$p = 1$	$0 < p < 1$
$\arg t$ on $\mathbb{C} \setminus (-\infty, 0]$	$(-\pi/p, \pi/p) \subset (-\pi, \pi)$	$(-\pi, \pi)$, open	wider than $(-\pi, \pi)$; poles of $\Gamma(t)$ on $\arg z = \pm\pi p$
where $\tilde{D} = e^{-j(t)}$ is used	slit plane	$\lvert\arg z\rvert < \vartheta_{\mathrm{w}}$	inner sector $S_{\mathrm{in}} = \{\lvert\arg z\rvert < \pi p/2\}$
outer scalar F (Lemma 5.1)	DBK, Binet-cut scalar B	DK, $D = e^{j(-n/z)}$, no cut	$DB_{\mathrm{in}}Ke^{\chi}$ in S_{in}, $B_{\mathrm{out}}Ke^{\chi}$ in S_{out}
$\lvert\log \frac{H_n(z)}{H_n(0^+)}\rvert$ near 0	$\le C\lvert z\rvert^{1/p}$	$\le C\lvert z\rvert$	$\le C\lvert z\rvert(1 + \log \frac{1}{\lvert z\rvert})$
lip angle ϑ_{lip} (default); Bessel rays $\pm(\pi - \vartheta_{\mathrm{lip}})$	$\pi/3$	$\pi/3$	$\pi p/4$
angle ϑ_{w} (default; weight evaluated)	$7\pi/16$	$7\pi/16$	$3\pi p/8$
super-small entries $\lvert e^{\mp 2i\theta}\rvert$ on the rays $\arg z = \pm\vartheta_{\mathrm{lip}}$	$e^{-2\pi n \sin(\vartheta_{\mathrm{lip}}/p)\lvert z\rvert^{-1/p}}$; default angles: $e^{-2\pi n \sin(\frac{\pi}{3p})\lvert z\rvert^{-1/p}}$, $e^{-\sqrt{3}\pi n/\lvert z\rvert}$, $e^{-\sqrt{2}\pi n\lvert z\rvert^{-1/p}}$		
angle ϑ of ℓ	any $\vartheta > 0$; default angles if $\vartheta \ge \frac{7\pi}{16p}$	any $\vartheta > 0$; default angles if $\vartheta \ge \frac{7\pi}{16}$	any $\vartheta > 0$; default angles if $\vartheta \ge \frac{3\pi}{8}$
Hölder exponent ν_p (Definition 4.10)	$\min(1/p, \frac{1}{2})$	$\frac{1}{2}$	$\frac{1}{2}$
classical quotient parametrix at 0	used for pure powers in Section 8	fails (Remark 8.5)	not used

TABLE 5.1. The ingredients of the Riemann–Hilbert analysis that depend on the geometry. Everything else in Sections 4–7 is the same for all $p > 0$. The angles ϑ_{lip} and ϑ_{w} are the default values ($\mathsf{m} = 1$ in (4.2)); for smaller ϑ both are multiplied by $\mathsf{m} = p\vartheta/\vartheta_{\mathrm{w}}^0 < 1$ (Remark 4.8). In the row for H_n, only the weaker bound $C\lvert z\rvert^{\nu_p}$ of Definition 4.10(c) is used.

Proof. For $p > 1$ and $p < 1$ this is Lemma A.1(iv), respectively Lemma A.2(iv). Let $p = 1$, so $t = n/z$.

The identity (5.4). Let $z \in \mathbb{C}_+$, so $\arg t \in (-\pi, 0)$ and $t \notin \mathbb{R}$. By DLMF 5.5.3 (§2.7), $1/\Gamma(t) = \Gamma(1-t)\sin(\pi t)/\pi$, and $\sin(\pi t)/(1-e^{-2\pi i t}) = e^{i\pi t}/(2i)$. Hence $D = \sqrt{2\pi}\, t^{t-1/2}e^{-t}\Gamma(1-t)e^{i\pi t}/(2\pi i)$. Put $u := -t = -n/z$; then $\arg u = \arg t + \pi \in (0, \pi)$, so $\ln u = \ln t + i\pi$ for the principal branches, and $t^{t-1/2}e^{i\pi t} = \exp((t - \frac{1}{2})(\ln u - i\pi) + i\pi t) = e^{i\pi/2}u^{-u-1/2}$. So $D = u^{-u-1/2}e^{u}\Gamma(1+u)/\sqrt{2\pi}$, and DLMF 5.5.1 ($\Gamma(1+u) = u\Gamma(u)$) together with the definition of j gives $D = e^{j(u)}$. For $z \in \mathbb{C}_-$ the same computation with $\sin(\pi t)/(1 - e^{2\pi i t}) = -e^{-i\pi t}/(2i)$

and $\ln u = \ln t - i\pi$ gives the same result. Since j is analytic on $\mathbb{C}\setminus(-\infty,0]$ and $u \in (-\infty,0]$ exactly when $z \in (0,\infty)$, $e^{j(-n/z)}$ is analytic on $\mathbb{C}\setminus[0,\infty)$; on $(-\infty,0)$, $u>0$ and $j(u)$ is real.

We check the properties (a)–(e) of Definition 4.10.

Property (a). $F = DK$ is analytic and zero-free off $[0,\infty)$, and $F_+/F_- = (-e^{2i\theta})(-1)$ on $(4c,\infty)$ off the nodes.

Property (b). $\tilde F = F/\upsilon_\pm = \tilde D K$ with $\tilde D = e^{-j(t)}$ analytic and zero-free on $\mathbb{C}\setminus(-\infty,0]$. So $H_n = \tilde D^{-2} = e^{2j(n/z)} > 0$ on $(0,\infty)$, and $H_n(0^+) = 1$.

Properties (c) and (d). On $\mathcal{Z}_n$, $|\arg t| = |\arg z| < \vartheta_{\mathrm{w}} < \pi/2$. For $|z|\le 1$, $|t|\ge n$ and (5.2) gives $|\log H_n| = 2|j(t)| \le C|z|/n$ and $|(\log H_n)'| \le C/n$; since $\nu_1 = \frac12$, this is (c). For $|z|\ge 1$, $|t|\ge e^{-c_0}$ and $j(t)$ is bounded, which is (d).

Property (e). Fix n. Near 0 outside the lens, $u = -n/z$ has $|u|\to\infty$ and $|\arg u| = \pi - |\arg z| \le \pi - \frac34\vartheta_{\mathrm{lip}}$, so $j(u)\to 0$ by (5.2) and $D\to 1$; K is bounded. In the lens $\tilde F = \tilde D K$ with $\tilde D \to 1$. For $|z|\ge n+\delta'$ we have $|u|\le n/(n+\delta') < 1$ and $|1+u|\ge \delta'/(2n)$, and $D = u^{-u-1/2}e^u\Gamma(1+u)/\sqrt{2\pi}$ is bounded above and below by $C_n|z|^{\pm1}$. Along $z = iy$, $D(iy) = (-iy/n)^{1/2}(2\pi)^{-1/2}(1+o(1))$ by (5.3), and $K(iy) = (-iy)^{-1/2}(1+o(1))$, so $F(iy)\to (2\pi n)^{-1/2}$. □

At $p=1$ the cut of D disappears, so this case is simpler than $p>1$; it does, however, break the classical construction of the hard-edge parametrix (Remark 8.5).

5.2. **Common steps: Szegő functions, normalisation, parametrices.** From now on F is the outer scalar of Lemma 5.1. Everything in this subsection uses only the properties listed in Definition 4.10.

Szegő functions. Let $\mathcal{R}(z) := (z(z-4c))^{1/2}$, analytic off $[0,4c]$ with $\mathcal{R}(z)\sim z$ at ∞, so that $\mathcal{R}_\pm(x) = \pm i\sqrt{x(4c-x)}$ on $(0,4c)$. Let q be a function on $(0,4c)$ that is a non-zero constant multiple of a positive function q_0 with $\log q_0 \in L^1(\mathrm{d}x/\sqrt{x(4c-x)})$, and fix a value of the logarithm of the constant. Put

$$D_q(z) := \exp\Big(\frac{\mathcal{R}(z)}{2\pi}\int_0^{4c}\frac{\log q(x)}{\sqrt{x(4c-x)}}\frac{\mathrm{d}x}{z-x}\Big), \qquad D_q(\infty) = \exp\Big(\frac{1}{2\pi}\int_0^{4c}\frac{\log q(x)\,\mathrm{d}x}{\sqrt{x(4c-x)}}\Big). \tag{5.5}$$

D_q is analytic and zero-free off $[0,4c]$, $D_{q_1q_2} = D_{q_1}D_{q_2}$, and $D_q = q^{1/2}$ for a constant q. Where $\log q_0$ is Hölder continuous, the Sokhotski–Plemelj formula and $\mathcal{R}_+ = -\mathcal{R}_-$ give $D_{q,+}D_{q,-} = q$.

We shall estimate D_q^2/q near the band by means of the following identity. Since

$$\frac{1}{\pi}\int_0^{4c}\frac{\mathrm{d}x}{\sqrt{x(4c-x)}\,(z-x)} = \frac{1}{\mathcal{R}(z)}$$

(the Stieltjes transform of the arcsine law, as in the proof of Lemma 4.4), we have for every $z\notin[0,4c]$ and every number $\mathfrak{c}$

$$\log D_q(z)^2 - \mathfrak{c} = \frac{\mathcal{R}(z)}{\pi}\int_0^{4c}\frac{(\log q(x)-\mathfrak{c})\,\mathrm{d}x}{\sqrt{x(4c-x)}\,(z-x)}. \tag{5.6}$$

We use it with $\mathfrak{c} = \log q(z)$, the value of an analytic continuation of $\log q$. Put $\eta_* := \min(1,2c)$ and $Z(r) := \{z:\ 0<|z|<r,\ |\arg z|<\vartheta_{\mathrm{w}}\}$.

The band weight and its Szegő function. Let W_0 be the band weight (4.24), $D_W := D_{W_0}$ and $D_\infty := D_W(\infty)$. Writing $H_n^\circ := H_n/H_n(0^+)$,

$$D_W = \big(\mathsf{w}_n H_n(0^+)\big)^{1/2}\, D_{x^\alpha}\, D_{4c-x}\, D_{H_n^\circ}\, D_{r_n}. \tag{5.7}$$

All factors are defined: by Definition 4.10(c), (d), $\log H_n^\circ$ is bounded on $(0,4c)$, and by Lemma 4.9(iii), $|\log r_n(x)| \le C\varepsilon_\Lambda(T)(1+|\log x|)$. The constant $\mathsf{w}_n H_n(0^+)$ cancels in the ratio

$$\omega(z) := \frac{D_W(z)^2}{W_0(z)} = \frac{D_{x^\alpha}^2}{z^\alpha}\cdot\frac{D_{4c-x}^2}{4c-z}\cdot\frac{D_{H_n^\circ}^2}{H_n^\circ}\cdot\frac{D_{r_n}^2}{r_n}, \tag{5.8}$$

defined wherever W_0 is, i.e. on $\mathcal{Z}_n\setminus[0,4c]$. In particular no bound on $H_n(0^+)$ is ever needed.

Lemma 5.2 (Szegő function estimates)**.** *The constants below depend only on c, the stated $\mathfrak{M}$, and ε_1 in (iv).*

(i) *Let $\Upsilon(z) := (z^{1/2}+(z-4c)^{1/2})^2 = 2z-4c+2\mathcal{R}(z)$, with principal roots. Then, with principal powers, $D_{x^\alpha}(z)^2 = (4cz/\Upsilon(z))^\alpha$ and $D_{4c-x}(z)^2 = 4c(z-4c)/\Upsilon(z)$. Υ is analytic and zero-free off $[0,4c]$, with continuous non-zero boundary values, $\Upsilon(z)\sim 4z$ at ∞, $\Upsilon(0)=-4c$, $\Upsilon(4c)=4c$, and $\Upsilon(z) = -4c(1+O(|z|^{1/2}))$ as $z\to 0$. In particular $|D_{x^\alpha}(z)^2| \asymp |z|^\alpha$ and $|D_{4c-x}(z)^2| \asymp |z-4c|$ for $|z|\le 8c$, $z\notin[0,4c]$.*

(ii) *(Hölder type at 0.) Let $\nu\in(0,1]$, $\mathfrak{M}\ge 0$, $\mathfrak{c}_0\in\mathbb{C}$. Suppose $\log q$ continues analytically to $Z(\eta_*)$ and continuously to $(0,\eta_*)$, with $|\log q(z)-\mathfrak{c}_0| \le \mathfrak{M}|z|^\nu$ and $|(\log q)'(z)| \le \mathfrak{M}|z|^{\nu-1}$ there, and $|\log q(x)-\mathfrak{c}_0| \le \mathfrak{M}(1+|\log(4c-x)|)$ for $\frac12\eta_* \le x < 4c$. Then for $z\in Z(\frac14\eta_*)\setminus(0,\infty)$,*

$$|\log D_q(z)^2 - \log q(z)| \le C\mathfrak{M}|z|^{\min(\nu,1/2)}\log(2/|z|).$$

(iii) *(Logarithmic type at 0.) Suppose $\log q$ continues analytically to $Z(\eta_*)$ and continuously to $(0,\eta_*)$, with $|(\log q)'(z)| \le \mathfrak{M}/|z|$ there and $|\log q(x)-\log q(z)| \le \mathfrak{M}(1+|\log(x/|z|)|)$ for $z\in Z(\eta_*)$, $x\in(0,4c)$. Then $|\log D_q(z)^2 - \log q(z)| \le C\mathfrak{M}$ for $z\in Z(\frac14\eta_*)\setminus(0,\infty)$.*

(iv) *(Near 4c.) Suppose $\log q$ continues analytically to $\mathbb{D}(4c,2\varepsilon_1)$, with $|\log q - \mathfrak{c}_1| \le \mathfrak{M}$ there and $|\log q(x)-\mathfrak{c}_1| \le \mathfrak{M}(1+|\log x|)$ on $(0,4c)$. Then $|\log D_q(z)^2 - \log q(z)| \le C\mathfrak{M}\min(1,|z-4c|^{1/2})$ for $z\in\mathbb{D}(4c,\varepsilon_1)\setminus[0,4c]$.*

(v) *(Away from the band.) If $|\log q(x)-\mathfrak{c}_1| \le \mathfrak{M}(1+|\log x|+|\log(4c-x)|)$ on $(0,4c)$, then $|\log D_q(z)^2 - \mathfrak{c}_1| \le C_d\mathfrak{M}$ for $\operatorname{dist}(z,[0,4c])\ge d$.*

(vi) *(The band weight.) For $n\ge n_0$, with C independent of n: (a) $|\omega(z)|^{\pm1}\le C$ for $z\in Z(\frac14\eta_*)\setminus(0,\infty)$, in particular on the lips in $\mathbb{D}(0,\delta)$; (b) in $\mathbb{D}(4c,2\varepsilon)$,*

$$-\omega = \frac{4c}{\Upsilon}e^{L_1}, \qquad L_1 := \log\frac{D_{q_1}^2}{q_1}, \quad q_1 := x^\alpha H_n^\circ r_n, \tag{5.9}$$

where L_1 is given by (5.6) with $\mathfrak{c}=\log q_1(z)$; L_1 is analytic in $\mathbb{D}(4c,2\varepsilon)\setminus[0,4c]$, real on $(4c,4c+2\varepsilon)$, satisfies $L_{1,+}+L_{1,-}=0$ on $(4c-2\varepsilon,4c)$ and $|L_1(z)|\le C\min(1,|z-4c|^{1/2})$; in particular $|\omega(z)|^{\pm1}\le C$ there; (c) $|\log\omega(z)| \le C_d\big(1+\log(1+|z|)\big)^2$ for $z\in\mathcal{Z}_n$ with $\operatorname{dist}(z,[0,4c])\ge d$. Moreover, for each fixed n, $|D_W(z)|^{\pm1}\le C_{n,\eta}|z|^{\pm\alpha/2-\eta}$ for every $\eta>0$ and $0<|z|<\frac14\eta_*$, $z\notin[0,\infty)$.

Proof. Part (i) identifies two explicit functions with the right jumps. Parts (ii)–(v) use the representation (5.6) and a splitting of $(0,4c)$ into the near zone $|x-z|<\frac12|z|$ (or a neighbourhood of $4c$) and the rest. Part (vi) applies (i)–(v) to the four factors of (5.8).

(i) Write $\Upsilon = z(1+\mathfrak{r})^2$ with $\mathfrak{r} := \mathcal{R}(z)/z = \sqrt{1-4c/z}$, which has $\operatorname{Re}\mathfrak{r}>0$ off $[0,4c]$. Then $4cz/\Upsilon = 4c/(1+\mathfrak{r})^2$ and $4c(z-4c)/\Upsilon = 4c\,(\mathfrak{r}/(1+\mathfrak{r}))^2$; both $1/(1+\mathfrak{r})$ and $\mathfrak{r}/(1+\mathfrak{r})$ lie in the disc $|\cdot-\frac12|<\frac12$, so their squares avoid $(-\infty,0]$ and the principal powers are analytic off $[0,4c]$. On the band, $\Upsilon_\pm = (\sqrt{x}\pm i\sqrt{4c-x})^2$, so $\Upsilon_+\Upsilon_- = 16c^2$, the two boundary values of each expression are complex conjugates, and the product of their principal powers is $x^{2\alpha}$, respectively $(4c-x)^2$. The expressions tend to c at ∞. Hence they are the Szegő functions

squared (the quotient of two functions with the same jump, bounded at ∞ and of at most logarithmic growth at 0 and $4c$, is constant). The values and the expansion of Υ at 0 and $4c$ are read off from $\Upsilon = (z^{1/2} + (z-4c)^{1/2})^2$.

(ii) Fix $z \in Z(\frac{1}{4}\eta_*) \setminus (0,\infty)$ and use (5.6) with $\mathfrak{c} = \log q(z)$; here $|\mathcal{R}(z)| \le C|z|^{1/2}$. Split $(0,4c)$ into $\mathcal{U}_z := \{x : |x-z| < \frac{1}{2}|z|\}$ and the rest. For $x \in \mathcal{U}_z$ the segment $[x,z]$ lies in $Z(\eta_*)$ (the sector is convex since $\vartheta_{\mathrm{w}} < \pi/2$) and at distance $\ge \frac{1}{2}|z|$ from 0, so $|\log q(x) - \log q(z)| \le \mathfrak{M}(|z|/2)^{\nu-1}|x-z|$, and the contribution is at most $C|z|^{1/2}\mathfrak{M}|z|^{\nu-1}\int_{\mathcal{U}_z} x^{-1/2}\,\mathrm{d}x \le C\mathfrak{M}|z|^{\nu}$. Off $\mathcal{U}_z$, $|x-z| \ge \frac{1}{8}(x+|z|)$ (if $x \le 3|z|$ then $|x-z| \ge \frac{1}{2}|z|$; otherwise $|x-z| \ge x - |z| \ge \frac{1}{2}x$). On $(0, \frac{1}{2}\eta_*)$ we have $|\log q(x) - \log q(z)| \le \mathfrak{M}(x^\nu + |z|^\nu)$, and

$$|z|^{1/2}\int_0^{\eta_*/2} \frac{\mathfrak{M}(x^\nu + |z|^\nu)\,\mathrm{d}x}{x^{1/2}(x+|z|)} \le C\mathfrak{M}|z|^{\min(\nu,1/2)}\log(2/|z|),$$

by the substitution $x = |z|u$: the term $|z|^\nu$ gives $C\mathfrak{M}|z|^\nu$; the term x^ν gives $C\mathfrak{M}|z|^\nu$ if $\nu < \frac{1}{2}$, $C\mathfrak{M}|z|^{1/2}\log(2/|z|)$ if $\nu = \frac{1}{2}$, and $C\mathfrak{M}|z|^{1/2}$ if $\nu > \frac{1}{2}$. On $[\frac{1}{2}\eta_*, 4c)$, $|x-z| \ge \frac{1}{4}\eta_*$ and $\int(1+|\log(4c-x)|)(4c-x)^{-1/2}\,\mathrm{d}x < \infty$ give $C\mathfrak{M}|z|^{1/2}$.

(iii) The same splitting. On $\mathcal{U}_z$ the derivative bound gives $|\log q(x) - \log q(z)| \le 2\mathfrak{M}|x-z|/|z|$ and a contribution $C\mathfrak{M}$. Off $\mathcal{U}_z$ the contribution is at most $C\mathfrak{M}|z|^{1/2}\int_0^{4c} \frac{(1+|\log(x/|z|)|)\,\mathrm{d}x}{\sqrt{x(4c-x)}\,(x+|z|)}$, and the substitution $x = |z|u$ bounds the part $x \le 2c$ by $C\mathfrak{M}\int_0^\infty (1+|\log u|)u^{-1/2}(1+u)^{-1}\,\mathrm{d}u$; the part $x \ge 2c$ is $O(\mathfrak{M}|z|^{1/2}\log(2/|z|))$.

(iv) By Cauchy's estimate $|(\log q)'| \le 2\mathfrak{M}/\varepsilon_1$ on $\mathbb{D}(4c, \frac{3}{2}\varepsilon_1)$. For $x \in (4c - \frac{3}{2}\varepsilon_1, 4c)$ and $z \in \mathbb{D}(4c,\varepsilon_1)$, $|\log q(x) - \log q(z)| \le C\mathfrak{M}|x-z|$, which cancels the kernel's singularity, and $\int(4c-x)^{-1/2}\,\mathrm{d}x < \infty$. For $x \le 4c - \frac{3}{2}\varepsilon_1$, $|z-x| \ge \frac{1}{2}\varepsilon_1$ and the integrand is $\le C\mathfrak{M}(1+|\log x|)x^{-1/2}$. So the integral in (5.6) is at most $C\mathfrak{M}$, and the prefactor satisfies $|\mathcal{R}(z)| \le C\min(1, |z-4c|^{1/2})$.

(v) In (5.6) with $\mathfrak{c} = \mathfrak{c}_1$, $|\mathcal{R}(z)/(z-x)| \le C_d$ for all $x \in [0,4c]$ when $\operatorname{dist}(z,[0,4c]) \ge d$, and the remaining integral is finite.

(vi)(a) Use the four factors of (5.8). The first two are bounded above and below by (i), since z^α and $(4cz/\Upsilon)^\alpha$ differ by a bounded factor $(4c/\Upsilon)^\alpha e^{2\pi i k\alpha}$. The third is bounded by (ii) with $q = H_n^\circ$, $\mathfrak{c}_0 = 0$, $\nu = \nu_p$ and $\mathfrak{M} = C$ from Definition 4.10(c), (d). The fourth is bounded by (iii) with $q = r_n$: by Lemma 4.9(iii) with $R = \max(4c,1)$, the hypotheses hold with $\mathfrak{M} = C\varepsilon_\Lambda(\log n - \frac{1}{p}\log\max(4c,1)) \le C$, and $|(\log r_n)'(z)| = |\Lambda'(\xi)|/(p|z|)$. (b) By (i), $D^2_{4c-x}/(4c-z) = -4c/\Upsilon$, and the three other factors of (5.8) combine, by multiplicativity, into $D^2_{q_1}/q_1$; this is (5.9). Near $4c$, H_n° and r_n are analytic and positive on the real axis, with $|\log H_n^\circ| \le C$ and $|\log r_n| \le C$ (Definition 4.10(d) and Lemma 4.9(iii)), so (iv) applies to q_1 with $\mathfrak{M} = C$. The boundary values of the exponent in (5.5) add up to $\log q_1$ on the band (this is $D_{q_1,+}D_{q_1,-} = q_1$), so $L_{1,+} + L_{1,-} = 2\log q_1 - 2\log q_1 = 0$; and L_1 is real on $(4c, 4c+2\varepsilon)$ because D_{q_1} and q_1 are positive there. (c) (v) with $q = W_0/(\mathsf{w}_n H_n(0^+))$ bounds $\log D_W^2 - \log(\mathsf{w}_n H_n(0^+))$, and $|\log(W_0(z)/(\mathsf{w}_n H_n(0^+)))| \le C(1+\log(1+|z|))^2$ on $\mathcal{Z}_n$ by Definition 4.10(c), (d) and Lemma 4.9(iii) with $R = e^{pc_0}n^p$ (here $|\log r_n(z)| \le \frac{1}{p}\varepsilon_\Lambda(-c_0)(\vartheta_{\mathrm{w}} + |\log|z||)$). For the last claim, fix n and $\eta > 0$. By Lemma 4.9(i), $|\log r_n(x)| \le C_{n,\eta} + \eta|\log x|$ for $x \in (0,4c)$, and $|r_n(z)|^{\pm 1} \le C_{n,\eta}|z|^{-\eta}$ for $z \in \mathcal{Z}_n$. For $|\arg z| < \vartheta_{\mathrm{w}}$ the claim follows from (a) and $|W_0| \asymp |z|^\alpha |r_n|$. For $|\arg z| \ge \frac{1}{2}\vartheta_{\mathrm{lip}}$ we have $|x-z| \ge (x+|z|)/C$ for $x > 0$, and (5.6) with $\mathfrak{c} = 0$ gives $|\log D_{H_n^\circ}(z)| \le C$ and

$$|\log D_{r_n}(z)^2| \le C|z|^{1/2}\int_0^{4c} \frac{C_{n,\eta} + \eta|\log x|}{\sqrt{x(4c-x)}(x+|z|)}\,\mathrm{d}x \le C_{n,\eta} + C\eta\log\frac{1}{|z|};$$

together with (i) for D_{x^α} and D_{4c-x} this gives the claim (after renaming η). □

The global parametrix. Let $a(z) := ((z-4c)/z)^{1/4}$, analytic off $[0,4c]$ with $a(\infty)=1$, and

$$P_0^\infty(z) := \begin{pmatrix} \frac{a+a^{-1}}{2} & \frac{a-a^{-1}}{2i} \\ -\frac{a-a^{-1}}{2i} & \frac{a+a^{-1}}{2} \end{pmatrix}, \qquad P^\infty := D_\infty^{\sigma_3} P_0^\infty D_W^{-\sigma_3}. \tag{5.10}$$

Then $\det P_0^\infty = 1$, $P_{0,+}^\infty = P_{0,-}^\infty J$ on $(0,4c)$ with $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ (because $a_+ = ia_-$ there), and $P_0^\infty(z) = I + O(1/z)$. Since $|a(z)|^{\pm 1} = |z-4c|^{\pm 1/4}|z|^{\mp 1/4}$,

$$|P_0^\infty(z)| \le C|z|^{-1/4}|z-4c|^{-1/4}(1+|z|)^{1/2}. \tag{5.11}$$

As $D_{W,+}D_{W,-} = W_0$, P^∞ has the band jump of V (Lemma 4.11(a)). We put

$$Q := D_\infty^{-\sigma_3} V D_W^{\sigma_3}; \tag{5.12}$$

in the frame of Q the global parametrix is P_0^∞, which does not involve the weight. A matrix X in the frame of V corresponds to $D_\infty^{-\sigma_3} X D_W^{\sigma_3}$ in the frame of Q.

Lemma 5.3 (Normalisation at infinity)**.** *Let $\mathsf{h}_\infty(z) := F(z)^{-1} e^{n(g(z)-\operatorname{Log} z)}$ for $|z| > n^p + \delta'$, $z \notin \mathbb{R}$. Then h_∞ extends to an analytic function on $\{|z| > n^p+\delta'\} \cup \{\infty\}$ with $\mathsf{h}_\infty(\infty) \neq 0$. With $C_\infty := \mathsf{h}_\infty(\infty)^{\sigma_3}$ in (4.20), $V(z) = I + O(1/z)$ and $Q(z) = I + O(1/z)$ as $z \to \infty$, and for $|z| > n^p + \delta' + h_0$*

$$Q(z) = k_\infty^{-\sigma_3}\big(U(z) z^{-n\sigma_3}\big) k(z)^{\sigma_3}, \qquad k := \frac{e^{n\gamma_p} D_W}{\mathsf{h}_\infty}, \tag{5.13}$$

where k is analytic at ∞ with $k_\infty := k(\infty) = e^{n\gamma_p} D_\infty / \mathsf{h}_\infty(\infty) \neq 0$.

Proof. On $(-\infty, -n^p - \delta')$, g and $\operatorname{Log} z$ both jump by $2\pi i$ and F has no jump (Definition 4.10(a)). On $(n^p+\delta', \infty)$, $e^{n(g-\operatorname{Log} z)}$ jumps by $e^{2\pi i n \mu^*((x,\infty))} = e^{2i\theta}$ (Lemma 4.4(i)) and so does F. So h_∞ has continuous boundary values that agree, and is analytic in $\{|z| > n^p+\delta'\}$ by Morera's theorem. By Lemma 4.4(ii) and Definition 4.10(e), $|\mathsf{h}_\infty(z)| \le C_n |z|^{N_n}$, so ∞ is at most a pole; since $\mathsf{h}_\infty(iy)$ is bounded and bounded away from 0 as $y \to \pm\infty$, it is a removable singularity and $\mathsf{h}_\infty(\infty) \neq 0$.

In Ω_∞, by (4.21), $V = C_\infty e^{-n\gamma_p\sigma_3}(Uz^{-n\sigma_3})(z^n e^{n\varphi} F)^{\sigma_3}$, and $z^n e^{n\varphi} F = e^{n\gamma_p} e^{-n(g-\operatorname{Log} z)} F = e^{n\gamma_p}/\mathsf{h}_\infty$. Hence $V = \mathsf{h}_\infty(\infty)^{\sigma_3} e^{-n\gamma_p\sigma_3}(Uz^{-n\sigma_3}) e^{n\gamma_p\sigma_3} \mathsf{h}_\infty^{-\sigma_3} = I + O(1/z)$, by Lemma 4.1. Multiplying by $D_\infty^{-\sigma_3}$ on the left and $D_W^{\sigma_3}$ on the right gives (5.13), and $D_W(z)/D_\infty = 1 + O(1/z)$ gives $Q = I + O(1/z)$; D_W is analytic at ∞ because the exponent in (5.5) is. □

Lemma 5.4 (Lip exponentials and super-small entries)**.** *Let $s_p := \sin(\frac{3}{4}\vartheta_{\mathrm{lip}}/p) > 0$. For $n \ge n_0$ and $z \in \Sigma_{\mathrm{lip}}^{\pm} \cap \mathbb{D}(0,\delta)$:*

(a) $|v_\pm(z) - 1| \le 2\exp(-2\pi s_p n |z|^{-1/p})$;
(b) $|e^{-2n\tilde\varphi(z)}| \le \exp(-\pi s_p n |z|^{-1/p})$;
(c) $|e^{2n\varphi(z)}| \le \exp\big(-2\sin(\vartheta_{\mathrm{lip}}/2)\, n |f(z)|^{1/2}\big) \le e^{-n|z|^{1/2}/C}$.

Proof. On these lips $\frac{3}{4}\vartheta_{\mathrm{lip}} \le |\arg z| \le \frac{5}{4}\vartheta_{\mathrm{lip}}$ (Lemma 4.6), and $\frac{5}{4}\vartheta_{\mathrm{lip}}/p < \pi/2$ in all cases, so $\sin(|\arg z|/p) \ge s_p$. (a) follows from (4.10) and $|v_\pm - 1| = |e^{\mp 2i\theta}|/|1 - e^{\mp 2i\theta}|$. (b) $e^{-2n\tilde\varphi} = e^{-2n\varphi} e^{\mp 2i\theta}$ and $|\operatorname{Re}\varphi| \le C|z|^{1/2}$ (Lemma 4.5(iii)), so $|e^{-2n\tilde\varphi}| \le \exp(2Cn|z|^{1/2} - 2\pi s_p n |z|^{-1/p})$, and $2C|z|^{1/2} \le \pi s_p |z|^{-1/p}$ for $|z| \le \delta$ small. (c) By Lemma 4.5(iii), $\operatorname{Re}\varphi = -2\operatorname{Re} f^{1/2}$, and on the lip $\arg f = \mp(\pi - \vartheta_{\mathrm{lip}})$, so $\operatorname{Re} f^{1/2} = |f|^{1/2}\sin(\vartheta_{\mathrm{lip}}/2)$; finally $|f(z)| \ge |z|/C$ on $\mathbb{D}(0,\delta)$. □

The soft edge. In $\mathbb{D}(4c,\varepsilon)$ the jumps of Q are, by Lemma 4.11(a), (c), (d) and the conjugation by $D_W^{\sigma_3}$,

(5.14)
$$J \text{ on } (4c-\varepsilon,4c), \qquad \begin{pmatrix} 1 & 0 \\ \omega e^{2n\tilde\varphi} & 1 \end{pmatrix} \text{ on } (4c,4c+\varepsilon), \qquad \begin{pmatrix} 1 & -\omega^{-1}e^{-2n\tilde\varphi} \\ 0 & 1 \end{pmatrix} \text{ on } \Sigma_v^{\pm}\cap\mathbb{D}(4c,\varepsilon),$$

the first two oriented from left to right and the last away from the real axis. This is the jump structure of [19, (6.4)]. We construct the local solution from Airy functions, in our variable and with our orientations, following the pattern of [19, (6.10)–(6.12)] (§2.4(c)); the facts about Ai that we use are those listed in §2.7.

Let $\mathsf{q} := e^{2\pi i/3}$ and define the vector functions

$$y_0(\zeta) := \begin{pmatrix} \mathrm{Ai}(\zeta) \\ \mathrm{Ai}'(\zeta) \end{pmatrix}, \qquad y_\pm(\zeta) := \begin{pmatrix} \mathrm{Ai}(\mathsf{q}^{\mp1}\zeta) \\ \mathsf{q}^{\mp1}\mathrm{Ai}'(\mathsf{q}^{\mp1}\zeta) \end{pmatrix}.$$

By DLMF 9.2.12 and its derivative, $y_0 + \mathsf{q}^{-1}y_+ + \mathsf{q}\,y_- = 0$. With $k_{\mathrm{Ai}} := \sqrt{2\pi}\,e^{i\pi/12}$ and $M = \frac{1}{\sqrt2}\left(\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right)$ (the matrix of §2.5(c)), put

$$\begin{aligned} \Psi^{\mathrm{I}} &:= k_{\mathrm{Ai}}e^{-i\pi\sigma_3/4}\big(y_+,\ \mathsf{q}\,y_0\big), & \Psi^{\mathrm{II}} &:= k_{\mathrm{Ai}}e^{-i\pi\sigma_3/4}\big(y_+,\ -\mathsf{q}^{-1}y_-\big), \\ \Psi^{\mathrm{III}} &:= k_{\mathrm{Ai}}e^{-i\pi\sigma_3/4}\big(-\mathsf{q}^{-1}y_-,\ -y_+\big), & \Psi^{\mathrm{IV}} &:= k_{\mathrm{Ai}}e^{-i\pi\sigma_3/4}\big(-\mathsf{q}^{-1}y_-,\ \mathsf{q}\,y_0\big). \end{aligned} \tag{5.15}$$

These are entire matrix functions of ζ, independent of n. The relation $y_0+\mathsf{q}^{-1}y_+ + \mathsf{q}y_- = 0$ gives

$$\Psi^{\mathrm{II}} = \Psi^{\mathrm{I}}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \Psi^{\mathrm{II}} = \Psi^{\mathrm{III}}J, \quad \Psi^{\mathrm{IV}} = \Psi^{\mathrm{III}}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \Psi^{\mathrm{I}} = \Psi^{\mathrm{IV}}\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \tag{5.16}$$

and $\det\Psi^{\mathrm{I}} = -k_{\mathrm{Ai}}^2\mathsf{q}\,\mathcal{W}\{\mathrm{Ai}(\zeta),\mathrm{Ai}(\mathsf{q}^{-1}\zeta)\} = -2\pi e^{i\pi/6}\mathsf{q}\,e^{i\pi/6}/(2\pi) = 1$ by DLMF 9.2.8, hence $\det\Psi^j\equiv1$ for all four. Let $\mathcal{S}_{\mathrm{I}} := \{0\le\arg\zeta\le\frac{7\pi}{12}\}$, $\mathcal{S}_{\mathrm{II}} := \{\frac{5\pi}{12}\le\arg\zeta\le\pi\}$ and $\mathcal{S}_{\mathrm{III}},\mathcal{S}_{\mathrm{IV}}$ their mirror images (with $\arg\zeta=\pm\pi$ meaning the boundary value from $\mathbb{C}_\pm$). Then, uniformly for $\zeta\in\mathcal{S}_j$, $|\zeta|\ge1$,

$$\Psi^j(\zeta) = \zeta^{-\sigma_3/4}\big(M + O(|\zeta|^{-3/2})\big)e^{\frac23\zeta^{3/2}\sigma_3}. \tag{5.17}$$

This holds because in each $\mathcal{S}_j$ the three arguments ζ, $\mathsf{q}^{\pm1}\zeta$ that occur have $|\mathrm{ph}|\le\frac{11\pi}{12}$, so the Airy asymptotics of §2.7 apply with relative errors $O(|\zeta|^{-3/2})$. For instance, for ζ in $\mathcal{S}_{\mathrm{I}}\cup\mathcal{S}_{\mathrm{II}}$ we have $(\mathsf{q}^{-1}\zeta)^{1/4} = e^{-i\pi/6}\zeta^{1/4}$ and $\frac23(\mathsf{q}^{-1}\zeta)^{3/2} = -\frac23\zeta^{3/2}$, so, entrywise with relative errors $O(|\zeta|^{-3/2})$, $y_+ = \frac{1}{2\sqrt\pi}e^{\frac23\zeta^{3/2}}\zeta^{-\sigma_3/4}\left(\begin{smallmatrix} e^{i\pi/6} \\ e^{i\pi/6}\end{smallmatrix}\right)$ and $y_0 = \frac{1}{2\sqrt\pi}e^{-\frac23\zeta^{3/2}}\zeta^{-\sigma_3/4}\left(\begin{smallmatrix} 1 \\ -1\end{smallmatrix}\right)$; and $k_{\mathrm{Ai}}e^{-i\pi\sigma_3/4}\frac{1}{2\sqrt\pi}\left(\begin{smallmatrix} e^{i\pi/6} & \mathsf{q} \\ e^{i\pi/6} & -\mathsf{q}\end{smallmatrix}\right) = M$. The other columns and sectors are the same computation with $(\mathsf{q}\zeta)^{1/4} = e^{-i\pi/3}\zeta^{1/4}$, $(\mathsf{q}\zeta)^{3/2} = \zeta^{3/2}$ in $\mathcal{S}_{\mathrm{II}}$, and the mirror relations in $\mathcal{S}_{\mathrm{III}}\cup\mathcal{S}_{\mathrm{IV}}$.

The disc $\mathbb{D}(4c,\varepsilon)$ minus the contour has four components: $\Omega_+^{\triangle}\cap\mathbb{D}(4c,\varepsilon)$, $\Omega_{\mathrm{b}}^+\cap\mathbb{D}(4c,\varepsilon)$, $\Omega_{\mathrm{b}}^-\cap\mathbb{D}(4c,\varepsilon)$, $\Omega_-^{\triangle}\cap\mathbb{D}(4c,\varepsilon)$; we label them I, II, III, IV. With f_r of Lemma 4.5(iv) put $\zeta := n^{2/3}f_r(z)$ and $\Psi_r(\zeta) := \Psi^j(\zeta)$ for z in component j. By Lemma 4.5(iv), $|\arg f_r(z) - \arg(z-4c)|\le\frac{\pi}{12}$, so $\zeta\in\mathcal{S}_j$ when z is in component j. By Lemma 5.2(vi)(b) let

$$(-\omega)^{1/2} := \frac{2\sqrt c}{z^{1/2}+(z-4c)^{1/2}}\,e^{L_1/2} \qquad (z\in\mathbb{D}(4c,2\varepsilon)\setminus[0,4c]), \tag{5.18}$$

a square root of $-\omega$ (as $4c/\Upsilon = (2\sqrt c/(z^{1/2}+(z-4c)^{1/2}))^2$). It is analytic in $\mathbb{D}(4c,2\varepsilon)\setminus(4c-2\varepsilon,4c]$, positive on $(4c,4c+2\varepsilon)$, bounded above and below uniformly in n, and its boundary

values on the band satisfy

$$(-\omega)_+^{1/2}(-\omega)_-^{1/2} = \frac{4c}{(\sqrt{x}+i\sqrt{4c-x})(\sqrt{x}-i\sqrt{4c-x})}\, e^{(L_{1,+}+L_{1,-})/2} = 1. \tag{5.19}$$

(The square root of ω itself would not do: $\omega(z)\to -1$ as $z\to 4c$, so a branch of $\omega^{1/2}$ that is continuous across $(4c, 4c+\varepsilon)$ has $\omega_+^{1/2}\omega_-^{1/2} = -1$ on the band.) Define

$$\begin{aligned} \hat P_r(z) &:= A_r(z)\, n^{\sigma_3/6} M_{r,n}(z), \qquad A_r := P_0^\infty\, (-\omega)^{-\sigma_3/2} M^{-1} f_r^{\sigma_3/4}, \\ M_{r,n}(z) &:= \Psi_r\big(n^{2/3} f_r(z)\big)\big((-\omega)^{1/2} e^{n\tilde\varphi}\big)^{\sigma_3}, \end{aligned} \tag{5.20}$$

with the principal root $f_r^{1/4}$ (cut along the band).

Lemma 5.5 (Airy parametrix)**.** *For $n\ge n_0$, with constants independent of n:*

(i) *A_r extends analytically to $\mathbb{D}(4c,\varepsilon)$, $\det A_r = 1$ and $|A_r^{\pm 1}|\le C$ there.*

(ii) *$\hat P_r$ is analytic in $\mathbb{D}(4c,\varepsilon)\setminus\Sigma$, $\det \hat P_r = 1$, and $\hat P_{r,+} = \hat P_{r,-} v$ on $\Sigma\cap\mathbb{D}(4c,\varepsilon)$ with the jumps (5.14).*

(iii) *$\hat P_r(z) P_0^\infty(z)^{-1} = I + O(1/n)$ uniformly for $\frac12\varepsilon\le |z-4c|<\varepsilon$, $z\notin\Sigma$.*

(iv) *(Two scales.) With $\zeta = n^{2/3} f_r(z)$: if $|\zeta|\ge 2$, then $M_{r,n}(z) = \zeta^{-\sigma_3/4}\Theta_\infty(z)$ with $|\Theta_\infty^{\pm1}|\le C$; if $|\zeta|\le 3$, then $M_{r,n}(z) = \Psi^{\mathrm{I}}(\zeta)\Theta_0(z)$ with $|\Theta_0^{\pm1}|\le C$, where Ψ^{I} is entire, independent of n, with $\det\Psi^{\mathrm{I}} = 1$. The same holds for the boundary values on Σ.*

Proof. Parts (ii) and (iv) follow from the relations (5.16) and (5.17) of the model; part (i) is a cancellation of jumps on the band; part (iii) combines (i) and (5.17). Put $h := (-\omega)^{1/2} e^{n\tilde\varphi}$. By (5.18), $h^{-2} = -\omega^{-1} e^{-2n\tilde\varphi}$. On the band, $\tilde\varphi_+ + \tilde\varphi_- = \varphi_+ + \varphi_- = 0$ (Lemma 4.4(i); θ is analytic across the band), so $h_+ h_- = 1$ by (5.19); on $(4c, 4c+\varepsilon)$ and on $\Sigma_v^\pm$, h has no jump.

(ii) $\hat P_r$ is analytic off Σ because A_r is analytic (part (i)) and the Ψ^j are entire, and $\det\hat P_r = 1$ because every factor has determinant 1. Since A_r is analytic, the jumps of $\hat P_r$ are those of $M_{r,n}$, and these are $h_-^{-\sigma_3}(\Psi^{j_-})^{-1}\Psi^{j_+} h_+^{\sigma_3}$, where $j_\pm$ are the components on the two sides. On Σ_v^+ (oriented upwards, + side II, − side I) and on Σ_v^- (oriented downwards, + side IV, − side III), (5.16) gives $h^{-\sigma_3}\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)h^{\sigma_3} = \left(\begin{smallmatrix}1&h^{-2}\\0&1\end{smallmatrix}\right)$, which is the third jump in (5.14). On $(4c, 4c+\varepsilon)$ (+ side I, − side IV) we get $h^{-\sigma_3}\left(\begin{smallmatrix}1&0\\-1&1\end{smallmatrix}\right)h^{\sigma_3} = \left(\begin{smallmatrix}1&0\\-h^2&1\end{smallmatrix}\right)$ and $-h^2 = \omega e^{2n\tilde\varphi}$. On the band (+ side II, − side III) we get $h_-^{-\sigma_3} J h_+^{\sigma_3} = h_+^{\sigma_3} J h_+^{\sigma_3} = J$, since $h_- = h_+^{-1}$ and $DJD = J$ for every diagonal D with $\det D = 1$.

(i) On $(4c, 4c+\varepsilon)$ all factors of A_r are analytic. On the band, $P_{0,+}^\infty = P_{0,-}^\infty J$; since f_r is negative there and maps $\mathbb{C}_\pm$ into $\mathbb{C}_\pm$, $f_{r,+}^{\sigma_3/4} = f_{r,-}^{\sigma_3/4}\mathrm{diag}(i,-i)$; and $(-\omega)_+^{-\sigma_3/2} = (-\omega)_-^{\sigma_3/2}$ by (5.19). Using $JD = D^{-1}J$ for diagonal D with $\det D = 1$, and the identity $MJM^{-1} = \mathrm{diag}(-i,i)$ (a direct computation),

$$A_{r,+} = P_{0,-}^\infty (-\omega)_-^{-\sigma_3/2}\, JM^{-1}\mathrm{diag}(i,-i)\, f_{r,-}^{\sigma_3/4} = P_{0,-}^\infty (-\omega)_-^{-\sigma_3/2} M^{-1} f_{r,-}^{\sigma_3/4} = A_{r,-}.$$

So A_r is analytic in $\mathbb{D}(4c, 2\varepsilon)\setminus\{4c\}$. Near $4c$, $|P_0^\infty|\le C|z-4c|^{-1/4}$ by (5.11), $(-\omega)^{\pm1/2}$ is bounded and $|f_r^{\pm1/4}|\le C|z-4c|^{\pm1/4}$, so $|A_r(z)|\le C|z-4c|^{-1/2}$, and an isolated singularity with this growth is removable (as in §2.5(c)). Each factor has determinant 1, and A_r^{-1} is the adjugate of A_r. On the circle $|z-4c| = \frac32\varepsilon$ all factors are bounded uniformly in n (Lemma 5.2(vi)(b)), so $|A_r^{\pm1}|\le C$ on $\mathbb{D}(4c,\varepsilon)$ by the maximum principle.

(iii) For $\frac12\varepsilon\le|z-4c|<\varepsilon$ we have $|\zeta|\ge n^{2/3}\varepsilon/C\ge 2$, and $\frac23\zeta^{3/2} = \frac23 n f_r^{3/2} = -n\tilde\varphi$ (principal powers; $n^{2/3}>0$). By (5.17) and $\zeta^{-\sigma_3/4} = n^{-\sigma_3/6} f_r^{-\sigma_3/4}$,

$$\hat P_r (P_0^\infty)^{-1} = P_0^\infty (-\omega)^{-\sigma_3/2} M^{-1}\big(M + O(n^{-1})\big)(-\omega)^{\sigma_3/2}(P_0^\infty)^{-1} = I + O(1/n),$$

since P_0^∞, $(P_0^\infty)^{-1}$ and $(-\omega)^{\pm 1/2}$ are bounded there uniformly in n.

(iv) For $|\zeta| \ge 2$ put

$$\Theta_\infty := \zeta^{\sigma_3/4}\Psi_r(\zeta)e^{-\frac{2}{3}\zeta^{3/2}\sigma_3}(-\omega)^{\sigma_3/2}.$$

By (5.17) it is $(M + O(|\zeta|^{-3/2}))(-\omega)^{\sigma_3/2}$, bounded with bounded inverse for $|\zeta| \ge R_0$; for $2 \le |\zeta| \le R_0$ it is bounded because the Ψ^j are continuous, and so is its inverse, as $\det \Theta_\infty = 1$. For $|\zeta| \le 3$, $\Psi_r = \Psi^{\mathrm{I}} S_j$ with S_j one of the constant matrices I, $\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)J^{-1}$, $\left(\begin{smallmatrix}1 & 0\\ 1 & 1\end{smallmatrix}\right)$ by (5.16); put $\Theta_0 := S_j h^{\sigma_3}$. Here $|e^{\pm n\tilde\varphi}| = |e^{\mp\frac{2}{3}\zeta^{3/2}}| \le e^{2\sqrt{3}}$ and $(-\omega)^{\pm 1/2}$ is bounded. □

5.3. **The properties of the frame.** We now collect what Sections 6–9 use.

Proposition 5.6 (Properties of the frame)**.** *Let $p > 0$, $s > 1$, $\alpha = (s-1-p)/p$, let ℓ be analytic of angle ϑ for some $\vartheta > 0$, let $\vartheta_{\mathrm{w}}, \vartheta_{\mathrm{lip}}$ be the angles (4.2), and let F be the outer scalar of Lemma 5.1. The radii δ, ε and the contour of §4.6 may be taken as small as we wish (but fixed, independent of n). Let Q be the matrix (5.12). There is n_0 such that for $n \ge n_0$:*

(i) *(Contour and jumps.) Q is analytic in $\mathbb{C} \setminus \Sigma$, $Q(z) = I + O(1/z)$ at ∞, Q has continuous boundary values on Σ minus its vertices, and $Q_+ = Q_- v$ with $\det v = 1$, where $v = J = \left(\begin{smallmatrix}0 & 1\\ -1 & 0\end{smallmatrix}\right)$ on the band $(0, 4c)$ (oriented from left to right), and on every other arc v is the jump v_V of Lemma 4.11 (same orientation) with W_0 replaced by ω^{-1}, where $\omega := D_W^2/W_0$ and W_0 on a lip is the analytic continuation of W_0 from the band. In particular, on the lips $\Sigma_{\mathrm{lip}}^{\pm}$ (oriented away from 0), $v = \left(\begin{smallmatrix}1 & 0\\ \omega e^{2n\varphi} & 1\end{smallmatrix}\right) + E$ with $E = \left(\begin{smallmatrix}\iota_\pm(v_\pm - 1) & \omega^{-1}e^{-2n\tilde\varphi}\\ (v_\pm - 1)\omega e^{2n\varphi} & \iota_\mp(v_\pm - 1)\end{smallmatrix}\right)$ ($\iota_+ = 1$, $\iota_- = 0$); there is no jump on $(-\infty, 0)$, on $(n^p + \delta', \infty)$, or (for $p < 1$) on the seams. $|v^{\pm 1}| \le C$ on Σ. Σ is a finite union of C^2 arcs; $\Sigma \cap (\mathbb{D}(0, 2\delta) \cup \mathbb{D}(4c, 2\varepsilon))$ does not depend on n; the length of Σ is $O(n^p)$; $|\Sigma \cap \mathbb{D}(z, \rho)| \le C\rho$ for all z, ρ; and $\|\mathcal{C}_\pm\|_{L^2(\Sigma)\to L^2(\Sigma)} \le C$.*

(ii) *(Band weight.) $W_0 = \mathsf{w}_n x^\alpha (4c - x) H_n(x) r_n(x)$ with $\mathsf{w}_n \ne 0$ constant, $r_n(z) = \exp(\Lambda(\log n - \frac{1}{p}\log z) - \Lambda(\log n))$, and $H_n > 0$ on $(0, 4c + \varepsilon]$. H_n and r_n are analytic on $\mathcal{Z}_n$, which contains the closed sectors $\{0 < |z| \le \delta,\ |\arg z| \le \frac{5}{4}\vartheta_{\mathrm{lip}}\}$ and a neighbourhood of $(0, 4c + \varepsilon]$, and, uniformly in n,*

$$|\log H_n(z) - \log H_n(0^+)| \le C|z|^{\nu_p}, \quad |(\log H_n)'(z)| \le C|z|^{\nu_p - 1} \qquad (z \in \mathcal{Z}_n,\ |z| \le 1),$$
$$|\log H_n(x) - \log H_n(0^+)| \le C \qquad (0 < x \le 4c + \varepsilon).$$

The constant $\mathsf{w}_n H_n(0^+)$ cancels in $\omega = D_W^2/W_0$; D_W is given by (5.5) with $q = W_0$.

(iii) *(Lips.) In $\mathbb{D}(0, \delta)$ the lips are the preimages under $\zeta = n^2 f(z)$ of the rays $\arg\zeta = \mp(\pi - \vartheta_{\mathrm{lip}})$ (the upper lip is mapped to the lower ray); they leave 0 at the angles $\pm\vartheta_{\mathrm{lip}}$ and satisfy $\frac{3}{4}\vartheta_{\mathrm{lip}} \le |\arg z| \le \frac{5}{4}\vartheta_{\mathrm{lip}}$; the Bessel rays are $\arg\zeta = \pm(\pi - \vartheta_{\mathrm{lip}})$.*

(iv) *(Super-small entries.) $|E(z)| \le C\exp(-n|z|^{-1/p}/C)$ on the lips in $\mathbb{D}(0, \delta)$.*

(v) *(Parametrices away from 0.) P_0^∞ is given by (5.10): $\det P_0^\infty = 1$, $P_{0,+}^\infty = P_{0,-}^\infty J$ on $(0, 4c)$, and (5.11) holds. On $\Sigma \setminus (\mathbb{D}(0, \delta) \cup \mathbb{D}(4c, \varepsilon) \cup [0, 4c])$, $|v - I| \le Ce^{-n/C}$. In $\mathbb{D}(4c, \varepsilon)$ the Airy parametrix $\hat P_r$ of (5.20) has exactly the jumps v, $\det \hat P_r = 1$, and $\hat P_r (P_0^\infty)^{-1} = I + O(1/n)$ uniformly on $\frac{1}{2}\varepsilon \le |z - 4c| < \varepsilon$. It has the* two-scale structure *$\hat P_r = A_r n^{\sigma_3/6} M_{r,n}$, where $A_r^{\pm 1}$ are analytic on $\mathbb{D}(4c, \varepsilon)$ and bounded uniformly in n, f_r is conformal on a neighbourhood of $\overline{\mathbb{D}(4c, \varepsilon)}$ and independent of n, $f_r(4c) = 0$, and, with $\zeta = n^{2/3} f_r(z)$: for $|\zeta| \ge 2$, $M_{r,n} = \zeta^{-\sigma_3/4}\Theta_\infty$ with $|\Theta_\infty^{\pm 1}| \le C$; for $|\zeta| \le 3$, $M_{r,n} = \Psi^{\mathrm{I}}(\zeta)\Theta_0$ with Ψ^{I} entire, independent of n, $\det \Psi^{\mathrm{I}} = 1$, and $|\Theta_0^{\pm 1}| \le C$ (also for the boundary values).*

(vi) *(Hard-edge map.) f is conformal on a neighbourhood of $\overline{\mathbb{D}(0,\delta)}$, independent of n and of the weight, real on $(-\delta,\delta)$, $f(0)=0$, $f'(0)<0$, $f=(\varphi\pm i\pi)^2/4$ in $\mathbb{D}(0,\delta)\cap\mathbb{C}_\pm$; $\operatorname{Re}\varphi<0$ on the lips and $|e^{2n\varphi}|\le e^{-n|z|^{1/2}/C}$ on the lips in $\mathbb{D}(0,\delta)$.*

(vii) *(A priori bound.) If moreover $|\alpha|<1$, then $Q_\pm$ and $(Q^{-1})_\pm$ belong to $I+\partial\mathcal{C}(L^2(\Sigma))$, and for each fixed n and every $\eta>0$, $|Q(z)|\le C_{n,\eta}(1+|z|^{-|\alpha|/2-\eta})$ for $z\notin\Sigma$.*

(viii) *(Extraction.) For $|z|>n^p+\delta'+h_0$, $Q=k_\infty^{-\sigma_3}(Uz^{-n\sigma_3})k^{\sigma_3}$ with a scalar k analytic at ∞, $k(\infty)=k_\infty\neq 0$ (5.13). Consequently, with $Q=I+Q_1/z+Q_2/z^2+\dots$, $\mathsf{b}(Q)=(Q_1)_{12}(Q_1)_{21}=n^{2p}b_n^2$ and $\mathsf{a}(Q)=(Q_2)_{12}/(Q_1)_{12}-(Q_1)_{22}=n^p a_n$.*

Proof. Items (i)–(vi) collect the lemmas of Sections 4 and 5; the only new arguments are the uniform Cauchy bound in (i) and the arc-by-arc bound in (v). None of them uses $\alpha<1$.

(i) The jumps are those of Lemma 4.11 conjugated by $D_W^{\sigma_3}$, which has no jump off $[0,4c]$; on the band $D_{W,-}^{-\sigma_3}\left(\begin{smallmatrix}0 & W_0\\ -1/W_0 & 0\end{smallmatrix}\right)D_{W,+}^{\sigma_3}=J$ because $D_{W,+}D_{W,-}=W_0$; on the seams there is no jump by Lemma A.2(iv). The behaviour at ∞ is Lemma 5.3. Boundedness of $v^{\pm1}$ (note $\det v=1$): on the band $v=J$; on the lips in $\mathbb{D}(0,\delta)$ by Lemma 5.2(vi)(a) and Lemma 5.4; on the remaining arcs it follows from the bounds in (v) below, from Lemma 5.2(vi)(b) in $\mathbb{D}(4c,\varepsilon)$, and from $\operatorname{Re}\tilde\varphi<0$ on $(4c,4c+\varepsilon)$, $\operatorname{Re}\tilde\varphi>0$ on Σ_v near $4c$ (Lemma 4.5(iv)). The arcs inside $\{\operatorname{Re}z\le 4c+1\}$ do not depend on n; the rest of Σ consists of pieces of the three lines $\operatorname{Im}z\in\{0,\pm h_0\}$ and the top segment. Hence $|\Sigma\cap\mathbb{D}(z,\rho)|\le C\rho$. For the Cauchy operator, split $\Sigma=\Sigma_1\cup\Sigma_2\cup\Sigma_3$ according to $\operatorname{Re}z\le 4c+1$, $4c+1<\operatorname{Re}z<4c+3$, $\operatorname{Re}z\ge 4c+3$ ($n\ge n_0$). The blocks between $\Sigma_1\cup\Sigma_2$ and itself are bounded because this curve is fixed and Carleson (§2.6(a)). $\Sigma_2\cup\Sigma_3$ is contained in a translate of the fixed infinite curve Γ_∞ consisting of three horizontal half-lines ending on a vertical segment of length $2h_0$. Γ_∞ is Carleson (a finite union of lines and segments), so its Cauchy operators are bounded by §2.6(a); the Cauchy operator is translation invariant, and restricting a bounded operator $\mathcal{C}_\pm$ to functions supported on, and evaluated on, a sub-curve is a compression, which does not increase the norm. So these blocks are bounded uniformly in n. The blocks between Σ_1 and Σ_3 have kernels bounded by $|s-z|^{-1}$ with $|s-z|\ge 2$, and $\int_{\Sigma_1}\int_{\Sigma_3}|s-z|^{-2}|\,\mathrm{d}s||\,\mathrm{d}z|\le C$, so they are Hilbert–Schmidt with a uniform bound.

(ii) This is (4.24) and Definition 4.10(b)–(d); the sectors lie in $\mathcal{Z}_n$ because $\frac54\vartheta_{\mathrm{lip}}<\vartheta_{\mathrm{w}}$, and $|\log H_n-\log H_n(0^+)|\le C$ on $[1,4c+\varepsilon]$ by Definition 4.10(d). The cancellation is (5.8).

(iii) This is Lemma 4.6(a).

(iv) By Lemma 5.4(a), (b), Lemma 5.2(vi)(a) and $|e^{2n\varphi}|\le 1$, every entry of E is at most $C\exp(-\pi s_p n|z|^{-1/p})$.

(v) The properties of P_0^∞ were shown above, and those of $\hat P_r$ are Lemma 5.5. We bound $|v-I|$ arc by arc, using Lemma 5.2(vi)(c) for ω (all these arcs have distance $\ge d>0$ from $[0,4c]$, except the parts near $4c$, where (vi)(b) applies). *Lips outside $\mathbb{D}(0,\delta)$:* this part of the lips is compact and does not meet $[0,4c]$, so $\operatorname{Re}\varphi\le -1/C$ there (Lemma 4.5(i)), $|\operatorname{Im}\theta|\ge n/C$ (by (4.10), since $|\arg z|$ is bounded below and $|z|\le 4c+h_0$), and $\operatorname{Re}\tilde\varphi\ge 1/C$ by Lemma 4.6(b) and compactness; so all entries of $v-I$ are $O(e^{-n/C})$. *Σ_v outside $\mathbb{D}(4c,\varepsilon)$:* $\operatorname{Re}\tilde\varphi\ge 1/C$ by Lemma 4.5(iv) and compactness. *Saturated segment beyond $4c+\varepsilon$:* $|e^{2n\tilde\varphi}|=e^{2n(\gamma_p-U^*(x))}$ is at most $e^{-n/C}$ for $x\le x_0$ and at most x^{-n} for $x\ge x_0$ (Lemma 4.4(iii)), while $|\omega|\le\exp(C(1+\log(1+x))^2)$ and $\log x\le C\log n$; so $|v-I|\le Ce^{-n/C}$. *$\Sigma^{\triangle\pm}$ and the top segment:* for $z=x\pm iy$ with $x>4c$, $\operatorname{Re}g(z)\ge U^*(x)$, so $\operatorname{Re}\varphi(z)\le\gamma_p-U^*(x)$, which is $\le -1/C$ for $x\ge 4c+\varepsilon$ and $\le -\frac12\log x$ for $x\ge x_0$; near $x=4c$ the bound $\operatorname{Re}\varphi\le -1/C$ on $\Sigma^{\triangle\pm}$ follows from Lemma 4.5(i) by compactness. Moreover $|\operatorname{Im}\theta(x\pm ih_0)|\ge h_0 n\,x^{-1-1/p}/C$, so $|v_\pm|\le Cn^p$ on these arcs (and on the top segment, where t is close to 1, $|1-e^{\mp 2i\theta}|\ge n^{-p}/C$).

With $|\omega| \le \exp(C\log^2 n)$ this gives $|v - I| \le Cn^p e^{C\log^2 n} e^{-n/C} \le Ce^{-n/(2C)}$, and $\le Cn^{-pn/4}$ on the top segment.

(vi) Lemma 4.5(i), (iii) and Lemma 5.4(c); f is built from μ^* alone.

(vii) This is Proposition B.4 and Corollary B.5 of Appendix B, which assume $|\alpha| < 1$. This is the only item that uses that assumption.

(viii) The first statement is Lemma 5.3; the second follows from (4.7) and Lemma 4.2. □

6. The hard edge for analytic factors: comparison with a Bessel model

After Sections 4 and 5 the orthogonal polynomials of $\sigma_{p,s,\ell}$ are encoded in a matrix Q, normalised at infinity, whose jumps are listed in Proposition 5.6. Away from the two endpoints 0 and $4c$ of the band these jumps are those of an explicit global parametrix, up to exponentially small errors, and at the soft edge $4c$ the Airy parametrix of Lemma 5.5 applies. This section treats the hard edge 0. We compare Q with an explicitly solvable comparison problem and show that the two have the same first two coefficients at infinity up to $O(1/n)$ (Proposition 6.6); Section 7 then reads off a_n and b_n.

Standing assumptions. Throughout this section, Section 7 and Appendices B–D we assume that $p > 0$ and $1 < s < 1 + 2p$, so that the Bessel order $\alpha = (s - 1 - p)/p$ lies in $(-1, 1)$, and that $\ell(k) = e^{\Lambda(\log k)}$ is analytic of angle ϑ for some $\vartheta > 0$ (Definition 1.1). The notation of Sections 4 and 5 is in force. Where a lemma of the appendices holds more generally, this is said there.

Why the classical parametrix fails. The slowly varying factor enters the band weight $W_0 = \mathsf{w}_n x^\alpha (4c - x) H_n(x) r_n(x)$ through $r_n(z) = \ell(nz^{-1/p})/\ell(n)$, that is, $\log r_n(z) = \Lambda(\xi) - \Lambda(\log n)$ with $\xi = \log n - (\log z)/p$. The point $z = 0$ corresponds to $\xi = +\infty$, where Λ has no expansion: r_n is neither analytic at 0 nor of Fisher–Hartwig type. The Bessel parametrix of §2.5 needs a square root $\tilde W$ of W_0 continued around 0 across $(-\delta, 0)$. Continuing $r_n^{1/2}$ from the two half-planes produces two values on $(-\delta, 0)$ whose ratio is $\exp\big(\frac12(\Lambda(\xi - i\pi/p) - \Lambda(\xi + i\pi/p))\big) = 1 + O(\varepsilon_\Lambda)$; this even presupposes that Λ is analytic on the strip $|\operatorname{Im}\xi| \le \pi/p$, which Definition 1.1 does not provide. The deviation from 1 tends to 0 with no rate in n, and after conjugation by the Bessel model at the scale $|z| \asymp n^{-2}$ it becomes $O(n\varepsilon_\Lambda)$, so a quotient argument does not close. A Bessel order that drifts with n removes only the power-like part $z^{-\Lambda'(\log n)/p}$ of r_n. This is the situation of [2, Remark 4.10].

Outline of the comparison argument. Following Conway and Deift (§2.6), we build the Bessel parametrix with $\tilde W := D_W$, the Szegő function of the whole band weight (§6.3). It has exactly the band jump of Q and no jump on $(-\delta, 0)$. Together with the global parametrix and the Airy parametrix it gives an explicit matrix $\tilde Q$, so the matching errors on the two circles belong to the comparison problem and contribute explicit integrals at order $1/n$. The comparison problem differs from the problem for Q only on the lips of the lens, where the true jump contains $\omega = D_W^2/W_0$ and the model jump contains the constant $\varrho_\pm = e^{\mp i\pi\alpha}$. The difference $\omega - \varrho_\pm$ is $o(1)$ uniformly on the lips, without a rate (§6.4). That suffices for a Neumann series once the singular integral operator of the comparison problem is boundedly invertible uniformly in n, which we prove for $|\alpha| < 1$ (§6.5 and Appendix D). The difference formula of Conway and Deift then turns the mismatch into a change of the coefficients at infinity (§6.6). The factor $1/n$ comes from *Bessel localisation*: the mismatch carries the factor $e^{2n\varphi}$, which confines it to $|z| \lesssim n^{-2}$ up to a tail, and there the comparison solution has size $n^{1/2}|n^2 z|^{-|\alpha|/2}$; so a mismatch $\varepsilon_n' = o(1)$ produces an error $O(\varepsilon_n'/n)$. The places where $|\alpha| < 1$ is used are collected in Remark 6.7.

6.1. **Conventions.** All matrices are 2×2, $|\cdot|$ is a fixed matrix norm, e_1, e_2 are the standard basis vectors and

$$J := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad M = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}, \qquad \varrho_\pm := e^{\mp i\pi\alpha}.$$

Constants C may depend on p, s, ℓ and on the radii δ (at 0) and ε (at $4c$), but never on n or on the point z; they may change from line to line. Exponentially small factors are written $e^{-n/C}$. Powers and logarithms are principal unless said otherwise. The Cauchy transform $\mathcal{C}$, its boundary values $\mathcal{C}_\pm$, the operator $\mathcal{C}_v f = \mathcal{C}_-(f(v-I))$ and the class $I + \partial\mathcal{C}(L^2)$ are as in §2.6.

We work in the frame of Q (see (5.12)) and use only the eight items of Proposition 5.6, for fixed radii δ, ε that may be taken as small as we wish. In $\mathbb{D}(0,\delta)$ the contour consists of the band and two lips, with $v = J$ on $(0,\delta)$ and $v = \left(\begin{smallmatrix} 1 & 0 \\ \omega e^{2n\varphi} & 1 \end{smallmatrix}\right) + E$ on the lips, where $\omega = D_W^2/W_0$ (Proposition 5.6(i)); there is no jump on $(-\delta, 0)$. Since $f'(0) < 0$ and f is real on $(-\delta,\delta)$ (Proposition 5.6(vi)), f maps $\mathbb{D}(0,\delta)\cap\mathbb{C}_\pm$ into $\mathbb{C}_\mp$, the band $(0,\delta)$ into $(-\infty, 0)$ and $(-\delta, 0)$ into $(0,\infty)$; and $|f(z)| \ge |z|/C$, $|f'| + |f'|^{-1} \le C$ on $\mathbb{D}(0,\delta)$. On the lips in $\mathbb{D}(0,\delta)$

$$\tfrac{3}{4}\vartheta_{\mathrm{lip}} \le |\arg z| \le \tfrac{5}{4}\vartheta_{\mathrm{lip}} \tag{6.1}$$

by Proposition 5.6(iii). Finally, with the Hölder exponent $\nu_p = \min(1/p, \frac12)$ of Proposition 5.6(ii), put $T_n := \log n - \frac{1}{p}\log\max(4c, 1)$ and

$$\varepsilon_n' := n^{-2\nu_p}\log n + \varepsilon_\Lambda(T_n) \;\longrightarrow\; 0. \tag{6.2}$$

6.2. **The Bessel model and its local form.** Let Ψ_α be the Bessel model of §2.5, built on the three rays $\arg\zeta = \pm(\pi - \vartheta_{\mathrm{lip}})$ and $\arg\zeta = \pi$ (angle freedom, §2.5(b); for $\vartheta_{\mathrm{lip}} = \pi/3$ these are the rays of [10, §6]). We orient them as in §2.5(a): all three rays point towards the origin, with the $+$ side on the left. Thus Ψ_α is given by [10, (6.23)] in the sector $|\arg\zeta| < \pi - \vartheta_{\mathrm{lip}}$ and by the Hankel formulas [10, (6.24)–(6.25)] in the two thin sectors, it has the jumps [10, (6.16)–(6.18)] on the three rays, and it satisfies the large-ζ expansion [10, (6.28)]. We have $\det\Psi_\alpha \equiv 1$: in the central sector this is the Wronskian identity [8, (10.28.2)], and the jump matrices have determinant 1.

In the central sector $|\arg 2\zeta^{1/2}| < \pi/2$, so all Bessel functions there are principal branches [8, §10.25(ii)]. The expansion [10, (6.28)] remains valid for the rays at $\pm(\pi - \vartheta_{\mathrm{lip}})$, since the Bessel and Hankel functions in [10, (6.23)–(6.25)] are then evaluated in closed subsectors of the sectors of validity of the large-argument expansions of §2.7: [8, (10.40.1)–(10.40.5)] for I_α, K_α in the central sector, where $|\arg 2\zeta^{1/2}| \le \frac12(\pi - \vartheta_{\mathrm{lip}}) < \frac12\pi$, and [8, (10.17.5)–(10.17.6)] for the Hankel functions $H_\alpha^{(1,2)}(2(-\zeta)^{1/2})$ in the thin sectors, where $\arg 2(-\zeta)^{1/2}$ lies in $(-\frac12\vartheta_{\mathrm{lip}}, 0)$, respectively $(0, \frac12\vartheta_{\mathrm{lip}})$, well inside both sectors $[-\pi+\delta, 2\pi-\delta]$ and $[-2\pi+\delta, \pi-\delta]$.

Lemma 6.1 (Local form of the Bessel model)**.** *Let $\alpha \in (-1, 1)$ and let $\mathcal{S}_\pm$ denote the thin sectors $\pi - \vartheta_{\mathrm{lip}} < \pm\arg\zeta < \pi$.*

(i) *If $\alpha \ne 0$, there is an entire matrix function $\hat\Psi_\alpha$ with $\det\hat\Psi_\alpha = -2i\sin\pi\alpha$ such that*

$$\Psi_\alpha(\zeta) = \hat\Psi_\alpha(\zeta)\,\zeta^{\alpha\sigma_3/2}\,\mathsf{S}, \qquad \mathsf{S} := \begin{pmatrix} 1 & -\frac{i}{2\sin\pi\alpha} \\ 0 & \frac{i}{2\sin\pi\alpha} \end{pmatrix},$$

in $|\arg\zeta| < \pi - \vartheta_{\mathrm{lip}}$, while in $\mathcal{S}_\pm$ the constant S is replaced by S times a constant jump matrix of Ψ_α.

(ii) *If $\alpha = 0$, there is an entire $\hat\Psi_0$ with $\det\hat\Psi_0 = 1$ such that*

$$\Psi_0(\zeta) = \hat\Psi_0(\zeta)\begin{pmatrix} 1 & \mathsf{k}_0\log\zeta \\ 0 & 1\end{pmatrix}, \qquad \mathsf{k}_0 := -\frac{i}{2\pi},$$

in $|\arg\zeta| < \pi - \vartheta_{\mathrm{lip}}$, while in $\mathcal{S}_\pm$ the right-hand side is multiplied on the right by a constant jump matrix.

(iii) *Consequently, for $0 < |\zeta| \le 3$ off the rays,*

$$|\Psi_\alpha(\zeta)^{\pm1}| \le C|\zeta|^{-|\alpha|/2}, \qquad |\Psi_\alpha(\zeta)e_2e_1^{\mathsf T}\Psi_\alpha(\zeta)^{-1}| \le C|\zeta|^{-|\alpha|} \qquad (\alpha \ne 0),$$

and the same with $|\zeta|^{-|\alpha|/2}$, $|\zeta|^{-|\alpha|}$ replaced by $1+|\log|\zeta||$ and $(1+|\log|\zeta||)^2$ if $\alpha = 0$. The same bounds hold for the boundary values on the rays.

Proof. We expand the entries of [10, (6.23)] by the power series [8, (10.25.2)] and read off the singular factors.

(i) Let $\nu = \pm\alpha$. By (10.25.2) with $x = 2\zeta^{1/2}$ we have $(\frac12 x)^{\nu+2k} = \zeta^{\nu/2+k}$, hence

$$I_\nu(2\zeta^{1/2}) = \zeta^{\nu/2}\mathfrak p_\nu(\zeta), \qquad 2\pi i\zeta^{1/2}I_\nu'(2\zeta^{1/2}) = \pi i\zeta^{\nu/2}\sum_{k\ge0}\frac{(\nu+2k)\zeta^k}{k!\,\Gamma(\nu+k+1)} =: \zeta^{\nu/2}\mathfrak q_\nu(\zeta),$$

with $\mathfrak p_\nu(\zeta) = \sum_k \zeta^k/(k!\Gamma(\nu+k+1))$; $\mathfrak p_\nu$ and $\mathfrak q_\nu$ are entire because $1/\Gamma$ is entire. Write $\mathbf u_\nu := (\mathfrak p_\nu, \mathfrak q_\nu)^{\mathsf T}$, so that the first column of [10, (6.23)] is $\zeta^{\alpha/2}\mathbf u_\alpha$. By [8, (10.27.4)], $\frac{i}{\pi}K_\alpha = \frac{i}{2\sin\pi\alpha}(I_{-\alpha}-I_\alpha)$, and differentiating, $-2\zeta^{1/2}K_\alpha'(2\zeta^{1/2}) = \frac{i}{2\sin\pi\alpha}\cdot 2\pi i\zeta^{1/2}(I_{-\alpha}'-I_\alpha')(2\zeta^{1/2})$ (both sides equal $-\frac{\pi}{\sin\pi\alpha}\zeta^{1/2}(I_{-\alpha}'-I_\alpha')$). So the second column of [10, (6.23)] is $\frac{i}{2\sin\pi\alpha}(\zeta^{-\alpha/2}\mathbf u_{-\alpha} - \zeta^{\alpha/2}\mathbf u_\alpha)$, and

$$\Psi_\alpha = \begin{pmatrix}\mathbf u_\alpha & \mathbf u_{-\alpha}\end{pmatrix}\begin{pmatrix}\zeta^{\alpha/2} & -\frac{i}{2\sin\pi\alpha}\zeta^{\alpha/2} \\ 0 & \frac{i}{2\sin\pi\alpha}\zeta^{-\alpha/2}\end{pmatrix} = \hat\Psi_\alpha\,\zeta^{\alpha\sigma_3/2}\mathsf S, \qquad \hat\Psi_\alpha := \begin{pmatrix}\mathbf u_\alpha & \mathbf u_{-\alpha}\end{pmatrix}.$$

Since $\det\Psi_\alpha = 1$ and $\det\mathsf S = i/(2\sin\pi\alpha)$, $\det\hat\Psi_\alpha = -2i\sin\pi\alpha \ne 0$. In a thin sector, Ψ_α is the continuation of the central formula multiplied on the right by a constant jump matrix, because the jumps of Ψ_α are constant and $\hat\Psi_\alpha\zeta^{\alpha\sigma_3/2}\mathsf S$ is analytic in $\mathbb C\setminus(-\infty,0]$.

(ii) By [8, (10.31.2)], $K_0(x) = -(\ln\frac{x}{2}+\gamma_{\mathrm E})I_0(x) + S_0(\frac14x^2)$ with S_0 entire. At $x = 2\zeta^{1/2}$, $\ln\frac{x}{2} = \frac12\log\zeta$, so $\frac{i}{\pi}K_0(2\zeta^{1/2}) = -\frac{i}{2\pi}\log\zeta\cdot\mathfrak p_0(\zeta) + (\text{entire})$. Differentiating the same identity, $-2\zeta^{1/2}K_0'(2\zeta^{1/2}) = I_0(2\zeta^{1/2}) + (\frac12\log\zeta+\gamma_{\mathrm E})\,2\zeta^{1/2}I_0'(2\zeta^{1/2}) - 2\zeta S_0'(\zeta)$, and $2\zeta^{1/2}I_0'(2\zeta^{1/2}) = \mathfrak q_0(\zeta)/(\pi i)$. Hence the second column equals $\mathsf k_0\log\zeta\cdot\mathbf u_0(\zeta) + \mathbf h(\zeta)$ with $\mathsf k_0 = -i/(2\pi)$ and $\mathbf h$ entire, and $\Psi_0 = (\mathbf u_0\ \ \mathbf h)\left(\begin{smallmatrix}1 & \mathsf k_0\log\zeta\\ 0 & 1\end{smallmatrix}\right)$; $\det(\mathbf u_0\ \mathbf h) = \det\Psi_0 = 1$. The thin sectors are treated as in (i).

(iii) On $|\zeta|\le3$ the entire factors and their inverses are bounded, and the constant matrices are bounded. The middle factors satisfy $|\zeta^{\pm\alpha\sigma_3/2}| \le C|\zeta|^{-|\alpha|/2}$, respectively $|\left(\begin{smallmatrix}1 & \mathsf k_0\log\zeta\\ 0 & 1\end{smallmatrix}\right)^{\pm1}| \le C(1+|\log|\zeta||)$. For the rank-one product write

$$\Psi_\alpha e_2e_1^{\mathsf T}\Psi_\alpha^{-1} = \hat\Psi_\alpha\,\zeta^{\alpha\sigma_3/2}(\mathsf S_je_2)(e_1^{\mathsf T}\mathsf S_j^{-1})\zeta^{-\alpha\sigma_3/2}\,\hat\Psi_\alpha^{-1};$$

the entries of the middle matrix are constants times $1, \zeta^\alpha, \zeta^{-\alpha}$, whence the bound $C|\zeta|^{-|\alpha|}$; for $\alpha = 0$ the same computation gives entries of size $(1+|\log|\zeta||)^2$. □

The model in the normalisation used below. Put

$$\Phi_\alpha(\zeta) := \sigma_3\Psi_\alpha(\zeta)\sigma_3\,e^{-2\zeta^{1/2}\sigma_3}. \tag{6.3}$$

This is the left-endpoint model of §2.5(c) with the exponential factor attached; it does not depend on n. Orient the three rays $\Gamma_\alpha := (-\infty,0]\cup\{\arg\zeta = \pm(\pi-\vartheta_{\mathrm{lip}})\}$ *away* from the origin.

Reversing the orientation of a ray interchanges its $\pm$ sides, and $\sigma_3 J^{-1}\sigma_3 = J$, $\sigma_3\left(\begin{smallmatrix}1&0\\x&1\end{smallmatrix}\right)^{-1}\sigma_3 = \left(\begin{smallmatrix}1&0\\x&1\end{smallmatrix}\right)$, and $e^{2w\sigma_3}\left(\begin{smallmatrix}1&0\\x&1\end{smallmatrix}\right)e^{-2w\sigma_3} = \left(\begin{smallmatrix}1&0\\xe^{-4w}&1\end{smallmatrix}\right)$. With [10, (6.16)–(6.18)] this gives:

$$\Phi_{\alpha,+} = \Phi_{\alpha,-}J \text{ on } (-\infty,0), \qquad \Phi_{\alpha,+} = \Phi_{\alpha,-}\begin{pmatrix}1&0\\e^{\mp i\pi\alpha}e^{-4\zeta^{1/2}}&1\end{pmatrix} \text{ on } \arg\zeta = \mp(\pi-\vartheta_{\mathrm{lip}}). \tag{6.4}$$

On $(-\infty,0)$ we used that the two boundary values $w_\pm$ of $\zeta^{1/2}$ satisfy $w_+ = -w_-$, so that $e^{2w_-\sigma_3}Je^{-2w_+\sigma_3} = J$. Moreover $\det\Phi_\alpha = 1$, and Φ_α has the two-scale structure

$$\Phi_\alpha(\zeta) = \zeta^{-\sigma_3/4}\Theta_\infty(\zeta), \quad \Theta_\infty = (2\pi)^{-\sigma_3/2}\big(M^{-1}+O(\zeta^{-1/2})\big), \quad |\Theta_\infty^{\pm1}| \le C \quad (|\zeta|\ge 2), \tag{6.5}$$

$$\Phi_\alpha(\zeta) = \hat A(\zeta)\,\mathsf{P}(\zeta)\,\Theta_0(\zeta), \quad |\Theta_0^{\pm1}| \le C \qquad (|\zeta|\le 3), \tag{6.6}$$

where $\hat A = \sigma_3\hat\Psi_\alpha\sigma_3$ is entire with entire inverse, $\Theta_0 = \sigma_3\mathsf{S}_j\sigma_3 e^{-2\zeta^{1/2}\sigma_3}$ in the j-th sector, and $\mathsf{P}(\zeta) = \zeta^{\alpha\sigma_3/2}$ for $\alpha\ne 0$, $\mathsf{P}(\zeta) = I - \mathsf{k}_0(\log\zeta)e_1e_2^{\mathsf T}$ for $\alpha = 0$. Here (6.5) follows from [10, (6.28)], since $\sigma_3M\sigma_3 = M^{-1}$ and the exponential factors cancel; the bound on Θ_∞^{-1} holds for large $|\zeta|$ by the expansion and for $2\le|\zeta|\le R$ because $\det\Theta_\infty = 1$ and Θ_∞ is bounded. (6.6) is Lemma 6.1 conjugated by σ_3.

6.3. **The comparison parametrix and the comparison problem.** We first relate $e^{n\varphi}$ to the Bessel variable $\zeta = n^2 f(z)$. In $\mathbb{D}(0,\delta)\cap\mathbb{C}_+$ both $(\varphi+i\pi)/2$ and the principal root $f^{1/2}$ are analytic (the latter because f avoids $(-\infty,0]$ there), and their squares agree by Proposition 5.6(vi). So $(\varphi+i\pi)/2 = \pm f^{1/2}$ with a fixed sign. On the upper lip $\operatorname{Re}\varphi < 0$ while $\operatorname{Re} f^{1/2} > 0$, so the sign is $-$. The lower half-disc is treated in the same way, and we obtain, with $\zeta^{1/2} = nf^{1/2}$,

$$e^{n\varphi(z)} = (-1)^n e^{-2\zeta^{1/2}}, \qquad e^{2n\varphi(z)} = e^{-4\zeta^{1/2}} \qquad (z\in\mathbb{D}(0,\delta)\setminus[0,\delta)). \tag{6.7}$$

On the lips $\arg\zeta = \mp(\pi-\vartheta_{\mathrm{lip}})$, so $\operatorname{Re}\zeta^{1/2} = |\zeta|^{1/2}\sin(\vartheta_{\mathrm{lip}}/2)$ and

$$|e^{2n\varphi(z)}| = e^{-4\sin(\vartheta_{\mathrm{lip}}/2)|\zeta|^{1/2}} \le e^{-n|z|^{1/2}/C} \qquad \text{on the lips in } \mathbb{D}(0,\delta). \tag{6.8}$$

Following §2.5(c) with $\tilde W$ replaced by D_W, define in $\mathbb{D}(0,\delta)$

$$\tilde P_0 := (-1)^n E_n\,\sigma_3\Psi_\alpha(n^2f)\sigma_3\,D_W^{-\sigma_3}e^{n\varphi\sigma_3}, \qquad E_n := D_\infty^{\sigma_3}P_0^\infty M f^{\sigma_3/4}(2\pi n)^{\sigma_3/2}. \tag{6.9}$$

In the frame of Q this becomes, by (6.7),

$$\hat P_0 := D_\infty^{-\sigma_3}\tilde P_0 D_W^{\sigma_3} = \hat E_n\,\Phi_\alpha(n^2f), \qquad \hat E_n := P_0^\infty M f^{\sigma_3/4}(2\pi n)^{\sigma_3/2} = A_0\,(2\pi n)^{\sigma_3/2}, \tag{6.10}$$

where $A_0 := P_0^\infty M f^{\sigma_3/4}$. Thus $\hat P_0$ contains neither D_W nor any other trace of the weight.

Lemma 6.2 (Comparison parametrix)**.** (i) *The matrix A_0 has a removable singularity at 0, so it is analytic on $\mathbb{D}(0,\delta)$; and $\det A_0 = 1$. It depends neither on n nor on the weight, and $|A_0^{\pm1}| \le C$ there. In particular E_n is analytic in $\mathbb{D}(0,\delta)$, $\det E_n = 1$, and $D_\infty^{-\sigma_3}E_n(2\pi n)^{-\sigma_3/2} = A_0$.*

(ii) *$\hat P_0$ is analytic in $\mathbb{D}(0,\delta)\setminus\Sigma$, $\det\hat P_0 = 1$, and*

$$\hat P_{0,+} = \hat P_{0,-}J \text{ on } (0,\delta), \qquad \hat P_{0,+} = \hat P_{0,-}\begin{pmatrix}1&0\\\varrho_\pm e^{2n\varphi}&1\end{pmatrix} \text{ on the lips in } \mathbb{D}(0,\delta)\cap\mathbb{C}_\pm;$$

it has no jump on $(-\delta,0)$. Equivalently, $\tilde P_0$ has the band jump of V, no jump on $(-\delta,0)$ (nor on the seams if $p<1$), and the lip jump $\left(\begin{smallmatrix}1&0\\\varrho_\pm e^{2n\varphi}/D_W^2&1\end{smallmatrix}\right)$.

(iii) *Uniformly for $\frac12\delta\le|z|\le\delta$ off Σ, $\hat P_0(z)P_0^\infty(z)^{-1} = I + O(1/n)$, and $|\hat P_0^{\pm1}(z)| \le C$.*

Proof. (i) On $(0,\delta)$, $P^\infty_{0,+} = P^\infty_{0,-}J$, and since f is negative there and maps $\mathbb{C}_+$ to $\mathbb{C}_-$, $f_+^{\sigma_3/4} = f_-^{\sigma_3/4}\mathrm{diag}(-i,i)$. A direct computation gives $M^{-1}JM = \mathrm{diag}(i,-i)$, so $A_{0,+} = P^\infty_{0,-}JMf_-^{\sigma_3/4}\mathrm{diag}(-i,i) = P^\infty_{0,-}M\,\mathrm{diag}(i,-i)\mathrm{diag}(-i,i)f_-^{\sigma_3/4} = A_{0,-}$. On $(-\delta,0)$ both P_0^∞ and $f^{1/4}$ are analytic. So A_0 is analytic in $\mathbb{D}(0,\delta)\setminus\{0\}$. At 0, $|P_0^\infty| \le C|z|^{-1/4}$ by Proposition 5.6(v) and $|f^{\pm1/4}| \le C|z|^{\pm1/4}$, so $|A_0(z)| \le C|z|^{-1/2}$; an isolated singularity with this growth is removable. This is the removability argument of §2.5(c), applied to a matrix that involves only P_0^∞, M and f, and therefore neither n nor the weight; in particular the hypothesis of [10] that the weight be analytic at the endpoint plays no role. $\det A_0 = 1$ since each factor has determinant 1, and A_0^{-1} is the adjugate of A_0. Shrinking δ slightly, $A_0^{\pm1}$ are bounded on $\mathbb{D}(0,\delta)$.

(ii) $\hat E_n$ is analytic, so the jumps of $\hat P_0$ are those of $\zeta \mapsto \Phi_\alpha(\zeta)$ at $\zeta = n^2 f(z)$. The map $z \mapsto n^2 f(z)$ is conformal, hence preserves the sides of an oriented curve; it maps the band (left to right) onto part of $(-\infty,0)$ traversed away from 0, the lips (away from 0) onto the rays at $\mp(\pi - \vartheta_{\mathrm{lip}})$ traversed away from 0, and $(-\delta,0)$ into $(0,\infty)$, where Φ_α is analytic. So (6.4) and (6.7) give the stated jumps. For $\tilde P_0 = D_\infty^{\sigma_3}\hat P_0 D_W^{-\sigma_3}$, conjugation by the diagonal matrix $D_W^{-\sigma_3}$ turns J into the band jump of V because $D_{W,+}D_{W,-} = W_0$, turns the lip jump into the stated one, and creates no jump on $(-\delta,0)$ because D_W is analytic off $[0,4c]$.

(iii) For $\frac12\delta \le |z| \le \delta$ we have $|\zeta| = n^2|f(z)| \ge n^2\delta/C \ge 2$. By (6.5) and $(n^2f)^{-\sigma_3/4} = n^{-\sigma_3/2}f^{-\sigma_3/4}$,

$$\begin{aligned}\hat P_0 &= A_0(2\pi n)^{\sigma_3/2}(n^2f)^{-\sigma_3/4}(2\pi)^{-\sigma_3/2}\big(M^{-1}+O(n^{-1})\big)\\ &= P_0^\infty M f^{\sigma_3/4}f^{-\sigma_3/4}\big(M^{-1}+O(n^{-1})\big) = P_0^\infty\big(I+O(n^{-1})\big),\end{aligned}$$

and P_0^∞, $(P_0^\infty)^{-1}$ are bounded on the annulus. Both claims follow. □

The comparison problem. Let $\Sigma_\circ := \partial\mathbb{D}(0,\delta)\cup\partial\mathbb{D}(4c,\varepsilon)$, with both circles oriented clockwise (so that their + side is the outside), and $\hat\Sigma := \Sigma\cup\Sigma_\circ$. Define

$$\tilde Q(z) := \begin{cases}\hat P_0(z), & z\in\mathbb{D}(0,\delta),\\ \hat P_r(z), & z\in\mathbb{D}(4c,\varepsilon),\\ P_0^\infty(z), & \text{otherwise,}\end{cases}\qquad z\notin\hat\Sigma. \tag{6.11}$$

Then $\tilde Q$ is analytic off $\hat\Sigma$, $\det\tilde Q = 1$, $\tilde Q \to I$ at infinity, and $\tilde Q_+ = \tilde Q_-\tilde v$ on $\hat\Sigma$ with

$$\tilde v = \begin{cases} J & \text{on } (0,4c),\\ \left(\begin{smallmatrix}1 & 0\\ \varrho_\pm e^{2n\varphi} & 1\end{smallmatrix}\right) & \text{on the lips in } \mathbb{D}(0,\delta)\cap\mathbb{C}_\pm,\\ v & \text{on } \Sigma\cap\mathbb{D}(4c,\varepsilon),\\ I & \text{on } \Sigma\setminus\big(\mathbb{D}(0,\delta)\cup\mathbb{D}(4c,\varepsilon)\cup[0,4c]\big),\\ \tilde Q_{\mathrm{in}}^{-1}\tilde Q_{\mathrm{out}} & \text{on } \Sigma_\circ,\end{cases} \tag{6.12}$$

by Lemma 6.2(ii), Proposition 5.6(v), and the fact that P_0^∞ has only the band jump J. Here $\tilde Q_{\mathrm{in}}$, $\tilde Q_{\mathrm{out}}$ are the values from inside and outside the circle. We extend v by I to $\Sigma_\circ$; Q has no jump there. We call the problem "$\tilde Q_+ = \tilde Q_-\tilde v$ on $\hat\Sigma$, $\tilde Q\to I$" the *comparison problem*; it is solved explicitly by (6.11).

The two problems differ in three places. On the circles, $v - \tilde v = I - \tilde Q_{\mathrm{in}}^{-1}\tilde Q_{\mathrm{out}} = O(1/n)$, by Lemma 6.2(iii) and Proposition 5.6(v). On $\Sigma\setminus(\mathbb{D}(0,\delta)\cup\mathbb{D}(4c,\varepsilon)\cup[0,4c])$, $|v-\tilde v| \le Ce^{-n/C}$

by Proposition 5.6(v). On the lips in $\mathbb{D}(0,\delta)$, $v-\tilde v = d_n e_2 e_1^{\mathsf T} + E$ with

$$d_n(z) := \Big(\frac{D_W(z)^2}{W_0(z)} - \varrho_\pm\Big) e^{2n\varphi(z)}. \tag{6.13}$$

This last difference is the subject of the next subsection.

6.4. **The lip mismatch.**

Lemma 6.3 (Lip mismatch)**.** *For $n \ge n_0$ and z on the lips in $\mathbb{D}(0,\delta)\cap\mathbb{C}_\pm$,*

$$\big|\omega(z) - \varrho_\pm\big| \le C\big(|z|^{\nu_p}\log(2/|z|) + \varepsilon_\Lambda(T_n)\big), \qquad \omega = \frac{D_W^2}{W_0}. \tag{6.14}$$

Consequently $\sup_{\text{lips}}|d_n| \le C\varepsilon_n'$, and

$$\|v-\tilde v\|_{L^\infty(\hat\Sigma)} \le C\big(\varepsilon_n' + n^{-1}\big) \longrightarrow 0. \tag{6.15}$$

The convergence in (6.15) has no rate beyond that of $\varepsilon_\Lambda(T_n)$, about which nothing is assumed.

The proof treats the four factors of ω in (5.8) separately, by Lemma 5.2. The first two are explicit; the third is Hölder at 0; the fourth, the slowly varying factor, is handled by the scale-invariant path bound of Lemma 4.9(iii), which is the point of the lemma: its right side depends on x and z only through $\arg z$ and $x/|z|$.

Proof. Let z be on a lip in $\mathbb{D}(0,\delta)\cap\mathbb{C}_\pm$. Since $\delta \le \frac14\eta_*$ and (6.1) holds, $z \in Z(\frac14\eta_*)$ (notation of Lemma 5.2). We write $\log\omega$ as the sum of the logarithms of the four factors in (5.8).

The factor x^α. By Lemma 5.2(i), $D_{x^\alpha}^2 = (4cz/\Upsilon)^\alpha$ and $4cz/\Upsilon = (-z)(1+O(|z|^{1/2}))$. On the lips $|\arg(-z)| \le \pi - \frac34\vartheta_{\text{lip}}$, so for δ small the principal powers multiply, $(4cz/\Upsilon)^\alpha = (-z)^\alpha(1+O(|z|^{1/2}))$, and $(-z)^\alpha = e^{\mp i\pi\alpha}z^\alpha$ in $\mathbb{C}_\pm$. Hence $D_{x^\alpha}(z)^2/z^\alpha = \varrho_\pm(1+O(|z|^{1/2}))$.

The factor $4c-x$. By Lemma 5.2(i), $D_{4c-x}^2/(4c-z) = -4c/\Upsilon = 1+O(|z|^{1/2})$.

The factor H_n°. By Definition 4.10(c)–(d), Lemma 5.2(ii) applies to $q = H_n^\circ$ with $\mathfrak{c}_0 = 0$, $\nu = \nu_p$ and $\mathfrak{M} = C$, and gives $|\log(D_{H_n^\circ}^2/H_n^\circ)| \le C|z|^{\nu_p}\log(2/|z|)$.

The slowly varying factor. By Lemma 4.9(iii) with $R = \max(4c,1)$, for $z' \in Z(\eta_*)$ and $x \in (0,4c)$,

$$|\log r_n(x) - \log r_n(z')| \le \frac{\varepsilon_\Lambda(T_n)}{p}\big(\vartheta_{\text{w}} + |\log(x/|z'|)|\big), \qquad |(\log r_n)'(z')| = \frac{|\Lambda'(\xi(z'))|}{p|z'|} \le \frac{\varepsilon_\Lambda(T_n)}{p|z'|},$$

since $\operatorname{Re}\xi \ge T_n$ and $|\operatorname{Im}\xi| \le \vartheta_{\text{w}}/p \le \vartheta$ on $Z(\eta_*)$. So Lemma 5.2(iii) applies to $q = r_n$ with $\mathfrak{M} = C\varepsilon_\Lambda(T_n)$ and gives $|\log(D_{r_n}^2/r_n)| \le C\varepsilon_\Lambda(T_n)$.

Conclusion. Hence $\omega = \varrho_\pm e^w$ with $|w| \le C(|z|^{1/2} + |z|^{\nu_p}\log(2/|z|) + \varepsilon_\Lambda(T_n)) \le C_0$; since $|e^w - 1| \le e^{C_0}|w|$, $|\varrho_\pm| = 1$ and $|z|^{1/2} \le |z|^{\nu_p}$, (6.14) follows. By (6.8),

$$|d_n(z)| \le C\big(|z|^{\nu_p}\log(2/|z|) + \varepsilon_\Lambda(T_n)\big)e^{-n|z|^{1/2}/C} \le C\varepsilon_n',$$

because with $t = n|z|^{1/2}$ the first term is $t^{2\nu_p}n^{-2\nu_p}(\log 2 + 2\log n - 2\log t)e^{-t/C} \le Cn^{-2\nu_p}\log n$. On the lips $|E| \le C\exp(-n|z|^{-1/p}/C) \le C\exp(-n\delta^{-1/p}/C)$ by Proposition 5.6(iv). Together with the bounds on the circles and on the rest of Σ stated at the end of §6.3 this gives (6.15). □

6.5. **Uniform invertibility.** The comparison problem is explicit, but its solution is unbounded near 0 at the Bessel scale: on $|\zeta| \le 2$ it has size $n^{1/2}|\zeta|^{-|\alpha|/2}$. The following proposition says that, nevertheless, the singular integral operator attached to it is boundedly invertible on L^2, uniformly in n.

Proposition 6.4 (Uniform resolvent bound for the comparison problem)**.** *There are n_0 and C such that for $n \ge n_0$:*

(i) $\tilde{Q}_\pm \in I + \partial\mathcal{C}(L^2(\hat{\Sigma}))$ *and* $\tilde{Q}_\pm^{-1} \in I + \partial\mathcal{C}(L^2(\hat{\Sigma}))$*;*

(ii) $\|\mathcal{C}_\pm\|_{L^2(\hat{\Sigma})\to L^2(\hat{\Sigma})} \le C$*, and $1 - \mathcal{C}_{\tilde{v}}$ is invertible on $L^2(\hat{\Sigma})$ with $\|(1-\mathcal{C}_{\tilde{v}})^{-1}\| \le C$.*

The proof is in Appendix D; we sketch it (the uses of $|\alpha| < 1$ are listed in Remark 6.7). By the easy half of §2.6(e) it suffices to solve $m_+ = m_-\tilde{v} + \mathfrak{g}$, $m_\pm \in \partial\mathcal{C}(L^2)$, with $\|m_\pm\| \le C\|\mathfrak{g}\|$ for $\mathfrak{g}$ in a dense subspace, and to show uniqueness, which follows from §2.6(b) once (i) is known. The explicit candidate is $m := \mathcal{C}(\mathfrak{g}\tilde{Q}_+^{-1})\tilde{Q}$. Away from the two discs $\tilde{Q}^{\pm 1}$ is bounded and the bound reduces to that of $\mathcal{C}_\pm$. Near 0 we write $\tilde{Q} = A_0(2\pi)^{\sigma_3/2}n^{\sigma_3/2}\Phi_\alpha(n^2 f)$ and pass to the variable ζ: the analytic prefactor contributes Hilbert–Schmidt commutators, and what remains is, on a star of three rays, the Cauchy operator conjugated by $|\zeta|^{\pm 1/4}$ for $|\zeta| \ge 2$ and by $|\zeta|^{\pm\alpha/2}$ for $|\zeta| \le 3$ (Lemma C.1 with $\beta = \pm\alpha/2$; at $\alpha = 0$ a logarithmic commutator, Lemma C.2). These operators are dilation invariant, so their norms do not depend on n. The soft edge is treated in the same way with the Airy model; no α is involved there. Membership of $m_\pm$ in $\partial\mathcal{C}(L^2)$ is checked for smooth data vanishing near the vertices by Cauchy's integral formula, and the uniform bound extends the inverse to all of L^2.

Proposition 6.5 (Uniform resolvent bound for the true problem)**.** *Extend v by I to $\Sigma_\circ$. There is n_1 such that for $n \ge n_1$ the operator $1 - \mathcal{C}_v$ is invertible on $L^2(\hat{\Sigma})$ and $\|(1-\mathcal{C}_v)^{-1}\| \le 2C$, with C from Proposition 6.4. Moreover $Q_\pm \in I + \partial\mathcal{C}(L^2(\hat{\Sigma}))$ and $Q_- = (1-\mathcal{C}_v)^{-1}I$.*

Proof. Since $\mathcal{C}_v f - \mathcal{C}_{\tilde{v}} f = \mathcal{C}_-(f(v-\tilde{v}))$, Proposition 6.4(ii) and (6.15) give $\|\mathcal{C}_v - \mathcal{C}_{\tilde{v}}\| \le C(\varepsilon_n' + n^{-1})$, which is at most $1/(2C)$ for large n. The Neumann-series step §2.6(d) of Conway and Deift then gives the bound. Here only the convergence $\varepsilon_n' \to 0$ is used, not its rate.

By Proposition 5.6(vii), $Q - I = \mathcal{C}_\Sigma(Q_+ - Q_-)$. On $\Sigma_\circ$ the function Q has no jump, and it is bounded there by the same item (the circles stay away from 0), so its values on $\Sigma_\circ$ are the boundary values of the same Cauchy integral, now regarded on $\hat{\Sigma}$ with density 0 on $\Sigma_\circ$. Hence $Q_\pm \in I + \partial\mathcal{C}(L^2(\hat{\Sigma}))$. Then $Q_- - I = \mathcal{C}_-(Q_+ - Q_-) = \mathcal{C}_-(Q_-(v-I))$, i.e. $(1-\mathcal{C}_v)Q_- = I$ in $L^2(\hat{\Sigma})$ ($\hat{\Sigma}$ has finite length, so $I \in L^2(\hat{\Sigma})$), and $Q_- = (1-\mathcal{C}_v)^{-1}I$. □

6.6. **The effect of the mismatch on the moments.** Write $Q = I + Q_1/z + Q_2/z^2 + O(z^{-3})$ and $P_0^\infty = I + P_1/z + P_2/z^2 + O(z^{-3})$ at infinity; outside the discs $\tilde{Q} = P_0^\infty$, so $\tilde{Q}_k = P_k$. The circle terms

$$\Xi_k := -\frac{1}{2\pi i}\oint_{\Sigma_\circ} s^{k-1}\big(\tilde{Q}_{\mathrm{in}}(s)\tilde{Q}_{\mathrm{out}}(s)^{-1} - I\big)\,\mathrm{d}s \qquad (k = 1, 2) \tag{6.16}$$

are $O(1/n)$ by Lemma 6.2(iii) and Proposition 5.6(v); they are the first-order corrections of the comparison problem itself.

Proposition 6.6 (Effect of the mismatch on the moments)**.** *For $n \ge n_1$,*

$$Q_1 = P_1 + \Xi_1 + O\big(\varepsilon_n'/n + n^{-2}\big),$$
$$Q_2 = P_2 + \Xi_2 + (Q_1 - P_1)P_1 + O\big(\varepsilon_n'/n + n^{-2}\big).$$

The lip mismatch d_n contributes $O(\varepsilon_n'/n)$; the remaining errors come from the interaction of the circle terms with themselves and are $O(n^{-2})$. In particular $Q_k = P_k + O(1/n)$ for $k = 1, 2$. Moreover

$$\|Q_- - \tilde{Q}_-\|_{L^2(\hat{\Sigma})} \le C\big(\varepsilon_n' n^{-1/2} + n^{-1}\big). \tag{6.17}$$

Proof. We split $v - \tilde{v}$ into the lip mismatch, the circle terms and an exponentially small rest; the lip mismatch is estimated pointwise and then integrated using Bessel localisation, and the correction term is bounded by Cauchy–Schwarz. We apply §2.6(b) and its consequence (2.7) with $A = Q$, $B = \tilde{Q}$, $v_A = v$, $v_B = \tilde{v}$ on $\hat{\Sigma}$, with exponents $r = r' = 2$. The hypotheses hold: $v, \tilde{v}^{\pm 1} \in L^\infty$, $Q_\pm \in I + \partial\mathcal{C}(L^2)$ by Proposition 6.5, and $\tilde{Q}_\pm^{\pm 1} \in I + \partial\mathcal{C}(L^2)$ by Proposition 6.4(i). With $\mathcal{E} := Q_-(v - \tilde{v})\tilde{v}^{-1}\tilde{Q}_-^{-1}$, (2.7) reads

$$Q_1 - P_1 = -\frac{1}{2\pi i}\int_{\hat{\Sigma}} \mathcal{E}\,\mathrm{d}s, \qquad Q_2 - P_2 = -\frac{1}{2\pi i}\int_{\hat{\Sigma}} s\,\mathcal{E}\,\mathrm{d}s + (Q_1 - P_1)P_1.$$

Split $(v - \tilde{v})\tilde{v}^{-1} = \Delta_1 + \Delta_2 + \Delta_3$, where $\Delta_1 := d_n e_2 e_1^{\mathsf{T}}$ on the lips in $\mathbb{D}(0, \delta)$ (we used $e_1^{\mathsf{T}}\tilde{v}^{-1} = e_1^{\mathsf{T}}$ there), $\Delta_2 := \tilde{v}^{-1} - I$ on $\Sigma_\circ$, and Δ_3 is the rest: $E\tilde{v}^{-1}$ on the lips and $v - I$ on $\Sigma \setminus (\mathbb{D}(0, \delta) \cup \mathbb{D}(4c, \varepsilon) \cup [0, 4c])$. Then $\|\Delta_3\|_{L^\infty} \le Ce^{-n/C}$, on a set of length $O(n^p)$. Write $Q_- = \tilde{Q}_- + (Q_- - \tilde{Q}_-)$.

Pointwise bounds on the lips. On the lips in $\mathbb{D}(0, \delta)$ we have, by (6.10) and (6.7),

$$\tilde{Q}_-\Delta_1\tilde{Q}_-^{-1} = \big(\omega - \varrho_\pm\big)e^{-4\zeta^{1/2}}\, A_0(2\pi)^{\sigma_3/2}\Big[n^{\sigma_3/2}\Phi_{\alpha,-}e_2e_1^{\mathsf{T}}\Phi_{\alpha,-}^{-1}n^{-\sigma_3/2}\Big](2\pi)^{-\sigma_3/2}A_0^{-1}.$$

If $|\zeta| \ge 2$, then $n^{\sigma_3/2}\Phi_\alpha = f^{-\sigma_3/4}\Theta_\infty$ by (6.5), so the bracket is at most $C|f|^{-1/2} \le C|z|^{-1/2}$, and $|e^{-4\zeta^{1/2}}| \le e^{-n|z|^{1/2}/C}$ by (6.8). If $|\zeta| \le 2$, the identity

$$\Phi_\alpha e_2 e_1^{\mathsf{T}}\Phi_\alpha^{-1} = -e^{4\zeta^{1/2}}\sigma_3\Psi_\alpha e_2 e_1^{\mathsf{T}}\Psi_\alpha^{-1}\sigma_3$$

and Lemma 6.1(iii) bound the bracket times $e^{-4\zeta^{1/2}}$ by $Cn|\zeta|^{-|\alpha|}$. With the right side of (6.14),

$$\mathfrak{m}_n(z) := |z|^{\nu_p}\log(2/|z|) + \varepsilon_\Lambda(T_n),$$

we obtain

$$|\tilde{Q}_-\Delta_1\tilde{Q}_-^{-1}| \le C\mathfrak{m}_n(z)\cdot\begin{cases} n|\zeta|^{-|\alpha|}, & |\zeta| \le 2,\\ |z|^{-1/2}e^{-n|z|^{1/2}/C}, & |\zeta| \ge 2.\end{cases} \tag{6.18}$$

In the same way, $\Phi_\alpha e_2 = \zeta^{-\sigma_3/4}\Theta_\infty e_2$ and $e_1^{\mathsf{T}}\Phi_\alpha^{-1} = e_1^{\mathsf{T}}\Theta_\infty^{-1}\zeta^{\sigma_3/4}$ for $|\zeta| \ge 2$, and $|\Phi_\alpha^{\pm 1}| \le C|\zeta|^{-|\alpha|/2}$ for $|\zeta| \le 2$, give

$$|d_n\tilde{Q}_-e_2| + |d_ne_1^{\mathsf{T}}\tilde{Q}_-^{-1}| \le C\mathfrak{m}_n(z)\cdot\begin{cases} n^{1/2}|\zeta|^{-|\alpha|/2}, & |\zeta| \le 2,\\ |z|^{-1/4}e^{-n|z|^{1/2}/C}, & |\zeta| \ge 2.\end{cases} \tag{6.19}$$

For $\alpha = 0$, replace $|\zeta|^{-|\alpha|}$ by $(1 + |\log|\zeta||)^2$ and $|\zeta|^{-|\alpha|/2}$ by $1 + |\log|\zeta||$.

Bessel localisation. We integrate (6.18) along the lips, on which $|\,\mathrm{d}z| \le C\,\mathrm{d}|z|$ and $|\zeta| \ge n^2|z|/C$. The region $|\zeta| \le 2$ lies in $|z| \le Cn^{-2}$, where $\mathfrak{m}_n \le C\varepsilon_n'$; its contribution is at most

$$C\varepsilon_n'\int_0^{Cn^{-2}} n\,(n^2r/C)^{-|\alpha|}\,\mathrm{d}r = \frac{C\varepsilon_n'}{n}\int_0^C u^{-|\alpha|}\,\mathrm{d}u = O(\varepsilon_n'/n),$$

because $|\alpha|<1$. In the region $|\zeta|\ge 2$ substitute $t=nr^{1/2}$:

$$\int_0^{\delta}\mathfrak{m}_n(r)r^{-1/2}e^{-nr^{1/2}/C}\,\mathrm{d}r=\frac{2}{n}\int_0^{n\delta^{1/2}}\mathfrak{m}_n(t^2/n^2)e^{-t/C}\,\mathrm{d}t$$
$$\le\frac{C}{n}\big(\varepsilon_\Lambda(T_n)+n^{-2\nu_p}\log n\big)=O(\varepsilon_n'/n).$$

So $\int|\tilde Q_-\Delta_1\tilde Q_-^{-1}|\,|\mathrm{d}s|=O(\varepsilon_n'/n)$; the same computation applied to the square of (6.19) gives

$$\|d_n\tilde Q_-e_2\|_{L^2}^2+\|d_ne_1^{\mathsf T}\tilde Q_-^{-1}\|_{L^2}^2=O(\varepsilon_n'^2/n). \tag{6.20}$$

(For $\alpha=0$ the factor $\int_0^C u^{-|\alpha|}\,\mathrm{d}u$ becomes $\int_0^C(1+|\log u|)^2\,\mathrm{d}u<\infty$.)

The other main terms. On $\Sigma_\circ$, $\tilde Q_-\Delta_2\tilde Q_-^{-1}=\tilde Q_{\mathrm{in}}\tilde Q_{\mathrm{out}}^{-1}-I$, whose integrals are exactly the circle terms Ξ_k. The term Δ_3 contributes $O(e^{-n/C})$: on $\Sigma\setminus(\mathbb{D}(0,\delta)\cup\mathbb{D}(4c,\varepsilon))$, $\tilde Q^{\pm1}=(P_0^\infty)^{\pm1}$ is bounded and $|s|\le Cn^p$, and on the lips $|\tilde Q_-^{\pm1}|\le Cn^{1/2}|z|^{-1/2}$ (by (6.5), (6.6) and Lemma 6.1(iii)), so $|E||\tilde Q_-||\tilde Q_-^{-1}|\le C\exp(-n|z|^{-1/p}/C)\,n|z|^{-1}$ by Proposition 5.6(iv), whose integral is $O(e^{-n/C})$.

The correction term. By Proposition 6.5 and §2.6(c) applied to both problems, $Q_-=I+\mathcal{C}_vQ_-$ and $\tilde Q_-=I+\mathcal{C}_{\tilde v}\tilde Q_-$, so $(1-\mathcal{C}_v)(Q_--\tilde Q_-)=\mathcal{C}_-(\tilde Q_-(v-\tilde v))$ and

$$\|Q_--\tilde Q_-\|_{L^2}\le 2C\,\|\mathcal{C}_-\|\,\|\tilde Q_-(v-\tilde v)\|_{L^2}\le C\big(\|d_n\tilde Q_-e_2\|_{L^2}+n^{-1}+e^{-n/C}\big)\le C\big(\varepsilon_n'n^{-1/2}+n^{-1}\big),$$

which is (6.17), using (6.20), the bound $O(1/n)$ for $v-\tilde v$ on the circles (where $\tilde Q_-$ is bounded), and Δ_3. Similarly $\|(v-\tilde v)\tilde v^{-1}\tilde Q_-^{-1}\|_{L^2}\le C(\varepsilon_n'n^{-1/2}+n^{-1})$. By the Cauchy–Schwarz inequality,

$$\Big|\int(Q_--\tilde Q_-)(v-\tilde v)\tilde v^{-1}\tilde Q_-^{-1}\,\mathrm{d}s\Big|\le C\big(\varepsilon_n'n^{-1/2}+n^{-1}\big)^2\le C\big(\varepsilon_n'^2/n+n^{-2}\big).$$

The factor s in the second moment is bounded on the lips and the circles, and costs at most n^p on the rest, where everything is exponentially small. Collecting terms gives the proposition. □

Remark 6.7 (Where $|\alpha|<1$ is used; the case $\alpha\ge1$). The restriction $|\alpha|<1$ is used in an essential way in the following three places, and nowhere else in Sections 6–7 and Appendices B–D.

(a) Square integrability at the Bessel scale. The bound $|z|^{-|\alpha|/2-\eta}$ is square integrable on the lips near 0 for small $\eta>0$ (any $\eta<\frac12(1-|\alpha|)$). This gives $Q_\pm\in I+\partial\mathcal{C}(L^2)$, that is, Proposition 5.6(vii) (Corollary B.5), and $\tilde Q_\pm^{\pm1}\in I+\partial\mathcal{C}(L^2)$ (Proposition 6.4(i)). These are the hypotheses of §2.6(b) with $r=r'=2$, used in the uniqueness step of Appendix D, in Proposition 6.5 and in Proposition 6.6.

(b) The power-weighted Cauchy bound of Lemma C.1 with $\beta=\pm\alpha/2$, which needs $|\beta|<\frac12$, in the uniform bound near 0 (Lemma D.1).

(c) The integrability of $|\zeta|^{-|\alpha|}$ on a ray near 0. It is used in the Hilbert–Schmidt bounds of Lemma D.1 and in the Bessel localisation in the proof of Proposition 6.6.

Three further uses are not essential. The vertex conditions in Appendix D need only $|\alpha|<2$; the bound on Δ_3 in the proof of Proposition 6.6 uses $|\alpha|\le1$ only to bound a power of $|z|$ that is multiplied by an exponentially small factor; and Lemma 6.1 is stated for $\alpha\in(-1,1)$, although its part (i) needs only $\alpha\notin\mathbb{Z}$. The a priori bound of Proposition B.4 itself holds for every $\alpha>-1$.

For $\alpha\ge1$ all three essential uses fail: $\tilde Q$ is of size $n^{1/2}|\zeta|^{-\alpha/2}$ at the Bessel scale and is not square integrable on the lips near 0; Lemma C.1 needs $|\beta|=\alpha/2<\frac12$; and $\int_0^2u^{-\alpha}\,\mathrm{d}u$ diverges. (Since $s>1$, always $\alpha>-1$.) This is why Theorem 7.1 is combined with the Christoffel shift (Theorem 3.1) in Section 7. It is consistent with the remark of Deift and Piorkowski [6,

§1] quoted in §1.2. For pure powers with $p > 1$ the difficulty does not arise: there the band weight continues across $(-\delta, 0)$ up to a super-exponentially small jump, and the classical Bessel parametrix of Section 8 works for every $\alpha > -1$. A direct analysis of a non-constant analytic factor for $\alpha \geq 1$ remains open (Section 11).

Remark 6.8 (The circle terms). Proposition 6.6 expresses Q_1 and Q_2, up to $O(\varepsilon_n'/n + n^{-2})$, through the coefficients P_k of the global parametrix and the circle terms Ξ_k, but it does not compute the Ξ_k. On $\partial\mathbb{D}(0,\delta)$, $\tilde{Q}_{\mathrm{in}}\tilde{Q}_{\mathrm{out}}^{-1} = \hat{P}_0(P_0^{\infty})^{-1}$ involves only α, f and c, whereas the Airy parametrix contains $\omega = D_W^2/W_0$ (see (5.20)); so the Ξ_k may depend on ℓ at order $1/n$, but only through the soft edge. We do not analyse this dependence, which is why the $O(1/n)$ of Theorem 7.1 is absolute and not relative (Remark 1.5).

7. Proof of Theorems 7.1 and 1.2 and Corollary 1.3

Proposition 6.6 determines the first two coefficients of Q at infinity up to $O(1/n)$. Reading the recurrence coefficients off them proves the following theorem, the base case of the argument, in the window $|\alpha| < 1$ to which Section 6 is restricted. The Christoffel shift of Section 3 then carries the result to every $s > 1$, and the case $\ell \equiv 1$ gives Corollary 1.3.

Theorem 7.1 (Base window). *Let $p > 0$ and $1 < s < 1+2p$, so that $\alpha = (s-1-p)/p \in (-1, 1)$, and let ℓ be analytic of angle ϑ for some $\vartheta > 0$ (Definition 1.1). Then the recurrence coefficients of $\sigma_{p,s,\ell}$ satisfy*

$$n^p b_n = c(p)\big(1 + O(1/n)\big), \qquad \frac{a_n}{2b_n} = 1 + O(1/n) \qquad (n \to \infty),$$

with constants depending on p, s and ℓ.

Theorem 7.1 rests on Sections 4–6, and through them on the external results of §§2.4–2.7.

7.1. **Extraction.** The recurrence coefficients are read off from the first two coefficients of Q at infinity through the functionals a and b of (4.4): by Proposition 5.6(viii), which rests on the invariance of these functionals (Lemma 4.2), $\mathsf{b}(Q) = n^{2p}b_n^2$ and $\mathsf{a}(Q) = n^p a_n$.

Proof of Theorem 7.1. For $n \geq n_1$, Proposition 6.6 gives $Q_1 = P_1 + O(1/n)$ and $Q_2 = P_2 + O(1/n)$, where P_1, P_2 are the coefficients of the global parametrix (5.10). We compute them. With $a(z) = (1 - 4c/z)^{1/4}$,

$$a = 1 - \frac{c}{z} - \frac{3c^2}{2z^2} + O(z^{-3}), \qquad a^{-1} = 1 + \frac{c}{z} + \frac{5c^2}{2z^2} + O(z^{-3}),$$

so $\frac{1}{2}(a + a^{-1}) = 1 + \frac{c^2}{2z^2} + O(z^{-3})$ and $\frac{1}{2i}(a - a^{-1}) = \frac{ic}{z} + \frac{2ic^2}{z^2} + O(z^{-3})$. Hence

$$(P_1)_{12} = ic, \quad (P_1)_{21} = -ic, \quad (P_1)_{22} = 0, \quad (P_2)_{12} = 2ic^2,$$

and $\mathsf{b}(P_0^{\infty}) = c^2$, $\mathsf{a}(P_0^{\infty}) = 2c$. By Proposition 5.6(viii),

$$n^{2p}b_n^2 = (Q_1)_{12}(Q_1)_{21} = \big(ic + O(\tfrac{1}{n})\big)\big(-ic + O(\tfrac{1}{n})\big) = c^2\big(1 + O(\tfrac{1}{n})\big),$$

so $n^p b_n = c(1 + O(1/n))$ since $b_n > 0$. Since $(Q_1)_{12} = ic + O(1/n) \neq 0$ for large n,

$$n^p a_n = \mathsf{a}(Q) = \frac{2ic^2 + O(1/n)}{ic + O(1/n)} - O(\tfrac{1}{n}) = 2c\big(1 + O(\tfrac{1}{n})\big),$$

and therefore $a_n/(2b_n) = 1 + O(1/n)$. The constants depend on p, s and ℓ. □

The limits come entirely from the global parametrix. The values $\mathsf{b}(P_0^\infty) = c^2$ and $\mathsf{a}(P_0^\infty) = 2c$ are the squared off-diagonal entry and the diagonal entry of the constant Jacobi matrix with diagonal $2c$ and off-diagonal c, whose spectrum is the band $[0, 4c]$, and the length $4c(p)$ of the band is fixed by the saturation of the constrained equilibrium problem (Proposition 2.3). The weight, including the slowly varying factor, affects only the $O(1/n)$ terms.

7.2. **From the base window to every $s > 1$.** For a sequence of measures with recurrence coefficients a_n, b_n and a constant $c > 0$, the two forms of the rate statement are equivalent:

$$\Big\{n^p b_n = c\big(1 + O(1/n)\big),\quad \frac{a_n}{2b_n} = 1 + O(1/n)\Big\} \iff |n^p a_n - 2c| + |n^p b_n - c| = O(1/n). \tag{7.1}$$

From the left side, $n^p a_n = 2n^p b_n \cdot a_n/(2b_n) = 2c(1 + O(1/n))^2$; from the right side, $n^p b_n = c + O(1/n)$ and $a_n/(2b_n) = (2c + O(1/n))/(2c + O(1/n))$. We write $c = c(p)$.

Proof of Theorem 1.2. Let ℓ be analytic of angle ϑ for some $\vartheta > 0$ and let $s > 1$. We show

$$|n^p a_n(\sigma_{p,s,\ell}) - 2c| + |n^p b_n(\sigma_{p,s,\ell}) - c| = O(1/n), \tag{7.2}$$

which is the claim by (7.1). Let $m \geq 0$ be the least integer with $s_0 := s - mp < 1 + 2p$. If $m = 0$ then $s_0 = s > 1$; if $m \geq 1$ then $s - (m-1)p \geq 1 + 2p$, so $s_0 \geq 1 + p$. In both cases $s_0 \in (1, 1 + 2p)$. Theorem 7.1 at s_0 and (7.1) give (7.2) for $\sigma_{p,s_0,\ell}$. By (3.1), $\sigma_{p,s,\ell} = x^m \sigma_{p,s_0,\ell}$, and each $x^j \sigma_{p,s_0,\ell} = \sigma_{p,s_0+jp,\ell}$ is a positive measure on $(0, 1]$ with infinite support and finite moments. Theorem 3.1, applied m times with the regular rate $\eta_n = 1/n$, gives (7.2) for $\sigma_{p,s,\ell}$. □

Proof of Corollary 1.3. The constant sequence $\ell \equiv 1$ is $e^{\Lambda(\log k)}$ with $\Lambda \equiv 0$, which is analytic of angle ϑ for every $\vartheta > 0$ (with $\varepsilon_\Lambda \equiv 0$). Theorem 1.2 gives the claim for $\sigma_{p,s}$, for every $p > 0$ and $s > 1$. For the moment sequence of zeta values, let $q > 0$ and $2q + r > 1$. The measure $\sigma_{q,2q+r} = \sum_k k^{-(2q+r)} \delta_{k^{-q}}$ has the moments

$$\int x^j \, \mathrm{d}\sigma_{q,2q+r} = \sum_{k \geq 1} k^{-(2q+r)} k^{-qj} = \zeta(qj + 2q + r) \qquad (j \geq 0),$$

so its Hankel determinants are $\det_{1 \leq i,j \leq n} \zeta(q(i + j) + r)$ [15, Corollary 1.4], and its recurrence coefficients are those of the moment sequence $(\zeta(qj + 2q + r))_{j \geq 0}$. The first part, with $p = q$ and $s = 2q + r > 1$, gives $n^q b_n = c(q)(1 + O(1/n))$ and $a_n/(2b_n) = 1 + O(1/n)$. □

Remark 7.2 (Which part of Theorem 7.1 is used). Any half-open s-window of length p inside $(1, 1+2p)$ can serve as the base in the proof of Theorem 1.2. For $s > 1+p/2$ one may take s_0 to be the unique point of $s - p\mathbb{N}_0$ in $(1+p/2, 1+3p/2]$, so that $\alpha = (s_0 - 1 - p)/p \in (-\frac{1}{2}, \frac{1}{2}]$. Thus for every $s > 1+p/2$ only the case $|\alpha| \leq \frac{1}{2}$ of Theorem 7.1 is used, and the range $\frac{1}{2} < |\alpha| < 1$ is needed only for $1 < s \leq 1+p/2$, where no shift takes place. The case $\alpha = 0$ cannot be avoided: for $s \in 1 + p + p\mathbb{N}_0$ every admissible base point has $s_0 \in 1 + p\mathbb{Z}$, that is, $\alpha \in \mathbb{Z} \cap (-1, 1) = \{0\}$.

8. Pure powers with $p > 1$: the classical hard-edge parametrix

The comparison argument of Section 6 bounds the hard-edge contribution by $O(1/n)$ but cannot compute its $1/n$ term, and it is confined to $|\alpha| < 1$. For $\ell \equiv 1$ and $p > 1$ neither limitation is necessary: the band weight continues across $(-\delta, 0)$ up to a super-exponentially small jump, so the classical left-endpoint Bessel parametrix of [10, §6] can be used at the hard edge for every Bessel order $\alpha > -1$. This section proves the first-order small-norm estimate in that setting (Proposition 8.4). It gives a second proof of Corollary 1.3 for $p > 1$, without the Christoffel shift, and it sets up the error matrix whose expansion to second order gives Theorem 1.4 in Section 9. The construction fails at $p = 1$, and we do not use it for

$p < 1$ (Remark 8.5). The last subsection states a localisation bound for the spectral weights (Proposition 8.6), used in [16]; its proof is in Appendix F.

Standing assumptions. In §§8.1–8.3 we fix $p > 1$, $s > 1$ and $\ell \equiv 1$, so that $\sigma = \sigma_{p,s}$, $r_n \equiv 1$ and $\alpha = (s-1-p)/p > -1$ is arbitrary. The function $\Lambda \equiv 0$ is analytic of every angle, so we may take $\vartheta \geq 7\pi/(16p)$ in (4.2); then $\mathsf{m} = 1$, $\vartheta_{\mathrm{lip}} = \pi/3$, $\vartheta_{\mathrm{w}} = 7\pi/16$, and the Bessel rays $\arg\zeta = \pm(\pi - \vartheta_{\mathrm{lip}}) = \pm 2\pi/3$ are those of [10, §6]. Sections 4 and 5 apply without change; we use the items (i)–(vi) and (viii) of Proposition 5.6, never item (vii), which needs $|\alpha| < 1$. The outer scalar is $F = DBK$ with the Binet-cut scalar B of Lemma A.1, the band factor is $H_n = (\tilde{D}B)^{-2}$, and $W_0 = \mathsf{w}_n z^\alpha (4c - z) H_n(z)$ is analytic on $\mathbb{C} \setminus (-\infty, 0]$. We write $H_n^\circ = H_n / H_n(0^+)$ as in (5.7), $\omega = D_W^2 / W_0$ as in (5.8), and

$$\omega^{\mathrm{mod}} := \frac{D_{x^\alpha}^2}{z^\alpha} \cdot \frac{D_{4c-x}^2}{4c - z}, \qquad \text{so that} \qquad \omega = \omega^{\mathrm{mod}} \, \frac{D_{H_n^\circ}^2}{H_n^\circ}; \tag{8.1}$$

ω^{mod} is the ratio (5.8) for the model weight $\mathsf{w}_n x^\alpha (4c - x)$, in which the band factor is replaced by a constant. Constants C depend on p, s and the contour parameters, never on n or z.

8.1. **The band factor near the negative axis.** Write $t_+(y)$, $\theta_+(y)$ for the boundary values from $\mathbb{C}_+$ at $y < 0$, and put

$$\begin{gathered} G := -j(t) + \log B \quad \text{on } \mathbb{C} \setminus (-\infty, 0], \\ \mathfrak{e}(y) := 2 \operatorname{Im} \operatorname{Log}\bigl(1 - e^{-2i\theta_+(y)}\bigr) \quad (y < 0), \qquad r_* := 4c + 2, \end{gathered} \tag{8.2}$$

with Binet's function j of Section 5. Then $H_n = e^{-2G}$ by Lemma A.1(iv), since $\tilde{D} = e^{-j(t)}$.

Lemma 8.1 (Band factor near the negative axis)**.** *Let $p > 1$. There are C and n_0 such that for $n \geq n_0$:*

(i) *G has continuous boundary values $G_\pm$ on $(-\infty, 0)$ and $G_+ - G_- = i\mathfrak{e}$ there. The function $\mathfrak{e}$ is real-analytic on $(-\infty, 0)$, and*

$$|\mathfrak{e}(y)| \leq \pi \exp\bigl(-2\pi \sin(\pi/p)\, n |y|^{-1/p}\bigr) \qquad (y < 0);$$

on $[-2r_*, 0)$ both $\mathfrak{e}$ and $\mathfrak{e}'$ are bounded by $Ce^{-n/C}$ and tend to 0 faster than every power of $|y|$ as $y \to 0^-$.

(ii) *For $z \in \mathbb{C} \setminus (-\infty, 0]$,*

$$G(z) - G(1) = \frac{1}{2\pi} \int_{-\infty}^{0} \mathfrak{e}(y) \, \frac{z - 1}{(y - z)(y - 1)} \, \mathrm{d}y. \tag{8.3}$$

(iii) *$|G(z) - G(1)| \leq Cn^{-p}$ for $|z| \leq r_*$, $z \notin (-\infty, 0]$, and for the boundary values on $(-r_*, 0)$; $|G'(z)| \leq Cn^{-p}$ for $|z| \leq \frac{1}{2} r_*$, with the same convention; and $|G'(z)| \leq Cn^{-p}/\eta$ whenever $0 < \eta \leq 1$, $|z| \leq r_* - \eta$ and $\operatorname{dist}(z, (-\infty, 0]) \geq \eta$.*

(iv) *$\bigl|\log\bigl(\omega(z)/\omega^{\mathrm{mod}}(z)\bigr)\bigr| \leq Cn^{-p}$ for $z \in \mathbb{D}(0, 2\delta) \cup \mathbb{D}(4c, 2\varepsilon)$, $z \notin [0, 4c]$, including the boundary values on $(-2\delta, 0)$.*

Part (iii) says that the band factor is constant up to $O(n^{-p})$ on a fixed neighbourhood of the band, although each of its two factors varies at order $1/n$. Here $-j(t(z)) = -z^{1/p}/(12n) + O(n^{-3})$ on compact sets by §2.7(a), while (iii) and $\log B(1) = 0$, $-j(n) = -1/(12n) + O(n^{-3})$ give $\log B(z) = (z^{1/p} - 1)/(12n) + O(n^{-p})$: the Binet-cut scalar removes exactly the cut of the Binet term, and the two $1/(12n)$ terms cancel up to a constant. The non-constant part of G comes from $|y| \asymp n^p$ in (8.3). At $p = 2$, $\theta_+(y) = -i\pi n |y|^{-1/2}$ is purely imaginary, so $e^{-2i\theta_+} \in (0, 1)$, $\mathfrak{e} \equiv 0$, and H_n is exactly constant.

Proof. (i) On $(-\infty,0)$, $t_- = \overline{t_+}$ and j is real on $(0,\infty)$, so $j(t_-) = \overline{j(t_+)}$; by Lemma A.1(ii)–(iii), $(\log B)_+ - (\log B)_- = -i\mathfrak{b}$ with $\mathfrak{b} = 2\operatorname{Im}(-j(t_+) - \operatorname{Log}(1-e^{-2i\theta_+}))$. Hence $G_+ - G_- = -2i\operatorname{Im} j(t_+) - i\mathfrak{b} = 2i\operatorname{Im}\operatorname{Log}(1-e^{-2i\theta_+}) = i\mathfrak{e}$. With $q := e^{-2i\theta_+(y)}$ we have $|q| = \exp(-2\pi\sin(\pi/p)n|y|^{-1/p}) < 1$ by (4.10), since $p>1$; and $|\operatorname{Im}\operatorname{Log}(1-q)| \le \arcsin|q| \le \frac{\pi}{2}|q|$, which is the bound on $\mathfrak{e}$. Real-analyticity follows from that of θ_+. For the derivative, $\mathrm{d}t_+/\mathrm{d}y = -t_+/(py)$ gives $|q'| \le Cn|y|^{-1-1/p}|q|$, and $|(\operatorname{Log}(1-q))'| \le 2|q'|$ when $|q| \le \frac12$; on $[-2r_*,0)$ we have $|y|^{-1/p} \ge (2r_*)^{-1/p}$, so $|q| \le e^{-n/C} \le \frac12$ for $n \ge n_0$, and the remaining claims follow.

(ii) Let K be $G(z) - G(1)$ minus the right-hand side of (8.3). The integral converges absolutely, because $\mathfrak{e}$ is bounded and the kernel is $O(|y|^{-2})$; by the Sokhotski–Plemelj formula its jump across $(-\infty,0)$ is $i\mathfrak{e}$, with continuous boundary values. So K has no jump, and it is analytic on $\mathbb{C}\setminus\{0\}$ by Morera's theorem. Near 0, $|\arg t| \le \pi/p < \pi$ and $|t| \to \infty$, so $j(t) \to 0$ by (5.2); $\log B$ is bounded near 0, together with its boundary values on $(-\infty,0)$, by the proof of Lemma A.1(iv) (the verification of Definition 4.10(e)); and the integral is bounded near 0 because $\mathfrak{e}$ vanishes to infinite order there. So 0 is a removable singularity. At infinity, for $|t| \le 1$ and $|\arg t| \le \pi/p$, §2.7(a) gives $\operatorname{Ln}\Gamma(t) = \operatorname{Ln}\Gamma(1+t) - \ln t$ with $|1+t| \ge \sin(\pi - \pi/p) > 0$, so $|j(t)| \le C(1+|\log|z||)$; $|\log B(z)| \le C_n(1+\log|z|)$ for $|z| \ge 2$ off the cut, again by the proof of Lemma A.1(iv); and the integral is $O(\log|z|)$. So K is entire with $K(z) = O(\log|z|)$, hence constant, and $K(1) = 0$.

(iii) Split (8.3) at $y = -2r_*$. For $|y| \ge 2r_*$ and $|z| \le r_*$ the kernel is at most $C|y|^{-2}$, and the substitution $|y| = n^p w$ gives

$$\int_{2r_*}^{\infty} e^{-\varrho n u^{-1/p}} u^{-2}\,\mathrm{d}u \le n^{-p}\int_0^{\infty} e^{-\varrho w^{-1/p}} w^{-2}\,\mathrm{d}w = Cn^{-p}, \qquad \varrho := 2\pi\sin(\pi/p),$$

the last integral being finite at both ends. On $[-2r_*,0)$ the density $\mathfrak{e}$ is C^1 with norm $O(e^{-n/C})$ and vanishes to infinite order at 0; the Cauchy integral of such a density, and its boundary values on $(-r_*,0)$, are bounded by $Ce^{-n/C}(1+|\log\operatorname{dist}(z,-2r_*)|) \le C'e^{-n/C}$ for $|z| \le r_*$ (subtract $\mathfrak{e}(\operatorname{Re} z)$ on a window around $\operatorname{Re} z$, as in the proof of Lemma A.1(iv)). This proves the first bound. Differentiating (8.3), the far part gives $\int_{|y|\ge 2r_*} |\mathfrak{e}||y|^{-2}\,\mathrm{d}y = O(n^{-p})$ and the near part, after an integration by parts, is the Cauchy integral of the C^1 density $\mathfrak{e}'$ plus a boundary term at $-2r_*$ that is bounded for $|z| \le \frac12 r_*$; this gives the second bound. The third is Cauchy's estimate applied to the first on the disc $\mathbb{D}(z,\eta) \subset \{|w| \le r_*\}\setminus(-\infty,0]$.

(iv) By (8.1) and (5.6) with $\mathfrak{c} = \log H_n^\circ(z)$,

$$\log\frac{\omega(z)}{\omega^{\mathrm{mod}}(z)} = \frac{\mathcal{R}(z)}{\pi}\int_0^{4c}\frac{\log H_n^\circ(x) - \log H_n^\circ(z)}{\sqrt{x(4c-x)}}\,\frac{\mathrm{d}x}{z-x}.$$

Since $\log H_n^\circ = -2(G - G(0^+))$ and $G(0^+) = \lim_{x\to 0^+} G(x)$, (iii) gives $|\log H_n^\circ| \le Cn^{-p}$ on $\{|z| \le r_*\}$, boundary values on the cut included. Let $z \in \mathbb{D}(0,2\delta)\setminus[0,2\delta)$. For $x \le \frac12 r_*$ the segment from x to z (from x to $z \pm i0$ if z lies on the cut) avoids $(-\infty,0)$, so $|\log H_n^\circ(x) - \log H_n^\circ(z)| \le Cn^{-p}|x-z|$ by the second bound of (iii); for $x > \frac12 r_*$ we have $|x-z| \ge c$ and use the sup bound. With $|\mathcal{R}(z)| \le C$ the integral is $O(n^{-p})$. For $z \in \mathbb{D}(4c,2\varepsilon)\setminus[0,4c]$ the same argument applies with the third bound of (iii) and $\eta = 1$ on $x \in (4c-4\varepsilon, 4c)$ (here $|w| \le r_* - 1$ and $\operatorname{dist}(w,(-\infty,0]) \ge 4c - 4\varepsilon \ge 1$ on the segment, as $c > \frac12$ for $p>1$ by (1.3)), and the sup bound with $|x-z| \ge \varepsilon$ elsewhere. □

8.2. **The classical Bessel parametrix.** We transcribe the left-endpoint construction of [10, (6.42)–(6.53)] (§2.5(c)) into the frame of Q. Fix a square root $\mathsf{w}_n^{1/2}$ and put, on $\mathbb{D}(0,2\delta)\setminus$

$(-2\delta, 2\delta)$,

$$\tilde W(z) := \mathsf{w}_n^{1/2}(-z)^{\alpha/2}(4c-z)^{1/2}e^{-G(z)}, \qquad \rho_n := \tilde W/D_W, \tag{8.4}$$

with principal powers, so that $\tilde W^2 = \mathsf{w}_n(-z)^\alpha(4c-z)H_n$. Since $(-z)^\alpha = e^{\mp i\pi\alpha}z^\alpha$ in $\mathbb{C}_\pm$, the identities

$$\begin{aligned} \tilde W^2 = e^{\mp i\pi\alpha}W_0, \qquad \rho_n^2 = \frac{e^{\mp i\pi\alpha}}{\omega} \quad \text{in } \mathbb{C}_\pm, \\ \rho_{n,+}\rho_{n,-} = 1 \text{ on } (0,2\delta), \qquad \rho_{n,+} = \rho_{n,-}e^{-i\mathfrak{e}} \text{ on } (-2\delta,0) \end{aligned} \tag{8.5}$$

hold: on the band $\tilde W_+\tilde W_- = W_0 = D_{W,+}D_{W,-}$, and on the negative axis D_W and $(-z)^{\alpha/2}$ are analytic while e^{-G} jumps by $e^{-i\mathfrak{e}}$ (Lemma 8.1(i)). This is the analogue of [10, (6.44)–(6.46)]. Let Ψ_α be the Bessel model of §2.5(a) on KMVV's rays, $\Phi_\alpha(\zeta) := \sigma_3\Psi_\alpha(\zeta)\sigma_3 e^{-2\zeta^{1/2}\sigma_3}$ (the matrix (6.3), now for every $\alpha > -1$), and define in $\mathbb{D}(0,2\delta)$

$$\check A := P_0^\infty\,\rho_n^{\sigma_3} M f^{\sigma_3/4}, \qquad \check P_0 := \check A\,(2\pi n)^{\sigma_3/2}\,\Phi_\alpha(n^2 f)\,\rho_n^{-\sigma_3}. \tag{8.6}$$

If $\tilde W$ is replaced by D_W in (8.4), then $\rho_n = 1$ and $\check A = A_0$, and $\check P_0$ becomes the comparison parametrix (6.10) of Section 6.

Lemma 8.2 (Classical Bessel parametrix)**.** *Let $p > 1$, $s > 1$, $\ell \equiv 1$. For $n \ge n_0$:*

(i) $|\rho_n^{\pm1}| \le C$ *on* $\mathbb{D}(0,2\delta)\setminus\mathbb{R}$, *with boundary values;* $\det\check A = 1$; *and* $\check A = \check A^{\mathrm a} + \check A^{\mathrm s}$ *on* $\mathbb{D}(0,\delta)$, *where* $\check A^{\mathrm a}$ *is analytic on* $\mathbb{D}(0,\delta)$ *and* $|\check A^{\mathrm s}| \le Ce^{-n/C}$. *In particular* $|\check A^{\pm1}| \le C$ *on* $\mathbb{D}(0,\delta)$.

(ii) $\check P_0$ *is analytic in* $\mathbb{D}(0,\delta)\setminus(\Sigma\cup(-\delta,0))$, $\det\check P_0 = 1$, *and*

$$\check P_{0,+} = \check P_{0,-}J \text{ on } (0,\delta), \qquad \check P_{0,+} = \check P_{0,-}\begin{pmatrix} 1 & 0 \\ \omega e^{2n\varphi} & 1\end{pmatrix} \text{ on the lips in } \mathbb{D}(0,\delta),$$

and $\check P_{0,+} = \check P_{0,-}v_{\mathrm h}$ *on* $(-\delta,0)$ *with* $|v_{\mathrm h} - I| \le Cn^C|x|^{-C}\exp(-2\pi\sin(\pi/p)\,n|x|^{-1/p})$.

(iii) $|\check P_0^{\pm1}(z)| \le Cn^C|z|^{-C}$ *on* $\mathbb{D}(0,\delta)$ *off the contour, with boundary values.*

(iv) *Uniformly for* $\frac12\delta \le |z| \le \delta$ *off the contour,* $\check P_0(z)P_0^\infty(z)^{-1} = I + O(1/n)$ *and* $|\check P_0^{\pm1}(z)| \le C$.

The lemma uses the Bessel model [10, (6.16)–(6.18), (6.23)–(6.28), Theorem 6.3] with its behaviour at 0 [10, (6.19)–(6.21)], and the removability argument of [10, proof of Proposition 6.5] (§2.5(a), (c)). Since $h(z)$ in [10, (1.1)] is not analytic at our endpoint, that argument is applied after the super-small jump of ρ_n on $(-\delta,0)$ has been split off, as in (i).

Proof. (i) By (8.5), $\rho_n^2 = e^{\mp i\pi\alpha}/\omega$, and by (8.1) and Lemma 8.1(iv) it suffices to bound ω^{mod} above and below on $\mathbb{D}(0,2\delta)\setminus\mathbb{R}$. This follows from Lemma 5.2(i): $|D_{x^\alpha}(z)^2| \asymp |z|^\alpha$ and $|D_{4c-x}(z)^2| \asymp |z-4c|$. Each factor of $\check A$ has determinant 1. On $(0,\delta)$, $P_{0,+}^\infty = P_{0,-}^\infty J$, $\rho_{n,+}^{\sigma_3} = \rho_{n,-}^{-\sigma_3}$ and $f_+^{\sigma_3/4} = f_-^{\sigma_3/4}\mathrm{diag}(-i,i)$ (the proof of Lemma 6.2(i)); with $J\rho^{-\sigma_3} = \rho^{\sigma_3}J$ and $JM\,\mathrm{diag}(-i,i) = M$ this gives $\check A_+ = \check A_-$. On $(-2\delta,0)$ the factors P_0^∞ and $f^{1/4}$ are analytic ($f > 0$ there), so by (8.5)

$$\check A_-^{-1}\check A_+ = f^{-\sigma_3/4}M^{-1}e^{-i\mathfrak{e}\sigma_3}Mf^{\sigma_3/4}, \qquad |\check A_-^{-1}\check A_+ - I| \le C|x|^{-1/2}|\mathfrak{e}(x)|. \tag{8.7}$$

The crude bound $|\check A| \le C|z|^{-1/2}$ on $\mathbb{D}(0,2\delta)$ follows from (5.11), the bound on ρ_n and $|f^{\pm1/4}| \le C|z|^{\pm1/4}$. Put $\check A^{\mathrm s}(z) := \frac{1}{2\pi i}\int_{-2\delta}^0 \check A_-(y)(\check A_-^{-1}\check A_+ - I)(y)\frac{\mathrm dy}{y-z}$. By (8.7) and Lemma 8.1(i),(iv) the density is C^1 on $[-2\delta,0)$ with norm $O(e^{-n/C})$ and vanishes to infinite order at 0, so $|\check A^{\mathrm s}| \le Ce^{-n/C}$ on $\mathbb{D}(0,\frac32\delta)$, boundary values included. Then $\check A^{\mathrm a} := \check A - \check A^{\mathrm s}$ has no jump on $(-\frac32\delta,0)$ and none on the band, so it is analytic in $\mathbb{D}(0,\frac32\delta)\setminus\{0\}$, with $|\check A^{\mathrm a}(z)| \le C|z|^{-1/2}$; an

isolated singularity with this growth is removable. On $|z| = \frac{5}{4}\delta$ we have $|\check{A}^{\mathrm{a}}| \le C$, by (5.11) and the bounds on ρ_n and f, so the maximum principle gives $|\check{A}^{\mathrm{a}}| \le C$ on $\mathbb{D}(0,\delta)$. Since $\det \check{A} = 1$, $\check{A}^{-1}$ is the adjugate of $\check{A}$ and obeys the same bound.

(ii) Each factor has determinant 1. On the band, $\check{A}$ has no jump, $\Phi_{\alpha,+} = \Phi_{\alpha,-}J$ by (6.4), whose derivation from [10, (6.16)–(6.18)] uses no restriction on α. Since $\rho_{n,+}^{-1} = \rho_{n,-}$ there, we also have $\rho_{n,-}^{\sigma_3} J \rho_{n,+}^{-\sigma_3} = \rho_{n,-}^{\sigma_3} J \rho_{n,-}^{\sigma_3} = J$. The upper lip in $\mathbb{D}(0,\delta)$ is mapped by $\zeta = n^2 f$ onto the ray $\arg\zeta = -2\pi/3$, where by (6.4) and (6.7) Φ_α jumps by

$$\begin{pmatrix} 1 & 0 \\ e^{-i\pi\alpha} e^{2n\varphi} & 1 \end{pmatrix};$$

conjugation by $\rho_n^{-\sigma_3}$ multiplies the lower-left entry by $\rho_n^{-2} = e^{i\pi\alpha}\omega$ by (8.5). The lower lip is the mirror image. On $(-\delta, 0)$, Φ_α is analytic ($\zeta > 0$), so

$$v_{\mathrm{h}} = \rho_{n,-}^{\sigma_3} \Phi_\alpha^{-1} (2\pi n)^{-\sigma_3/2} \big(\check{A}_-^{-1} \check{A}_+\big) (2\pi n)^{\sigma_3/2} \Phi_\alpha \, \rho_{n,-}^{-\sigma_3} e^{i\epsilon\sigma_3},$$

and the bound follows from (8.7), Lemma 8.1(i), the bounds on ρ_n and the bound on $\Phi_\alpha^{\pm 1}$ in (iii) below.

(iii) For $|\zeta| \ge 2$, [10, (6.28)] gives $\Phi_\alpha(\zeta) = \zeta^{-\sigma_3/4} (2\pi)^{-\sigma_3/2} (M^{-1} + O(\zeta^{-1/2}))$, so $|\Phi_\alpha^{\pm 1}| \le C|\zeta|^{1/4}$; this is (6.5), whose derivation is valid for every $\alpha > -1$. For $|\zeta| \le 3$, [10, (6.19)–(6.21)] give $|\Psi_\alpha(\zeta)| \le C|\zeta|^{-|\alpha|/2}$ for $\alpha \ne 0$ and $\le C(1 + |\log|\zeta||)$ for $\alpha = 0$; since $\det \Psi_\alpha = 1$ the same holds for Ψ_α^{-1}, and $e^{\pm 2\zeta^{1/2}}$ is bounded there. With $\zeta = n^2 f$, $|f| \asymp |z|$, part (i), and the factor $(2\pi n)^{\pm 1/2}$, this gives (iii).

(iv) For $\frac{1}{2}\delta \le |z| \le \delta$, $|\zeta| \ge n^2\delta/C \ge 2$, and, since $(n^2 f)^{-\sigma_3/4} = n^{-\sigma_3/2} f^{-\sigma_3/4}$,

$$\begin{aligned} \check{P}_0 &= P_0^\infty \rho_n^{\sigma_3} M f^{\sigma_3/4} (2\pi n)^{\sigma_3/2} (n^2 f)^{-\sigma_3/4} (2\pi)^{-\sigma_3/2} \big(M^{-1} + O(n^{-1})\big) \rho_n^{-\sigma_3} \\ &= P_0^\infty \rho_n^{\sigma_3} \big(I + O(n^{-1})\big) \rho_n^{-\sigma_3}. \end{aligned}$$

Since P_0^∞, $(P_0^\infty)^{-1}$ and $\rho_n^{\pm 1}$ are bounded on the annulus uniformly in n, both claims follow. □

Lemma 8.3 (Hard-edge mismatch)**.** *For $n \ge n_0$ put $\mathsf{R} := Q\check{P}_0^{-1}$ in $\mathbb{D}(0,\delta)$. Then R has no jump on $(0,\delta)$, and $\mathsf{R}_+ = \mathsf{R}_- J_{\mathsf{R}}$ on the lips in $\mathbb{D}(0,\delta)$ and on $(-\delta,0)$, where for every $K \ge 1$*

$$|J_{\mathsf{R}}(z) - I| \le C_K n^{-K} \min(1, |z|^K).$$

Proof. On the band Q and $\check{P}_0$ both jump by J (Proposition 5.6(i), Lemma 8.2(ii)), and the boundary values of R are continuous, so R is analytic across $(0,\delta)$. On a lip, $Q_+ = Q_- v$ with $v = v_{\mathrm{m}} + E$, $v_{\mathrm{m}} := \left(\begin{smallmatrix} 1 & 0 \\ \omega e^{2n\varphi} & 1 \end{smallmatrix}\right)$ (Proposition 5.6(i)), and $\check{P}_{0,+} = \check{P}_{0,-} v_{\mathrm{m}}$; so $J_{\mathsf{R}} = \check{P}_{0,-}(I + E v_{\mathrm{m}}^{-1}) \check{P}_{0,-}^{-1}$. Here $|\omega e^{2n\varphi}| \le C$ on the lips (Lemma 5.2(vi)(a), Proposition 5.6(vi)) and $|E| \le C \exp(-n|z|^{-1/p}/C)$ (Proposition 5.6(iv)). On $(-\delta,0)$, Q has no jump, so $J_{\mathsf{R}} = \check{P}_{0,-} v_{\mathrm{h}}^{-1} \check{P}_{0,-}^{-1}$. In both cases Lemma 8.2(ii), (iii) give $|J_{\mathsf{R}} - I| \le C n^C |z|^{-C} \exp(-n|z|^{-1/p}/C)$, and $e^{-u} \le C_M u^{-M}$ with $u = n|z|^{-1/p}$ and M large gives the claim, since $|z| < 1$. □

8.3. **The small-norm estimate.** Let $U_0 := \mathbb{D}(0,\delta)$ and $U_r := \mathbb{D}(4c, \varepsilon)$, and define the *error matrix*

$$\mathsf{R} := Q\check{P}_0^{-1} \text{ in } U_0, \qquad \mathsf{R} := Q\hat{P}_r^{-1} \text{ in } U_r, \qquad \mathsf{R} := Q(P_0^\infty)^{-1} \text{ elsewhere,} \tag{8.8}$$

with the Airy parametrix $\hat{P}_r$ of (5.20). (The letter R is not the matrix R of (4.19).) Orient ∂U_0 and ∂U_r clockwise, so that their + side is the outside, and let Σ_{R} be the union of the two circles, of $\Sigma \setminus (\overline{U_0} \cup \overline{U_r} \cup [0, 4c])$, and of the lips and the segment $(-\delta, 0)$ inside U_0. Then $\mathsf{R}_+ = \mathsf{R}_- J_{\mathsf{R}}$ on Σ_{R} with $J_{\mathsf{R}} = \check{P}_0 (P_0^\infty)^{-1}$ on ∂U_0, $\hat{P}_r (P_0^\infty)^{-1}$ on ∂U_r, $P_0^\infty v (P_0^\infty)^{-1}$ on the arcs of Σ outside the discs, and J_{R} of Lemma 8.3 inside U_0.

Proposition 8.4 (First-order small norm)**.** *Let $p>1$, $s>1$ and $\ell\equiv 1$. There are n_0 and C such that for $n\ge n_0$:*

(i) R *is analytic off* Σ_{R}, $\mathsf{R}(z)\to I$ *as* $z\to\infty$, *and* R *has no jump on* $\Sigma\cap U_r$ *or on the band outside* U_0*;*

(ii) $|J_{\mathsf{R}}-I|\le C/n$ *on the circles,* $|J_{\mathsf{R}}-I|\le C_K n^{-K}\min(1,|z|^K)$ *on* $\Sigma_{\mathsf{R}}\cap U_0$ *for every* K*, and* $|J_{\mathsf{R}}-I|\le Ce^{-n/C}$ *on the remaining arcs, whose total length is* $O(n^p)$*; in particular* $\|J_{\mathsf{R}}-I\|_{L^1\cap L^2\cap L^\infty(\Sigma_{\mathsf{R}})}\le C/n$*;*

(iii) $\mathsf{R}_\pm - I\in L^2(\Sigma_{\mathsf{R}})$ *and* $\mathsf{R}=I+\mathcal{C}\big(\mathsf{R}_-(J_{\mathsf{R}}-I)\big)$ *off* Σ_{R}*;*

(iv) $\|\mathsf{R}_- - I\|_{L^2(\Sigma_{\mathsf{R}})}\le C/n$, $|\mathsf{R}(z)-I|\le C_d/n$ *for* $\operatorname{dist}(z,\Sigma_{\mathsf{R}})\ge d>0$, *and, with* $\mathsf{R}=I+\mathsf{R}_1/z+\mathsf{R}_2/z^2+O(z^{-3})$ *at infinity,* $|\mathsf{R}_1|+|\mathsf{R}_2|\le C/n$.

Consequently $n^p b_n=c(p)(1+O(1/n))$ *and* $a_n/(2b_n)=1+O(1/n)$ *for* $\sigma_{p,s}$, $p>1$, $s>1$.

Besides Lemmas 8.2 and 8.3 (and so [10, §6]), the proof uses Proposition 5.6(i)–(vi), (viii), the Airy parametrix (Lemma 5.5, which rests on §2.7(b)), the sum bounds of Lemmas B.2 and B.3 (whose proofs use the Bessel order only through $\alpha-\eta>-1$), the behaviour of the Bessel model at 0 [10, (6.19)–(6.21)], the series (10.25.2) of §2.7(c), and the boundedness of the Cauchy operator on Carleson curves (§2.6(a)). It does not use Proposition 5.6(vii) or Corollary B.5.

Proof. Outline. Step 1 collects the jumps. Step 2 is the identification: for fixed n it shows that R is not too singular at 0 and is the Cauchy integral of its jump; this replaces the a priori bound of Appendix B, which needs $|\alpha|<1$, by an argument at the level of R, whose columns are balanced by the parametrix. Step 3 is the small-norm estimate and Step 4 the extraction.

Step 1: jumps. Inside U_r, $\hat P_r$ has exactly the jumps of Q (Proposition 5.6(v)). On the band outside U_0 both Q and P_0^∞ jump by J. Inside U_0 the jumps are those of Lemma 8.3. On the circles, Lemma 8.2(iv) and Proposition 5.6(v) give $J_{\mathsf{R}}=I+O(1/n)$. On the remaining arcs $|v-I|\le Ce^{-n/C}$ (Proposition 5.6(v)), and P_0^∞ is bounded with bounded inverse there by (5.11). The lengths are those of Proposition 5.6(i). At infinity $\mathsf{R}=Q(P_0^\infty)^{-1}\to I$. This proves (i) and (ii).

Step 2: behaviour at 0 for fixed n, and identification. Fix n; constants in this step may depend on n. We first show that for every $\eta>0$

$$|\mathsf{R}(z)|\le C_{n,\eta}|z|^{\min(0,\alpha)-\eta}\qquad (0<|z|<\delta_{\mathrm s}/2,\ z\notin\Sigma_{\mathsf{R}}), \tag{8.9}$$

with boundary values, where $\delta_{\mathrm s}$ is that of Lemma B.3. By (4.21), (5.12), (6.7) and (8.6), outside the lens $Q=c_n\,U\,(e^{n\varphi}FD_W)^{\sigma_3}$ with a constant diagonal c_n, and therefore

$$\mathsf{R}=\pm c_n\,U\,(F\tilde W)^{\sigma_3}\,\sigma_3\Psi_\alpha(n^2f)^{-1}\sigma_3\,(2\pi n)^{-\sigma_3/2}\check A^{-1},$$

because $e^{n\varphi\sigma_3}\Phi_\alpha^{-1}=(-1)^n\sigma_3\Psi_\alpha^{-1}\sigma_3$ by (6.7) and $D_W\rho_n=\tilde W$. Here $F^{\pm1}$ is bounded near 0 outside the lens (Definition 4.10(e)), $|\tilde W|\asymp|z|^{\alpha/2}$, $\check A^{-1}$ is bounded (Lemma 8.2(i)), the first column of U is polynomial and its second column is $O(|z|^{-\eta}(1+|z|^\alpha))$ (Lemma B.2). Outside the lens $\zeta=n^2f$ lies in the sector $|\arg\zeta|<2\pi/3$, where [10, (6.19)–(6.21)] give, for $\alpha>0$, columns of Ψ_α of sizes $|\zeta|^{\alpha/2}$ and $|\zeta|^{-\alpha/2}$, hence rows of $\Psi_\alpha^{-1}=\operatorname{adj}\Psi_\alpha$ of sizes $|\zeta|^{-\alpha/2}$ and $|\zeta|^{\alpha/2}$; for $\alpha<0$ all entries are $O(|\zeta|^{\alpha/2})$, and for $\alpha=0$ they are logarithmic. Multiplying out gives (8.9): for $\alpha\ge0$ the powers of $|z|$ cancel column by row, and for $\alpha<0$ the worst product is $|z|^{\alpha-\eta}$. Inside the lens, by (4.21) and (4.19), $Q=Q^{\mathrm u}L$ with $Q^{\mathrm u}:=c_n\,(UL_\pm)(e^{n\varphi}\tilde FD_W)^{\sigma_3}$ and $L:=\left(\begin{smallmatrix}1&0\\ \mp\omega e^{2n\varphi}&1\end{smallmatrix}\right)$, and by Lemma 8.2(ii) $\check P_0=\check P^{\mathrm u}L$ with the same L, where $\check P^{\mathrm u}$ is the continuation of $\check P_0$ from outside the lens. So $\mathsf{R}=Q^{\mathrm u}(\check P^{\mathrm u})^{-1}$ there. The continuation of Ψ_α from $|\arg\zeta|<2\pi/3$ into the lens is the formula [10, (6.23)], analytic in $|\arg\zeta|<\pi$; its first

column is $\zeta^{\alpha/2}$ times an entire function by (10.25.2), and its second column is that of Ψ_α in the thin sectors, because the jumps [10, (6.16), (6.18)] are lower triangular; so [10, (6.19)–(6.21)] give the same column sizes as outside. With Lemma B.3 for the second column of $UL_\pm$ and Definition 4.10(e) for $\tilde F$, the same computation gives (8.9) in the lens.

Next, let $\delta_2 < \delta_s/2$ and $\Gamma := \Sigma_{\mathsf R} \cap \mathbb{D}(0,\delta_2)$, the two lips and $(-\delta_2, 0)$, which meet only at 0 and leave it at fixed angles. Since $\|J_{\mathsf R}^{-1} - I\|_{L^2\cap L^\infty(\Gamma)} \le C_K n^{-K}$ by Lemma 8.3 and the Cauchy operators are bounded on $L^2(\Gamma)$ (§2.6(a)), for $n \ge n_0$ there is, by a Neumann series for the transposed problem, a matrix $\tilde X$, analytic in $\mathbb{C}\setminus\Gamma$, with $\tilde X - I = \mathcal{C}_\Gamma(\mathfrak h)$ for some $\mathfrak h \in L^2(\Gamma)$ and $\tilde X_+ = J_{\mathsf R}^{-1}\tilde X_-$ on Γ. By Cauchy–Schwarz and $|J_{\mathsf R} - I| \le C_K|s|^K$, $|\tilde X(z)| \le C(1+|z|^K \operatorname{dist}(z,\Gamma)^{-1/2})$ near 0. The determinant of $\tilde X$ has no jump, and the sub-mean-value property of $|\det\tilde X - 1|^{1/2}$ on discs of radius comparable to $|z|$ shows that its singularity at 0 is removable and that $\det\tilde X = 1 + O(n^{-K})$ near 0. Hence $\mathsf G := \mathsf R\tilde X$ has no jump on $\Gamma\setminus\{0\}$, and its Laurent coefficients at 0 satisfy

$$|c_{-k}| \le \frac{r^{k-1}}{2\pi}\int_{|z|=r} |\mathsf R||\tilde X|\,|\mathrm dz| \le C_{n,\eta}\, r^{k-1} r^{\min(0,\alpha)-\eta}\big(r + r^{K+1/2}\big)$$

by (8.9), since $\int_{|z|=r}\operatorname{dist}(z,\Gamma)^{-1/2}|\,\mathrm dz| \le Cr^{1/2}$. As $\alpha > -1$, choosing $\eta < 1 + \min(0,\alpha)$ and letting $r\to 0$ gives $c_{-k} = 0$ for all $k \ge 1$; so $\mathsf G$ is analytic near 0. Then $\mathsf R = \mathsf G\operatorname{adj}(\tilde X)/\det\tilde X$, where $\operatorname{adj}(\tilde X) - I$ is a Cauchy integral over Γ of an L^2 density (the adjugate of a 2×2 matrix is affine in its entries) and $\mathsf G/\det\tilde X$ is analytic near 0. Since $\mathfrak g\,\mathcal{C}_\Gamma(\mathfrak h) - \mathcal{C}_\Gamma(\mathfrak g\mathfrak h)$ is analytic near 0 for analytic $\mathfrak g$, $\mathsf R$ is, near 0, the sum of an analytic function and a Cauchy integral over Γ of an L^2 density. Away from 0 the matrix $\mathsf R$ is continuous up to $\Sigma_{\mathsf R}$ off the vertices and bounded near them (Proposition B.4(ii)–(iii), whose proof does not involve α, and the boundedness of the parametrices there), and $\mathsf R = I + O(1/z)$ at infinity. Subtracting the local Cauchy integral near 0 and applying Lemma B.1(c) to the rest, we obtain $\mathsf R_\pm - I \in L^2(\Sigma_{\mathsf R})$ and $\mathsf R = I + \mathcal{C}(\mathsf R_+ - \mathsf R_-)$, which is (iii) because $\mathsf R_+ - \mathsf R_- = \mathsf R_-(J_{\mathsf R} - I)$.

Step 3: small norm. The proof of Proposition 5.6(i) applies without change to $\Sigma\cup\partial U_0\cup\partial U_r\cup[-\delta,0]$, since the added arcs are fixed and lie in $\{\operatorname{Re} z \le 4c+1\}$; and $\Sigma_{\mathsf R}$ is a subset, so $\|\mathcal{C}_-\|_{L^2(\Sigma_{\mathsf R})} \le C$. Taking boundary values in (iii), $(1-\mathcal{C}_{J_{\mathsf R}})(\mathsf R_- - I) = \mathcal{C}_-(J_{\mathsf R} - I)$, and $\|\mathcal{C}_{J_{\mathsf R}}\| \le C\|J_{\mathsf R} - I\|_{L^\infty} \le \frac12$ for $n \ge n_0$ by (ii); so $\|\mathsf R_- - I\|_{L^2} \le C\|J_{\mathsf R} - I\|_{L^2} \le C/n$. Then $\mathsf R - I = \mathcal{C}(J_{\mathsf R} - I) + \mathcal{C}((\mathsf R_- - I)(J_{\mathsf R} - I))$ gives the pointwise bound, and $\mathsf R_k = -\frac{1}{2\pi i}\int_{\Sigma_{\mathsf R}} \mathsf R_-(J_{\mathsf R} - I)s^{k-1}\,\mathrm ds$ $(k = 1, 2)$ gives the bound on the moments: on the far arcs $|s| \le Cn^p$ is absorbed by $e^{-n/C}$.

Step 4: extraction. For $|z|$ large, $Q = \mathsf R P_0^\infty$, so $Q_1 = P_1 + \mathsf R_1$ and $Q_2 = P_2 + \mathsf R_1 P_1 + \mathsf R_2$, where $P_0^\infty = I + P_1/z + P_2/z^2 + O(z^{-3})$. Expanding $a(z) = (1-4c/z)^{1/4}$ in (5.10),

$$P_1 = \begin{pmatrix} 0 & ic \\ -ic & 0\end{pmatrix}, \qquad P_2 = \begin{pmatrix} c^2/2 & 2ic^2 \\ -2ic^2 & c^2/2\end{pmatrix}, \tag{8.10}$$

so $\mathsf b(P_0^\infty) = c^2$ and $\mathsf a(P_0^\infty) = 2c$. By Proposition 5.6(viii) and (iv), $n^{2p}b_n^2 = \mathsf b(Q) = c^2 + O(1/n)$ and $n^p a_n = \mathsf a(Q) = 2c + O(1/n)$, as $(Q_1)_{12} = ic + O(1/n) \ne 0$. □

Remark 8.5 (The classical hard-edge parametrix at $p = 1$). For pure powers and $p \ne 1$ the hard-edge parametrix can be built in the classical way, from a square root $\tilde W$ of the band weight continued to a full neighbourhood of 0; for $p > 1$ its jump on $(-\delta, 0)$ is super-exponentially close to I (Lemma 8.1(i)), and this is the construction of the present section. At $p = 1$ it fails, already for $\ell \equiv 1$: on $(-\delta, 0)$, $\theta = \pi n/x$ is real, the continuations of the lens scalar from the two half-planes differ by the unimodular factor $\tilde F_+/\tilde F_- = -e^{-2i\theta}$, which is not close to 1, and the continuation of $H_n = e^{2j(n/z)}$ (Lemma 5.1) has double poles at $z = -n/m$, $m \in \mathbb{N}$,

accumulating at 0. The comparison parametrix of Section 6 never continues anything across $(-\delta, 0)$: neither V nor Q has a jump there, and it uses only D_W. This is why Theorem 1.2 at $p = 1$ is proved by that route. (The obstruction concerns this classical construction, not every quotient construction.) For $p < 1$ we do not use the present module either: the expansion of Section 9 to order $1/n$ also needs the band factor to be constant to $o(1/n)$ near 0, as Lemma 8.1(iii) provides for $p > 1$, and for the seam construction of Appendix A the available bound near 0 is only $|\log H_n^\circ(z)| \le C|z|(1 + \log(1/|z|))$ (Table 5.1), which does not give this.

8.4. **Localisation of the spectral weights.** For $\sigma_{p,s}$ let $p_n = P_n/\sqrt{h_n}$ be the orthonormal polynomials and

$$m_n(k) := k^{-s} p_n(k^{-p})^2 = \frac{w_k P_n(x_k)^2}{h_n} \qquad (k \ge 1),$$

so that $\sum_k m_n(k) = 1$. The following proposition says that the spectral weights of degree n live on indices k of order n, with a power tail beyond. In this subsection $p > 0$ is arbitrary.

Proposition 8.6 (Localisation of the spectral weights)**.** *Let $p > 0$ and $s > 1$, and assume that* (a) $1 < s < 1 + 2p$ *or* (b) $p > 1$. *There are C, $\gamma_* > 0$ and n_0 such that for $n \ge n_0$, $j \in \{n-1, n\}$ and $1 \le M \le \gamma_* n^{2/p}$,*

$$\sum_{k > Mn} m_j(k) \le C\, M^{-p/2}. \tag{8.11}$$

In case (b), moreover, for every τ with $0 < \tau < \min(p/2, s-1)$ there is C_τ with $\sum_{k>Mn} m_j(k) \le C_\tau M^{-\tau}$ for all $n \ge n_0$, $j \in \{n-1, n\}$ and $M \ge 1$.

In case (a) the proof uses the comparison frame of Section 6 (so [2, Theorem 4.4] through §2.6(b), and [10, §6]); in case (b) it uses Proposition 8.4 and the integral representations (10.32.4) and (10.32.9) of [8] for I_α and K_α. In case (a) the bound is stated only up to the Bessel scale $M \asymp n^{2/p}$; in [16] it is combined with a polynomial-inequality argument to reach larger M. The proof is given in Appendix F.

9. The $1/n$ term for $p > 1$

This section proves Theorem 1.4 by expanding the error matrix R of §8.3 to second order in $1/n$. The standing assumptions and the notation of Section 8 are in force: $p > 1$, $s > 1$, $\ell \equiv 1$, and $\alpha > -1$ is arbitrary.

Outline. The global parametrix P_0^∞ does not depend on n or on the weight, and it contributes exactly c^2 and $2c$ to the two extraction functionals. So the $1/n$ terms of b_n and a_n come only from the first corrections of the two local models, the Bessel model at 0 and the Airy model at $4c$, on the two circles (Lemmas 9.1 and 9.2). These corrections depend on n only through the band factor H_n, and H_n is constant up to $O(n^{-p})$ because the two $1/(12n)$ Binet terms cancel (Lemma 8.1); so at order $1/n$ the corrections are those of the fixed model weight $x^\alpha(4c - x)$, whose local data are explicit (Lemma 9.3). The corrections are meromorphic in the two discs, and the residues enter the coefficients at infinity (Proposition 9.5). The residue at the hard edge is a multiple of $\left(\begin{smallmatrix} 1 & i \\ i & -1 \end{smallmatrix}\right)$, which both extraction functionals annihilate; so the $1/n$ terms come entirely from the Airy correction, through two combinations of its Laurent coefficients that we compute by hand (Lemma 9.4).

Throughout, $\mathsf{T} := \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ i & -i \end{smallmatrix}\right)$, so that $P_0^\infty = \mathsf{T} a^{\sigma_3} \mathsf{T}^{-1}$ with the function $a(z) = ((z-4c)/z)^{1/4}$ of (5.10) (it is never indexed, and is not to be confused with the coefficients a_n). For a matrix

Z with $Z_{22}=-Z_{11}$, a direct computation gives

$$P_0^\infty Z (P_0^\infty)^{-1} = \mathsf{T}\begin{pmatrix} Y_{11} & a^2 Y_{12} \\ a^{-2}Y_{21} & -Y_{11}\end{pmatrix}\mathsf{T}^{-1}, \qquad \begin{aligned} Y_{11} &= \tfrac{i}{2}(Z_{12}-Z_{21}),\\ Y_{12} &= Z_{11}-\tfrac{i}{2}(Z_{12}+Z_{21}),\\ Y_{21} &= Z_{11}+\tfrac{i}{2}(Z_{12}+Z_{21}),\end{aligned} \tag{9.1}$$

and, for any numbers y_{11}, y_{12}, y_{21},

$$\mathsf{T}\begin{pmatrix} y_{11} & y_{12} \\ y_{21} & -y_{11}\end{pmatrix}\mathsf{T}^{-1} = \begin{pmatrix} \frac{1}{2}(y_{12}+y_{21}) & \frac{i}{2}(y_{12}-y_{21}-2y_{11}) \\ \frac{i}{2}(y_{12}-y_{21}+2y_{11}) & -\frac{1}{2}(y_{12}+y_{21})\end{pmatrix}. \tag{9.2}$$

9.1. **The first corrections.**

Lemma 9.1 (First corrections of the two models)**.** (a) *Let $\alpha > -1$ and*

$$\mathsf{Z}_{\mathrm{B}} := \begin{pmatrix} -(1+4\alpha^2) & 2i \\ 2i & 1+4\alpha^2\end{pmatrix}.$$

Uniformly as $\zeta\to\infty$ off the rays of Ψ_α,

$$\Psi_\alpha(\zeta) = (2\pi\zeta^{1/2})^{-\sigma_3/2} M\Big[I + \frac{\sigma_3 \mathsf{Z}_{\mathrm{B}}\sigma_3}{16\zeta^{1/2}} + O(\zeta^{-1})\Big] e^{2\zeta^{1/2}\sigma_3},$$

equivalently $\Phi_\alpha(\zeta) = (2\pi\zeta^{1/2})^{-\sigma_3/2} M^{-1}\big[I + \mathsf{Z}_{\mathrm{B}}/(16\zeta^{1/2}) + O(\zeta^{-1})\big]$.

(b) *Let $\mathsf{Z}_{\mathrm{Ai}} := \left(\begin{smallmatrix} -1/72 & -i/12 \\ -i/12 & 1/72\end{smallmatrix}\right)$ and $\xi := \frac{2}{3}\zeta^{3/2}$. Uniformly for $\zeta\in\mathcal{S}_j$, $|\zeta|\ge 1$, the Airy model (5.15) satisfies*

$$\Psi^j(\zeta) = \zeta^{-\sigma_3/4} M\big(I + \mathsf{Z}_{\mathrm{Ai}}/\xi + O(\xi^{-2})\big) e^{\xi\sigma_3} \qquad (j=\mathrm{I},\dots,\mathrm{IV}).$$

Part (a) uses [10, (6.23)–(6.25)] and the large-argument expansions (10.40.1)–(10.40.4) with the coefficients (10.17.1), (10.17.8), and (10.17.5), (10.17.6), (10.17.11), (10.17.12) for the Hankel functions (§2.7(d)); these hold for every fixed order, so no restriction on α arises. Part (b) uses (9.7.2), (9.7.5) and (9.7.6) (§2.7(b)). Both are the computations behind [10, (6.28)] and (5.17), carried one term further.

Proof. (a) In $|\arg\zeta| < 2\pi/3$, [10, (6.23)] expresses Ψ_α through I_α, K_α and their derivatives at $w = 2\zeta^{1/2}$, and $|\mathrm{ph}\, w| < \pi/3$, inside the sectors of (10.40.1)–(10.40.4). With the first coefficients $a_1(\alpha) = (4\alpha^2-1)/8$ and $b_1(\alpha) = (4\alpha^2+3)/8$ these give, entrywise with relative errors $O(w^{-2})$, $\Psi_\alpha e^{-w\sigma_3} = (2\pi\zeta^{1/2})^{-\sigma_3/2}\big[M + w^{-1}\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} -a_1 & ia_1 \\ -ib_1 & b_1\end{smallmatrix}\right)\big]$. Since

$$M^{-1}\frac{1}{\sqrt{2}}\begin{pmatrix} -a_1 & ia_1 \\ -ib_1 & b_1\end{pmatrix} = \frac{1}{2}\begin{pmatrix} -(a_1+b_1) & i(a_1-b_1) \\ i(a_1-b_1) & a_1+b_1\end{pmatrix}, \qquad a_1+b_1 = \frac{4\alpha^2+1}{4}, \quad a_1-b_1 = -\frac{1}{2},$$

the correction is $\sigma_3\mathsf{Z}_{\mathrm{B}}\sigma_3/(16\zeta^{1/2})$. In the sectors $2\pi/3 < \pm\arg\zeta < \pi$, [10, (6.24)–(6.25)] use $H_\alpha^{(1,2)}$ and their derivatives at $2(-\zeta)^{1/2}$, whose phase lies in $(-\pi/6, \pi/6)$, inside the sectors of (10.17.5)–(10.17.6) and (10.17.11)–(10.17.12); the same computation with the first two terms gives the same formula. The form for Φ_α follows from $\sigma_3 M\sigma_3 = M^{-1}$.

(b) In $\mathcal{S}_{\mathrm{I}}$ the columns of Ψ^{I} are $k_{\mathrm{Ai}}e^{-i\pi\sigma_3/4}y_+$ and $k_{\mathrm{Ai}}e^{-i\pi\sigma_3/4}\mathsf{q}y_0$. By (9.7.5)–(9.7.6) with $u_1 = \frac{5}{72}$, $v_1 = -\frac{7}{72}$ from (9.7.2), applied at ζ and at $\mathsf{q}^{-1}\zeta$ (where $(\mathsf{q}^{-1}\zeta)^{1/4} = e^{-i\pi/6}\zeta^{1/4}$ and $\frac{2}{3}(\mathsf{q}^{-1}\zeta)^{3/2} = -\xi$), with relative errors $O(\xi^{-2})$,

$$y_+ = \frac{e^{\xi}e^{i\pi/6}}{2\sqrt{\pi}}\zeta^{-\sigma_3/4}\begin{pmatrix} 1+u_1/\xi \\ 1+v_1/\xi\end{pmatrix}, \qquad y_0 = \frac{e^{-\xi}}{2\sqrt{\pi}}\zeta^{-\sigma_3/4}\begin{pmatrix} 1-u_1/\xi \\ -(1-v_1/\xi)\end{pmatrix}.$$

So $\Psi^{\mathrm{I}} = \zeta^{-\sigma_3/4}(N_0 + N_1/\xi + O(\xi^{-2}))e^{\xi\sigma_3}$ with $N_0 = M$ (this is (5.17)) and $M^{-1}N_1 = \left(\begin{smallmatrix} (u_1+v_1)/2 & -i(u_1-v_1)/2 \\ -i(u_1-v_1)/2 & -(u_1+v_1)/2 \end{smallmatrix}\right) = \mathsf{Z}_{\mathrm{Ai}}$, using $\mathsf{q}e^{-i\pi/6} = i$. In each of the other sectors the two columns of Ψ^j are evaluated at points with the same leading asymptotics as in $\mathcal{S}_{\mathrm{I}}$ (this is the computation behind (5.17); for instance, in $\mathcal{S}_{\mathrm{II}}$, $-\mathsf{q}^{-1}y_-$ has the same two-term expansion as $\mathsf{q}y_0$, since $(\mathsf{q}\zeta)^{1/4} = e^{-i\pi/3}\zeta^{1/4}$ and $(\mathsf{q}\zeta)^{3/2} = \zeta^{3/2}$ there), so the correction is the same. □

Let $f^{1/2}$ be the principal root; it is analytic and non-zero on ∂U_0. Put, on ∂U_0 and ∂U_r respectively,

(9.3)
$$\Delta_0 := \frac{1}{16 f^{1/2}}\, P_0^\infty \rho_n^{\sigma_3} \mathsf{Z}_{\mathrm{B}} \rho_n^{-\sigma_3} (P_0^\infty)^{-1}, \qquad \Delta_r := -\frac{1}{\tilde\varphi}\, P_0^\infty (-\omega)^{-\sigma_3/2} \mathsf{Z}_{\mathrm{Ai}} (-\omega)^{\sigma_3/2} (P_0^\infty)^{-1},$$

with ρ_n of (8.4) and the root (5.18). Here Δ_0 depends on ρ_n only through $\rho_n^{\pm 2} = (e^{\mp i\pi\alpha}/\omega)^{\pm 1}$, and Δ_r only through $(-\omega)^{\pm 1}$. Let Δ_0^{mod} and Δ_r^{mod} be the same expressions with ω replaced by ω^{mod} of (8.1), on the whole punctured discs $\mathbb{D}(0, 2\delta) \setminus \{0\}$ and $\mathbb{D}(4c, 2\varepsilon) \setminus \{4c\}$ (off $[0, 4c]$ to begin with).

Lemma 9.2 (Jumps on the two circles)**.** *Uniformly on the circles, with boundary values where they cross the contour,*

$$\check P_0 (P_0^\infty)^{-1} = I + \frac{\Delta_0}{n} + O(n^{-2}) \text{ on } \partial U_0, \qquad \hat P_r (P_0^\infty)^{-1} = I + \frac{\Delta_r}{n} + O(n^{-2}) \text{ on } \partial U_r,$$

and $\Delta_0 - \Delta_0^{\mathrm{mod}} = O(n^{-p})$, $\Delta_r - \Delta_r^{\mathrm{mod}} = O(n^{-p})$ *there.*

Proof. Bessel. On ∂U_0, $\zeta = n^2 f$ and $\zeta^{1/2} = n f^{1/2}$, with $|\zeta| \asymp n^2$. By (8.6) and Lemma 9.1(a), since $f^{\sigma_3/4}(2\pi n)^{\sigma_3/2}(2\pi n f^{1/2})^{-\sigma_3/2} = I$,

$$\check P_0 = P_0^\infty \rho_n^{\sigma_3} M M^{-1} \Big[I + \frac{\mathsf{Z}_{\mathrm{B}}}{16 n f^{1/2}} + O(n^{-2}) \Big] \rho_n^{-\sigma_3},$$

which is the first claim, because P_0^∞, $(P_0^\infty)^{-1}$ and $\rho_n^{\pm 1}$ are bounded on ∂U_0 uniformly in n (Lemma 8.2(i)).

Airy. On ∂U_r, by (5.20), $\hat P_r = P_0^\infty (-\omega)^{-\sigma_3/2} M^{-1} f_r^{\sigma_3/4} n^{\sigma_3/6} \Psi_r(\zeta) ((-\omega)^{1/2} e^{n\tilde\varphi})^{\sigma_3}$ with $\zeta = n^{2/3} f_r$, so $\zeta^{-\sigma_3/4} = n^{-\sigma_3/6} f_r^{-\sigma_3/4}$ and $\xi = \frac{2}{3} n f_r^{3/2} = -n\tilde\varphi$ (Lemma 4.5(iv)). Lemma 9.1(b) and $e^{\xi\sigma_3} e^{n\tilde\varphi\sigma_3} = I$ give

$$\hat P_r = P_0^\infty (-\omega)^{-\sigma_3/2} \Big(I - \frac{\mathsf{Z}_{\mathrm{Ai}}}{n\tilde\varphi} + O(n^{-2}) \Big) (-\omega)^{\sigma_3/2},$$

uniformly on ∂U_r because $|\xi| \asymp n$ there and P_0^∞, $(P_0^\infty)^{-1}$, $(-\omega)^{\pm 1/2}$ are bounded (Lemmas 5.2(vi)(b) and 5.5).

Dependence on n. The functions f, $\tilde\varphi$ and P_0^∞ do not depend on n. By Lemma 8.1(iv), $\omega/\omega^{\mathrm{mod}} = 1 + O(n^{-p})$ on both circles, and ω, ω^{mod} are bounded above and below there; since Δ_0, Δ_r are smooth functions of $\omega^{\pm 1}$ with these bounds, the last claim follows. □

9.2. **The local data and the Laurent coefficients.**

Lemma 9.3 (Local data of the model)**.** (i) $f(z) = f'(0) z\big(1 + f_2 z + O(z^2)\big)$ *near* 0, *with*

$$f'(0) = -\frac{1}{c(p+2)^2}, \qquad f_2 = \frac{p+2}{12c(3p+2)}.$$

(ii) *With* $\mathsf{u} := (z-4c)^{1/2}$ *(principal root),* $\tilde\varphi = -\mathsf{g}_3\mathsf{u}^3 - \mathsf{g}_5\mathsf{u}^5 + O(\mathsf{u}^7)$ *near* $4c$*, where* $\tilde\varphi/\mathsf{u}^3$ *is analytic at* $4c$ *and*

$$\mathsf{g}_3 = \frac{1}{6pc^{3/2}}, \qquad \mathsf{g}_5 = -\frac{7p+4}{240p^2c^{5/2}}.$$

(iii) *Near* $4c$ *and near* 0 *(in* $\mathbb{C}_\pm$*), respectively,*

$$(-\omega^{\mathrm{mod}})^{1/2} = \Big(\frac{2\sqrt{c}}{z^{1/2}+\mathsf{u}}\Big)^{1+\alpha}, \qquad \frac{e^{\mp i\pi\alpha}}{\omega^{\mathrm{mod}}} = \Big(\frac{((-z)^{1/2}+(4c-z)^{1/2})^2}{4c}\Big)^{1+\alpha};$$

equivalently, with $\mathsf{y} := (-z)^{1/2}$*,*

$$L_r := (1+\alpha)\operatorname{arsinh}\frac{\mathsf{u}}{2\sqrt{c}}, \quad L_0 := (1+\alpha)\operatorname{arsinh}\frac{\mathsf{y}}{2\sqrt{c}}, \quad (-\omega^{\mathrm{mod}})^{\mp 1} = e^{\pm 2L_r}, \quad \frac{e^{\mp i\pi\alpha}}{\omega^{\mathrm{mod}}} = e^{2L_0}.$$

The lemma uses only the equilibrium measure (Proposition 2.3, from [15, Theorem 1.3]), through the formulas for f and $\tilde\varphi$ in the proof of Lemma 4.5, and Lemma 5.2(i).

Proof. Put $a' := \frac{1}{p}+\frac{1}{2}$; recall $(4c)^{-1/p} = 2I(p)/\pi$ and $B(a',\frac12) = 2I(p)$ by (1.3) and (5.12.1).

(i) By Proposition 2.3, on $(0,4c)$, $\psi^*(x) = \lambda_p(x)\mathcal{B}_{x/4c}(a',\frac12)$, and $\mathcal{B}_y(a',\frac12) = \frac{y^{a'}}{a'B(a',\frac12)}\big(1 + \frac{a'y}{2(a'+1)} + O(y^2)\big)$. Hence $\psi^*(x) = C_*x^{-1/2}(1+\tau_*x+O(x^2))$ with $C_* = (4c)^{-1/2}/(\pi(1+p/2))$ (Proposition 2.3(iii)) and $\tau_* = a'/(8c(a'+1)) = (p+2)/(8c(3p+2))$. So $\mu^*((0,x]) = x^{1/2}\mathfrak{a}(x)$ with $\mathfrak{a}(z) = 2C_*(1+\tau_*z/3+O(z^2))$, and $f = -\frac{\pi^2}{4}z\mathfrak{a}^2$ (proof of Lemma 4.5(iii)) gives $f'(0) = -\pi^2C_*^2 = -1/(c(p+2)^2)$ and $f_2 = 2\tau_*/3$.

(ii) By the proof of Lemma 4.5(iv), $\tilde\varphi_+(x) = i\pi\int_x^{4c}(\lambda_p - \psi^*)$ on the band. With $y = 4c(1-e)$, $1-\mathcal{B}_{1-e}(a',\frac12) = \frac{1}{B(a',1/2)}\big(2e^{1/2} - \frac23(a'-1)e^{3/2} + O(e^{5/2})\big)$ and $\lambda_p(y) = \frac1p(4c)^{-1-1/p}(1+(1+\frac1p)e+O(e^2))$; integrating, with $v := 4c-x$,

$$\int_x^{4c}(\lambda_p-\psi^*) = \frac{1}{\pi p}\Big[\frac43\Big(\frac{v}{4c}\Big)^{3/2} + \frac25\Big(\frac73+\frac{4}{3p}\Big)\Big(\frac{v}{4c}\Big)^{5/2} + O(v^{7/2})\Big].$$

On the band, approached from $\mathbb{C}_+$, $\mathsf{u} = iv^{1/2}$, so $-\mathsf{g}_3\mathsf{u}^3 - \mathsf{g}_5\mathsf{u}^5 = i\mathsf{g}_3v^{3/2} - i\mathsf{g}_5v^{5/2}$; comparing coefficients gives the values. The quotient $\tilde\varphi/\mathsf{u}^3$ is analytic at $4c$ because $\tilde\varphi = -\frac23 f_r^{3/2}$ with f_r conformal and $f_r(4c) = 0$ (Lemma 4.5(iv)), so the band values determine the expansion.

(iii) By Lemma 5.2(i), $\omega^{\mathrm{mod}} = (4cz/\Upsilon)^\alpha z^{-\alpha}\cdot(-4c/\Upsilon)$. Near $4c$ all arguments are close to 0, so $-\omega^{\mathrm{mod}} = (4c/\Upsilon)^{1+\alpha}$ and, as in (5.18), its root is the stated one; $z^{1/2} = 2\sqrt{c}(1+\mathsf{u}^2/(4c))^{1/2}$ gives the form with arsinh. Near 0, in $\mathbb{C}_\pm$, $z^{1/2} = \pm i(-z)^{1/2}$ and $(z-4c)^{1/2} = \pm i(4c-z)^{1/2}$, so $\Upsilon = -\hat\Upsilon$ with $\hat\Upsilon := ((-z)^{1/2}+(4c-z)^{1/2})^2$; and $(4cz/\Upsilon)^\alpha = (-z)^\alpha(4c/\hat\Upsilon)^\alpha$, $z^{-\alpha} = (-z)^{-\alpha}e^{\mp i\pi\alpha}$. Hence $\omega^{\mathrm{mod}} = e^{\mp i\pi\alpha}(4c/\hat\Upsilon)^{1+\alpha}$. □

Lemma 9.4 (Laurent coefficients)**.** Δ_0^{mod} *is meromorphic in* $\mathbb{D}(0,2\delta)$ *with at most a simple pole at* 0*, and* Δ_r^{mod} *is meromorphic in* $\mathbb{D}(4c,2\varepsilon)$ *with at most a double pole at* $4c$*:*

$$\Delta_0^{\mathrm{mod}}(z) = \frac{\mathfrak{A}_0}{z} + O(1), \qquad \Delta_r^{\mathrm{mod}}(z) = \frac{\mathfrak{B}_r}{(z-4c)^2} + \frac{\mathfrak{A}_r}{z-4c} + O(1).$$

Here

$$\mathfrak{A}_0 = \frac{c(p+2)(4\alpha^2-1)}{16}\begin{pmatrix}1 & i\\ i & -1\end{pmatrix}, \qquad \mathfrak{B}_r = \frac{5c^2p}{12}\begin{pmatrix}1 & -i\\ -i & -1\end{pmatrix},$$

and

$$(\mathfrak{A}_r)_{21} - (\mathfrak{A}_r)_{12} = ipc(1+\alpha), \qquad (\mathfrak{A}_r)_{11} - (\mathfrak{A}_r)_{22} - 2i(\mathfrak{A}_r)_{12} - \frac{i}{c}(\mathfrak{B}_r)_{12} = -pc(\alpha+2). \tag{9.4}$$

The full matrix $\mathfrak{A}_r$ is given in Lemma E.1.

Proof. Meromorphy. On $(0,2\delta)$, $f_+^{1/2} = -f_-^{1/2}$ (as $f<0$ there and f maps $\mathbb{C}_\pm$ into $\mathbb{C}_\mp$), $P_{0,+}^\infty = P_{0,-}^\infty J$, and the model value of ρ^2 in Lemma 9.3(iii) satisfies $\rho_+^2\rho_-^2=1$; since $J\mathsf{Z}_{\mathrm{B}}J^{-1} = -\mathsf{Z}_{\mathrm{B}}$ and $Jd^{-\sigma_3} = d^{\sigma_3}J$ for scalars d, $\Delta_{0,+}^{\mathrm{mod}} = \Delta_{0,-}^{\mathrm{mod}}$. On $(-2\delta,0)$ every factor is analytic. So Δ_0^{mod} is analytic in the punctured disc, and $|\Delta_0^{\mathrm{mod}}| \le C|z|^{-1}$ by (5.11) and $|f| \asymp |z|$; hence 0 is at most a simple pole. On $(4c-2\varepsilon, 4c)$, $\tilde\varphi_+ = -\tilde\varphi_-$ (Lemma 4.4(i)), $(-\omega)_+^{1/2}(-\omega)_-^{1/2} = 1$ by (5.19) (in particular for the model), and $J\mathsf{Z}_{\mathrm{Ai}}J^{-1} = -\mathsf{Z}_{\mathrm{Ai}}$, so again there is no jump; on $(4c, 4c+2\varepsilon)$ all factors are analytic. Since $|\tilde\varphi| \asymp |z-4c|^{3/2}$ and $|P_0^\infty| \le C|z-4c|^{-1/4}$ near $4c$, $|\Delta_r^{\mathrm{mod}}| \le C|z-4c|^{-2}$.

The coefficient $\mathfrak{A}_0$. By (9.1) with $Z = \rho^{\sigma_3}\mathsf{Z}_{\mathrm{B}}\rho^{-\sigma_3}$ and $\rho^2 = e^{2L_0}$,

$$Y_{11} = -2\sinh 2L_0, \qquad Y_{12} = 1-4\alpha^2+(\rho-\rho^{-1})^2, \qquad Y_{21} = -(3+4\alpha^2)-(\rho-\rho^{-1})^2.$$

Let $z\to 0$ along $(-2\delta,0)$, where $f>0$, $L_0 = O(|z|^{1/2})$, $a^2 = (4c-z)^{1/2}(-z)^{-1/2}$ and $f^{1/2} = |f'(0)|^{1/2}(-z)^{1/2}(1+O(z))$. Then $zY_{11}/f^{1/2}\to 0$, $za^{-2}Y_{21}/f^{1/2}\to 0$, and $za^2Y_{12}/f^{1/2} \to -2\sqrt{c}|f'(0)|^{-1/2}(1-4\alpha^2) = -2c(p+2)(1-4\alpha^2)$ by Lemma 9.3(i). Since the pole is at most simple, $\mathfrak{A}_0 = \lim_{z\to 0} z\Delta_0^{\mathrm{mod}}(z)$, and (9.2) with $y_{12}=1$, $y_{11}=y_{21}=0$ gives $\mathsf{T}e_1e_2^{\mathsf{T}}\mathsf{T}^{-1} = \frac12\left(\begin{smallmatrix}1 & i\\ i & -1\end{smallmatrix}\right)$; so $\mathfrak{A}_0 = \frac{1}{16}\cdot(-2c(p+2)(1-4\alpha^2))\cdot\frac12\left(\begin{smallmatrix}1 & i\\ i & -1\end{smallmatrix}\right)$.

The Airy coefficients. By (9.1) with $Z = (-\omega^{\mathrm{mod}})^{-\sigma_3/2}\mathsf{Z}_{\mathrm{Ai}}(-\omega^{\mathrm{mod}})^{\sigma_3/2}$, whose off-diagonal entries are $-\frac{i}{12}e^{2L_r}$ and $-\frac{i}{12}e^{-2L_r}$ by Lemma 9.3(iii),

$$Y_{11} = \frac{\sinh 2L_r}{12}, \qquad Y_{12} = -\frac{1}{72}-\frac{\cosh 2L_r}{12}, \qquad Y_{21} = -\frac{1}{72}+\frac{\cosh 2L_r}{12}, \tag{9.5}$$

and $\Delta_r^{\mathrm{mod}} = -\tilde\varphi^{-1}\mathsf{T}\left(\begin{smallmatrix}Y_{11} & a^2Y_{12}\\ a^{-2}Y_{21} & -Y_{11}\end{smallmatrix}\right)\mathsf{T}^{-1}$. Take $z\in(4c,4c+2\varepsilon)$, so $\mathsf{u}>0$, and put $\epsilon := \mathsf{u}/(2\sqrt{c})$. Then $a^2 = \epsilon(1+\epsilon^2)^{-1/2}$ is odd in u, $L_r = (1+\alpha)\epsilon + O(\epsilon^3)$, Y_{11} is odd and Y_{12}, Y_{21} are even, and $\tilde\varphi^{-1}$ is u^{-3} times an even series. A function meromorphic at $4c$ is an even Laurent series in u, and its residue is the coefficient of u^{-2}. Since the pole is at most double, $\mathfrak{B}_r = \lim(z-4c)^2\Delta_r^{\mathrm{mod}} = \mathsf{g}_3^{-1}\lim \mathsf{u}\,\mathsf{T}(\cdots)\mathsf{T}^{-1}$, and $\mathsf{u}a^{-2}Y_{21}\to 2\sqrt{c}\cdot\frac{5}{72}$ while the other entries times u tend to 0; with $\mathsf{T}e_2e_1^{\mathsf{T}}\mathsf{T}^{-1} = \frac12\left(\begin{smallmatrix}1 & -i\\ -i & -1\end{smallmatrix}\right)$ and $\mathsf{g}_3^{-1} = 6pc^{3/2}$ this gives $\mathfrak{B}_r$.

For the first combination in (9.4), (9.2) gives $(\Delta_r^{\mathrm{mod}})_{21} - (\Delta_r^{\mathrm{mod}})_{12} = -2iY_{11}/\tilde\varphi$, and

$$\frac{Y_{11}}{\tilde\varphi} = \frac{(1+\alpha)\mathsf{u}/(12\sqrt{c})+O(\mathsf{u}^3)}{-\mathsf{g}_3\mathsf{u}^3(1+O(\mathsf{u}^2))} = -\frac{1+\alpha}{12\sqrt{c}\,\mathsf{g}_3}\,\mathsf{u}^{-2}+O(1);$$

so the difference has no double pole and its residue is $\frac{i(1+\alpha)}{6\sqrt{c}\,\mathsf{g}_3} = ipc(1+\alpha)$. For the second, $(\mathfrak{A}_r)_{11}-(\mathfrak{A}_r)_{22}-2i(\mathfrak{A}_r)_{12}-\frac{i}{c}(\mathfrak{B}_r)_{12}$ is the residue of $g := (\Delta)_{11}-(\Delta)_{22}-2i(\Delta)_{12}-\frac{i}{c}(z-4c)(\Delta)_{12}$, $\Delta = \Delta_r^{\mathrm{mod}}$, and (9.2) gives

$$g = -\frac{2}{\tilde\varphi}\left(a^2Y_{12}-Y_{11}\right) - \frac{\mathsf{u}^2}{2c\,\tilde\varphi}\left(a^2Y_{12}-a^{-2}Y_{21}-2Y_{11}\right).$$

In the first term $a^2Y_{12}-Y_{11} = \left(-\frac{7}{144\sqrt{c}}-\frac{1+\alpha}{12\sqrt{c}}\right)\mathsf{u}+O(\mathsf{u}^3)$, an odd series, which contributes $\frac{2}{\mathsf{g}_3}\cdot\left(-\frac{12\alpha+19}{144\sqrt{c}}\right) = -\frac{pc(12\alpha+19)}{12}$ to the coefficient of u^{-2}. In the second, the bracket is $-\frac{2\sqrt{c}}{\mathsf{u}}\cdot\frac{5}{72}+O(\mathsf{u})$, which contributes $\frac{1}{2c\,\mathsf{g}_3}\cdot\left(-\frac{10\sqrt{c}}{72}\right) = -\frac{5pc}{12}$. The sum is $-pc(\alpha+2)$. □

The higher Taylor coefficients f_2 and g_5 do not enter (9.4): only the leading terms of $a^{\pm 2}$, Y and $\tilde\varphi$ were used. They do enter $\mathfrak{A}_r$ itself and the constant term at 0 (Lemma E.1).

9.3. **The second-order small-norm estimate.**

Proposition 9.5 (Second-order small norm)**.** *Let $p>1$, $s>1$, $\ell\equiv 1$. For $n\ge n_0$,*

$$\mathsf{R}(z)=I+\frac{\mathsf{R}^{(1)}(z)}{n}+O(n^{-2})$$

uniformly for $\operatorname{dist}(z,\Sigma_{\mathsf{R}})\ge d>0$, *and the coefficients at infinity satisfy* $\mathsf{R}_k=\mathsf{R}_k^{(1)}/n+O(n^{-2})$, $k=1,2$, *where*

$$\mathsf{R}^{(1)}(z)=\frac{\mathfrak{A}_0}{z}+\frac{\mathfrak{A}_r}{z-4c}+\frac{\mathfrak{B}_r}{(z-4c)^2}\quad(z\notin\overline{U_0}\cup\overline{U_r}),\qquad \mathsf{R}_1^{(1)}=\mathfrak{A}_0+\mathfrak{A}_r,\qquad \mathsf{R}_2^{(1)}=4c\,\mathfrak{A}_r+\mathfrak{B}_r.$$

Proof. By Proposition 8.4(iii),

$$\mathsf{R}(z)-I=\frac{1}{2\pi i}\int_{\Sigma_{\mathsf{R}}}\frac{(J_{\mathsf{R}}-I)(s)}{s-z}\,\mathrm{d}s+\frac{1}{2\pi i}\int_{\Sigma_{\mathsf{R}}}\frac{(\mathsf{R}_-\!-I)(J_{\mathsf{R}}-I)(s)}{s-z}\,\mathrm{d}s.$$

By Cauchy–Schwarz and Proposition 8.4(ii), (iv) the second integral is at most $\|\mathsf{R}_--I\|_{L^2}\|J_{\mathsf{R}}-I\|_{L^2}/(2\pi d)=O(n^{-2})$. In the first, the arcs inside U_0 contribute $O(n^{-K})$ and the remaining arcs off the circles $O(e^{-n/C})$; by Lemma 9.2 the circles contribute

$$\frac{1}{n}\cdot\frac{1}{2\pi i}\oint_{\mathrm{cw}}\frac{\Delta^{\mathrm{mod}}(s)}{s-z}\,\mathrm{d}s+O(n^{-2}+n^{-1-p}),$$

where Δ^{mod} is Δ_0^{mod} on ∂U_0 and Δ_r^{mod} on ∂U_r, both circles oriented clockwise; and $n^{-1-p}\le n^{-2}$ since $p>1$. For z outside the discs, by Lemma 9.4 the clockwise integrals equal minus the sums of the residues of $\Delta^{\mathrm{mod}}(s)/(s-z)$ at $s=0$ and $s=4c$, that is, $\mathfrak{A}_0/z+\mathfrak{A}_r/(z-4c)+\mathfrak{B}_r/(z-4c)^2$. For the coefficients at infinity, $\mathsf{R}_k=-\frac{1}{2\pi i}\int_{\Sigma_{\mathsf{R}}}\mathsf{R}_-(J_{\mathsf{R}}-I)s^{k-1}\,\mathrm{d}s$ with the same splitting; on the far arcs $|s|^{k-1}\le Cn^p$ is absorbed by $e^{-n/C}$, and $-\frac{1}{2\pi i}\oint_{\mathrm{cw}}\Delta^{\mathrm{mod}}(s)s^{k-1}\,\mathrm{d}s$ is the sum of the residues of $\Delta^{\mathrm{mod}}(s)s^{k-1}$, which is $\mathfrak{A}_0+\mathfrak{A}_r$ for $k=1$ and $4c\mathfrak{A}_r+\mathfrak{B}_r$ for $k=2$ (the simple pole at 0 is removed by the factor s). □

9.4. **Proof of Theorem 1.4.** By Proposition 5.6(viii) and Lemma 4.2, $n^{2p}b_n^2=\mathsf{b}(Q)$ and $n^pa_n=\mathsf{a}(Q)$. For $|z|$ large, $Q=\mathsf{R}P_0^\infty$, so $Q_1=P_1+\mathsf{R}_1$ and $Q_2=P_2+\mathsf{R}_1P_1+\mathsf{R}_2$ with P_1,P_2 of (8.10). Write $\mathsf{R}_1=\mathcal{A}/n+O(n^{-2})$ and $\mathsf{R}_2=\mathcal{B}/n+O(n^{-2})$, with $\mathcal{A}:=\mathfrak{A}_0+\mathfrak{A}_r$ and $\mathcal{B}:=4c\mathfrak{A}_r+\mathfrak{B}_r$ (Proposition 9.5).

The coefficient b_n. We have

$$n^{2p}b_n^2=\Big(ic+\frac{\mathcal{A}_{12}}{n}\Big)\Big(-ic+\frac{\mathcal{A}_{21}}{n}\Big)+O(n^{-2})=c^2\Big(1+\frac{i(\mathcal{A}_{21}-\mathcal{A}_{12})}{cn}\Big)+O(n^{-2}).$$

The off-diagonal entries of $\mathfrak{A}_0$ are equal, so by (9.4) $i(\mathcal{A}_{21}-\mathcal{A}_{12})/c=i\cdot ipc(1+\alpha)/c=-p(1+\alpha)=-(s-1)$. Since $b_n>0$, $n^pb_n=c(1-(s-1)/(2n)+O(n^{-2}))$.

The coefficient a_n. Since $(Q_1)_{12}=ic+O(1/n)$ and a is smooth near (P_1,P_2), expanding $\mathsf{a}(Q)=(Q_2)_{12}/(Q_1)_{12}-(Q_1)_{22}$ to first order, with $(\mathsf{R}_1P_1)_{12}=ic(\mathsf{R}_1)_{11}$, gives

$$n^pa_n=2c+\frac{1}{n}\Big(\mathcal{A}_{11}-\mathcal{A}_{22}+2i\mathcal{A}_{12}-\frac{i}{c}\mathcal{B}_{12}\Big)+O(n^{-2}).$$

For $\mathfrak{A}_0=k\left(\begin{smallmatrix}1 & i\\ i & -1\end{smallmatrix}\right)$ the bracket vanishes ($k(1+1+2i\cdot i)=0$), and $\mathfrak{A}_0$ does not enter $\mathcal{B}$. The Airy part of the bracket is $(\mathfrak{A}_r)_{11}-(\mathfrak{A}_r)_{22}+2i(\mathfrak{A}_r)_{12}-4i(\mathfrak{A}_r)_{12}-\frac{i}{c}(\mathfrak{B}_r)_{12}=-pc(\alpha+2)$ by (9.4); the terms $12\alpha^2p$ of $\mathfrak{A}_r$ (Lemma E.1) cancel between the contributions of R_1 and R_2. Since $p(\alpha+2)=s-1+p$, $n^pa_n=2c(1-(s-1+p)/(2n)+O(n^{-2}))$.

Consequences. Dividing, $a_n/(2b_n)=(1-\frac{s-1+p}{2n})(1-\frac{s-1}{2n})^{-1}+O(n^{-2})=1-\frac{p}{2n}+O(n^{-2})$. Finally $c(n+\frac{s-1}{2p})^{-p}=cn^{-p}(1-\frac{s-1}{2n}+O(n^{-2}))$, which gives the last form of b_n. □

Remark 9.6 (The hard edge at order $1/n$; the case $p=2$). The hard edge contributes nothing at order $1/n$: its residue $\mathfrak{A}_0$ is a multiple of $\mathsf{T}e_1e_2^{\mathsf{T}}\mathsf{T}^{-1}$, because only the entry a^2Y_{12} in (9.1) has a pole at 0, and this direction is annihilated by both functionals. Hence the $1/n$ terms of b_n and a_n are those of the Airy correction at the soft edge, and of these only the residue combinations (9.4) enter; the coefficient of b_n comes from the simple pole alone. At $p=2$ the band factor is exactly constant (the remark after Lemma 8.1), so $\Delta_0=\Delta_0^{\mathrm{mod}}$ and $\Delta_r=\Delta_r^{\mathrm{mod}}$ exactly, and the closed forms of Remark 1.6 for $(p,s)=(2,2)$ give the values of Theorem 1.4 there.

Derivation of the closed form of Remark 1.6. Put $\beta_{n+1}:=[2\sqrt{(n+\nu)(n+\nu+1)}]^{-1}$. Van Assche [18, §3] records that the orthonormal Lommel polynomials satisfy $xp_n=\beta_{n+1}p_{n+1}+\beta_np_{n-1}$. For $\nu=\frac{3}{2}$ the orthogonality measure μ is identified as follows. By the Bessel recurrence [8, (10.6.1)], $\mathsf{u}_n:=\sqrt{n+\nu}\,J_{\nu+n}(1/x)$ solves $x\mathsf{u}_n=\beta_{n+1}\mathsf{u}_{n+1}+\beta_n\mathsf{u}_{n-1}$ for $n\ge 0$ (with $\beta_0:=[2\sqrt{\nu(\nu-1)}]^{-1}$), and it is square summable by [8, (10.2.2)]. For $x\notin\mathbb{R}$ the resolvent vector $(\mathcal{J}-x)^{-1}e_0$ of the Jacobi matrix $\mathcal{J}$ solves the same recurrence for $n\ge 1$ and is square summable, and such solutions form a line, since the Wronskian $\beta_n(\mathsf{u}_n\tilde{\mathsf{u}}_{n-1}-\mathsf{u}_{n-1}\tilde{\mathsf{u}}_n)$ of two of them is constant and tends to 0. Hence $\int(t-x)^{-1}\,\mathrm{d}\mu(t)$, the first entry of the resolvent, equals $-\mathsf{u}_0/(\beta_0\mathsf{u}_{-1})=-2\nu J_\nu(1/x)/J_{\nu-1}(1/x)$. For $\nu=\frac{3}{2}$ this is $-3(z^{-1}-\cot z)$ with $z=1/x$ [8, (10.16.1), (10.6.1)], and the partial fractions of cot [8, (4.22.3)] show that μ has mass $3/(\pi k)^2$ at $\pm1/(\pi k)$, $k\ge1$. Its image under $t\mapsto\pi^2t^2$ is $(6/\pi^2)\sigma_{2,2}$. The quadratic transformation (the monic orthogonal polynomials of μ of degree $2n$ are $q_n(t^2)$, where q_n are those of the image of μ under $t\mapsto t^2$) gives $a_n=\pi^2(\beta_{2n}^2+\beta_{2n+1}^2)$ and $b_n=\pi^2\beta_{2n-1}\beta_{2n}$ for $\sigma_{2,2}$, which are the formulas of Remark 1.6.

10. Numerical illustrations

This section compares the theorems with computed recurrence coefficients. Table 10.1 shows the normalised deviations controlled by Theorem 1.2 and Corollary 1.3 for a factor ℓ that grows faster than every power of $\log k$, and Table 10.2 compares fitted $1/n$ coefficients with Theorem 1.4 for $p>1$ and with Conjecture 11.1 for $p\le1$. The computations are floating-point illustrations; no proof in this paper uses them.

Method. The atoms $k\le K$ alone do not suffice. The spectral weight of P_n near the hard edge sits at indices $k\asymp n^{1+2/p}$, about 10^{16} for $p=\frac12$ and $n=1600$, and a truncated measure has a soft edge at its smallest atom and obeys a different law. For $p=\frac12$, $s=\frac32$, the factor ℓ of Table 10.1 and the atoms $k\le2\cdot10^5$ alone, $n(n^pb_n/c(p)-1)$ equals -1.81, -13.3 and -105.9 at $n=100$, 400 and 1600. We therefore replaced the atoms $k>K$ by a quadrature of the continuous tail $\int_{K+1/2}^{\infty}t^{-s}\ell(t)\,\delta_{t^{-p}}\,\mathrm{d}t$ and computed the Jacobi matrix of the resulting discrete measure by the updating algorithm that adds one atom at a time and restores tridiagonal form by plane rotations, in extended precision (the precision and the truncation checks are given with each table). The scripts and their outputs are available as described in the statement on data and code availability at the end of the paper.

10.1. **The rate $O(1/n)$.** Table 10.1 compares the factor $\ell(k)=\exp\sqrt{\log(k+e)}$ with $\ell\equiv1$ for $p=\frac12$, 1, 2 and $s=2$. Here $\ell(k)=\exp\Lambda(\log k)$ with $\Lambda(\xi)=\sqrt{\log(e^\xi+e)}$, whose singularities lie on the lines $|\operatorname{Im}\xi|=\pi$, so ℓ is analytic of angle ϑ for every $\vartheta<\pi$ (Definition 1.1). By Theorem 1.2 and Corollary 1.3 the normalised deviations $n(n^pb_n/c(p)-1)$ and $n(a_n/(2b_n)-1)$ stay bounded. The Bessel orders $\alpha=(s-1-p)/p$ are $\alpha=1$, which lies outside the window of Theorem 7.1 and is reached through the shift (Theorem 3.1) from $s=\frac32$; $\alpha=0$, the logarithmic case of the Bessel model; and $\alpha=-\frac12$.

The tail was integrated by Gauss–Legendre quadrature on logarithmic panels (ratio 1.02, eight nodes per panel) up to $T = 10^{48}$ for $p = \frac{1}{2}$ and $T = 10^{16}$ for $p = 1, 2$, and the recurrence was run in long double precision. We repeated each computation with $K = 10^5$, with $K = 2\cdot 10^5$, and with $K = 2\cdot 10^5$ and T reduced by a factor 10^4. The largest change of any tabulated entry under these variations is $9.5\cdot 10^{-8}$ for $p = \frac{1}{2}$, $1.9\cdot 10^{-8}$ for $p = 1$ and $1.7\cdot 10^{-11}$ for $p = 2$. The column $\ell \equiv 1$ at $p = 2$ agrees in every digit shown with the closed form of Remark 1.6.

	$n\big(n^p b_n/c(p)-1\big)$		$n\big(a_n/(2b_n)-1\big)$		
n	ℓ	$\ell\equiv 1$	ℓ	$\ell\equiv 1$	$\frac{1}{2}\Lambda'(\log n)$
$p=\frac{1}{2}$, $s=2$, $\alpha=1$ (reached through the shift)					
100	−0.4039	−0.5069	−0.2442	−0.2408	0.1131
200	−0.4069	−0.5052	−0.2470	−0.2453	0.1070
400	−0.4101	−0.5039	−0.2484	−0.2477	0.1014
800	−0.4133	−0.5028	−0.2492	−0.2488	0.0963
1600	−0.4163	−0.5020	−0.2496	−0.2494	0.0919
$p=1$, $s=2$, $\alpha=0$					
50	−0.3821	−0.4934	−0.4954	−0.4950	0.1191
100	−0.3893	−0.4967	−0.4978	−0.4975	0.1131
200	−0.3958	−0.4983	−0.4989	−0.4987	0.1070
400	−0.4015	−0.4992	−0.4994	−0.4994	0.1014
800	−0.4065	−0.4996	−0.4997	−0.4997	0.0963
$p=2$, $s=2$, $\alpha=-\frac{1}{2}$					
50	−0.3794	−0.4938	−0.9756	−0.9780	0.1191
100	−0.3868	−0.4969	−0.9878	−0.9889	0.1131
200	−0.3936	−0.4984	−0.9939	−0.9944	0.1070
400	−0.3995	−0.4992	−0.9970	−0.9972	0.1014
800	−0.4046	−0.4996	−0.9985	−0.9986	0.0963

TABLE 10.1. Normalised deviations from the pointwise law for $\sigma_{p,2,\ell}$, with $\ell(k) = e^{\sqrt{\log(k+e)}}$, and for $\ell \equiv 1$, computed in extended precision from the atoms $k \le K$ and a quadrature of the continuous tail. All entries are stable under doubling of K and under a change of the tail cut-off (see the text). Boundedness is what Theorem 1.2 and Corollary 1.3 assert. The limits $-(s-1)/2$ and $-p/2$ in the columns $\ell \equiv 1$ are proved for $p = 2$ (Theorem 1.4) and observed, not proved, for $p = \frac{1}{2}$ and $p = 1$. The slow drift of the first column is the effect of $\Lambda'(\log n)$ (Conjecture 11.2).

All entries stay bounded, as the theorems require. For the pure powers the two normalised deviations approach $-(s-1)/2 = -\frac{1}{2}$ and $-p/2$ in all three geometries. At $p = 1$ and $p = 2$ the distance of the column $\ell \equiv 1$ for b_n from $-\frac{1}{2}$ roughly halves when n doubles, as a remainder $O(n^{-2})$ in $n^p b_n/c(p)$ requires; at $p = \frac{1}{2}$ it shrinks only by a factor close to $2^{-1/2}$, which suggests a remainder of order n^{-1-p} there. For ℓ the deviation of $n^p b_n$ is smaller in absolute value, and its difference from the pure power drifts slowly towards 0, roughly like $\frac{1}{2}\Lambda'(\log n)$ (last column); see Conjecture 11.2. The factor ℓ has almost no effect on $a_n/(2b_n)$ at order $1/n$.

A factor analytic only in a thin strip. Theorem 1.2 needs ℓ to be analytic of some angle $\vartheta > 0$ only. As a test we took $p = 2$ and

$$\Lambda(\xi) = \sqrt{\xi+1} + (\xi+2)^{-1/2}\log(1 - q\cos 2\xi), \qquad q = 1/\cosh(0.6).$$

Here $1 - q\cos 2\xi$ vanishes at $\xi = \pi m \pm 0.3i$ for every integer m, so Λ is analytic of angle ϑ exactly for $\vartheta < 0.3$, and the singularities are not confined to a bounded part of the strip. The factor $\ell(k) = e^{\Lambda(\log k)}$ grows like $e^{\sqrt{\log k}}$ and carries a log-periodic oscillation of relative size $(\log k)^{-1/2}$. With the same numerical scheme, at $n = 50 \cdot 2^{j/4}$, $0 \le j \le 24$ (so $50 \le n \le 3200$), the deviation $n(n^2 b_n/c(2) - 1)$ oscillates log-periodically between -0.6031 and -0.0293 for $s = 2$ ($\alpha = -\frac{1}{2}$) and between -1.1014 and -0.5494 for $s = 3$ ($\alpha = 0$). Its maximum modulus is 0.598 on $50 \le n < 400$ and 0.589 on $400 < n \le 3200$ for $s = 2$, and 1.097, respectively 1.088, for $s = 3$, while $n(a_n/(2b_n) - 1)$ stays between -1.0091 and -0.9505. The deviations stay bounded, as Theorem 1.2 asserts, but the range of n covers only about 1.3 periods of the oscillation (period π in $\log n$), so this is weak evidence. The oscillation of the $1/n$ term reflects that of Λ', which here has no sign.

10.2. **The $1/n$ coefficients.** Write $n^p b_n/c(p) = 1 + d/n + \cdots$ and $a_n/(2b_n) = 1 + e/n + \cdots$ for the pure powers $\sigma_{p,s}$. Theorem 1.4 gives $d = -(s-1)/2$ and $e = -p/2$ for $p > 1$, and Conjecture 11.1 asserts the same for $p \le 1$. Table 10.2 lists d and e fitted to the computed coefficients with $n \le 1600$. The tail was integrated by a Gauss–Jacobi rule in the variable $x = t^{-p}$ that is exact on all polynomials of the degree needed, so that the remaining error comes from replacing the atoms $k > K$ by the integral; the recurrence was run in quadruple precision, with $K = 1.6 \cdot 10^5$ for $p \ge 1$ and $K = 3.2 \cdot 10^5$ for $p = \frac{1}{2}$, and repeated with $2K$. At $(p, s) = (2, 2)$ the computed b_n and a_n agree with the closed form of Remark 1.6 for $n \le 400$ to a relative error $6.6 \cdot 10^{-15}$ (at $K = 4 \cdot 10^4$). The fits are least-squares fits in the powers $n^{-1-i-jp}$, $i, j \ge 0$, with several numbers of terms and fitting windows; the table gives the median of the estimates and half their range as an error estimate. These estimates are not bounds.

For $p > 1$ every fitted value agrees with Theorem 1.4 in all nine digits shown. For $p = 1$ the same holds, and for $p = \frac{1}{2}$ the discrepancy is within the error estimate; at $p = \frac{1}{2}$ the fits need the non-integer powers n^{-1-jp}, in line with the slower convergence seen in Table 10.1. At $(p, s) = (2, 2)$ the exact values $d = -\frac{1}{2}$ and $e = -1$ also follow from the closed form of Remark 1.6. That closed form was checked in interval (ball) arithmetic for $1 \le n < 300$: the coefficients computed by the Chebyshev algorithm from the moments $\zeta(2j + 2)$ have rigorous enclosures that agree with it to a relative accuracy better than 10^{-2000}.

11. Further remarks and open problems

Theorem 1.2 gives the pointwise law with rate $O(1/n)$ for every $p > 0$, and Theorem 1.4 identifies the $1/n$ term for pure powers with $p > 1$. This section records what the computations of Section 10 suggest beyond these results, states two conjectures, and lists the directions in which the method stops.

11.1. **The $1/n$ term for $p \le 1$.** The fits of Table 10.2 for $p = \frac{1}{2}$ and $p = 1$ agree with the values of Theorem 1.4 as well as the error estimates allow.

Conjecture 11.1. *Let $0 < p \le 1$ and $s > 1$. Then the pure powers satisfy*

$$n^p b_n(\sigma_{p,s}) = c(p)\Big(1 - \frac{s-1}{2n} + o\Big(\frac{1}{n}\Big)\Big), \qquad \frac{a_n(\sigma_{p,s})}{2b_n(\sigma_{p,s})} = 1 - \frac{p}{2n} + o\Big(\frac{1}{n}\Big).$$

The evidence consists of the rows $p \le 1$ of Table 10.2 ($s \in \{\frac{3}{2}, 2, 3, 4\}$, $n \le 1600$) and the columns $\ell \equiv 1$ of Table 10.1 at $p = \frac{1}{2}$ and $p = 1$. At $p = \frac{1}{2}$ the remainder does not appear to be $O(n^{-2})$: the fits need the powers n^{-1-jp}, and in Table 10.1 the deviation from $-\frac{1}{2}$ decays

		coefficient d of $n^p b_n/c(p)$			coefficient e of $a_n/(2b_n)$		
p	s	fitted	estimate	$-\frac{s-1}{2}$	fitted	estimate	$-\frac{p}{2}$
$p>1$: Theorem 1.4							
$\frac{3}{2}$	$\frac{3}{2}$	-0.250000000	$3.4\cdot10^{-9}$	-0.25	-0.750000000	$1.9\cdot10^{-8}$	-0.75
$\frac{3}{2}$	2	-0.500000000	$4.8\cdot10^{-9}$	-0.5	-0.750000000	$6.9\cdot10^{-9}$	-0.75
$\frac{3}{2}$	3	-1.000000000	$1.6\cdot10^{-8}$	-1	-0.750000000	$1.4\cdot10^{-8}$	-0.75
$\frac{3}{2}$	4	-1.500000000	$1.4\cdot10^{-8}$	-1.5	-0.750000000	$1.6\cdot10^{-7}$	-0.75
2	$\frac{3}{2}$	-0.250000000	$6.0\cdot10^{-10}$	-0.25	-1.000000000	$1.6\cdot10^{-9}$	-1
2	2	-0.500000000	$3.4\cdot10^{-11}$	-0.5	-1.000000000	$1.1\cdot10^{-10}$	-1
2	3	-1.000000000	$3.2\cdot10^{-11}$	-1	-1.000000000	$1.2\cdot10^{-11}$	-1
2	4	-1.500000000	$8.6\cdot10^{-11}$	-1.5	-1.000000000	$2.6\cdot10^{-10}$	-1
3	$\frac{3}{2}$	-0.250000000	$2.9\cdot10^{-12}$	-0.25	-1.500000000	$4.5\cdot10^{-10}$	-1.5
3	2	-0.500000000	$3.9\cdot10^{-12}$	-0.5	-1.500000000	$3.2\cdot10^{-10}$	-1.5
3	3	-1.000000000	$2.2\cdot10^{-11}$	-1	-1.500000000	$1.0\cdot10^{-10}$	-1.5
3	4	-1.500000000	$8.1\cdot10^{-11}$	-1.5	-1.500000000	$1.7\cdot10^{-11}$	-1.5
4	$\frac{3}{2}$	-0.250000000	$1.8\cdot10^{-12}$	-0.25	-2.000000000	$1.1\cdot10^{-9}$	-2
4	2	-0.500000000	$1.5\cdot10^{-12}$	-0.5	-2.000000000	$8.8\cdot10^{-10}$	-2
4	3	-1.000000000	$1.3\cdot10^{-11}$	-1	-2.000000000	$4.6\cdot10^{-10}$	-2
4	4	-1.500000000	$5.7\cdot10^{-11}$	-1.5	-2.000000000	$1.7\cdot10^{-10}$	-2
$p\le1$: support for Conjecture 11.1							
$\frac{1}{2}$	$\frac{3}{2}$	-0.249999998	$3.8\cdot10^{-7}$	-0.25	-0.250000021	$2.7\cdot10^{-7}$	-0.25
$\frac{1}{2}$	2	-0.500000014	$5.7\cdot10^{-7}$	-0.5	-0.249999950	$6.3\cdot10^{-6}$	-0.25
$\frac{1}{2}$	3	-1.000000247	$2.9\cdot10^{-5}$	-1	-0.249999294	$9.3\cdot10^{-5}$	-0.25
$\frac{1}{2}$	4	-1.500001361	$1.3\cdot10^{-4}$	-1.5	-0.249996155	$3.4\cdot10^{-4}$	-0.25
1	$\frac{3}{2}$	-0.250000000	$1.3\cdot10^{-10}$	-0.25	-0.500000000	$3.7\cdot10^{-10}$	-0.5
1	2	-0.500000000	$5.6\cdot10^{-10}$	-0.5	-0.500000000	$1.1\cdot10^{-9}$	-0.5
1	3	-1.000000000	$5.6\cdot10^{-10}$	-1	-0.500000000	$1.1\cdot10^{-9}$	-0.5
1	4	-1.500000000	$2.1\cdot10^{-9}$	-1.5	-0.500000000	$8.5\cdot10^{-9}$	-0.5

TABLE 10.2. Fitted $1/n$ coefficients d and e for the pure powers $\sigma_{p,s}$, from the coefficients with $n\le1600$, against the values of Theorem 1.4 ($p>1$) and of Conjecture 11.1 ($p\le1$). The estimate is half the range of the fits; the change under doubling of the truncation index K is of the same order or smaller. Neither is a bound.

like a term of order n^{-1-p} in $n^p b_n/c(p)$. This is why the conjecture is stated with $o(1/n)$ only. For $p=1$ the averaged form of the statement for b_n is proved in [16]: for every $s>1$,

$$\sum_{j=1}^{n}\log\frac{j\,b_j(\sigma_{1,s})}{c(1)}+\frac{s-1}{2}\log n$$

converges as $n\to\infty$. This is what the conjecture predicts after summation, but it does not imply the pointwise statement.

The method of Sections 8 and 9 does not reach $p\le1$, for the reasons given in Remark 8.5, and the comparison argument of Section 6 bounds the hard-edge contribution by $O(1/n)$ without computing its $1/n$ term. At $p=1$ the band factor equals $1+z/(6n)+O(n^{-2})$, a defect that alone enters the coefficients only at order n^{-2}; the obstruction is the hard-edge parametrix. For $p<1$ the numerics show remainders of order n^{-1-p}, which a proof would

have to produce. What is missing is a hard-edge parametrix with matching error $O(n^{-2})$ at $p = 1$, and one adapted to the seams of Appendix A for $p < 1$.

11.2. **Slowly varying factors at order $1/n$.** For an analytic factor ℓ the columns ℓ of Table 10.1 drift away from the pure-power values. The drift is described by the following conjecture.

Conjecture 11.2. *Let $p > 0$, $s > 1$, and let ℓ be analytic of angle ϑ for some $\vartheta > 0$, with $\ell(k) = \exp \Lambda(\log k)$. Then*

$$\frac{n^p b_n(\sigma_{p,s,\ell})}{c(p)} = 1 - \frac{s - 1 - \Lambda'(\log n)}{2n} + o\Big(\frac{\Lambda'(\log n)}{n}\Big) \qquad \text{whenever} \quad n\Lambda'(\log n) \to \infty.$$

The conjecture says that, at order $1/n$, a slowly varying factor acts like a change of the exponent s by the local logarithmic derivative $\Lambda'(\log n)$, that is, like a hard-edge Bessel order that drifts with n. In Table 10.1 the difference between the first two columns is 0.91 to 0.99 times $\frac{1}{2}\Lambda'(\log n)$, and this ratio increases with n in all three geometries. In the thin-strip example of Section 10, where Λ' oscillates without a sign, the $1/n$ term oscillates with it. Within our method the conjecture amounts to computing the main term of the comparison estimate of Section 6, as Conway and Deift do for their weight [2, (5.5)]; this would need a Bessel parametrix with a drifting order.

11.3. **Other Bessel orders, other nodes, rougher weights.** For analytic factors ℓ, and for pure powers with $p \le 1$, Bessel orders $\alpha \ge 1$ are reached only through the Christoffel shift. A direct Riemann–Hilbert analysis for $\alpha \ge 1$ would need a different comparison frame (Remark 6.7); it would be needed to compute the $1/n$ term in that range.

At $p = 1$ one can also work in the symmetric variable $y = \pm x^{1/2}$ of [19], where 0 is an interior Fisher–Hartwig point of the band. The comparison method then covers only $1 < s < 2$ directly, and $1 < s < 3$ after folding; we have not pursued this.

Our nodes are exact powers k^{-p}. For nodes $x_k = f(k)$ with $f'(\xi) \sim -pC\xi^{-p-1}$ the averaged law is known [15, Corollary 1.2], but the pointwise law $n^p b_n \to c(p)C$ is open. One expects it to hold when f is smooth in a sense analogous to Definition 1.1, and to fail under rough displacements of the nodes, as it does for rough weights: bounded rough weights $k^{-s}e^{\epsilon_k}$, $|\epsilon_k| \le \varepsilon$, can break the pointwise law although the averaged law holds for them, as is proved in [16].

Appendix A. The outer scalar for $p \neq 1$

This appendix proves Lemma 5.1 for $p > 1$ (§A.1) and for $0 < p < 1$ (§A.2), that is, it constructs an admissible outer scalar (Definition 4.10) in these two geometries. The standing assumptions of §4.1 hold; nothing here depends on s. We use the notation of Section 5: Binet's function j, $\tilde{D} = e^{-j(t)}$, $D = \upsilon_{\pm}\tilde{D}$, $K = (4c - z)^{-1/2}$, and the bounds (5.2), (5.3). In both cases the difficulty is the cut of $t(z)$ on $(-\infty, 0)$, where D has a jump that is not of the form required by Definition 4.10(a). For $p > 1$ the jump is removed exactly by a Cauchy integral; for $p < 1$, D is used only in an inner sector and glued to an elementary function outside.

A.1. **The case $p > 1$: the Binet-cut scalar.** For $p > 1$, $|\arg t| < \pi/p < \pi$ on the slit plane, so $\Gamma(t)$ has no poles there, and the jump of D on $(-\infty, 0)$ is a bounded phase $e^{i\mathfrak{b}}$.

Lemma A.1 (Binet-cut scalar)**.** *Let $p > 1$.*

(i) *$\tilde{D}$ is analytic and zero-free on $\mathbb{C} \setminus (-\infty, 0]$, and D is analytic and zero-free in $\mathbb{C}_{\pm}$, with continuous boundary values on $\mathbb{R} \setminus (\{X_k\} \cup \{0\})$. On $(-\infty, 0)$, $D_- = \overline{D_+}$.*

(ii) *Let* $\mathfrak{b}(y) := 2\operatorname{Im}\big(-j(t_+(y)) - \operatorname{Log}(1-e^{-2i\theta_+(y)})\big)$ *for* $y<0$, *where* t_+, θ_+ *are boundary values from* $\mathbb{C}_+$. *Then* $D_+/D_- = e^{i\mathfrak{b}}$ *on* $(-\infty,0)$; $\mathfrak{b}$ *is real-analytic,* $|\mathfrak{b}(y)| \le C\min(1,|y|^{1/p})$ *with* C *independent of* n, $|\mathfrak{b}'(y)| \le C_n|y|^{1/p-1}$ *for* $-1 \le y < 0$, $\mathfrak{b}'$ *is bounded on* $(-\infty,-\frac12]$, *and* $\mathfrak{b}(y) \to \pi/p - \pi$ *as* $y\to-\infty$, *with* $\int_{-\infty}^{-1} |\mathfrak{b}(y) - \pi/p + \pi|\,|y|^{-1}\,\mathrm{d}y < \infty$.

(iii) *The function*

$$B(z) := \exp\Big(-\frac{1}{2\pi}\int_{-\infty}^{0} \mathfrak{b}(y)\Big[\frac{1}{y-z} - \frac{1}{y-1}\Big]\,\mathrm{d}y\Big) \tag{A.1}$$

is analytic and zero-free on $\mathbb{C}\setminus(-\infty,0]$, *positive on* $(0,\infty)$, *has continuous boundary values on* $(-\infty,0)$ *with* $B_+ = B_- e^{-i\mathfrak{b}}$, *and* DB *extends analytically across* $(-\infty,0)$. *For* $z\in\mathcal{Z}_n$, *with* C *independent of* n,

$$\begin{gathered}|\log B(z) - \log B(0)| \le C|z|^{1/p}, \quad |(\log B)'(z)| \le C|z|^{1/p-1} \quad (|z|\le 1),\\ |\log B(z)| \le C(1+\log(1+|z|)).\end{gathered} \tag{A.2}$$

(iv) $F := DBK$ *is an admissible outer scalar, with* $\tilde F = \tilde D B K$ *and* $H_n = (\tilde D B)^{-2}$, $H_n(0^+) = B(0)^{-2}$.

Proof. Part (i) is the absence of poles of Γ in the sector; (ii) estimates the jump $\mathfrak{b}$ by Stirling's formula for large $|t|$ and by (5.3) for small $|t|$; (iii) is the Sokhotski–Plemelj formula together with a splitting of the Cauchy integral at $|y| = |z|$; (iv) checks the properties (a)–(e) of Definition 4.10 one by one.

(i) On the slit plane $|\arg t| < \pi/p < \pi$, so $\Gamma(t)$ has neither zeros nor poles and $\tilde D$ is analytic and zero-free. By (4.10), $|e^{\mp 2i\theta}| < 1$ in $\mathbb{C}_\pm$ (here $\pi p > \pi$), so D is analytic and zero-free in $\mathbb{C}_\pm$. On $(-\infty,0)$, $t_- = \overline{t_+}$ and Γ, j are real on the real axis, so $D_- = \overline{D_+}$.

(ii) Since $|e^{-2i\theta_+}| < 1$ on $(-\infty,0)$, $\log D_+ := -j(t_+) - \operatorname{Log}(1-e^{-2i\theta_+})$ is a continuous logarithm, real-analytic in y, and $D_+/D_- = e^{2i\operatorname{Im}\log D_+} = e^{i\mathfrak{b}}$. With $|t_+| = n|y|^{-1/p}$ and $\arg t_+ = -\pi/p$: for $|t_+|\ge 1$, (5.2) and $|\operatorname{Log}(1-e^{-2i\theta_+})| \le 2\exp(-2\pi\sin(\pi/p)|t_+|)$ give $|\mathfrak{b}| \le C/|t_+| = C|y|^{1/p}/n$. For $|t_+|\le 1$ the expansion (5.3) and $\operatorname{Log}(1-e^{-2\pi i t}) = \log(2\pi|t|) + i(\frac{\pi}{2} + \arg t) + O(t)$ (note $\frac{\pi}{2} + \arg t_+ \in (-\frac{\pi}{2},\frac{\pi}{2})$) show that $\mathfrak{b}$ is bounded and

$$\mathfrak{b}(y) = 2\big(\tfrac12\arg t_+ - \tfrac{\pi}{2} - \arg t_+\big) + O(|t_+|\log(2/|t_+|)) = \frac{\pi}{p} - \pi + O\big(n|y|^{-1/p}\log|y|\big).$$

This gives the bounds on $\mathfrak{b}$ and the integrability. For the derivative, $\mathrm{d}t_+/\mathrm{d}y = -t_+/(py)$, and $\frac{\mathrm{d}}{\mathrm{d}t}\big(j(t) + \operatorname{Log}(1-e^{-2\pi i t})\big)$ is $O(|t|^{-2})$ for $|t|\ge 1$ and $O(|t|^{-1})$ for $|t|\le 1$; multiplied by $|t|/|y|$ this gives $|\mathfrak{b}'(y)| \le C|y|^{1/p-1}/n$ for $|y|\le n^p$ and $|\mathfrak{b}'(y)| \le C/|y|$ otherwise.

(iii) The integral converges absolutely, since $\mathfrak{b}$ is bounded and the bracket is $(z-1)/((y-z)(y-1))$. By the Sokhotski–Plemelj formula (the density is real-analytic on $(-\infty,0)$), $\log B$ has continuous boundary values and $(\log B)_+ - (\log B)_- = -i\mathfrak{b}$. So $(DB)_+ = (DB)_-$ on $(-\infty,0)$, and DB, being continuous across, is analytic there by Morera's theorem. B is positive on $(0,\infty)$ because the integrand is real there.

For (A.2), let $z\in\mathcal{Z}_n$. Then $|y-z|^2 = y^2 + |z|^2 + 2|y||z|\cos(\arg z) \ge \frac12(|y|+|z|)^2$ for $y<0$, since $|\arg z| < \vartheta_{\mathrm{w}} < \pi/2$. We have $\log B(z) - \log B(0) = -\frac{z}{2\pi}\int_{-\infty}^0 \frac{\mathfrak{b}(y)\,\mathrm{d}y}{y(y-z)}$, so by (ii), for $|z|\le 1$,

$$|\log B(z) - \log B(0)| \le C|z|\int_0^\infty \frac{\min(1,u^{1/p})}{u(u+|z|)}\,\mathrm{d}u \le C|z|\Big(\int_0^1 \frac{u^{1/p-1}\,\mathrm{d}u}{u+|z|} + 1\Big) \le C|z|^{1/p},$$

because $1/p - 1 \in (-1,0)$. The same splitting applied to $(\log B)'(z) = -\frac{1}{2\pi}\int \mathfrak{b}(y)(y-z)^{-2}\,\mathrm{d}y$ gives $C|z|^{1/p-1}$, and applied to the original integral gives $C(1+\log(1+|z|))$.

(iv) *Property (a).* F is analytic and zero-free off $[0,\infty)$ by (i), (iii) and the choice of K. On $(4c,\infty)$ off the nodes, $F_+/F_- = (D_+/D_-)(K_+/K_-) = (-e^{2i\theta})(-1)$, since B is analytic there.

Property (b). $\tilde F = F/\upsilon_\pm = \tilde D B K$ is analytic and zero-free on $\mathbb{C}\setminus\big((-\infty,0]\cup[4c,\infty)\big)$, and $\tilde F_+ = -\tilde F_-$ on $(4c,\infty)$ because of K. Thus $H_n = 1/((4c-z)\tilde F^2) = (\tilde D B)^{-2}$, which is positive on $(0,\infty)$, and $H_n(0^+) = B(0)^{-2}$ because $\tilde D(z)\to 1$ as $z\to 0$ in $\mathcal{Z}_n$ by (5.2).

Properties (c) and (d). $\log H_n - \log H_n(0^+) = 2j(t) - 2(\log B - \log B(0))$. For $z\in\mathcal{Z}_n$, $|z|\le 1$, we have $|t|\ge n$ and $|\arg t|\le \vartheta_{\mathrm w}/p$, so $|j(t)|\le C|z|^{1/p}/n$ and $|\frac{\mathrm d}{\mathrm dz} j(t)| = |j'(t)||t|/(p|z|) \le C|z|^{1/p-1}/n$. Together with (A.2) and $\nu_p\le 1/p$, this is (c). For $|z|\ge 1$ in $\mathcal{Z}_n$ we have $|t|\ge e^{-c_0}$, so $j(t)$ is bounded (by (5.2) for $|t|\ge 1$ and by continuity on the compact set $\{e^{-c_0}\le|t|\le 1,\ |\arg t|\le\vartheta_{\mathrm w}/p\}$), and (A.2) gives (d).

Property (e). Fix n. Near 0 and outside the lens, $|\arg z|\ge \frac34\vartheta_{\mathrm{lip}}$, so $\upsilon_\pm\to 1$ by (4.10) and $\tilde D\to 1$; K is bounded; and $\log B$ is bounded on $\mathbb{D}(0,\delta)\setminus(-\delta,0]$ uniformly up to the cut. For the last point split $\int \mathfrak b(y)/(y-z)\,\mathrm dy$ into the window $|y-\operatorname{Re} z|\le|\operatorname{Re} z|/2$ and the rest. On the window write $\mathfrak b(y) = \mathfrak b(\operatorname{Re} z) + O(|y-\operatorname{Re} z|\,|\operatorname{Re} z|^{1/p-1})$ by (ii); the constant part integrates to $\mathfrak b(\operatorname{Re} z)\operatorname{Log}(\cdot)$ with a bounded logarithm (the end points of the window are at comparable distances from z), and the rest is bounded; outside the window $|y-z|\ge(|y|+|z|)/C$ and $|\mathfrak b(y)|\le C|y|^{1/p}$ give a bounded contribution. In the lens, $\tilde F = \tilde D B K$ is bounded above and below by the same facts.

Near ∞, the same window argument on $(-\infty,-\frac12]$, where $\mathfrak b'$ is bounded, gives $|\log B(z)|\le C_n(1+\log|z|)$ for $|z|\ge 2$ off the cut. For $|z|\ge n^p+\delta'$ we have $|t|<1$, and t stays away from 1, so D is bounded above and below by a power of $|z|$; hence $|F|^{\pm1}\le C_n|z|^{N_n}$. Finally, along $z=iy$, $y\to\pm\infty$: by (5.3), $|D(iy)| = |t|^{-1/2}(2\pi)^{-1/2}(1+o(1))\asymp|y|^{1/(2p)}$; $|K(iy)|\asymp|y|^{-1/2}$; and splitting $\mathfrak b = (\pi/p-\pi)\mathbf 1_{(-\infty,-1)} + \mathfrak e$, the first part contributes $-\frac{\pi/p-\pi}{2\pi}\operatorname{Log}\frac{z+1}{2}$ to $\log B$ (an explicit integral), while the second contributes a quantity with a finite limit, by dominated convergence and the integrability in (ii). So $|B(iy)|\asymp|y|^{\frac12-\frac{1}{2p}}$, and $|F(iy)|\asymp 1$. □

A.2. **The case $0<p<1$: seams.** For $p<1$ the poles $t=-1,-2,\dots$ of Γ lie on the rays $\arg z=\pm\pi p$, inside the slit plane, and D has no useful continuation to the negative axis. We use D only in the inner sector, where $|\arg t|<\pi/2$, and an elementary function outside; a Cauchy integral χ over the two separating rays glues them together exactly. Put

$$\vartheta_{\mathrm{seam}} := \tfrac{\pi p}{2},\qquad S_{\mathrm{in}} := \{z\ne 0:\ |\arg z|<\vartheta_{\mathrm{seam}}\},\qquad S_{\mathrm{out}} := \mathbb{C}\setminus(\overline{S_{\mathrm{in}}}\cup\{0\}),$$
$$\Sigma^{\mathrm s}_\pm := \{re^{\pm i\vartheta_{\mathrm{seam}}}:\ r>0\},$$

both seams oriented away from 0, and

$$\varpi := \tfrac12 - \tfrac{1}{2p} < 0,\qquad B_{\mathrm{in}}(z) := (1+z)^{\varpi},\qquad B_{\mathrm{out}}(z) := \Big(1-\frac{z}{2\pi n}\Big)^{1/2},$$

with principal powers; the cut $(-\infty,-1]$ of B_{in} lies in S_{out}, and the cut $[2\pi n,\infty)$ of B_{out} lies in S_{in}. The lens, the saturated regions and the top segment lie in $\mathcal{Z}_n\subset S_{\mathrm{in}}$, since $\vartheta_{\mathrm w}<\vartheta_{\mathrm{seam}}$ by (4.3).

Lemma A.2 (Seam scalar)**.** *Let $0<p<1$.*

(i) *On S_{in}, $|\arg t|<\pi/2$; $\tilde D = e^{-j(t)}$ is analytic and zero-free there, and $D=\upsilon_\pm\tilde D$ is analytic and zero-free in $S_{\mathrm{in}}\cap\mathbb{C}_\pm$.*

(ii) *On $\Sigma^{\mathrm s}_\pm$ put, with principal logarithms,*

$$\chi_\pm(\zeta) := -j(t(\zeta)) - \operatorname{Log}\big(1-e^{\mp 2\pi i t(\zeta)}\big) + \varpi\operatorname{Log}(1+\zeta) - \tfrac12\operatorname{Log}\Big(1-\frac{\zeta}{2\pi n}\Big). \tag{A.3}$$

Then $\chi_\pm$ is real-analytic on the open ray, $e^{\chi_\pm} = DB_{\rm in}/B_{\rm out}$ there, $\chi_\pm(\zeta) \to 0$ as $\zeta \to 0$, and $\chi_\pm(\zeta) = O\big(n|\zeta|^{-1/p}\log|\zeta| + n|\zeta|^{-1}\big)$ as $\zeta \to \infty$. In particular $\chi_\pm$ has no winding: it is the continuous logarithm of $DB_{\rm in}/B_{\rm out}$ that vanishes at both ends. Moreover $\chi_-(\bar\zeta) = \overline{\chi_+(\zeta)}$.

(iii) *The* seam scalar

$$\chi(z) := \frac{1}{2\pi i}\Big[\int_{\Sigma_+^{\rm s}} \frac{\chi_+(\zeta)\,\mathrm{d}\zeta}{\zeta - z} - \int_{\Sigma_-^{\rm s}} \frac{\chi_-(\zeta)\,\mathrm{d}\zeta}{\zeta - z}\Big] \tag{A.4}$$

is analytic off $\Sigma_+^{\rm s} \cup \Sigma_-^{\rm s}$, real on $\mathbb{R}\setminus\{0\}$, tends to 0 at ∞, has continuous boundary values on the open seams, and $\chi_{\rm out} - \chi_{\rm in} = \chi_\pm$ on $\Sigma_\pm^{\rm s}$, where "out" and "in" refer to the boundary values from $S_{\rm out}$ and $S_{\rm in}$.

(iv) *The function*

$$F := DB_{\rm in}Ke^{\chi} \text{ in } S_{\rm in}, \qquad F := B_{\rm out}Ke^{\chi} \text{ in } S_{\rm out}, \tag{A.5}$$

extends analytically across $\Sigma_\pm^{\rm s}$ and is an admissible outer scalar, with $\tilde F = \tilde D B_{\rm in}Ke^{\chi}$ and $H_n = (\tilde D B_{\rm in}e^{\chi})^{-2}$, $H_n(0^+) = e^{-2\chi(0)}$.

Proof. Part (i) is (4.10); (ii) expands each term of (A.3) at both ends of the seam; (iii) is the Sokhotski–Plemelj formula; (iv) checks the properties (a)–(e) of Definition 4.10, with the bounds on χ taken from Lemma A.3 below.

(i) $|\arg t| = |\arg z|/p < \pi/2$ on $S_{\rm in}$, so $\Gamma(t)$ has no poles or zeros there; and $\pm\operatorname{Im}\theta < 0$ in $S_{\rm in}\cap\mathbb{C}_\pm$ by (4.10).

(ii) Each term of (A.3) is continuous on the open ray $\Sigma_+^{\rm s}$: j is analytic off $(-\infty,0]$; $\arg t = -\pi/2$, so $e^{-2\pi i t} = e^{-2\pi|t|} \in (0,1)$; $1+\zeta \notin (-\infty,0]$ since $\arg\zeta \in (0,\pi)$; and $1-\zeta/(2\pi n)$ has non-zero imaginary part. The exponential of the sum is $\tilde D v_+ B_{\rm in}/B_{\rm out}$. As $\zeta\to 0$, $|t|\to\infty$ and each term tends to 0 (by (5.2) for j). As $\zeta\to\infty$, $t\to 0$ and, by (5.3), using $\arg t = -\pi/2$, $\arg\zeta = \vartheta_{\rm seam}$ and $\operatorname{Log} t = \log n - \frac{1}{p}\operatorname{Log}\zeta$:

$$\begin{aligned}
-j(t) &= \tfrac12\operatorname{Log} t + \tfrac12\log(2\pi) + O(|t|\log(2/|t|)),\\
-\operatorname{Log}(1-e^{-2\pi i t}) &= -\log(2\pi) - \operatorname{Log} t - \tfrac{i\pi}{2} + O(|t|),\\
\varpi\operatorname{Log}(1+\zeta) &= \varpi\operatorname{Log}\zeta + O(|\zeta|^{-1}),\\
-\tfrac12\operatorname{Log}\big(1-\tfrac{\zeta}{2\pi n}\big) &= -\tfrac12\operatorname{Log}\zeta + \tfrac12\log(2\pi n) + \tfrac{i\pi}{2} + O(n|\zeta|^{-1});
\end{aligned}$$

the second line uses $1-e^{-2\pi i t} = 2\pi i t(1+O(t))$ and $\arg(it) = 0$, the last uses $\arg(-\zeta/(2\pi n)) = \vartheta_{\rm seam} - \pi \in (-\pi,0)$. The sum of the main terms is $-\frac12\operatorname{Log} t + \frac12\log n + (\varpi - \frac12)\operatorname{Log}\zeta = \big(\frac{1}{2p} + \varpi - \frac12\big)\operatorname{Log}\zeta = 0$. The symmetry follows from $\overline{t(\bar\zeta)} = t(\zeta)$ and the reality of j.

(iii) By (ii), $\chi_\pm(\zeta)/\zeta$ is integrable on $\Sigma_\pm^{\rm s}$ near 0 and at ∞ (since $1/p > 1$), so the integrals converge and $\chi\to 0$ at ∞. The Sokhotski–Plemelj formula gives the jumps: $\Sigma_+^{\rm s}$ is oriented outwards with $S_{\rm out}$ on its left, so $\chi_{\rm out} - \chi_{\rm in} = \chi_+$; on $\Sigma_-^{\rm s}$ the left side is $S_{\rm in}$ and the density carries a minus sign, so again $\chi_{\rm out} - \chi_{\rm in} = \chi_-$. Reality on $\mathbb{R}$ follows from the symmetry in (ii).

(iv) On $\Sigma_\pm^{\rm s}$, $F_{\rm in}/F_{\rm out} = DB_{\rm in}e^{\chi_{\rm in}}/(B_{\rm out}e^{\chi_{\rm out}}) = e^{\chi_\pm}e^{-\chi_\pm} = 1$, so F is continuous, hence analytic, across the seams.

Property (a) of Definition 4.10. In $S_{\rm out}$ every factor is analytic and zero-free, and $(-\infty,0)\subset S_{\rm out}$. In $S_{\rm in}\setminus[0,\infty)$ the same holds by (i). On $(4c,\infty)$, $B_{\rm in}$ and χ are analytic, so $F_+/F_- = (-e^{2i\theta})(-1)$.

Property (b). $\tilde F = \tilde D B_{\rm in}Ke^{\chi}$ on $S_{\rm in}\supset\mathcal{Z}_n$, analytic and zero-free off $[4c,\infty)$, with $\tilde F_+ = -\tilde F_-$ there. $H_n = (\tilde D B_{\rm in}e^{\chi})^{-2} > 0$ on $(0,\infty)$ by (iii), and $H_n(0^+) = e^{-2\chi(0)}$, where $\chi(0) := \lim_{z\to 0}\chi(z)$ exists by Lemma A.3.

Properties (c) and (d). Points of $\mathcal{Z}_n$ satisfy $|\arg z| < \vartheta_{\mathrm{w}}$, hence $\operatorname{dist}(z, \Sigma^{\mathrm{s}}_+ \cup \Sigma^{\mathrm{s}}_-) \geq \sin(\vartheta_{\mathrm{seam}} - \vartheta_{\mathrm{w}})|z|$ (here $0 < \vartheta_{\mathrm{seam}} - \vartheta_{\mathrm{w}} < \pi/2$). So Lemma A.3(ii) applies with $\tau_0 = \sin(\vartheta_{\mathrm{seam}} - \vartheta_{\mathrm{w}})$. For $|z| \leq 1$,

$$\begin{aligned} |\log H_n - \log H_n(0^+)| &\leq 2\big(|j(t)| + |\varpi||\operatorname{Log}(1+z)| + |\chi - \chi(0)|\big) \\ &\leq C\big(|z|^{1/p}/n + |z| + |z|(1 + \log \tfrac{1}{|z|})\big) \leq C|z|^{1/2}, \end{aligned}$$

and similarly $|(\log H_n)'| \leq C(|z|^{1/p-1}/n + 1 + \log \frac{1}{|z|}) \leq C|z|^{-1/2}$. Since $\nu_p = \frac{1}{2}$, this is (c). For $|z| \geq 1$ in $\mathcal{Z}_n$, $j(t)$ is bounded, $|\operatorname{Log}(1+z)| \leq C(1 + \log|z|)$, and $|\chi - \chi(0)| \leq C(1+\log|z|)^2$ by Lemma A.3(ii); this is (d).

Property (e). Fix n. Near 0: in S_{in} outside the lens, $\upsilon_\pm \to 1$ (as $\frac{3}{4}\vartheta_{\mathrm{lip}}/p \leq |\arg t| < \frac{\pi}{2}$ there, so that $\sin|\arg t| \geq \sin(\frac{3}{4}\vartheta_{\mathrm{lip}}/p) > 0$ in (4.10)) and $\tilde{D}, B_{\mathrm{in}} \to 1$; in S_{out}, $B_{\mathrm{out}} \to 1$; K is bounded; and χ is bounded near 0 uniformly up to the seams by Lemma A.3(iii). In the lens, $\tilde{F}$ is bounded above and below by the same facts. Near ∞: in S_{out} (which contains the imaginary axis, as $\vartheta_{\mathrm{seam}} < \pi/2$), $F = B_{\mathrm{out}} K e^{\chi} \to (2\pi n)^{-1/2}$, because $B_{\mathrm{out}}(z)K(z) = \big(\frac{-z}{2\pi n}\big)^{1/2}(-z)^{-1/2}(1+o(1))$ with consistent principal branches in S_{out}. In $S_{\mathrm{in}} \cap \{|z| \geq n^p + \delta'\}$, $|t| < 1$ and t stays away from 1, so D, B_{in}, K and e^{χ} are bounded above and below by powers of $|z|$. □

The seams carry no weight: $\chi_\pm$ and χ are built from j, $e^{\mp 2\pi i t}$, B_{in} and B_{out} only, and F has no jump across them. Neither V nor Q has a jump on the seams, and the comparison parametrix of Section 6 has none either, since it is built from D_W. The only trace of the seams in the hard-edge analysis is the factor $e^{-2\chi}$ in H_n, which is controlled by the following lemma. The constant $\chi(0)$ depends on n, but no bound on it is needed: it enters only through $H_n(0^+) = e^{-2\chi(0)}$, hence only through the constant $\mathsf{w}_n H_n(0^+)$, which cancels in $\omega = D_W^2/W_0$ (see (5.7)).

Lemma A.3 (Corner modulus of the seam scalar)**.** *Let $0 < p < 1$ and $\tau_0 \in (0,1)$. There are $C = C(p, \tau_0)$ and n_0 such that for $n \geq n_0$:*

(i) *$|\chi_\pm(\zeta)| \leq C|\zeta|$ for $\zeta \in \Sigma^{\mathrm{s}}_\pm$, $|\zeta| \leq 1$; $|\chi_\pm(\zeta)| \leq C(1+\log|\zeta|)$ for $|\zeta| \geq 1$; and $|\chi'_\pm(\zeta)| \leq C$ for $|\zeta| \leq 2$.*

(ii) *For z with $\operatorname{dist}(z, \Sigma^{\mathrm{s}}_+ \cup \Sigma^{\mathrm{s}}_-) \geq \tau_0|z|$: $|\chi(z) - \chi(0)| \leq C|z|(1 + \log\frac{1}{|z|})$ and $|\chi'(z)| \leq C(1 + \log\frac{1}{|z|})$ if $|z| \leq 1$, and $|\chi(z) - \chi(0)| \leq C(1+\log|z|)^2$ if $|z| \geq 1$. Here $\chi(0) := \frac{1}{2\pi i}\big[\int_{\Sigma^{\mathrm{s}}_+} \chi_+(\zeta)\frac{\mathrm{d}\zeta}{\zeta} - \int_{\Sigma^{\mathrm{s}}_-} \chi_-(\zeta)\frac{\mathrm{d}\zeta}{\zeta}\big]$.*

(iii) *For each fixed n, χ is bounded on $\mathbb{D}(0,1) \setminus (\Sigma^{\mathrm{s}}_+ \cup \Sigma^{\mathrm{s}}_-)$, uniformly up to the seams.*

Proof. (i) On Σ^{s}_+, $t = -i|t|$ with $|t| = n|\zeta|^{-1/p}$. If $|\zeta| \leq 1$, then $|t| \geq n$; by (5.2) with $\phi = \pi/2$, $|j(t)| \leq \frac{1}{6|t|} \leq |\zeta|^{1/p}/n \leq |\zeta|$; $|\operatorname{Log}(1 - e^{-2\pi|t|})| \leq 2e^{-2\pi n|\zeta|^{-1/p}} \leq C|\zeta|$; $|\varpi \operatorname{Log}(1+\zeta)| \leq C|\zeta|$; $|\operatorname{Log}(1 - \zeta/(2\pi n))| \leq C|\zeta|/n$. If $|\zeta| \geq 1$ and $|t| \geq 1$, then $|j(t)| \leq \frac{1}{6}$; if $|t| < 1$, then (5.3) (with a bounded remainder on $\{|t| \leq 1, \operatorname{Re} t \geq 0\}$) and $|\log|t|| \leq \frac{1}{p}\log|\zeta|$ give $|j(t)| \leq C(1 + \log|\zeta|)$. Next $|\operatorname{Log}(1 - e^{-2\pi|t|})| \leq C(1 + |\log|t||) \leq C(1 + \log|\zeta|)$, since $1 - e^{-y} \geq (1 - e^{-2\pi})y/(2\pi)$ for $0 < y \leq 2\pi$. Finally $|1 - \zeta/(2\pi n)|$ lies between $\sin\vartheta_{\mathrm{seam}}$ and $1 + |\zeta|$, because $\zeta/(2\pi n)$ lies on the ray of angle $\vartheta_{\mathrm{seam}} \in (0, \pi/2)$. For the derivative on $|\zeta| \leq 2$ (where $|t| \geq n2^{-1/p} \geq 1$): $\frac{\mathrm{d}}{\mathrm{d}\zeta} j(t) = -j'(t)t/(p\zeta) = O(|t|^{-1}|\zeta|^{-1}) = O(|\zeta|^{1/p-1}/n)$, which is bounded since $1/p > 1$; the derivative of $\operatorname{Log}(1 - e^{-2\pi i t})$ is $O(|t||\zeta|^{-1}e^{-2\pi|t|})$, also bounded; the other two terms have bounded derivatives. Σ^{s}_- follows by conjugation.

(ii) For $z \notin \Sigma^{\mathrm{s}}_{\pm}$, $\chi(z)-\chi(0) = \frac{z}{2\pi i}\sum_{\pm}(\pm 1)\int_{\Sigma^{\mathrm{s}}_{\pm}} \frac{\chi_{\pm}(\zeta)\,\mathrm{d}\zeta}{\zeta(\zeta-z)}$ and $\chi'(z) = \frac{1}{2\pi i}\sum_{\pm}(\pm 1)\int_{\Sigma^{\mathrm{s}}_{\pm}} \frac{\chi_{\pm}(\zeta)\,\mathrm{d}\zeta}{(\zeta-z)^2}$. Under the hypothesis, $|\zeta-z| \ge \tau_0|z|$, and $|\zeta-z| \ge |\zeta|/2$ if $|\zeta| \ge 2|z|$; hence $|\zeta-z| \ge \frac{\tau_0}{3}(|\zeta|+|z|)$. With (i), for $|z| \le 1$,

$$|\chi(z)-\chi(0)| \le C|z|\Big[\int_0^1 \frac{\mathrm{d}\rho}{\rho+|z|} + \int_1^\infty \frac{1+\log\rho}{\rho^2}\,\mathrm{d}\rho\Big] \le C|z|\big(1+\log\tfrac{1}{|z|}\big),$$

and $|\chi'(z)| \le C[\int_0^1 \rho(\rho+|z|)^{-2}\,\mathrm{d}\rho + \int_1^\infty (1+\log\rho)\rho^{-2}\,\mathrm{d}\rho] \le C(1+\log\frac{1}{|z|})$. For $|z| \ge 1$ the same splitting gives $|z|\int_0^1 \frac{\mathrm{d}\rho}{|z|} + |z|\int_1^\infty \frac{(1+\log\rho)\,\mathrm{d}\rho}{\rho(\rho+|z|)} \le C(1+\log|z|)^2$.

(iii) Let $z \in \mathbb{D}(0,1)$ be close to Σ^{s}_+, with nearest point $z^* \in \Sigma^{\mathrm{s}}_+$, $r := |z^*|$ (other z are covered by (ii)). Split $\int_{\Sigma^{\mathrm{s}}_+} \chi_+(\zeta)\frac{\mathrm{d}\zeta}{\zeta-z}$ into the part $|\zeta-z^*| \le r/2$ and the rest. On the first part write $\chi_+(\zeta) = \chi_+(z^*) + O(|\zeta-z^*|)$ by (i): the constant contributes $\chi_+(z^*)$ times a logarithm whose modulus is at most $\pi + \log C$, because the two end points $z^* \pm \frac{r}{2}e^{i\vartheta_{\mathrm{seam}}}$ are at comparable distances from z; since $|\chi_+(z^*)| \le Cr$, this is bounded, and so is the Lipschitz part. On the rest, $|\zeta-z| \ge (|\zeta|+r)/C$, and the contribution is at most $C\int_{\Sigma^{\mathrm{s}}_+} |\chi_+(\zeta)|(|\zeta|+r)^{-1}|\,\mathrm{d}\zeta|$, which is finite for fixed n because $\chi_+(\zeta) = O(|\zeta|)$ at 0 and $\chi_+(\zeta)/\zeta$ is integrable at ∞ (Lemma A.2(ii)). The integral over Σ^{s}_- is bounded in the same way, since $|\zeta-z| \ge (|\zeta|+|z|)/C$ for $\zeta \in \Sigma^{\mathrm{s}}_-$. □

Appendix B. The a priori bound at the hard edge

The comparison argument of Section 6 applies [2, Theorem 4.4] (§2.6(b)) to Q, and its hypothesis is that the boundary values of $Q-I$ are Cauchy integrals of L^2 functions. This appendix verifies it: it proves item (vii) of Proposition 5.6, that $Q_\pm$ and $(Q^{-1})_\pm$ belong to $I + \partial\mathcal{C}(L^2(\Sigma))$, together with a pointwise bound on Q near 0. The a priori bound of Proposition B.4 holds for every $\alpha > -1$; the restriction $|\alpha| < 1$ enters item (vii) only through Corollary B.5, where the bound has to be square integrable at 0.

We never solve a Riemann–Hilbert problem for Q; Q is built from the orthogonal polynomials, and we only have to check that it is not too singular at the vertices of Σ. The only delicate vertex is the hard edge 0, where infinitely many nodes accumulate. There we bound the Cauchy sums in Y directly (Lemma B.2), except in thin sectors around the band, where the poles are removed by R and a contour argument is needed (Lemma B.3). Throughout this appendix $n \ge n_0$ is *fixed*, and constants may depend on n.

B.1. **Three facts about Cauchy integrals.** The following facts hold for general contours and are used here and in Appendix D. In this subsection Γ is a bounded, finite union of oriented C^2 arcs with $|\Gamma \cap \mathbb{D}(z,\rho)| \le C_\Sigma\rho$ for all $z \in \mathbb{C}$, $\rho > 0$, and $\mathcal{V}$ is the finite set of endpoints and intersection points of the arcs. We say that a function $\mathfrak{F}$ analytic in $\mathbb{C}\setminus\Gamma$ is *continuous up to* Γ if near every point of $\Gamma\setminus\mathcal{V}$ its restriction to each side of Γ extends continuously to the arc; $\mathfrak{F}_\pm$ then denote these extensions. We use only Cauchy's integral formula, Green's formula and the facts of §2.6(a).

Lemma B.1 (Cauchy integrals on a union of arcs)**.** (a) *For $\mathfrak{h} \in L^2(\Gamma)$ and $z \notin \Gamma$,*

$$|\mathcal{C}\mathfrak{h}(z)| \le \pi^{-1}C_\Sigma^{1/2}\|\mathfrak{h}\|_{L^2}\operatorname{dist}(z,\Gamma)^{-1/2}.$$

(b) *Let $\mathfrak{k}$ be C^1 (in arc length) on each arc and vanish near $\mathcal{V}$. Then $\mathcal{C}\mathfrak{k}$ is analytic near each point of $\mathcal{V}$ and continuous up to Γ, and $(\mathcal{C}\mathfrak{k})_+ - (\mathcal{C}\mathfrak{k})_- = \mathfrak{k}$ on $\Gamma\setminus\mathcal{V}$. In particular $(\mathcal{C}\mathfrak{k})_\pm = \mathcal{C}_\pm\mathfrak{k}$.*

(c) *Let $\mathfrak{F}$ be analytic in $\mathbb{C}\setminus\Gamma$ and continuous up to Γ, with $\mathfrak{F}_\pm \in L^2(\Gamma)$, $\mathfrak{F}(z) = O(1/z)$ as $z \to \infty$, and $\rho\sup\{|\mathfrak{F}(z)| : |z-z_0| = \rho,\ z \notin \Gamma\} \to 0$ as $\rho \to 0$ for every $z_0 \in \mathcal{V}$. Then $\mathfrak{F} = \mathcal{C}(\mathfrak{F}_+ - \mathfrak{F}_-)$ in $\mathbb{C}\setminus\Gamma$; hence $\mathfrak{F}_\pm = \mathcal{C}_\pm(\mathfrak{F}_+ - \mathfrak{F}_-)$, i.e. $\mathfrak{F}_\pm \in \partial\mathcal{C}(L^2(\Gamma))$.*

Proof. (a) By Cauchy–Schwarz it suffices to show $\int_\Gamma |s-z|^{-2}|\,\mathrm{d}s| \le 4C_\Sigma/\mathfrak{d}$ with $\mathfrak{d} = \operatorname{dist}(z,\Gamma)$. Split Γ into the shells $\{2^k\mathfrak{d} \le |s-z| < 2^{k+1}\mathfrak{d}\}$, $k \ge 0$; the k-th contributes at most $C_\Sigma 2^{k+1}\mathfrak{d}\,(2^k\mathfrak{d})^{-2}$, and the sum is $4C_\Sigma/\mathfrak{d}$.

(b) Analyticity near $\mathcal{V}$ is clear, since $\mathfrak{k}$ vanishes there. By a partition of unity we may assume that $\mathfrak{k}$ is supported in a compact sub-arc of the interior of one arc γ. Choose an open disc D which meets Γ only in an open sub-arc of γ containing that support and which γ cuts into two parts, D_+ on the left and D_- on the right. Using normal coordinates along γ (which are C^1 because γ is C^2), extend $\mathfrak{k}$ to a C^1 function K on $\mathbb{C}$ with compact support in D and $K|_\gamma = \mathfrak{k}$. Green's formula on D_+ (the Cauchy–Pompeiu formula) gives, since the positively oriented boundary of D_+ runs along γ in its own direction and $K = 0$ on ∂D,

$$\mathcal{C}\mathfrak{k}(z) = K(z)\mathbf{1}_{D_+}(z) + \mathfrak{P}(z), \qquad \mathfrak{P}(z) := \frac{1}{\pi}\iint_{D_+} \frac{\bar\partial K(w)}{w-z}\,\mathrm{d}A(w), \qquad z \notin \Gamma.$$

Since $\bar\partial K$ is bounded with compact support and $|w-z|^{-1}$ is locally integrable uniformly in z, $\mathfrak{P}$ is continuous on $\mathbb{C}$. So $\mathcal{C}\mathfrak{k}$ is continuous up to γ from both sides, with $(\mathcal{C}\mathfrak{k})_+ - (\mathcal{C}\mathfrak{k})_- = K = \mathfrak{k}$. As continuous extensions, these are the non-tangential limits $\mathcal{C}_\pm\mathfrak{k}$ of §2.6.

(c) Fix $z \notin \Gamma$. For small ρ and large R apply Cauchy's integral formula to $\mathfrak{F}$ on the open set $\mathbb{D}(0,R) \setminus \big(\Gamma \cup \bigcup_{z_0\in\mathcal{V}} \overline{\mathbb{D}(z_0,\rho)}\big)$, whose boundary consists of the two sides of the arcs of Γ outside the small discs, the circles $|s-z_0| = \rho$ and the circle $|s| = R$; $\mathfrak{F}$ is continuous on the closure (taken separately on the two sides of each arc). The two sides of an arc contribute $\frac{1}{2\pi i}\int (\mathfrak{F}_+ - \mathfrak{F}_-)(s)\,\mathrm{d}s/(s-z)$, the left side being traversed in the direction of the arc. As $\rho \to 0$ the small circles contribute at most $\sum_{z_0} 2\rho \sup_{|s-z_0|=\rho} |\mathfrak{F}(s)|/|z-z_0| \to 0$ (for $\rho \le |z-z_0|/2$), and the arc integrals converge to $\mathcal{C}(\mathfrak{F}_+ - \mathfrak{F}_-)(z)$ because $\mathfrak{F}_+ - \mathfrak{F}_- \in L^2(\Gamma) \subset L^1(\Gamma)$. As $R \to \infty$ the large circle contributes $O(1/R)$. Hence $\mathfrak{F}(z) = \mathcal{C}(\mathfrak{F}_+ - \mathfrak{F}_-)(z)$. Taking boundary values from either side, $\mathfrak{F}_\pm = \mathcal{C}_\pm(\mathfrak{F}_+ - \mathfrak{F}_-)$ wherever the non-tangential limits $\mathcal{C}_\pm$ exist, i.e. almost everywhere. □

B.2. **The bound at the hard edge.** Recall from Lemma 4.1 that

$$U_{j2}(z) = \sum_{k\ge 1} \frac{w(X_k)\,U_{j1}(X_k)}{z - X_k} \qquad (j = 1,2), \qquad w(X_k) = n^p w_k, \tag{B.1}$$

and that U_{11}, U_{21} are polynomials. Put $\varphi_1 := \frac{1}{2}\vartheta_{\mathrm{lip}}$.

Lemma B.2 (Sum bound)**.** *For every $\eta > 0$ there is C (depending on n and η) such that $|U_{j2}(z)| \le C|z|^{-\eta}(1+|z|^\alpha)$ for $0 < |z| \le 1$, $\varphi_1 \le |\arg z| \le \pi$.*

Proof. For such z and $X > 0$, $|z - X| \ge \sin\varphi_1 \max(|z|, X)$. By Lemma 4.9(i), $w_k \le C_\eta k^{-s+\eta}$, and U_{j1} is bounded on $[0, n^p]$. Let $t_z := |t(z)| = n|z|^{-1/p}$, so that $X_k \ge |z|$ if and only if $k \le t_z$. Then

$$|U_{j2}(z)| \le C\sum_{k\le t_z} \frac{k^{-s+\eta}}{(n/k)^p} + \frac{C}{|z|}\sum_{k>t_z} k^{-s+\eta} \le C\big(1 + t_z^{1+p-s+\eta} + \log t_z\big) + C|z|^{-1}t_z^{1-s+\eta}.$$

Since $t_z^{1+p-s} = n^{1+p-s}|z|^\alpha$ and $|z|^{-1}t_z^{1-s} = n^{1-s}|z|^\alpha$ (recall $(s-1)/p - 1 = \alpha$), and $t_z^\eta = n^\eta|z|^{-\eta/p}$, the claim follows after renaming η. □

Lemma B.3 (Lens sectors)**.** *For every $\eta > 0$ there are $\delta_{\mathrm{s}} \in (0,\delta)$ and C such that $|R_{j2}(z)| \le C|z|^{-\eta}(1+|z|^\alpha)$ for $0 < |z| \le \delta_{\mathrm{s}}/2$, $0 < |\arg z| < \varphi_1$, and also for $z \in \Omega_{\mathrm{b}} \cap \mathbb{D}(0,\delta_{\mathrm{s}}/2)$.*

Proof. Inside the lens the Cauchy sum (B.1) has poles that R removes; we move the sum onto the boundary of a sector by the residue theorem and bound the boundary integral. Take z

with $0 < \arg z < \varphi_1$; the case $\arg z < 0$ is the mirror image. In Ω_{b}^+, $R_{j2} = U_{j2} - U_{j1} W e^{-2i\theta} \upsilon_+$ by (4.19). Put

$$\mathcal{K}_j(s) := \frac{U_{j1}(s) W(s)}{e^{2i\theta(s)} - 1}, \qquad s \in \mathcal{G} := \{s : \ r < |s| < \delta_{\mathrm{s}}, \ |\arg s| < \vartheta_{\mathrm{lip}}\},$$

where δ_{s} and r are chosen with $t(\delta_{\mathrm{s}}), t(r) \in \mathbb{N} + \frac{1}{2}$, and $r < \frac{1}{2}|z|$, $|z| < \frac{1}{2}\delta_{\mathrm{s}}$. Since $e^{-2i\theta}\upsilon_+ = 1/(e^{2i\theta} - 1)$, we have $R_{j2}(z) = U_{j2}(z) - \mathcal{K}_j(z)$. The function $\mathcal{K}_j$ is meromorphic on a neighbourhood of $\overline{\mathcal{G}}$: W is analytic there (Lemma 4.7), and $e^{2i\theta} = 1$ only where $t \in \mathbb{Z}$, i.e. at the nodes (the points with $t \in -\mathbb{N}$ have $|\arg s| = \pi p > \vartheta_{\mathrm{lip}}$). At a node, $e^{2i\theta(s)} - 1 = 2i\theta'(X_k)(s - X_k)(1 + O(s - X_k))$, so $\operatorname{Res}_{X_k} \mathcal{K}_j = U_{j1}(X_k) W(X_k)/(2i\theta'(X_k)) = w(X_k) U_{j1}(X_k)$. The residue theorem on $\partial\mathcal{G}$ (positively oriented) and (B.1) give

$$R_{j2}(z) = \sum_{X_k \notin \mathcal{G}} \frac{w(X_k) U_{j1}(X_k)}{z - X_k} - \frac{1}{2\pi i} \oint_{\partial\mathcal{G}} \frac{\mathcal{K}_j(s)\, \mathrm{d}s}{s - z}. \tag{B.2}$$

We bound the terms. (1) The nodes with $X_k \geq \delta_{\mathrm{s}}$ are finitely many and at distance $\geq \delta_{\mathrm{s}}/2$ from z; those with $X_k \leq r$ contribute at most $\frac{2}{|z|} \sum_{X_k \leq r} w(X_k)|U_{j1}(X_k)|$, which tends to 0 as $r \to 0$. (2) On the ray $\arg s = \vartheta_{\mathrm{lip}}$, $|e^{2i\theta} - 1|^{-1} = |e^{-2i\theta}|/|1 - e^{-2i\theta}| \leq 2\exp(-2\pi n |s|^{-1/p} \sin(\vartheta_{\mathrm{lip}}/p))$ by (4.10), which beats $|W(s)| \leq C|s|^{\alpha-\eta}$ (Lemmas 4.3, 4.9(ii)); and $|s - z| \geq |s| \sin(\vartheta_{\mathrm{lip}} - \varphi_1)$. So this part is bounded. (3) On the ray $\arg s = -\vartheta_{\mathrm{lip}}$ we write $1/(e^{2i\theta} - 1) = -1 - e^{2i\theta}/(1 - e^{2i\theta})$, where now $|e^{2i\theta}| < 1$ is super-small, so $\mathcal{K}_j = -U_{j1} W + (\text{bounded})$. Since $|s - z| \geq (|s| + |z|)/C$,

$$\int_0^{\delta_{\mathrm{s}}} \frac{|W(\rho e^{-i\vartheta_{\mathrm{lip}}})|\, \mathrm{d}\rho}{\rho + |z|} \leq C \int_0^{\delta_{\mathrm{s}}} \frac{\rho^{\alpha-\eta}\, \mathrm{d}\rho}{\rho + |z|} \leq C|z|^{-\eta}(1 + |z|^{\alpha}),$$

because $\alpha - \eta > -1$. (4) On the arcs $|s| = r$ and $|s| = \delta_{\mathrm{s}}$, $|t| \in \mathbb{N} + \frac{1}{2}$. For such $t = u + iv$ we have $|e^{2\pi i t} - 1| \geq 1/C > 0$: if $|v| \geq \frac{1}{4}$ this is clear, and if $|v| < \frac{1}{4}$ then u is at distance at least $\frac{3}{8}$ from $\mathbb{Z}$, so $|e^{2\pi i t} - 1| = 2e^{-\pi v}|\sin \pi t| \geq 2e^{-\pi/4} \sin(3\pi/8)$. Hence $|\mathcal{K}_j| \leq C|W| \leq C|s|^{\alpha-\eta}$ there. The arc $|s| = r$ has length $\leq 2\vartheta_{\mathrm{lip}} r$ and $|s - z| \geq \frac{1}{2}|z|$, so its contribution is $\leq C r^{1+\alpha-\eta}/|z| \to 0$ as $r \to 0$; the arc $|s| = \delta_{\mathrm{s}}$ contributes a bounded amount. Letting $r \to 0$ in (B.2) proves the bound for $0 < \arg z < \varphi_1$. For $z \in \Omega_{\mathrm{b}} \cap \mathbb{D}(0, \delta_{\mathrm{s}}/2)$ with $\varphi_1 \leq |\arg z| < \frac{5}{4}\vartheta_{\mathrm{lip}}$, $R_{j2} = U_{j2} - U_{j1} W e^{\mp 2i\theta} \upsilon_{\pm}$ is bounded by Lemma B.2 plus a super-small term. □

Proposition B.4 (A priori bound). *Fix $n \geq n_0$. Then $\det Q \equiv 1$, and:*

(i) *for every $\eta > 0$ there is $C_{n,\eta}$ with $|Q(z)| \leq C_{n,\eta}|z|^{-|\alpha|/2-\eta}$ for $z \in \mathbb{D}(0, \delta) \setminus \Sigma$;*
(ii) *Q is bounded on every bounded set at positive distance from 0, and $Q(z) \to I$ as $z \to \infty$;*
(iii) *Q has continuous boundary values on Σ minus its vertices.*

The same holds for Q^{-1}.

Proof. We follow the transformations from Y to Q factor by factor; every factor except the Cauchy sums in U is explicit, and those sums are controlled by Lemmas B.2 and B.3. First, $\det U = \det Y = 1$ (Lemma 4.1), and every other factor in (4.21) and in $Q = D_\infty^{-\sigma_3} V D_W^{\sigma_3}$ has determinant 1; so $\det Q = 1$ and Q^{-1} has the same entries up to sign and order.

(i) Let $\eta > 0$ and $|z| < \delta_{\mathrm{s}}/2$ (the annulus $\delta_{\mathrm{s}}/2 \leq |z| < \delta$ is covered by (ii)). By (4.21) and (5.12), Q is a constant diagonal matrix times $U\, M\, D_W^{\sigma_3}$. We use: $|e^{\pm n\varphi}| \leq e^{Cn|z|^{1/2}} \leq C$ (Lemma 4.5(iii)); $|F|^{\pm 1} \leq C$ outside the lens and $|\tilde{F}|^{\pm 1} \leq C$ in the lens near 0 (Definition 4.10(e)); $|D_W(z)|^{\pm 1} \leq C|z|^{\pm\alpha/2-\eta}$ (Lemma 5.2(vi)); $|1/W(z)| \leq C|z|^{-\alpha-\eta}$ in the lens (Lemmas 4.3, 4.9(ii)); and U_{j1} bounded.

Outside the lens ($z \in \Omega_\infty$), $|\arg z| \geq \frac{3}{4}\vartheta_{\mathrm{lip}} > \varphi_1$ and $M = e^{n\varphi\sigma_3}F^{\sigma_3}$. The first column of Q is a constant times $U_{\cdot 1}e^{n\varphi}FD_W$, of size $\leq C|z|^{\alpha/2-\eta}$; the second is a constant times $U_{\cdot 2}e^{-n\varphi}F^{-1}D_W^{-1}$, of size $\leq C|z|^{-\eta}(1+|z|^\alpha)|z|^{-\alpha/2-\eta}$ by Lemma B.2. Both are $\leq C|z|^{-|\alpha|/2-2\eta}$.

In the lens $\Omega_{\mathrm{b}}^{\pm}$, $UM = Re^{n\varphi\sigma_3}A_\pm\tilde{F}^{\sigma_3}$, and $e^{n\varphi\sigma_3}A_\pm = \left(\begin{smallmatrix} e^{n\varphi} & 0 \\ \mp e^{n\varphi}/W & e^{-n\varphi}\end{smallmatrix}\right)$. The first column of R is that of U, and the second is bounded by Lemma B.3. Hence the first column of Q is at most $C(1+|z|^{-\alpha-2\eta}(1+|z|^\alpha))|z|^{\alpha/2-\eta} \leq C|z|^{-|\alpha|/2-3\eta}$, and the second is at most $C|z|^{-\eta}(1+|z|^\alpha)|z|^{-\alpha/2-\eta} \leq C|z|^{-|\alpha|/2-2\eta}$. Renaming η gives (i).

(ii) Away from 0 each of U (meromorphic, with finitely many nodes in any compact set not containing 0), the triangular and diagonal factors in (4.19)–(4.20), F, $\tilde{F}$ and D_W is analytic in each region up to its boundary, except for the poles of U and of $v_\pm$ at the nodes, which cancel in R (Step 1 of §4.7). Near $4c$ the only singular factors are $\tilde{F} \sim (4c-z)^{-1/2}$ and $D_W \sim (4c-z)^{1/2}$, and they occur only in the product $\tilde{F}D_W$, whose square $(\tilde{F}D_W)^2 = \mathsf{w}_n z^\alpha r_n\omega$ is bounded above and below on $\mathbb{D}(4c,\varepsilon)$ (Lemma 5.2(vi)(b)). At the remaining vertices ($4c \pm ih_0$, $n^p+\delta' \pm ih_0$, $n^p+\delta'$) all factors are analytic and bounded. At ∞, $Q \to I$ by Lemma 5.3. (iii) follows from the same description and Lemma 4.11. □

Corollary B.5 (Q is an admissible solution)**.** *For each fixed $n \geq n_0$, $Q_\pm - I \in L^2(\Sigma)$ and $Q = I + \mathcal{C}(Q_+ - Q_-)$ off Σ. Thus $Q_\pm \in I + \partial\mathcal{C}(L^2(\Sigma))$ in the sense of §2.6, and the same holds for Q^{-1}.*

Proof. We apply Lemma B.1(c) to $\mathfrak{F} := Q - I$ on $\Gamma = \Sigma$, which for fixed n is bounded and satisfies $|\Sigma \cap \mathbb{D}(z,\rho)| \leq C\rho$ (Proposition 5.6(i)). By Proposition B.4(iii), $\mathfrak{F}$ is continuous up to Σ. By Proposition B.4(i) with $\eta = (1-|\alpha|)/4$, which is positive because $|\alpha| < 1$, we have $|Q(z)| \leq C|z|^{-(1+|\alpha|)/4}$ near 0, and $(1+|\alpha|)/4 < \frac{1}{2}$; with (ii) this gives $\mathfrak{F}_\pm \in L^2(\Sigma)$ and $\rho\sup_{|z|=\rho}|\mathfrak{F}(z)| = O(\rho^{1-(1+|\alpha|)/4}) \to 0$ at the vertex 0, while at every other vertex the corresponding quantity is $O(\rho)$. Finally $\mathfrak{F} = O(1/z)$ at ∞ (Lemma 5.3). The argument for Q^{-1} is identical. □

Proposition B.4 and Corollary B.5 together are item (vii) of Proposition 5.6. Any $\eta \in (0, \frac{1}{2}(1-|\alpha|))$ in Proposition B.4(i) makes the boundary values of Q square integrable at 0; this is the use (a) of $|\alpha| < 1$ in Remark 6.7.

Appendix C. Two L^2 bounds on a star of arcs

The uniform resolvent bound of Appendix D reduces, at the two edges, to two operators on a star of arcs through a point: a Cauchy operator conjugated by a power weight, and a Cauchy operator with a logarithmic commutator. Both are dilation invariant, which is what makes the bounds uniform in n. We prove them here. The two lemmas are statements about a fixed geometric configuration; they involve none of the standing assumptions of Section 6, and $|\alpha| < 1$ enters only when they are applied with $\beta = \pm\alpha/2$.

Setting. A *star* is a set $\Gamma = \bigcup_{j=1}^m \gamma_j([0,R_j])$ of finitely many C^2 arcs γ_j, parametrised by arc length, with $\gamma_j(0) = 0$, pairwise disjoint except at 0, and with distinct tangent directions $\gamma_j'(0) = e^{i\theta_j}$. Each arc carries an orientation, and $\mathcal{C}_\pm$ are the boundary values of $\mathcal{C}$ on Γ as in §2.6. By compactness and transversality there is $C_\Gamma \geq 1$ with

(C.1)
$$|\gamma_j'(\rho) - e^{i\theta_j}| \leq C_\Gamma\rho, \qquad |\gamma_j(\rho) - \gamma_i(r)| \geq (\rho+r)/C_\Gamma \ (i \neq j), \qquad |\gamma_j(\rho) - \gamma_j(r)| \geq |\rho - r|/C_\Gamma.$$

A finite union of C^2 arcs is a Carleson curve, so $\mathcal{C}_\pm$ are bounded on $L^2(\Gamma)$ by §2.6(a). For $\lambda > 0$ the dilation $(\mathsf{D}_\lambda\mathfrak{h})(z) := \lambda^{-1/2}\mathfrak{h}(z/\lambda)$ is unitary from $L^2(\Gamma)$ onto $L^2(\lambda\Gamma)$, and since $\mathrm{d}s/(s-z)$ is invariant under $(s,z) \mapsto (\lambda s, \lambda z)$, D_λ intertwines the Cauchy operators of Γ and

$\lambda\Gamma$. The same holds for the operators below, whose kernels are $\mathrm{d}s/(s-z)$ times a function of $(s/|s|,\, z/|z|,\, |z|/|s|)$.

We use one elementary bound repeatedly. Let $\mathfrak{k}$ be measurable on $(0,\infty)$ with

$$\|\mathfrak{k}\|_* := \int_0^\infty |\mathfrak{k}(u)|u^{-1/2}\,\mathrm{d}u < \infty, \qquad (\mathcal{M}_{\mathfrak{k}}\mathfrak{h})(r) := \int_0^\infty \mathfrak{k}(\rho/r)\mathfrak{h}(\rho)\frac{\mathrm{d}\rho}{r}.$$

Then

$$\|\mathcal{M}_{\mathfrak{k}}\mathfrak{h}\|_{L^2(0,\infty)} \le \|\mathfrak{k}\|_*\,\|\mathfrak{h}\|_{L^2(0,\infty)}. \tag{C.2}$$

Indeed $(\mathcal{M}_{\mathfrak{k}}\mathfrak{h})(r) = \int_0^\infty \mathfrak{k}(u)\mathfrak{h}(ru)\,\mathrm{d}u$, and $\|\mathfrak{h}(\cdot\,u)\|_{L^2} = u^{-1/2}\|\mathfrak{h}\|_{L^2}$, so (C.2) is Minkowski's integral inequality. Operators with kernels that are bounded on a bounded set are Hilbert–Schmidt, and we use that without comment.

Lemma C.1 (Power-weighted Cauchy operator on a star)**.** *Let Γ be a star and $|\beta| < \frac12$. Then the operators*

$$\mathfrak{h} \;\longmapsto\; |z|^{\beta}\,\mathcal{C}_{\pm}\big(|s|^{-\beta}\mathfrak{h}\big)(z)$$

are bounded on $L^2(\Gamma)$, with a norm N_β that depends only on β and Γ. The same bound holds on $\lambda\Gamma$ for every $\lambda > 0$, on every subset $\Gamma' \subset \lambda\Gamma$ (functions supported on, and evaluated on, Γ'), and for the weights z^{β}, $s^{-\beta}$ (any fixed continuous branches of the argument on each arc) in place of $|z|^{\beta}$, $|s|^{-\beta}$, with N_β multiplied by $e^{4\pi|\beta|}$.

Proof. We split off the plain Cauchy operator, which is bounded, and compare the remainder, which has a kernel regular on the diagonal, with its analogue on the star of tangent rays.

Write $T_{\pm}\mathfrak{h} := |z|^{\beta}\mathcal{C}_{\pm}(|s|^{-\beta}\mathfrak{h}) = \mathcal{C}_{\pm}\mathfrak{h} + R\mathfrak{h}$, where

$$(R\mathfrak{h})(z) = \frac{1}{2\pi i}\int_\Gamma \Big(\frac{|z|^{\beta}}{|s|^{\beta}} - 1\Big)\frac{\mathfrak{h}(s)\,\mathrm{d}s}{s-z}.$$

The kernel of R is bounded near the diagonal of each arc (see below), so $R\mathfrak{h}$ has no boundary-value ambiguity, and it suffices to bound R. In arc-length coordinates, $z = \gamma_i(r)$, $s = \gamma_j(\rho)$, the kernel of R is

$$\mathfrak{r}_{ij}(\rho, r) = \frac{1}{2\pi i}\Big(\frac{|\gamma_i(r)|^{\beta}}{|\gamma_j(\rho)|^{\beta}} - 1\Big)\frac{\gamma_j'(\rho)}{\gamma_j(\rho) - \gamma_i(r)}.$$

Its flat analogue, on the star of rays $\{\rho e^{i\theta_j}\}$, is $\mathfrak{k}_{ij}(\rho/r)/r$ with

$$\mathfrak{k}_{ij}(u) := \frac{1}{2\pi i}\,\frac{(u^{-\beta}-1)\,e^{i\theta_j}}{ue^{i\theta_j} - e^{i\theta_i}}.$$

For $i \ne j$ we have $|ue^{i\theta_j} - e^{i\theta_i}| \ge \sin(|\theta_i - \theta_j|/2)(1+u)$, and for $i = j$ the quotient $(u^{-\beta}-1)/(u-1)$ is continuous at $u = 1$. So $|\mathfrak{k}_{ij}(u)| \le C(1+u^{-\beta})$ for $u \le 2$ and $|\mathfrak{k}_{ij}(u)| \le C(u^{-\beta}+1)u^{-1}$ for $u \ge 2$, and $\|\mathfrak{k}_{ij}\|_* < \infty$ precisely because $-\frac12 < \beta < \frac12$: near 0 the integrand is $O(u^{-1/2-\max(\beta,0)})$, near infinity $O(u^{-3/2-\min(\beta,0)})$. By (C.2) the flat operator is bounded on L^2 of the rays, hence on L^2 of the segments $[0, R_j]$.

The difference $\mathfrak{r}_{ij}(\rho, r) - \mathfrak{k}_{ij}(\rho/r)/r$ is a sum of two terms. Put $w_j(\rho) := (|\gamma_j(\rho)|/\rho)^{\beta}$; by (C.1) and $\gamma_j \in C^2$, w_j is C^1 on $[0, R_j]$ and bounded above and below. The first term is

$$\frac{1}{2\pi i}\Big(\frac{r}{\rho}\Big)^{\beta}\Big(\frac{w_i(r)}{w_j(\rho)} - 1\Big)\frac{\gamma_j'(\rho)}{\gamma_j(\rho) - \gamma_i(r)};$$

since $|w_i(r)-w_j(\rho)| \le C|\rho-r|$ for $i=j$ and $\le C(\rho+r)$ for $i\neq j$, (C.1) bounds it by $C(r/\rho)^\beta$. The second term is $\frac{1}{2\pi i}((r/\rho)^\beta-1)\,\mathfrak{e}_{ij}(\rho,r)$ with

$$\mathfrak{e}_{ij}(\rho,r) := \frac{\gamma_j'(\rho)}{\gamma_j(\rho)-\gamma_i(r)} - \frac{e^{i\theta_j}}{\rho e^{i\theta_j}-re^{i\theta_i}}.$$

We claim $|\mathfrak{e}_{ij}| \le C$. For $i\neq j$ both denominators are at least $(\rho+r)/C$, and the numerator of the difference, $\gamma_j'(\rho)(\rho e^{i\theta_j}-re^{i\theta_i}) - e^{i\theta_j}(\gamma_j(\rho)-\gamma_i(r))$, is $O(\rho(\rho+r))+O(\rho^2+r^2)$ by (C.1). For $i=j$ the flat term is $1/(\rho-r)$, and $\gamma_j(\rho)-\gamma_j(r) = (\rho-r)(\gamma_j'(\rho)+O(\rho-r))$, so $\gamma_j'(\rho)/(\gamma_j(\rho)-\gamma_j(r)) = (\rho-r)^{-1}(1+O(\rho-r))$, which gives $\mathfrak{e}_{jj}=O(1)$. Here the factor $\gamma_j'(\rho)$, i.e. the arc-length element $\mathrm{d}s=\gamma_j'(\rho)\,\mathrm{d}\rho$, must be kept: without it the difference is only $O(r/|\rho-r|)$ on a single arc. Hence the difference kernel is at most $C(1+(r/\rho)^\beta)$, and $\iint_{[0,R]^2}(1+(r/\rho)^{2\beta})\,\mathrm{d}\rho\,\mathrm{d}r<\infty$ for $|\beta|<\frac12$: it is Hilbert–Schmidt.

So R is bounded on $L^2(\Gamma)$ (arc length makes the parametrisation unitary), and so is $T_\pm$. Dilation invariance gives the bound on $\lambda\Gamma$. Restriction to Γ' is a compression. Finally $z^\beta = |z|^\beta e^{i\beta\arg z}$ with $|\arg z|\le 2\pi$, so the complex weights differ from the absolute ones by multiplication operators of norm at most $e^{2\pi|\beta|}$ on either side. □

Lemma C.2 (Logarithmic commutator on a star)**.** *Let Γ be a star and fix on each arc a continuous branch $\mathrm{l}(z)=\log|z|+i\arg z$ of the logarithm. Then the operator*

$$(L\mathfrak{h})(z) := \frac{1}{2\pi i}\int_\Gamma \big(\mathrm{l}(z)-\mathrm{l}(s)\big)\frac{\mathfrak{h}(s)\,\mathrm{d}s}{s-z}$$

is bounded on $L^2(\Gamma)$, with the same norm on $\lambda\Gamma$ for every $\lambda>0$ and on subsets.

Proof. The kernel is regular on the diagonal, since l is Lipschitz in arc length away from 0 and the factor $\mathrm{l}(z)-\mathrm{l}(s)$ vanishes at $s=z$; we split it into the part with $\log|z|-\log|s|$ and the part with $\arg z-\arg s$.

The modulus part. On the star of rays its kernel is $\mathfrak{k}_{ij}(\rho/r)/r$ with

$$\mathfrak{k}_{ij}(u) = -\frac{1}{2\pi i}\,\frac{\log u\; e^{i\theta_j}}{ue^{i\theta_j}-e^{i\theta_i}}.$$

This has no singularity at $u=1$ even for $i=j$ ($\log u/(u-1)\to 1$), and $|\mathfrak{k}_{ij}(u)|u^{-1/2}$ is $O(|\log u|u^{-1/2})$ at 0 and $O(|\log u|u^{-3/2})$ at infinity, so $\|\mathfrak{k}_{ij}\|_*<\infty$ and (C.2) applies. On the curved star the difference from the flat kernel is

$$\frac{1}{2\pi i}\Big[(\lambda_i(r)-\lambda_j(\rho))\frac{\gamma_j'(\rho)}{\gamma_j(\rho)-\gamma_i(r)} + \log\frac{r}{\rho}\;\mathfrak{e}_{ij}(\rho,r)\Big],$$

with $\lambda_j(\rho):=\log(|\gamma_j(\rho)|/\rho)$, a C^1 function on $[0,R_j]$. As in the proof of Lemma C.1 this is $O(1+|\log(r/\rho)|)$, and $\iint_{[0,R]^2}(1+\log^2(r/\rho))\,\mathrm{d}\rho\,\mathrm{d}r<\infty$.

The argument part. Its kernel is $\frac{i}{2\pi i}(\arg\gamma_i(r)-\arg\gamma_j(\rho))\gamma_j'(\rho)/(\gamma_j(\rho)-\gamma_i(r))$. For $i=j$, $\arg\gamma_j(\rho)=\theta_j+O(\rho)$ is C^1 on $[0,R_j]$, so by (C.1) the kernel is bounded. For $i\neq j$ the numerator is bounded and the denominator is at least $(\rho+r)/C_\Gamma$, so the kernel is at most $C/(\rho+r)=C\,\mathfrak{k}(\rho/r)/r$ with $\mathfrak{k}(u)=(1+u)^{-1}$, $\|\mathfrak{k}\|_*=\pi$.

Both parts are bounded on $L^2(\Gamma)$. The kernel depends on (s,z) only through $s/|s|$, $z/|z|$ and $|z|/|s|$, so the operator commutes with D_λ and the bound holds on $\lambda\Gamma$. □

Remark C.3 (The condition $|\beta|<\frac12$). For Γ a union of rays the curvature terms vanish, and Lemma C.1 reduces to the boundedness of the Mellin multiplier $\int_0^\infty u^{w-1}\,\mathrm{d}u/(u-e^{i\phi_{ij}})$ on the line $\operatorname{Re} w=\frac12-\beta$; on a single ray its principal value is $-\pi\cot(\pi w)$ (by [8, (5.12.3), (5.5.3)],

see §2.7), which is bounded on that line exactly when $0 < \frac{1}{2} - \beta < 1$. This explains the role of the condition $|\beta| < \frac{1}{2}$.

Appendix D. Uniform invertibility of the comparison problem

The Neumann-series step of Proposition 6.5 rests on the uniform invertibility of the singular integral operator of the comparison problem, Proposition 6.4, which we prove here. The difficulty is that $\tilde{Q}$ is unbounded near 0 at the Bessel scale, with a size that grows with n; the bound must therefore come from the structure of $\tilde{Q}$, not from its size. We use the conjugated Cauchy operators of Appendix C, which are dilation invariant, and the Cauchy-integral criterion of Lemma B.1.

We use the notation of Section 6. In particular $\hat{\Sigma} = \Sigma \cup \Sigma_\circ$, $\tilde{Q}$ is the explicit solution (6.11) of the comparison problem with jump $\tilde{v}$, and $n \geq n_0$. We use the three facts about Cauchy integrals of §B.1, prove a uniform bound for the singular integral operator near an edge where $\tilde{Q}$ has a two-scale structure (§D.1), and assemble the proof (§D.2).

D.1. The model block.

The following lemma covers both edges. At the hard edge it is applied with $\mathfrak{t}_n = n^2$, at the soft edge with $\mathfrak{t}_n = n^{2/3}$.

Lemma D.1 (Uniform bound near an edge)**.** *Let $z_0 \in \mathbb{C}$, $0 < r_1 < r_0$, $\mathfrak{t}_n \geq 1$, and let f_0 be conformal on a neighbourhood of $\overline{\mathbb{D}(z_0, r_0)}$, independent of n, with $f_0(z_0) = 0$. Assume that $\hat{\Sigma} \cap \mathbb{D}(z_0, r_0)$ consists of finitely many C^2 arcs from z_0 with distinct tangents and is independent of n, and put $\Sigma_1 := \hat{\Sigma} \cap \mathbb{D}(z_0, r_1)$, $\Gamma_1 := f_0(\Sigma_1)$ (a star with vertex 0, in the sense of Appendix C). Assume that for $z \in \mathbb{D}(z_0, r_0) \setminus \hat{\Sigma}$, with $\zeta := \mathfrak{t}_n f_0(z)$,*

$$\tilde{Q}(z) = \mathcal{A}(z)\, \mathfrak{t}_n^{\sigma_3/4} M_n(z),$$

where, with a constant C_1 independent of n, and with the same statements for the boundary values on $\hat{\Sigma}$:

- (B1) *$\mathcal{A}^{\pm 1}$ are analytic on $\mathbb{D}(z_0, r_0)$ and $|\mathcal{A}^{\pm 1}| \leq C_1$;*
- (B2) *if $|\zeta| \geq 2$, then $M_n(z) = \zeta^{-\sigma_3/4} \Theta_\infty(z)$ with $|\Theta_\infty^{\pm 1}(z)| \leq C_1$;*
- (B3) *if $|\zeta| \leq 3$, then $M_n(z) = \hat{A}(\zeta) \mathsf{P}(\zeta) \Theta_0(z)$, where $\hat{A}$ does not depend on n, $\hat{A}^{\pm 1}$ are analytic on $|\zeta| < 4$ with $|\hat{A}^{\pm 1}| \leq C_1$, $|\Theta_0^{\pm 1}(z)| \leq C_1$, and either $\mathsf{P}(\zeta) = \zeta^{\beta \sigma_3}$ with $|\beta| < \frac{1}{2}$, or $\mathsf{P}(\zeta) = I + \mathsf{k}(\log \zeta) e_1 e_2^{\mathsf{T}}$ with a constant k.*

Then the operators $\mathfrak{g} \mapsto 1_{\Sigma_1} \mathcal{C}_\pm (\mathfrak{g} 1_{\Sigma_1} \tilde{Q}_+^{-1}) \tilde{Q}_\pm$ are bounded on $L^2(\Sigma_1)$ uniformly in n, and $\int_{\Sigma_1} (|\tilde{Q}_\pm|^2 + |\tilde{Q}_\pm^{-1}|^2) |\,\mathrm{d}s| \leq C$.

Proof. We remove the analytic prefactor $\mathcal{A}$ at the cost of a Hilbert–Schmidt error, pass to the variable ζ, and split the resulting operator according to $|\zeta| \leq 3$ and $|\zeta| \geq 2$. Throughout, C depends on C_1, f_0, r_0, r_1, β and the geometry, but not on n.

Weights. Put $\mathfrak{W}(z) := |f_0(z)|^{-1/4}$ if $|\zeta| \geq 2$ and $\mathfrak{W}(z) := \mathfrak{t}_n^{1/4} \mathfrak{w}(\zeta)$ if $|\zeta| < 2$, where $\mathfrak{w}(\zeta) = |\zeta|^{-|\beta|}$, respectively $\mathfrak{w}(\zeta) = 1 + |\log|\zeta||$ in the logarithmic case. Since $\mathfrak{t}_n^{\sigma_3/4} \zeta^{-\sigma_3/4} = f_0^{-\sigma_3/4}$, (B2)–(B3) give

$$|\mathfrak{t}_n^{\sigma_3/4} M_n(z)| + |M_n(z)^{-1} \mathfrak{t}_n^{-\sigma_3/4}| \leq C\, \mathfrak{W}(z), \qquad |M_n(z)^{\pm 1}| \leq C \mathfrak{w}(\zeta) \ (|\zeta| \leq 3). \tag{D.1}$$

On Σ_1 we have $|\,\mathrm{d}z| \leq C |\,\mathrm{d}\zeta| / \mathfrak{t}_n$, $|f_0(z)| \geq |z - z_0|/C$, and Γ_1 satisfies an Ahlfors bound $|\Gamma_1 \cap \mathbb{D}(\zeta, t)| \leq Ct$, as does $\mathfrak{t}_n \Gamma_1$ with the same constant. Hence

$$\int_{\Sigma_1} \mathfrak{W}^2 |\,\mathrm{d}z| \leq C \int_{\Sigma_1} |z - z_0|^{-1/2} |\,\mathrm{d}z| + C \mathfrak{t}_n^{1/2} \mathfrak{t}_n^{-1} \int_{\mathfrak{t}_n \Gamma_1 \cap \{|\zeta| \leq 2\}} \mathfrak{w}(\zeta)^2 |\,\mathrm{d}\zeta| \leq C, \tag{D.2}$$

since $2|\beta|<1$. With (D.1) and (B1) this gives the bound on $\int_{\Sigma_1}|\tilde{Q}_\pm^{\pm1}|^2$.

Step 1: the prefactor. For $s,z\in\mathbb{D}(z_0,r_1)$ write $\mathcal{A}(s)^{-1}\mathcal{A}(z)=I+(s-z)\mathfrak{k}_A(s,z)$; by Cauchy's estimates on the convex set $\mathbb{D}(z_0,r_1)$, $|\mathfrak{k}_A|\le C$. Then

$$\mathcal{C}(\mathfrak{g}\tilde{Q}_+^{-1})(z)\tilde{Q}(z)=\mathcal{C}(\mathfrak{g}M_{n,+}^{-1})(z)M_n(z)+\frac{1}{2\pi i}\int_{\Sigma_1}\mathfrak{g}(s)\big[M_{n,+}(s)^{-1}\mathfrak{t}_n^{-\sigma_3/4}\big]\mathfrak{k}_A(s,z)\big[\mathfrak{t}_n^{\sigma_3/4}M_n(z)\big]\,\mathrm{d}s.$$

By (D.1) the kernel of the second term is at most $C\,\mathfrak{W}(s)\mathfrak{W}(z)$, so by (D.2) it is Hilbert–Schmidt with norm at most C.

Step 2: the variable ζ. Let $\mathfrak{k}_0(s,z):=f_0'(s)/(f_0(s)-f_0(z))-1/(s-z)$, which is analytic and bounded on $\mathbb{D}(z_0,r_1)^2$. Then $\mathrm{d}s/(s-z)=\mathrm{d}\zeta_s/(\zeta_s-\zeta_z)-\mathfrak{k}_0(s,z)\,\mathrm{d}s$. The term with $\mathfrak{k}_0$ is a scalar kernel times $M_{n,+}(s)^{-1}M_n(z)=[M_{n,+}(s)^{-1}\mathfrak{t}_n^{-\sigma_3/4}][\mathfrak{t}_n^{\sigma_3/4}M_n(z)]$, hence Hilbert–Schmidt with norm at most C as in Step 1. The main term is $\mathcal{S}\mathfrak{h}:=\mathcal{C}^{\mathfrak{t}_n\Gamma_1}(\mathfrak{h}\mathbb{M}_+^{-1})\mathbb{M}$ with $\mathbb{M}(\zeta):=M_n(z)$ and $\mathfrak{h}(\zeta)=\mathfrak{g}(z)$; as $|f_0'|^{\pm1}\le C$, the map $\mathfrak{g}\mapsto\mathfrak{h}$ changes L^2 norms by the factor $\mathfrak{t}_n^{1/2}$ up to constants, the same on input and output, so it suffices to bound $\mathcal{S}$ on $L^2(\mathfrak{t}_n\Gamma_1)$ uniformly. The conformal map preserves the sides of the contour, so $\pm$ boundary values correspond.

Step 3: splitting in ζ. Let $\mathcal{B}_{\le3}:=\{|\zeta|\le3\}$, $\mathcal{B}_{\ge2}:=\{|\zeta|\ge2\}$, $\mathcal{B}_{2,3}:=\{2\le|\zeta|\le3\}$. The pairs (ζ',ζ) (source, target) are partitioned into $\mathcal{B}_{\le3}\times\mathcal{B}_{\le3}$; $(\mathcal{B}_{\ge2}\times\mathcal{B}_{\ge2})\setminus(\mathcal{B}_{2,3}\times\mathcal{B}_{2,3})$; $\{|\zeta'|<2,|\zeta|>3\}$; and $\{|\zeta'|>3,|\zeta|<2\}$. Accordingly $\mathcal{S}=1_{\mathcal{B}_{\le3}}\mathcal{S}1_{\mathcal{B}_{\le3}}+\big(1_{\mathcal{B}_{\ge2}}\mathcal{S}1_{\mathcal{B}_{\ge2}}-1_{\mathcal{B}_{2,3}}\mathcal{S}1_{\mathcal{B}_{2,3}}\big)+1_{\{|\zeta|>3\}}\mathcal{S}1_{\{|\zeta|<2\}}+1_{\{|\zeta|<2\}}\mathcal{S}1_{\{|\zeta|>3\}}$, and we bound each part.

$\mathcal{B}_{\le3}\times\mathcal{B}_{\le3}$. Here $\mathbb{M}=\hat{A}\mathsf{P}\Theta_0$. Write $\hat{A}(\zeta')^{-1}\hat{A}(\zeta)=I+(\zeta'-\zeta)\hat{\mathfrak{k}}(\zeta',\zeta)$ with $|\hat{\mathfrak{k}}|\le C$ on $\{|\zeta|,|\zeta'|\le3\}$. The resulting commutator kernel is $\Theta_{0,+}(\zeta')^{-1}\mathsf{P}_+(\zeta')^{-1}\hat{\mathfrak{k}}\,\mathsf{P}(\zeta)\Theta_0(\zeta)$, of size at most $C\mathfrak{w}(\zeta')\mathfrak{w}(\zeta)$, and $\int_{\mathfrak{t}_n\Gamma_1\cap\mathcal{B}_{\le3}}\mathfrak{w}^2|\,\mathrm{d}\zeta|\le C$: Hilbert–Schmidt with norm $\le C$. The remaining part is $\mathfrak{h}\mapsto\big[\mathcal{C}(\mathfrak{h}'\mathsf{P}_+^{-1})\mathsf{P}\big]\Theta_0$ with $\mathfrak{h}':=\mathfrak{h}\Theta_{0,+}^{-1}$. If $\mathsf{P}=\zeta^{\beta\sigma_3}$, its first column is $\zeta^\beta\mathcal{C}(\zeta'^{-\beta}\mathfrak{h}'e_1)$ and its second column is $\zeta^{-\beta}\mathcal{C}(\zeta'^{\beta}\mathfrak{h}'e_2)$; both are bounded on $L^2(\mathfrak{t}_n\Gamma_1\cap\mathcal{B}_{\le3})$ by Lemma C.1 with exponents $\pm\beta$, uniformly in n by dilation invariance. If P is logarithmic, $\mathsf{P}_+(\zeta')^{-1}\mathsf{P}(\zeta)=I+\mathsf{k}(\log\zeta-\log_+\zeta')e_1e_2^{\mathsf{T}}$, and the second term is the logarithmic commutator of Lemma C.2 applied to $\mathfrak{h}'e_1$ (the branches $\log_\pm$ on the arc $(-\infty,0)$ differ by the constant $2\pi i$, which adds a multiple of the Cauchy operator).

$(\mathcal{B}_{\ge2}\times\mathcal{B}_{\ge2})\setminus(\mathcal{B}_{2,3}\times\mathcal{B}_{2,3})$. On $\mathcal{B}_{\ge2}$, $\mathbb{M}=\zeta^{-\sigma_3/4}\Theta_\infty$, so $1_{\mathcal{B}_{\ge2}}\mathcal{S}1_{\mathcal{B}_{\ge2}}$ is Θ_∞ times the operator $\mathfrak{h}''\mapsto\mathcal{C}(\mathfrak{h}''\zeta'^{\sigma_3/4})\zeta^{-\sigma_3/4}$, with $\mathfrak{h}''=\mathfrak{h}\Theta_{\infty,+}^{-1}$; this is Lemma C.1 with $\beta=\mp\frac14$. On $\mathcal{B}_{2,3}$ the matrices $\mathbb{M}^{\pm1}$ are bounded, so $1_{\mathcal{B}_{2,3}}\mathcal{S}1_{\mathcal{B}_{2,3}}$ is bounded by $\|\mathcal{C}_\pm\|$ on $\mathfrak{t}_n\Gamma_1$ (Lemma C.1 with $\beta=0$).

The separated parts. If $|\zeta'|<2$ and $|\zeta|>3$, then $|\zeta-\zeta'|\ge|\zeta|/3$, $|\mathbb{M}(\zeta')^{-1}|\le C\mathfrak{w}(\zeta')$ and $|\mathbb{M}(\zeta)|\le C|\zeta|^{1/4}$. The kernel is at most $C\mathfrak{w}(\zeta')|\zeta|^{-3/4}$, and $\int_{|\zeta'|<2}\mathfrak{w}^2\int_{|\zeta|>3}|\zeta|^{-3/2}|\,\mathrm{d}\zeta||\,\mathrm{d}\zeta'|\le C$ by the Ahlfors bound. The other part is symmetric. □

The two instances. At the hard edge, $z_0=0$, $r_0=\delta$, $r_1=\frac34\delta$, $f_0=f$, $\mathfrak{t}_n=n^2$, $\mathcal{A}=A_0(2\pi)^{\sigma_3/2}$ (Lemma 6.2(i)) and $M_n(z)=\Phi_\alpha(n^2f(z))$; (B2) and (B3) are (6.5) and (6.6), with $\beta=\alpha/2$, or the logarithmic case if $\alpha=0$. Here $|\beta|<\frac12$ is exactly $|\alpha|<1$. At the soft edge, $z_0=4c$, $r_0=\varepsilon$, $r_1=\frac34\varepsilon$, $f_0=f_r$, $\mathfrak{t}_n=n^{2/3}$, $\mathcal{A}=A_r$ and $M_n=M_{r,n}$: by the two-scale structure in Proposition 5.6(v) (Lemma 5.5(i), (iv)), (B1)–(B3) hold with $\mathsf{P}=I$ and $\hat{A}=\Psi^{\mathrm{I}}$, the entire Airy model matrix. No α is involved.

D.2. **Proof of Proposition 6.4.** Step 0 bounds the Cauchy operator on $\hat{\Sigma}$, and Step 1 proves part (i) for each fixed n. Part (ii) is then reduced to a solvability statement for the

inhomogeneous problem (Step 2), whose uniqueness half follows from part (i) (Step 3) and whose existence half, with a bound uniform in n, comes from an explicit solution (Steps 4–5).

Step 0: the contour. $\hat\Sigma$ is a finite union of C^2 arcs with $|\hat\Sigma\cap\mathbb{D}(z,\rho)|\le C_\Sigma\rho$, C_Σ independent of n: this holds for Σ by Proposition 5.6(i) and for the two fixed circles. We show $\|\mathcal{C}_\pm\|_{L^2(\hat\Sigma)}\le C$. The operator splits into blocks according to source and target in Σ or $\Sigma_\circ$. The block $\Sigma\to\Sigma$ is bounded by Proposition 5.6(i), and $\Sigma_\circ\to\Sigma_\circ$ is the Cauchy operator of two fixed circles. For the mixed blocks let $\Sigma_{\mathrm{loc}}:=\Sigma\cap(\mathbb{D}(0,2\delta)\cup\mathbb{D}(4c,2\varepsilon))$, which is independent of n; on the fixed curve $\Sigma_{\mathrm{loc}}\cup\Sigma_\circ$ the Cauchy operator is bounded by §2.6(a). The rest of Σ has distance at least $\min(\delta,\varepsilon)$ from $\Sigma_\circ$, and there the kernel is Hilbert–Schmidt: $\int_{\Sigma_\circ}\int_{\Sigma\setminus\Sigma_{\mathrm{loc}}}|s-z|^{-2}|\,\mathrm{d}s||\,\mathrm{d}z|\le|\Sigma_\circ|\,4C_\Sigma/\min(\delta,\varepsilon)$ by the proof of Lemma B.1(a).

Step 1: part (i). Fix n. By (6.6), $|\tilde Q^{\pm1}(z)|\le C_n|z|^{-|\alpha|/2}$ near 0 (with $C_n(1+|\log|z||)$ if $\alpha=0$). By Proposition 5.6(v) and (B3) with $\mathsf{P}=I$, $\tilde Q^{\pm1}$ is bounded near $4c$. Elsewhere $\tilde Q^{\pm1}$ is bounded, and it is continuous up to $\hat\Sigma$ in the sense of §B.1, being built from explicit functions that are continuous up to the arcs from each side. Since $|\alpha|<1$, the boundary values of $\tilde Q^{\pm1}-I$ lie in $L^2(\hat\Sigma)$; and $\rho\sup_{|z|=\rho}|\tilde Q^{\pm1}(z)-I|=O(\rho^{1-|\alpha|/2})\to0$, while at the other vertices the corresponding quantity is $O(\rho)$. Finally $\tilde Q^{\pm1}-I=O(1/z)$ at infinity. Lemma B.1(c), applied for fixed n, gives $\tilde Q_\pm^{\pm1}\in I+\partial\mathcal{C}(L^2(\hat\Sigma))$. This is part (i).

Step 2: reduction. Let $\mathcal{D}$ be the space of functions on $\hat\Sigma$ that are C^1 on each arc and vanish near the vertices; it is dense in $L^2(\hat\Sigma)$, and $\mathfrak{f}(\tilde v-I)\in\mathcal{D}$ for $\mathfrak{f}\in\mathcal{D}$, because $\tilde v$ is smooth on the arcs away from the vertices. Suppose that

(R1) for every $\mathfrak{g}\in\mathcal{D}$ there is m with $m_\pm\in\partial\mathcal{C}(L^2)$, $m_+=m_-\tilde v+\mathfrak{g}$ and $\|m_\pm\|\le C_2\|\mathfrak{g}\|$, with C_2 independent of n and $\mathfrak{g}$;

(R2) if $m_\pm\in\partial\mathcal{C}(L^2)$ and $m_+=m_-\tilde v$, then $m=0$.

Then $1-\mathcal{C}_{\tilde v}$ is invertible on $L^2(\hat\Sigma)$ with $\|(1-\mathcal{C}_{\tilde v})^{-1}\|\le1+C_2\|\tilde v-I\|_\infty$. This is the easy half of §2.6(e). Indeed, for $\mathfrak{f}\in\mathcal{D}$ solve (R1) with $\mathfrak{g}:=\mathfrak{f}(\tilde v-I)$ and put $\mu:=m_-+\mathfrak{f}$. By (R2) (proved in Step 3, which uses only Step 1) the solution m of (R1) is unique, so $\mathfrak{f}\mapsto\mu$ is linear. Then $\mu(\tilde v-I)=m_-\tilde v-m_-+\mathfrak{f}(\tilde v-I)=m_+-m_-$; as $m_\pm\in\partial\mathcal{C}(L^2)$, $m_-=\mathcal{C}_-(m_+-m_-)=\mathcal{C}_-(\mu(\tilde v-I))$, so $\mu-\mathcal{C}_{\tilde v}\mu=\mathfrak{f}$ and $\|\mu\|\le(1+C_2\|\tilde v-I\|_\infty)\|\mathfrak{f}\|$. The map $\mathfrak{f}\mapsto\mu$ extends by continuity to a bounded operator $\mathcal{R}$ on $L^2(\hat\Sigma)$ with $(1-\mathcal{C}_{\tilde v})\mathcal{R}=1$, since $1-\mathcal{C}_{\tilde v}$ is bounded. Conversely, if $\mu=\mathcal{C}_{\tilde v}\mu$, then $m:=\mathcal{C}(\mu(\tilde v-I))$ satisfies $m_\pm\in\partial\mathcal{C}(L^2)$, $m_-=\mu$ and $m_+=\mu\tilde v=m_-\tilde v$, so $m=0$ by (R2) and $\mu=0$. Hence $1-\mathcal{C}_{\tilde v}$ is injective with right inverse $\mathcal{R}$, and $(1-\mathcal{C}_{\tilde v})^{-1}=\mathcal{R}$.

Step 3: uniqueness. Let $m_\pm\in\partial\mathcal{C}(L^2)$ with $m_+=m_-\tilde v$. Put $X:=\tilde Q+m$. Then $X_+=X_-\tilde v$, $X_\pm\in I+\partial\mathcal{C}(L^2)$, $\tilde Q_\pm^{-1}\in I+\partial\mathcal{C}(L^2)$ by Step 1, and $\tilde v^{\pm1}\in L^\infty$. By §2.6(b) with $A=X$, $B=\tilde Q$, $v_A=v_B=\tilde v$ and exponents $r=r'=2$, $X\tilde Q^{-1}=I$, i.e. $m=0$. This is (R2). Step 3 uses only Step 1.

Step 4: the explicit solution and the uniform bound. Let $\mathfrak{g}\in\mathcal{D}$. Then $\mathfrak{k}:=\mathfrak{g}\tilde Q_+^{-1}\in\mathcal{D}$, since $\tilde Q_+^{-1}$ is smooth on the arcs away from the vertices. Put $m:=\mathcal{C}\mathfrak{k}\cdot\tilde Q$. By Lemma B.1(b), m is continuous up to $\hat\Sigma$ and $m_+-m_-\tilde v=\big((\mathcal{C}\mathfrak{k})_+-(\mathcal{C}\mathfrak{k})_-\big)\tilde Q_+=\mathfrak{g}$.

To bound $\|m_\pm\|_{L^2}$ uniformly, let $\Sigma_A:=\hat\Sigma\cap\mathbb{D}(0,\frac34\delta)$, $\Sigma_B:=\hat\Sigma\cap\mathbb{D}(4c,\frac34\varepsilon)$ and $\Sigma_R:=\hat\Sigma\setminus(\mathbb{D}(0,\frac12\delta)\cup\mathbb{D}(4c,\frac12\varepsilon))$. On Σ_R the matrices $\tilde Q^{\pm1}$ are bounded uniformly in n: outside the discs $\tilde Q=P_0^\infty$ by (5.10); in $\frac12\delta\le|z|\le\delta$ by Lemma 6.2(iii); in $\frac12\varepsilon\le|z-4c|\le\varepsilon$ by (B2) for the Airy instance, since $|\zeta|\ge2$ there. Partition the source–target pairs into $\Sigma_A\times\Sigma_A$, $\Sigma_B\times\Sigma_B$, $(\Sigma_R\times\Sigma_R)\setminus(\Sigma_A\times\Sigma_A\cup\Sigma_B\times\Sigma_B)$ and the rest. The first two blocks are bounded by Lemma D.1. The third is $1_{\Sigma_R}T1_{\Sigma_R}-1_{\Sigma_R\cap\Sigma_A}T1_{\Sigma_R\cap\Sigma_A}-1_{\Sigma_R\cap\Sigma_B}T1_{\Sigma_R\cap\Sigma_B}$

with $T\mathfrak{g} = \mathcal{C}_\pm(\mathfrak{g}\tilde{Q}_+^{-1})\tilde{Q}_\pm$, and each term is bounded by Step 0 since $\tilde{Q}^{\pm1}$ is bounded on Σ_R. In the remaining pairs one point lies in $\mathbb{D}(0,\frac12\delta)$ (or $\mathbb{D}(4c,\frac12\varepsilon)$) and the other outside $\mathbb{D}(0,\frac34\delta)$ (resp. $\mathbb{D}(4c,\frac34\varepsilon)$), so $|s-z|\geq\frac14\min(\delta,\varepsilon)$. The kernel is at most $C|\tilde{Q}_+^{-1}(s)||\tilde{Q}(z)||s-z|^{-1}$, and its Hilbert–Schmidt norm is bounded by the L^2 bounds of Lemma D.1 on the discs, the boundedness of $\tilde{Q}^{\pm1}$ on Σ_R, and $\int_{|s-z|\geq t}|s-z|^{-2}|\,\mathrm{d}s|\leq 4C_\Sigma/t$. Hence $\|m_\pm\|\leq C_2\|\mathfrak{g}\|$ with C_2 independent of n and of $\mathfrak{g}\in\mathcal{D}$.

Step 5: $m_\pm\in\partial\mathcal{C}(L^2)$. We apply Lemma B.1(c) to $\mathfrak{F}:=m$, for fixed n. Since $\mathfrak{k}$ vanishes near the vertices, $\mathcal{C}\mathfrak{k}$ is analytic and bounded near each vertex, and $\mathcal{C}\mathfrak{k}=O(1/z)$ at infinity; hence $|m|\leq C_n|\tilde{Q}|$ near each vertex, so that $\rho\sup_{|z-z_0|=\rho}|m|=O(\rho^{1-|\alpha|/2})\to0$ (by Step 1), $m=O(1/z)$ at infinity, and $m_\pm\in L^2(\hat{\Sigma})$ by Step 4. Lemma B.1(c) gives $m_\pm\in\partial\mathcal{C}(L^2)$. This proves (R1).

Working with data in $\mathcal{D}$ avoids any analysis of m at the vertex 0, where $\tilde{Q}$ is unbounded: for such data the singular factor $\tilde{Q}$ only multiplies a function that is analytic near 0, and the uniform estimate of Step 4, which is where the structure at 0 enters, extends the inverse to all of L^2.

Steps 2–5 (with (R2) from Step 3) prove part (ii) of Proposition 6.4, with $\|(1-\mathcal{C}_{\tilde{v}})^{-1}\|\leq 1+C_2\sup_n\|\tilde{v}-I\|_\infty$, which is finite because $\tilde{v}^{\pm1}$ is bounded uniformly in n (on the lips $|\varrho_\pm e^{2n\varphi}|\leq1$, on the circles $\tilde{v}=I+O(1/n)$, elsewhere $\tilde{v}\in\{I,J\}$ or $\tilde{v}=v$). □

Appendix E. The first corrections and their Laurent coefficients

This appendix gives the full Laurent data of the model corrections Δ_0^{mod} and Δ_r^{mod} of Section 9. Only the coefficient $\mathfrak{A}_0$, the coefficient $\mathfrak{B}_r$ and the two combinations (9.4) of $\mathfrak{A}_r$ enter the proof of Theorem 1.4, and these are computed in the proof of Lemma 9.4; the full matrices below are recorded for completeness, and $\mathfrak{C}_0$ is not used in this paper.

Lemma E.1 (Laurent data of the first corrections)**.** *With the notation of Lemmas 9.3 and 9.4,*

$$\mathfrak{A}_r=\frac{c}{48}\begin{pmatrix}12\alpha^2p+24\alpha p+11p+2 & -i(12\alpha^2p+48\alpha p+49p+2)\\ -i(12\alpha^2p+p+2) & -(12\alpha^2p+24\alpha p+11p+2)\end{pmatrix},$$

and $\Delta_0^{\mathrm{mod}}(z)=\mathfrak{A}_0/z+\mathfrak{C}_0+O(z)$ *near* 0 *with*

$$\mathfrak{C}_0=\frac{p+2}{96(3p+2)}\begin{pmatrix}-\mathsf{c}_{11} & i(26\alpha^2p+16\alpha^2+72\alpha p+48\alpha+70p+47)\\ i(26\alpha^2p+16\alpha^2-2p-1) & \mathsf{c}_{11}\end{pmatrix},$$

where $\mathsf{c}_{11}:=10\alpha^2p+8\alpha^2-36\alpha p-24\alpha-7p-5$.

Proof. The soft edge. Put $\lambda:=1+\alpha$ and $\epsilon:=\mathsf{u}/(2\sqrt{c})$. By Lemma 9.3(iii) and (9.5), using $\operatorname{arsinh}\epsilon=\epsilon-\epsilon^3/6+O(\epsilon^5)$,

$$Y_{11}=\frac{\lambda\epsilon}{6}+\frac{4\lambda^3-\lambda}{36}\epsilon^3+O(\epsilon^5),\qquad a^2Y_{12}=-\frac{7}{72}\epsilon+\Big(\frac{7}{144}-\frac{\lambda^2}{6}\Big)\epsilon^3+O(\epsilon^5),$$
$$a^{-2}Y_{21}=\frac{5}{72}\epsilon^{-1}+\Big(\frac{5}{144}+\frac{\lambda^2}{6}\Big)\epsilon+O(\epsilon^3),$$

since $a^2=\epsilon-\frac12\epsilon^3+O(\epsilon^5)$ and $\cosh 2L_r=1+2\lambda^2\epsilon^2+O(\epsilon^4)$. By Lemma 9.3(ii), $-\tilde{\varphi}^{-1}=\mathsf{g}_3^{-1}\mathsf{u}^{-3}\big(1-(\mathsf{g}_5/\mathsf{g}_3)\mathsf{u}^2+O(\mathsf{u}^4)\big)$ with $\mathsf{g}_5/\mathsf{g}_3=-(7p+4)/(40pc)$. Insert these into (9.2) with $y_{11}=Y_{11}$, $y_{12}=a^2Y_{12}$, $y_{21}=a^{-2}Y_{21}$: each entry of the matrix in (9.2) is $e_{-1}\mathsf{u}^{-1}+e_1\mathsf{u}+O(\mathsf{u}^3)$, and the corresponding entry of Δ_r^{mod} has the coefficient e_{-1}/g_3 at u^{-4} (this is $\mathfrak{B}_r$) and $(e_1-(\mathsf{g}_5/\mathsf{g}_3)e_{-1})/\mathsf{g}_3$ at u^{-2}, which is $\mathfrak{A}_r$. The odd powers of u cancel, as they must by Lemma 9.4.

The hard edge. Put $\eta := \mathsf{y}/(2\sqrt{c})$, $z = -\mathsf{y}^2$. By Lemma 9.3(iii) and the formulas for Y in the proof of Lemma 9.4,

$$Y_{11} = -4\lambda\eta + O(\eta^3), \qquad Y_{12} = 1 - 4\alpha^2 + 4\lambda^2\eta^2 + O(\eta^4), \qquad Y_{21} = -(3+4\alpha^2) - 4\lambda^2\eta^2 + O(\eta^4),$$

with $a^2 = \eta^{-1} + \frac{1}{2}\eta + O(\eta^3)$, $a^{-2} = \eta - \frac{1}{2}\eta^3 + O(\eta^5)$, and, by Lemma 9.3(i), $f^{-1/2} = \sqrt{c}(p+2)\,\mathsf{y}^{-1}(1 + \frac{1}{2}f_2\mathsf{y}^2 + O(\mathsf{y}^4))$. Inserting into (9.2) and multiplying by $1/(16f^{1/2})$ gives a series in y whose odd powers cancel, whose coefficient of y^{-2} is $-\mathfrak{A}_0$ (as $1/z = -\mathsf{y}^{-2}$), and whose constant term is $\mathfrak{C}_0$ after $f_2 = (p+2)/(12c(3p+2))$ is inserted. □

The coefficients $\mathfrak{A}_0$, $\mathfrak{A}_r$, $\mathfrak{B}_r$ and $\mathfrak{C}_0$ were verified by exact symbolic series expansion, and the resulting coefficients at infinity also by contour integration over the two circles of corrections constructed numerically from quadratures of the g-function and of the Szegő integral.

Appendix F. Proof of the localisation bound

This appendix proves Proposition 8.6. The spectral weights $m_n(k)$ form a probability measure on the rescaled nodes, and the mass it gives to a short interval $[0, r]$ at the hard edge is controlled by its Stieltjes transform at one point off the contour (Lemma F.1). That transform is the product of two entries of the first row of Q (Lemma F.2), so the bound reduces to an estimate of Q at a single point at distance $\asymp r$ from all contours: in case (a) from the comparison argument of Section 6, in case (b) from the error matrix of §8.3 and explicit bounds on Bessel functions. The notation is that of §8.4.

Lemma F.1 (Stieltjes bound)**.** *Let ν be a probability measure on $[0,\infty)$, $r > 0$ and $z = re^{5\pi i/6}$. Then $\nu([0,r]) \le \frac{8}{\sqrt{3}}\, r \left| \int \mathrm{d}\nu(X)/(X-z) \right|$.*

Proof. For $X \ge 0$, $\operatorname{Re}\frac{1}{X-z} = \frac{X + r|\cos(5\pi/6)|}{|X-z|^2} \ge 0$, and for $0 \le X \le r$ also $|X-z|^2 \le 4r^2$, so $\operatorname{Re}\frac{1}{X-z} \ge \frac{\sqrt{3}/2}{4r}$. Hence $\nu([0,r])\frac{\sqrt{3}}{8r} \le \operatorname{Re}\int \frac{\mathrm{d}\nu(X)}{X-z}$. □

Lemma F.2 (Row product)**.** *Let $\nu_n := \sum_k m_n(k)\delta_{X_k}$, with the nodes $X_k = (n/k)^p$ of (4.6), and $S_n(z) := \int \mathrm{d}\nu_n(X)/(X-z)$. Suppose that $Q = \mathsf{c}\, U\, \mathsf{k}^{\sigma_3}$ on a connected open set containing $z \notin \{X_k\} \cup \{0\}$ and a neighbourhood of infinity, where $\mathsf{c} = \operatorname{diag}(\mathsf{c}_1, \mathsf{c}_2)$ is constant, k is scalar, and $Q = I + Q_1/z + O(z^{-2})$. Then $S_n(z) = -Q_{11}(z)Q_{12}(z)/(Q_1)_{12}$.*

Proof. By (B.1), $U_{12}(z) = \sum_k n^p w_k U_{11}(X_k)/(z - X_k)$, and $U_{11}(X) = n^{pn}P_n(Xn^{-p})$ is monic of degree n in X. Since $(U_{11}(z) - U_{11}(X))/(z-X)$ is a polynomial of degree $n-1$ in X, orthogonality gives $U_{11}(z)U_{12}(z) = \sum_k n^p w_k U_{11}(X_k)^2/(z - X_k) = -\mathsf{h}S_n(z)$ with $\mathsf{h} := \sum_k n^p w_k U_{11}(X_k)^2$, because $m_n(k) = n^p w_k U_{11}(X_k)^2/\mathsf{h}$. Expanding U_{12} at infinity, orthogonality kills the terms z^{-j-1}, $j < n$, and the coefficient of z^{-n-1} is h; so the $(1,2)$ entry of the first coefficient of $Uz^{-n\sigma_3}$ is h. Now $Q_{11}Q_{12} = \mathsf{c}_1^2 U_{11}U_{12}$, and $Q_{11} \to 1$ forces $\mathsf{c}_1 z^n \mathsf{k}(z) \to 1$, so $(Q_1)_{12} = \lim_{z\to\infty} zQ_{12} = \mathsf{c}_1^2\mathsf{h}$. □

The identity of Lemma F.2 expresses the diagonal of the resolvent of the Jacobi matrix through the product of the two solutions of the recurrence; we use it only in the form stated.

Proof of Proposition 8.6. Reduction. Take $j = n$ first. Since $k > Mn$ exactly when $X_k < M^{-p}$, we have $\sum_{k>Mn} m_n(k) \le \nu_n([0,r])$ with $r := M^{-p} \in (0,1]$. Let $z := re^{5\pi i/6}$. For $r \le \frac{1}{4}\delta$ the point z lies in Ω_∞ (the lips stay in $|\arg z| \le \frac{5}{4}\vartheta_{\mathrm{lip}} < \pi/2$) and off all contours, and there $Q = \mathsf{c}\, U(e^{n\varphi}FD_W)^{\sigma_3}$ by (4.21) and (5.12), as also near infinity; so Lemmas F.1 and F.2 give

$$\nu_n([0,r]) \le \frac{8}{\sqrt{3}}\, r\, \frac{|Q_{11}(z)Q_{12}(z)|}{|(Q_1)_{12}|}, \qquad (Q_1)_{12} = ic + O(1/n), \tag{F.1}$$

the latter by Proposition 6.6 in case (a) and by Proposition 8.4 in case (b), together with (8.10). For $r_0 < r \le 1$, where $r_0 := \frac14\delta$ in case (a) and $r_0 := \frac12\delta_1$ in case (b) (δ_1 is fixed below), we use $\nu_n([0,r]) \le 1 \le r_0^{-1/2} r^{1/2}$. For $j = n-1$ apply the result at degree $n-1$ with $M' := Mn/(n-1) \ge M$; this costs a factor $\frac12$ in γ_*. The geometric fact used below is that for s' on the band or on a lip in $\mathbb{D}(0,\delta)$ the angle between s' and z is at least $\pi/3$, so $|s'-z| \ge \frac12(|s'|+|z|)$; and that $|f(z)| \asymp r$, $\zeta_z := n^2 f(z)$ satisfies $|\arg\zeta_z| < \pi/6 + \pi/24$.

Case (a). Here $1 < s < 1+2p$, $|\alpha| < 1$, $\ell \equiv 1$, so $\varepsilon_\Lambda \equiv 0$ and $\varepsilon_n' = n^{-2\nu_p}\log n$. Put $y := n^2 r$, $D_z := (2\pi n)^{\sigma_3/2}\zeta_z^{-\sigma_3/4}$ and let $y_0 \ge 1$ be such that $|\zeta_z| \ge 2$ for $y \ge y_0$. We show

$$|Q(z)| \le C r^{-1/4} \qquad (y_0 n^{-2} \le r \le \tfrac14\delta), \tag{F.2}$$

which with (F.1) gives $\nu_n([0,r]) \le C r^{1/2} = C M^{-p/2}$ for $1 \le M \le (n^2/y_0)^{1/p}$.

By Proposition 6.4(i), Proposition 6.5 and §2.6(b), as in the proof of Proposition 6.6, $\mathsf{K} := Q\tilde Q^{-1} = I + \mathcal{C}(\mathcal{E})$ off $\hat\Sigma$, with $\mathcal{E} = Q_-(v-\tilde v)\tilde v^{-1}\tilde Q_-^{-1}$. For $|\zeta_z| \ge 2$, $\tilde Q(z) = A_0(z) D_z \Theta_\infty(\zeta_z)$ by (6.10) and (6.5), and $|A_0(z)D_z| \le C r^{-1/4}$ since the entries of D_z have moduli $\asymp r^{\mp 1/4}$. Since $Q = [A_0 D_z + (\mathsf{K}-I)A_0 D_z]\Theta_\infty$, it suffices to show

$$|(\mathsf{K}(z) - I)A_0(z)D_z| \le C(\varepsilon_n' + n^{-1})\, r^{-1/4}. \tag{F.3}$$

The right factor $A_0(z)D_z$ is the scale of the comparison solution at z; without it the bound would not be small at the Bessel scale. We split $(v-\tilde v)\tilde v^{-1} = \Delta_1 + \Delta_2 + \Delta_3$ and $Q_- = \tilde Q_- + (Q_- - \tilde Q_-)$ as in the proof of Proposition 6.6, and put $\mathfrak{m}_n(s') := |s'|^{\nu_p}\log(2/|s'|)$, so that $|d_n(s')| \le C\mathfrak{m}_n(s') e^{-n|s'|^{1/2}/C}$ by (6.14) and (6.8). Write $\varrho := |s'|$, $t := n\varrho^{1/2}$ and $B(s') := A_0(s')^{-1}A_0(z)$; since $A_0^{\pm1}$ are bounded and A_0 is Lipschitz on $\mathbb{D}(0,\delta/2)$, $|B(s') - I| \le C(\varrho + r)$.

The lips, $|\zeta_{s'}| \ge 2$. The integrand is the rank-one matrix $d_n(\tilde Q(s')e_2)(e_1^{\mathsf T}\Theta_\infty(\zeta_{s'})^{-1}N)$, $N := D_{s'}^{-1} B D_z$. Here $|\tilde Q(s')e_2| \le C\varrho^{-1/4}$, and the entries of $D_w^{\pm1}$ have moduli $\asymp |w|^{\mp1/4}, |w|^{\pm1/4}$, which with $|B - I| \le C(\varrho + r)$ gives $|N| \le C((\varrho/r)^{1/4} + (r/\varrho)^{1/4})$. So, using $|\,\mathrm{d}s'| \le C\,\mathrm{d}\varrho$ and $|s'-z| \ge \frac12(\varrho + r)$, this part of (F.3) is at most

$$\begin{aligned} &C r^{-1/4} \int \mathfrak{m}_n e^{-t/C}\big[1 + (r/\varrho)^{1/2}\big]\frac{\mathrm{d}\varrho}{\varrho + r} \\ &\quad = C r^{-1/4} \int_{t \ge 1/C} \mathfrak{m}_n\big(\tfrac{t^2}{n^2}\big) e^{-t/C}\Big[\frac{2t}{t^2+y} + \frac{2\sqrt{y}}{t^2+y}\Big]\,\mathrm{d}t \le C\varepsilon_n' r^{-1/4}, \end{aligned}$$

since $2t \le t^2 + y$ (as $y \ge 1$), $2\sqrt{y} \le (t^2+y)/t$, and $\mathfrak{m}_n(t^2/n^2) \le C t^{2\nu_p} n^{-2\nu_p}(\log n + |\log t| + 1)$.

The lips, $|\zeta_{s'}| \le 2$. Then $\varrho \le C n^{-2} \le C r$ and $\mathfrak{m}_n(s') \le C\varepsilon_n'$. By (6.6) and Lemma 6.1(iii), $|\tilde Q(s')e_2| \le C n^{1/2} |\zeta_{s'}|^{-|\alpha|/2}$, and

$$e_1^{\mathsf T}\tilde Q(s')^{-1} A_0(z) D_z = e_1^{\mathsf T}\Phi_\alpha(\zeta_{s'})^{-1}(2\pi n)^{-\sigma_3/2} B (2\pi n)^{\sigma_3/2}\zeta_z^{-\sigma_3/4},$$

whose norm is at most $C|\zeta_{s'}|^{-|\alpha|/2}|\zeta_z|^{1/4} \le C|\zeta_{s'}|^{-|\alpha|/2} n^{1/2} r^{1/4}$ (the lower-left entry of B is $O(r)$ and $nr|\zeta_z|^{-1/4} \asymp |\zeta_z|^{1/4} r^{1/2}$). With $|s'-z| \ge r/2$ and $\int_0^{Cn^{-2}} (n^2\varrho)^{-|\alpha|}\,\mathrm{d}\varrho \le C n^{-2}$ ($|\alpha| < 1$; logarithms at $\alpha = 0$), this part is at most $C\varepsilon_n' n\, r^{1/4} r^{-1} n^{-2} = C\varepsilon_n' r^{-1/4} y^{-1/2}$.

The circles and the rest. On $\Sigma_\circ$, $|\Delta_2| \le C/n$, $|\tilde Q_-^{\pm1}| \le C$ and $|s'-z| \ge \frac34\delta$, which gives $C n^{-1} r^{-1/4}$. On the lips, $|E| \le C e^{-n|s'|^{-1/p}/C}$ and $|\tilde Q_-^{\pm1}| \le C n^{1/2}|s'|^{-1/2}$, which gives $C r^{-1/4} e^{-n/C}$ (as $r \ge y_0 n^{-2}$); the rest of $\hat\Sigma$, of length $O(n^p)$, gives $C n^p e^{-n/C} r^{-1/4}$.

The correction term. By Cauchy–Schwarz and (6.17) it is at most $C(\varepsilon_n' n^{-1/2} + n^{-1})\,\|(v-\tilde v)\tilde v^{-1}\tilde Q_-^{-1} A_0(z) D_z/(s'-z)\|_{L^2(\hat\Sigma)}$. In the last norm, the part $|\zeta_{s'}| \le 2$ of Δ_1 is at most

$C\varepsilon'_n n^{1/2} r^{1/4} r^{-1} n^{-1} = C\varepsilon'_n n^{-1/2} r^{-3/4}$; the part $|\zeta_{s'}| \ge 2$ has square at most

$$C\int \mathfrak{m}_n^2 e^{-2t/C}\Big[\Big(\frac{\varrho}{r}\Big)^{1/2} + \Big(\frac{r}{\varrho}\Big)^{1/2}\Big]\frac{\mathrm{d}\varrho}{(\varrho+r)^2}$$
$$= C\int \mathfrak{m}_n^2 e^{-2t/C}\Big(\frac{2t^2}{\sqrt{y}} + 2\sqrt{y}\Big)\frac{n^2\,\mathrm{d}t}{(t^2+y)^2} \le C\varepsilon_n'^2 n^2 y^{-3/2},$$

that is, the part is at most $C\varepsilon'_n n^{-1/2} r^{-3/4}$; Δ_2 gives $Cn^{-1}r^{-1/4}$ and Δ_3 an exponentially small multiple of $r^{-1/4}$. Since $nr^{1/2} = \sqrt{y} \ge 1$, the product is at most $Cr^{-1/4}(\varepsilon'_n + n^{-1})$. Summing the four parts gives (F.3), hence (F.2), and (8.11) follows with $\gamma_* = \frac{1}{2}y_0^{-1/p}$.

Case (b). Here $p > 1$ and $\alpha > -1$ is arbitrary, and we work in the frame of §8.3. First, $|\mathsf{R}(z) - I| \le C/n$ for $0 < r \le \frac{1}{2}\delta_1$, with a fixed $\delta_1 \le \delta$. To see this, note $\mathsf{R}(z) - I = \mathcal{C}(\mathsf{R}_-(J_{\mathsf{R}} - I))(z)$ by Proposition 8.4(iii); the part of Σ_{R} outside $\mathbb{D}(0,\delta_1)$ is at distance $\ge \frac{1}{2}\delta_1$ from z and contributes $O(1/n)$ by Proposition 8.4(ii), (iv); on $\Sigma_{\mathsf{R}} \cap \mathbb{D}(0,\delta_1)$ (two lips leaving 0 at angles $\pm\pi/3$ and the segment $(-\delta_1, 0)$) we have $|s' - z| \ge \frac{1}{4}(|s'| + |z|)$ for δ_1 small, and by Lemma 8.3 and Cauchy–Schwarz the contribution is $O(n^{-K})$. Next, in U_0, $Q = \mathsf{R}\check{A}(2\pi n)^{\sigma_3/2}\sigma_3\Psi_\alpha(\zeta)\sigma_3\,(e^{-2\zeta^{1/2}}\rho_n^{-1})^{\sigma_3}$, and the last factor is diagonal with determinant 1, so it does not change $Q_{11}Q_{12}$. With $(\mathsf{r}_1, \mathsf{r}_2) := e_1^{\mathsf{T}}\mathsf{R}(z)\check{A}(z)$, which is bounded by Lemma 8.2(i), and $u := 2\zeta_z^{1/2}$, [10, (6.23)] gives

$$Q_{11} = \sqrt{2\pi n}\Big[\mathsf{r}_1 I_\alpha(u) - \frac{i\,\mathsf{r}_2}{2n}uI'_\alpha(u)\Big], \qquad Q_{12} = -\sqrt{2\pi n}\Big[\frac{i\,\mathsf{r}_1}{\pi}K_\alpha(u) + \frac{\mathsf{r}_2}{2\pi n}uK'_\alpha(u)\Big].$$

Here $|\operatorname{ph} u| \le \pi/3$. We use the following bounds, for $|\operatorname{ph} u| \le \pi/3$ and with C depending on α: $|I_\alpha(u)| + |uI'_\alpha(u)| \le C|u|^\alpha$, $|K_\alpha(u)| \le C(1 + \log(4/|u|))|u|^{-|\alpha|}$ and $|uK'_\alpha(u)| \le C|u|^{-|\alpha|}$ for $|u| \le 2$; and, with $\mathfrak{r} := \operatorname{Re} u \ge |u|/2$, $|I_\alpha(u)| + |I'_\alpha(u)| \le C(\mathfrak{r}^{-1/2}e^{\mathfrak{r}} + 1)$ and $|K_\alpha(u)| + |K'_\alpha(u)| \le C\mathfrak{r}^{-1/2}e^{-\mathfrak{r}}$ for $\mathfrak{r} \ge 1$. The first pair follows from the series (10.25.2), with $\Gamma(\alpha + k + 1)$ bounded below for $\alpha > -1$; the others from (10.32.9), $K_\alpha(u) = \int_0^\infty e^{-u\cosh t}\cosh(\alpha t)\,\mathrm{d}t$, which gives $|K_\alpha(u)| \le K_\alpha(\mathfrak{r})$ and the same for the derivative, and from (10.32.4) for I_α, by elementary estimates of the integrals (for small $\mathfrak{r}$ use $\cosh t \ge e^t/2$, for large $\mathfrak{r}$ use $\cosh t \ge 1 + t^2/2$ and $1 - \cos\theta \ge 2\theta^2/\pi^2$). Since $|u| \le Cn$, these give $|Q_{11}Q_{12}| \le Cn\,b(|u|)$ with $b(\varrho) := (1 + \log(4/\varrho))\varrho^{\alpha-|\alpha|}$ for $\varrho \le 2$ and $b(\varrho) := 1/\varrho$ for $\varrho \ge 2$. By (F.1), with $|u| \asymp \sqrt{y}$, $y = n^2 r$: if $|u| \ge 2$ then $\nu_n([0,r]) \le Crn/(n\sqrt{r}) = Cr^{1/2}$; if $|u| \le 2$ then $\nu_n([0,r]) \le Cn^{-1}y^{1+\min(\alpha,0)}(1 + \log(C/y))$. With $r = M^{-p}$ and $p(1 + \min(\alpha, 0)) = \min(p, s-1) =: \beta$, the first case gives (8.11) for $M \le Cn^{2/p}$; the second gives it for the remaining $M \le \gamma_* n^{2/p}$, where y is bounded below and above, so that $n^{-1} \le CM^{-p/2}$; and the second gives $\sum_{k>Mn} m_n(k) \le Cn^{-1}(M/n^{2/p})^{-\beta}(1 + \log(M/n^{2/p}))$ for $M \ge n^{2/p}$. For $\tau < \min(p/2, s-1) \le \beta$ we have $n^{-1} = (n^{2/p})^{-p/2} \le (n^{2/p})^{-\tau}$ and $(M/n^{2/p})^{-\beta}(1 + \log(M/n^{2/p})) \le C_\tau(M/n^{2/p})^{-\tau}$, so the second bound is at most $C_\tau M^{-\tau}$; and $M^{-p/2} \le M^{-\tau}$ in the first range. □

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Data and code availability

The code, the numerical data of Section 10 and the Lean formalisation are available at https://github.com/speagleg/jacobi-asymptotics (tag [to be fixed at submission]). The Lean declarations `christoffel_shift_law` and `christoffel_shift_rate` (with the variants

`christoffel_shift_law_pow` and `christoffel_shift_rate_pow` at the scale n^p) are the formal versions of Theorem 3.1, without and with a rate, and `chain_limit` and `chain_rate_core` are those of Lemmas 3.4 and 3.5.

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